Country Variable Suppression	
SC018Q09JA01	SCHSIZE
SC018Q09JA02	TOTAT
SC018Q10JA01	TOTMATH
SC018Q10JA02	TOTSTAFF
SC182Q01WA01	WB151Q01HA
SC182Q01WA02	WB152Q01HA
Norway	
CLSIZE	SC018Q08JA01
GRADE	SC018Q08JA02
LANGTEST_COG	SC018Q09JA01
LANGTEST_QQQ	SC018Q09JA02
MCLSIZE	SC018Q10JA01
PRIVATESCH	SC018Q10JA02
PROADMIN	SC168Q01JA
PROATCE	SC168Q02JA
PROMGMT	SC168Q03JA
PROOSTAF	SC168Q04JA
PROPAT6	SC182Q01WA01
PROPAT7	SC182Q01WA02
PROPAT8	SC182Q06WA01
PROPMATH	SC182Q06WA02
PROPSUPP	SC182Q07JA01
RATCMP1	SC182Q07JA02
RATCMP2	SC182Q08JA01
RATTAB	SC182Q08JA02
SC002Q01TA	SC182Q09JA01
SC002Q02TA	SC182Q09JA02
SC004Q01TA	SC182Q10JA01
SC004Q02TA	SC182Q10JA02
SC004Q03TA	SCHLTYPE
SC004Q05NA	SCHSIZE
SC004Q06NA	SMRATIO
SC004Q07NA	ST001D01T
SC004Q08JA	ST003D02T
SC012Q03TA	ST003D03T
SC013Q01TA	STRATIO
SC014Q01TA	TOTAT
SC016Q01TA	TOTMATH
SC016Q02TA	TOTSTAFF
SC016Q03TA	
SC016Q04TA	
SC018Q01TA01	
SC018Q01TA02	
SC018Q02TA01	
SC018Q02TA02	
Singapore	
LANGN	
OCOD1 (2-digit)	
OCOD2 (2-digit)	
SC211Q02JA	
SC211Q03JA	
Sweden	
GRADE	SC182Q06WA02
SC001Q01TA	SC182Q07JA01
SC002Q01TA	SC182Q07JA02
SC002Q02TA	SC182Q08JA01
SC004Q01TA	SC182Q08JA02
SC004Q02TA	SC182Q09JA01

Country Variable Suppression	
SC004Q03TA	SC182Q09JA02
SC004Q08JA	SC182Q10JA01
SC013Q01TA	SC182Q10JA02
SC014Q01TA	SC211Q01JA
SC018Q01TA01	SC211Q02JA
SC018Q01TA02	SC211Q03JA
SC018Q02TA01	SC211Q04JA
SC018Q02TA02	SC211Q05JA
SC018Q08JA01	SC211Q06JA
SC018Q08JA02	ST001D01T
SC018Q09JA01	ST003D02T
SC018Q09JA02	ST003D03T
SC018Q10JA01	ST021Q01TA
SC018Q10JA02	ST022Q01TA
SC182Q01WA01	ST126Q01TA
SC182Q01WA02	ST226Q01JA
SC182Q06WA01	
Thailand	
STRATUM	

Note: 1. Cyprus data are suppressed from the public use files. Information on data for Cyprus: https://oe.cd/cyprus-disclaimer.

13 Sampling Outcomes

This chapter reports on the PISA 2022 sampling outcomes. Details of the sample design and selection are provided in Chapter 6 of this Technical Report.

Population coverage

Quality indicators for population coverage and the information used to develop them are presented in Annex Web Table 13.A.2 and Annex Web Table 13.A.3 (please refer Annex Table 13.A.1 for access to these tables), for participating countries/economies and adjudicated regions, respectively. The following notes explain the meaning of each coverage index and how the data in each column of the table were used.

Coverage indices 1, 2 and 3 are intended to measure PISA population coverage. Coverage indices 4 and 5 are intended to be diagnostic in cases where indices 1, 2 or 3 have unexpected values. Many references are made in this chapter to the various sampling tasks on which National Project Managers (NPMs) documented statistics and other information needed in undertaking the sampling of schools and students. Note that although no comparison is made between the total population of 15-year-olds and the enrolled population of 15-year-old students, generally the enrolled population was expected to be less than or equal to the total population. Occasionally this was not the case due to differing data sources for these two values.

The components used for the coverage indices are the following:

- ST7a_1: National population of all 15-year-olds based on national statistics.
- ST7a_2.1: Enrolled 15-year-old students in grades 7 and above based on national statistics.
- ST7b_1: Target population that includes all enrolled 15-year-old students in grades 7 and above that omits schools based on national statistics such as schools located in unsafe areas.
- ST7b_3: Target population that includes all enrolled 15-year-old students in grades 7 and above, minus school-level exclusions, based on national statistics.
- P: Weighted number of participating students calculated from the PISA sample.
- E: Weighted estimate of within-school excluded students calculated from the PISA sample.
- S: Estimate of enrolled students from school sampling frame calculated as the sum over all sampled schools of the product of each school's sampling weight and its number of 15-year-old students.

Coverage Index 1: Coverage of the national *target* population, $P/(P+E)$ × (ST7b_3/ST7b_1). This estimates the extent to which the weighted participants covered the final *target* population after all exclusions. It indicates the overall proportion of the *target* population covered by the non-excluded portion of the student sample.

Coverage Index 2: Coverage of the national *enrolled* population, $P/(P+E)$ × (ST7b_3/ST7a_2.1). This estimates the extent to which the weighted participants covered the population of all *enrolled* students in grades 7 and above. Thus, this index may be somewhat lower than Index 1.

Coverage Index 3: Coverage of the national *15-year-old* population, *P*/ST7a_1. This estimates the proportion of the national population of 15-year-olds covered by the non-excluded portion of the student sample. It is below 1.0 to the extent that 15-year-olds were excluded, or not enrolled in grade 7 or higher.

Coverage Index 4: Coverage of the estimated school population, (*P*+*E*)/*S.* This estimates the proportion of the estimated school 15-year-old population that is represented by the weighted student sample of all PISA-eligible 15-year-old students. Its purpose is to assess whether the enrolment data on the sampling frame is a reliable measure of the number of enrolled 15-year-olds. As the enrolment data on the frame was often inaccurate, this index usually differed noticeably from 1.0. In such cases, Indexes 1 and 2 may be suspect, as they rely on national enrolment data for their denominators, often derived from the same source as the school-level enrolment data.

Coverage Index 5: Coverage of the school sampling frame population, *S*/ST7b_3. This estimate provides a check as to whether the data on enrolment obtained from national statistics is consistent with the enrolment on the sampling frame. However, in most cases for PISA, the enrolment data based on national statistics were derived using data from the sampling frame by the NPM, and so this ratio was close to 1.0 for most countries/economies, even when the enrolment data on the school sampling frame were poor.

Annex Table 13.A.4, Annex Table 13.A.5, Annex Table 13.A.6, Annex Table 13.A.7, Annex Table 13.A.8 and Annex Table 13.A.9 (please refer Annex Table 13.A.1 for access to these tables) present school and student-level response rates at the national and regional levels. Response rates are all presented separately by participating country/economy, and by adjudicated regions.

When calculating school response rates before replacement, the numerator consisted of all original sample schools with enrolled age-eligible students who participated (i.e. assessed a sample of PISA-eligible students and obtained a student response rate of at least 33%). The denominator consisted of all the schools in the numerator, plus those original sample schools with enrolled age-eligible students that either did not participate or failed to assess at least 33% of PISA-eligible sample students. Schools that were included in the sampling frame but were found to have no age-eligible students, or which were excluded in the field were omitted from the calculation of response rates. Replacement schools do not figure in these calculations.

When calculating school response rates after replacement, the numerator consisted of all sampled schools (original plus replacement) with enrolled age-eligible students that participated (i.e. assessed a sample of PISA-eligible students and obtained a student response rate of at least 33%). The denominator consisted of all the schools in the numerator, plus those original sample schools that had age-eligible students enrolled, but that failed to assess at least 33% of PISA-eligible sample students and for which no replacement school participated. Schools that were included in the sampling frame but were found to contain no age-eligible students, were omitted from the calculation of response rates. Replacement schools were included in rates only when they participated and were replacing a refusing school that had age-eligible students.

When calculating weighted school response rates, each school received a weight equal to the product of its base weight (the reciprocal of its selection probability) and the number of age-eligible students enrolled in the school, as indicated on the school sampling frame.

With the use of probability proportional to size sampling, where there are no certainty or small schools, the product of the initial weight and the enrolment will be a constant, so in participating countries/economies with few certainty school selections and no oversampling or undersampling of any explicit strata, weighted and unweighted rates are very similar. The weighted school response rate before replacement is given by the formula:

$$\begin{array}{c}\textit{weighted school response rate}\\ \textit{before replacement}\end{array} = \frac{\sum_{i \in Y} W_i E_i}{\sum_{i \in (Y \cup N)} W_i E_i} \qquad \textbf{Equation 13.1}$$

where *Y* denotes the set of responding original sample schools with age-eligible students, *N* denotes the set of eligible non-responding original sample schools, *Wi* denotes the base weight for school *i*, *Wi* = 1/*Pi* where *Pi* denotes the school selection probability for school *i*, and *Ei* denotes the enrolment size of age-eligible students, as indicated on the sampling frame. The weighted school response rate, after replacement, is given by the formula:

$$\begin{array}{c}\textit{weighted school response rate}\\ \textit{after replacement}\end{array} = \frac{\sum_{i \in (Y \cup R)} W_i E_i}{\sum_{i \in (Y \cup R \cup N)} W_i E_i} \qquad \textbf{Equation 13.2}$$

where *Y* denotes the set of responding original sample schools, *R* denotes the set of responding replacement schools, for which the corresponding original sample school was eligible but was non-responding, *N* denotes the set of eligible refusing original sample schools, *Wi* denotes the base weight for school *i*, *Wi* = 1/*Pi*, where *Pi* denotes the school selection probability for school *i*, and for weighted rates, *Ei* denotes the enrolment size of age-eligible students, as indicated on the sampling frame.

For unweighted student response rates, the numerator is the number of students for whom assessment data were included in the results. The denominator is the number of sampled students who were age-eligible, and not explicitly excluded as student exclusions.

For weighted student response rates, the same students appear in the numerator and denominator as for unweighted rates, but each student is weighted by its student base weight. This is given as the product of the school base weight – for the school in which the student was enrolled – and the reciprocal of the student selection probability within the school.

In countries/economies with no oversampling of any explicit strata, weighted and unweighted student response rates are very similar.

Overall response rates are calculated as the product of school and student response rates. Although overall weighted and unweighted rates can be calculated, there is little value in presenting overall unweighted rates. The weighted rates indicate the proportion of the student population represented by the sample prior to making the school and student non-response adjustments.

Teacher response rates

Unweighted response rates for teachers were created using similar methods to those for unweighted student and school response rates – that is, ineligible teachers are not used in the denominator for the rate calculation.

For weighted teacher response rates, the same teachers appear in the numerator and denominator as for unweighted rates, but each teacher is weighted by its teacher base weight. This is given as the product of the school base weight – for the school in which the teacher was working – and the reciprocal of the teacher selection probability within the school (Annex Table 13.A.10, please refer to Annex Table 13.A.1 for access to this table).

Design effects and effective sample sizes

Surveys in education and especially international surveys rarely sample students by simply selecting a random sample of students (known as a simple random sample, or SRS). Rather, a sampling design is used where schools are first selected and, within each selected school, classes or students are randomly sampled. Sometimes, geographic areas are first selected before sampling schools and students. This sampling design is usually referred to as a cluster sample or a multi-stage sample.

Selected students attending the same school cannot be considered as independent observations as assumed with a simple random sample because they are usually more similar to one another than to students attending other schools. For instance, the students are offered the same school resources, may have the same teachers and therefore are taught a common implemented curriculum, and so on. School differences are also larger if different educational programmes are not available in all schools. One expects to observe greater differences between a vocational school and an academic school than between two comprehensive schools.

Furthermore, it is well known that within a country/economy, within sub-national entities and within a city, people tend to live in areas according to their financial resources. As children usually attend schools close to their home, it is likely that students attending the same school come from similar social and economic backgrounds.

Therefore, a simple random sample of 4 000 students within a country/economy is thus likely to cover the diversity of the population better than a sample of 100 schools with 40 students observed within each school. It follows that the uncertainty associated with any population parameter estimate (i.e. standard error) will be larger for a clustered sample estimate than for a simple random sample estimate of the same size.

In the case of a simple random sample, the standard error of a mean estimate is equal to:

$$\sigma_{(\hat{\mu})} = \sqrt{\frac{\sigma^2}{n}}$$

Equation 13.3

where σ^2 denotes the variance of the whole student population and *n* is the student sample size.

For an infinite population of schools and infinite populations of students within schools, the standard error of a mean estimate from a cluster sample is equal to:

$$\sigma_{(\hat{\mu})} = \sqrt{\frac{\sigma^2_{schools}}{n_{schools}} + \frac{\sigma^2_{within}}{n_{schools} n_{students}}}$$

Equation 13.4

where $\sigma^2_{schools}$ denotes the variance of the school means, σ^2_{within} denotes the variances of students within schools, $n_{schools}$ denotes the sample size of schools, and $n_{students}$ denotes the sample size of students within each school.

The standard error for the mean from a simple random sample is inversely proportional to the square root of the number of selected students. The standard error for the mean from a cluster sample is proportional to the variance that lies between clusters (i.e. schools) and within clusters, and inversely proportional to the square root of the number of selected schools and is also a function of the number of students selected per school.

It is usual to express the decomposition of the total variance into the between-school variance and the within-school variance by the coefficient of intraclass correlation, also denoted *Rho*. Mathematically, this index is equal to:

$$Rho = \frac{\sigma^2_{schools}}{\sigma^2_{schools} + \sigma^2_{within}}$$

Equation 13.5

This index provides an indication of the percentage of variance that lies between schools. A low intraclass *correlation* indicates that schools are performing similarly while higher values point towards large differences between school performance.

To limit the reduction of precision in the population parameter estimate, multi-stage sample designs usually use supplementary information to improve coverage of the population diversity. In PISA the following techniques were implemented to limit the increase in the standard error: (i) explicit and implicit stratification of the school sampling frame and (ii) selection of schools with probabilities proportional to their size. Complementary information generally cannot compensate totally for the increase in the standard error due to the multi-stage design however but will greatly reduce it.

It is usual to express the effect of the sampling design on the standard errors by a statistic referred to as the design effect. This corresponds to the ratio of the variance of the estimate obtained from the (more complex) sample to the variance of the estimate that would be obtained from a simple random sample of the same number of sampling units. The design effect has two primary uses – in sample size estimation and in appraising the efficiency of more complex sampling plans (Cochran, 1977[1]).

In PISA, as sampling variance has to be estimated by using the 80 *BRR* replicates, a design effect can be computed for a statistic *t* using:

$$Deff(t) = \frac{Var_{BRR}(t)}{Var_{SRS}(t)}$$

Equation 13.6

where $Var_{BRR}(t)$ is the sampling variance for the statistic *t* computed by the *BRR* replication method, and $VAR_{SRS}(t)$ is the sampling variance for the same statistic t on the same data but considering the sample as a simple random sample.

Based on a hypothetical country/economy, where the unbiased BRR standard error on the mean proficiency estimate is equal to 1.46, and the standard deviation is equal to *102.29*, on a sample of 14 530 students, the design effect for the mean proficiency estimate is therefore calculated as:

$$Deff(t) = \frac{Var_{BRR}(t)}{Var_{SRS}(t)} = \frac{(1.46)^2}{[102.29^2/14\,530]} = 2.96$$

Equation 13.7

This means the sampling variance on the proficiency estimate is about 2.96 times larger than it would have been with a simple random sample of the same sample size.

Another way to express the reduction of precision due to the complex sampling design is through the effective sample size, which expresses the simple random sample size that would give the same sampling variance as the one obtained from the actual complex sample design. The effective sample size for a statistic *t* is equal to:

$$Effn(t) = \frac{n}{Deff(t)} = \frac{n \times Var_{SRS}(t)}{Var_{BRR}(t)}$$

Equation 13.8

where n is equal to the actual number of units in the sample. The effective sample size in our example would then be equal to:

$$Effn(t) = \frac{n}{Deff(t)} = \frac{14\ 530}{2.96} = 4\ 909$$

Equation 13.9

In other words, a simple random sample of about *4 909* students in this hypothetical country/economy would have been as precise as the actual sample for the national proficiency estimate.

Variability of the design effect

Neither the design effect nor the effective sample size is a definitive characteristic of a sample. Both the design effect and the effective sample size vary with the variable and statistic of interest.

As previously stated, the sampling variance for estimates of the mean from a cluster sample is proportional to the intraclass correlation. In some countries/economies, student performance varies between schools. Students in academic schools usually tend to perform well while on average student performance in vocational schools is lower. Let us now suppose that the height of the students was also measured, and there are no reasons why students in academic schools should be of different height than students in vocational schools. For this particular variable, the expected value of the between-school variance should be equal to zero and therefore, the design effect should tend to one. As the segregation effect differs according to the variable, the design effect will also differ according to the variable.

The second factor that influences the size of the design effect is the choice of requested statistics. It tends to be large for means, proportions, and sums but substantially smaller for bivariate or multivariate statistics such as correlation and regression coefficients.

Design effects in PISA for performance variables

The notion of design effect as given earlier can be extended and gives rise to five different design effect formulae to describe the influence of the sampling and test designs on the standard errors for statistics.

The total errors computed for population estimates based on performance variables (scale scores) in the international PISA reports consist of two components: sampling variance (Var_{BRR}) and measurement variance. The measurement variance is approximated by means of the imputation variance ($MVar$) which is calculated from the statistics calculated from imputed plausible values assigned to the participating students.

The standard error of proficiency estimates in PISA are inflated because the students were not sampled according to a simple random sample and because the estimation of student proficiency includes some amount of measurement error.

Therefore, the variance of a statistic calculated using plausible values is then calculated as the sum of the sampling and the imputation variances, or $Var_{BRR} + MVar$.

The five design effects and their respective effective sample sizes can then be defined as follows:

Design Effect 1: This design effect shows the inflation of the total variance that would have occurred due to measurement error if in fact the samples were considered as simple random samples.

$$Deff_1(r) = \frac{Var_{SRS}(r) + MVar(r)}{Var_{SRS}(r)}$$

Equation 13.10

Design Effect 2: shows the inflation of the *total* variance due only to the use of a complex sampling design.

$$Deff_2\,(r) = \frac{Var_{BRR}(r) + MVar(r)}{Var_{SRS}(r) + MVar(r)}$$

Equation 13.11

Design Effect 3: shows the inflation of the sampling variance due to the use of a complex design. This is the same as Formula 7 introduced above.

$$Deff_3\,(r) = \frac{Var_{BRR}(r)}{Var_{SRS}(r)}$$

Equation 13.12

Design Effect 4: shows the inflation of the total variance due to measurement variance.

$$Deff_4\,(r) = \frac{Var_{BRR}(r) + MVar(r)}{Var_{BRR}(r)}$$

Equation 13.13

Design Effect 5: shows the inflation of the total variance due to the measurement variance and due to the complex sampling design.

$$Deff_5\,(r) = \frac{Var_{BRR}(r) + MVar(r)}{Var_{SRS}(r)}$$

Equation 13.14

Table 13.10, Table 13.11., Table 13.12. Table 13.13, Table 13.14. Table 13.15. Tale 13.16 present the values of the different design effects and the effective sample size using $Deff_5$, for each of the main PISA domains.

To better understand the design effect for a country/economy, some information related to the design effects and their respective effective sample sizes are presented in Annex C.

References

Cochran, W. (1977), *Sampling Techniques (3rd ed.)*, John Wiley and Sons. [1]

Annex 13.A. Sampling outcomes

Web tables can be accessed via the StatLink.

Annex Table 13.A.1. Chapter 13: Population Characteristics, Response Rates, and Proficiency Errors Across Regions and Domains

Tables	Title
Web Table 13.A.2	Population characteristics, sample characteristics, exclusions and coverage indices for participating countries/economies
Web Table 13.A.3	Population characteristics, sample characteristics, exclusions and coverage indices for participating adjudicated regions
Table 13.A.4	Response rates for participating countries/economies calculated by using only original schools and no replacement schools
Table 13.A.5	Response rates for adjudicated regions calculated by using only original schools and no replacement schools
Table 13.A.6	Response rates for participating countries/economies when first and second replacement schools were accounted for in the rates
Table 13.A.7	Response rates for adjudicated regions when first and second replacement schools were accounted for in the rates
Table 13.A.8	Student response rates among the full set of participating schools in participating countries/economies
Table 13.A.9	Student response rates among the full set of participating schools in adjudicated regions
Table 13.A.10	Teacher response rates among the full set of participating schools in participating countries
Web Table 13.A.11	Standard errors and related statistics for the average mathematics proficiency
Web Table 13.A.12	Standard errors and related statistics for the average reading proficiency
Web Table 13.A.13	Standard errors and related statistics for the average science proficiency
Web Table 13.A.14	Standard errors and related statistics for the average creative thinking proficiency
Web Table 13.A.15	Standard errors and related statistics for the average financial literacy proficiency (Financial Literacy sample)
Web Table 13.A.16	Standard errors and related statistics for the average mathematics proficiency (Financial Literacy sample)
Web Table 13.A.17	Standard errors and related statistics for the average reading proficiency (Financial Literacy sample)

StatLink https://stat.link/mou0ea

14 Scaling outcomes

This chapter reports on the outcomes of implementing the item response theory (IRT) and population modelling methods described in Chapter 11 for the PISA 2022 main survey cognitive assessment data. It provides results of the assessments of the invariance of the IRT item parameters within and across countries/economies, estimates of the reliability and correlations across assessments domains, and estimates of the linking errors between the 2022 and prior PISA cycles. The location of the items across the full range of proficiencies based on their common international parameters are also reported. Finally, the correlations between scales and the percentage of students in each country at each proficiency level are presented for each cognitive domain.

IRT scaling outcomes

IRT scaling outcomes include the proportions of item were invariant across countries and PISA cycles, as well as the common and unique items parameters and the dropped items used for the population modelling of each country/economy data. The international (common) item parameters are provided in this technical report's Annex A and unique country/economy's item parameters are provided in Annex F. The next section provides an assessment of item parameter invariance across countries/economies supporting that the comparability of the PISA scales across cycles and countries was achieved in each domain by reaching a desirable proportion of invariant item parameters across countries/economies and cycles. The following section describes the international characteristics of each domain's item pool and shows the item maps that locate the items on the reporting scales.

Invariance of item parameters

The item parameters for all the items used in the assessment were obtained through IRT scaling. In PISA 2022, IRT scaling was implemented through a multi-group (i.e., country-by-language groups) IRT concurrent calibration using the 2022 main survey data, using the trend items as fixed linking items and setting the scale to the PISA scale established in 2015 and 2018. That is, item parameters for trend items were fixed to the ones used in PISA 2018 (either common international or unique to a specific country-by-language group or groups), unless there was evidence that the 2018 parameters did not fit the 2022 data (see Chapter 11 for details).

In most cases the international item parameters fitted data for all country-by-language groups. When they did not fit a particular country-by-language group, unique or group-specific parameters were estimated and used, unless it was found that the unique parameters could not be estimated, still did not fit the data well enough, or were extreme. That is, an item was dropped if in the end, its RMSD fit could not be reduced to 0.15 or below, its slope parameter was below 0.1 or its difficulty parameter was larger than 5.0 in absolute value. These criteria were not applied to reading fluency items because they typically are very easy items. In rare cases, items were also be dropped when, despite being checked in the field trial, content and/or translation issues were nonetheless found in the main survey—given feedback from countries/economies, content and psychometric reviews. In even rarer cases, items were dropped entirely (in all countries/economies) if analyses indicated that it did not fit the data collected in the majority of the countries/economies. In this PISA 2022 cycle: one mathematics, one reading, one financial literacy and six creative thinking items were dropped from all groups [1].

To assess the invariance of item parameters across country-by-language groups and cycles, items were categorized as:

- *invariant* when common international parameters could be used;
- *group-specific invariant* when the same unique parameters could be used across cycles (applies only to trend items);
- *variant* for all other cases where unique item parameters were estimated (new items) or when unique parameters were estimated that are different from the 2018 parameters (trend items); and
- *dropped* when the item could not be fitted to the data and was dropped for one or more country-by-language groups.

For countries with multiple language groups, the number of invariant, variant, or dropped items were averaged across the different language groups within the country to calculate the proportion of unique item parameters used. Sample weights were used for this calculation. Annex Table 14.A.2 shows the proportions of items categorized as invariant, variant, and dropped, averaged across countries participating in the 2022 computer-based assessment (CBA). The proportion of invariant items was large for all domains, ranging from 76.4% for the reading MSAT items to 93.7% for the reading fluency items. A large proportion of invariant items is critical for ensuring the comparability of scores across countries and cycles. Group-specific invariant items also contribute to the comparability of scores across cycles. The proportion of invariant total (invariant and group-specific invariant) was above 98.5% for all domains but creative thinking at 77.4%. Regarding the dropped category, the proportions were small for all domains (less than 2%).

Annex Table 14.A.3 shows the proportion of items categorized as invariant, variant, and dropped, averaged across countries participating in the new 2022 paper-based assessment (new PBA). The results across the three new PBA participating countries showed somewhat lower proportions of invariance than with CBA. Nevertheless, proportions of total invariant items were above 80% for all domains and few items were dropped for any country.

An overview of the frequencies of invariant, variant, and dropped items for each domain is presented in Figure 14.1, Figure 14.2, Figure 14.3, Figure 14.4 and Figure 14.5 for CBA, new PBA and PBA participating countries/economies. Each country is represented by stacked vertical bars: above the horizontal line at zero, dark green represents the number of items classified as invariant and light green represents the number of group-specific invariant items (only trend items); below the 0 horizontal line, yellow represents the number of variant items[2] and red represents the number of items dropped from scaling. The frequencies of variant and dropped items are shown using negative values to highlight differences between the number of items that contribute to ensuring the comparability of the PISA scales (invariant) and the number of items that do not (variant). The countries are sorted from left to right by increasing number of invariant items, first CBA, new PBA, and PBA countries.

These plots show that while there is some variability across countries, the numbers of invariant item parameters and group-specific invariant item parameters are large enough to ensure the comparability of the proficiency estimates across countries/economies and across cycles.

Figure 14.1. Frequency of invariant, variant, and dropped items for mathematics, by country/economy

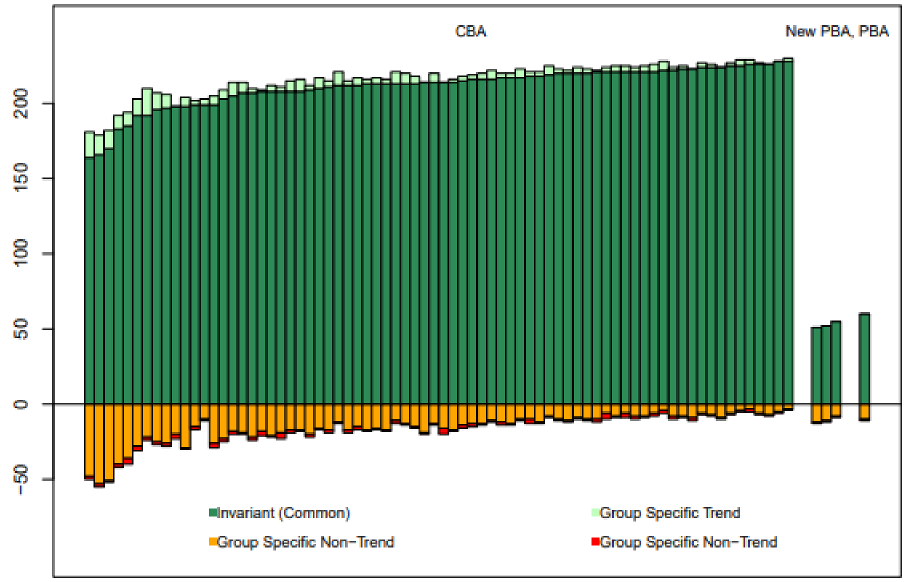
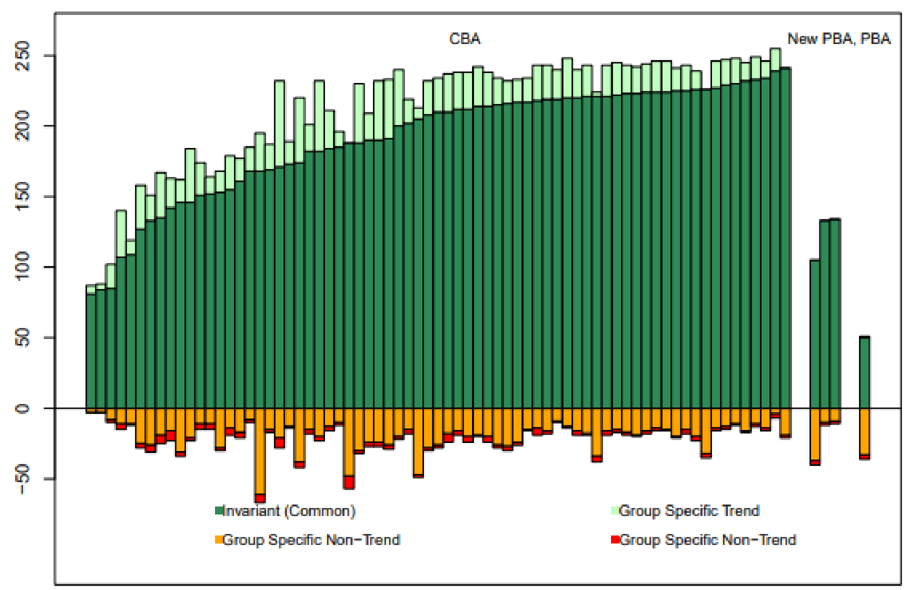


Figure 14.2. Frequency of invariant, variant, and dropped items for reading, by country/economy



Note: Because reading is a minor domain in 2022, in some countries, sample size was not enough to assess fit with the 2022 data. This cases are not included in this plot, resulting in fewer than the number of items used being displayed in these cases. However all items were evaluated for fit in 2018 when reading was the major domain--see PISA 2018 Technical report, Chapter 12 for these results).

Figure 14.3. Frequency of invariant, variant, and dropped items for science, by country/economy

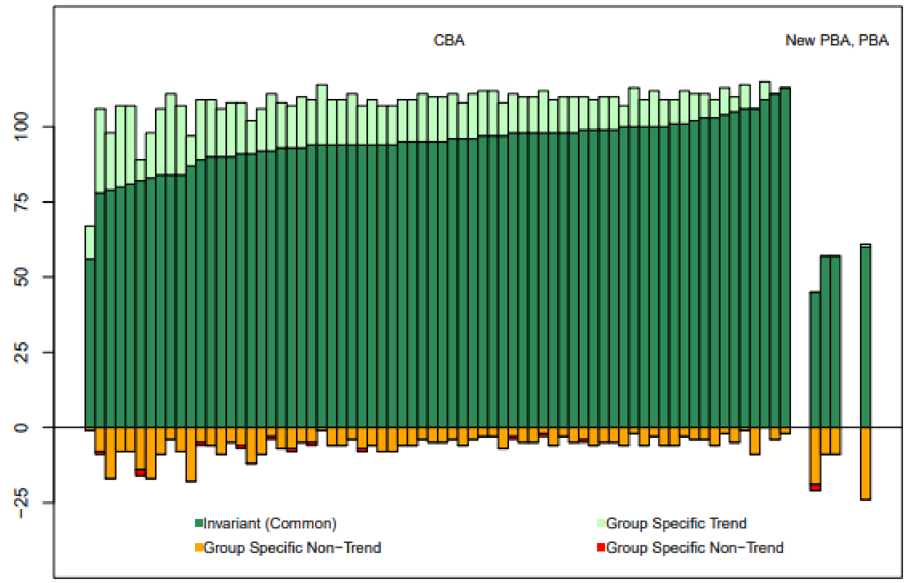


Figure 14.4. Frequency of invariant, variant, and dropped items for creative thinking, by country/economy

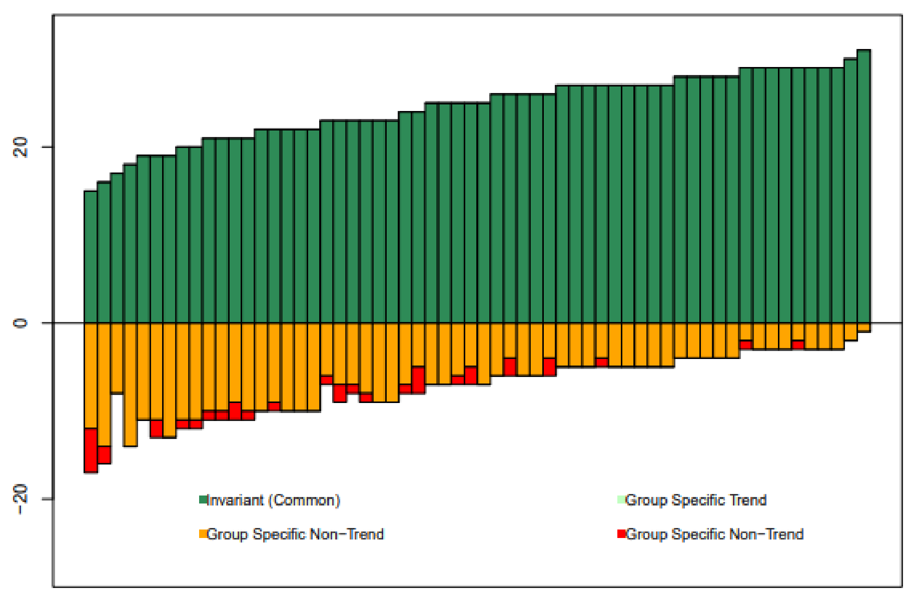
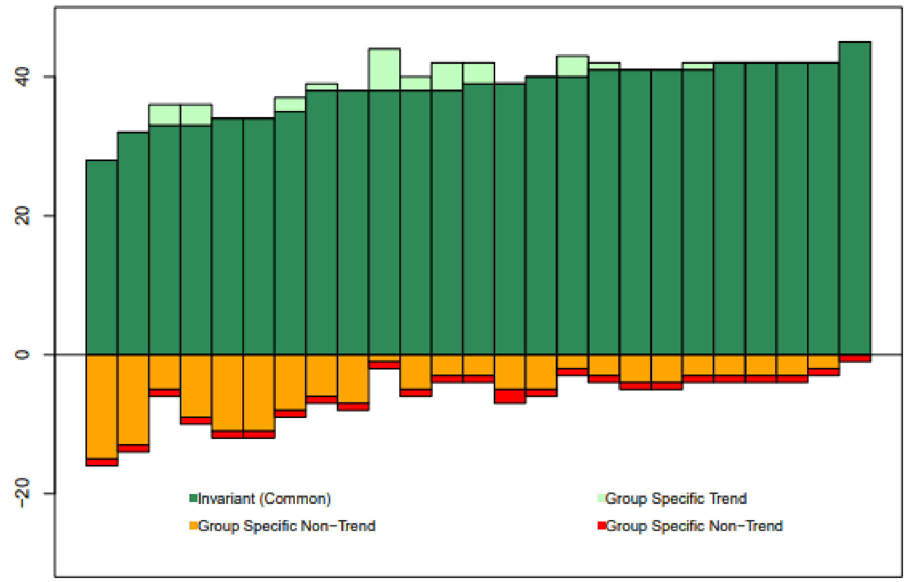


Figure 14.5. Frequency of invariant, variant, and dropped items for financial literacy, by country/economy



International characteristics of the item pools

This section provides an overview of the test targeting, the domain inter-correlations, and the correlations among the mathematics subscales.

Test targeting

Similar to assigning a specific score on a scale to students according to their performance on an assessment (OECD, 2022[1]), each item in PISA 2022 was assigned a specific value on a scale based on response probability (RP) calculated using the item's IRT parameters (discrimination and difficulty). Chapter 17 describes how items can be placed along a scale based on their RP values and how these values can be used to classify items into proficiency levels.

Historically in PISA, a response probability of 0.62 (RP62) has been used to classify items into levels. Students with a proficiency located at or below this point have a probability of 0.62 or less of getting the item correct, while students with a proficiency above this point have a higher probability of getting the item correct higher than 0.62. The RP62 values for all items and their performance level classification are presented in Annex A, together with the final international/common item parameter estimates obtained from the IRT scaling. Note that for polytomous items, the RP62 value is provided for partial credit as well as full credit responses. The partial credit RP62 has been defined as the minimum proficiency level a student need to have an expected score that is 62% of the full credit.

Annex Table 14.A.4, Annex Table 14.A.5, Annex Table 14.A.6, Annex Table 14.A.7 and Annex Table 14.A.8 show the proficiency levels defined for each cognitive domain, along with the percentage of items and the percentages of students classified at each level of proficiency, using the first plausible value. Note that although polytomous items have two RP62 levels (partial credit and full credit), they were classified according to the full credit RP62 only for all domains but creative thinking. For creative thinking, most of the items are polytomous items (28 out of 32), therefore we describe both partial- and full-credit RP62 levels.

Since RP62 values and the plausible values are on the same PISA scale, the distribution of students' latent ability and the items' RP62 values can be compared and contrasted. In Figure 14.6, Figure 14.7, Figure 14.8, Figure 14.9 and Figure 14.10, the left side of each figure illustrates the distribution of the first plausible values (PV1) across countries. In each figure, the blue line indicates the empirical density of the first plausible values across all countries, and the red line indicates the theoretical normal distribution with the mean and the variance of plausible values across all countries. The figures show that the distribution of the plausible values for each domain are approximately normal. On the right side of each figure, the RP62 value for each of the items is plotted. As with the tables above, in all domains but creative thinking, only the RP62 values for full-credit are shown.

Figure 14.6. Distribution of the first plausible values and item RP62 values in mathematics

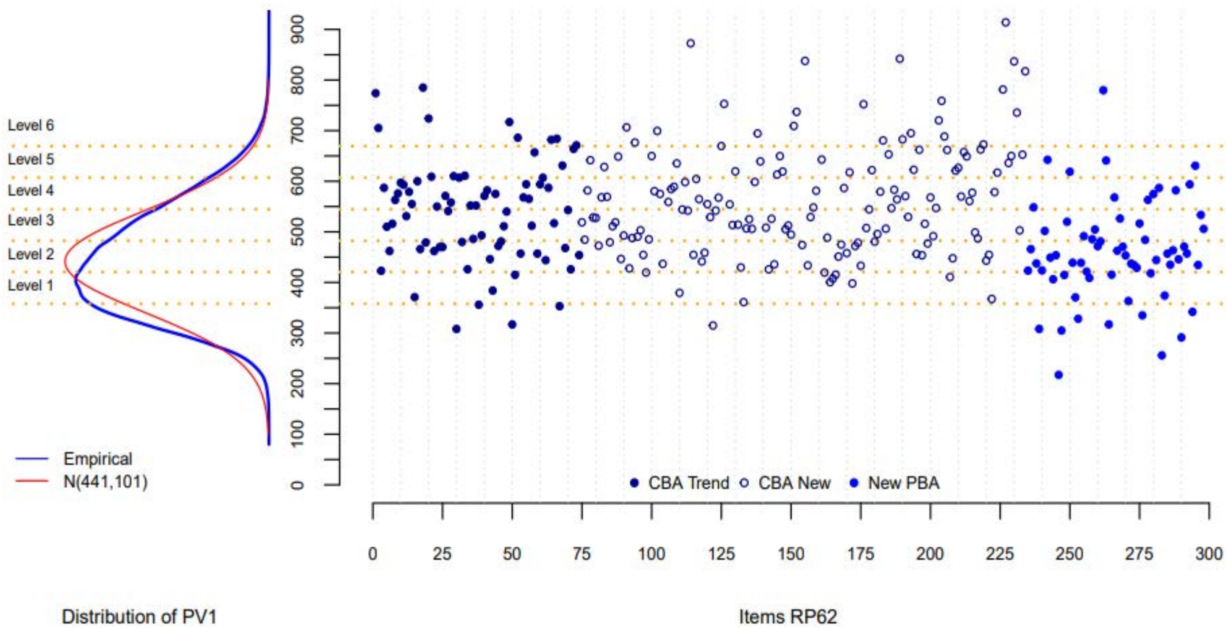


Figure 14.7. Distribution of the first plausible values and item RP62 values in reading

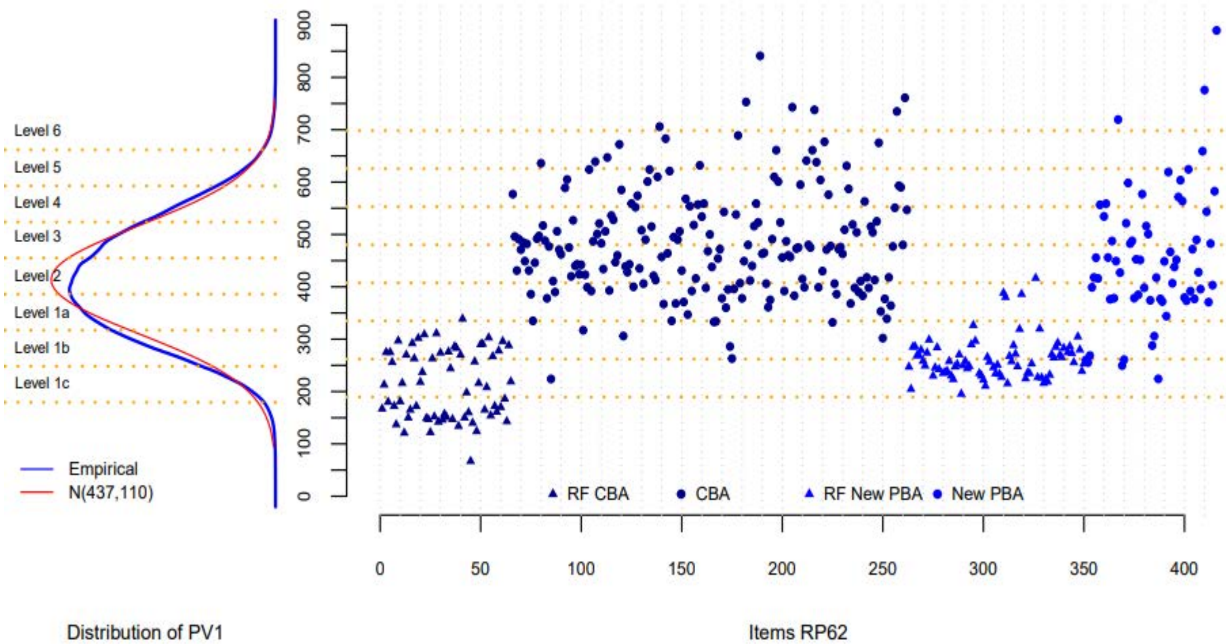


Figure 14.8. Distribution of the first plausible values and item RP62 values in science

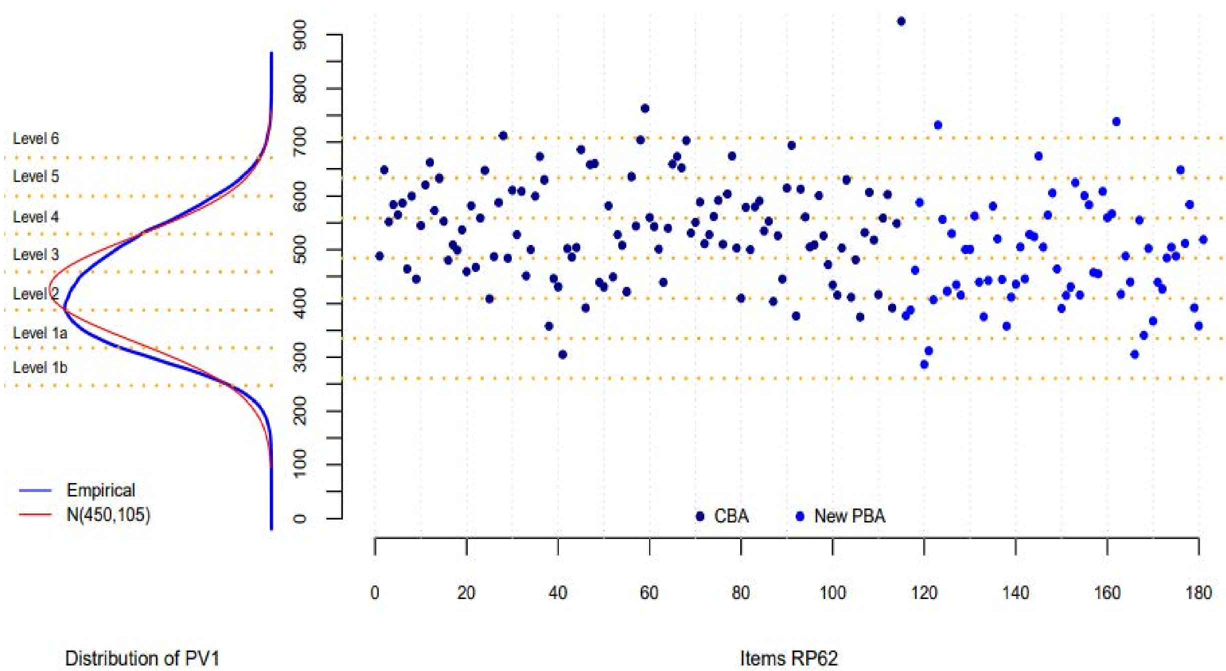


Figure 14.9. Distribution of the first plausible values and item RP62 values in creative thinking

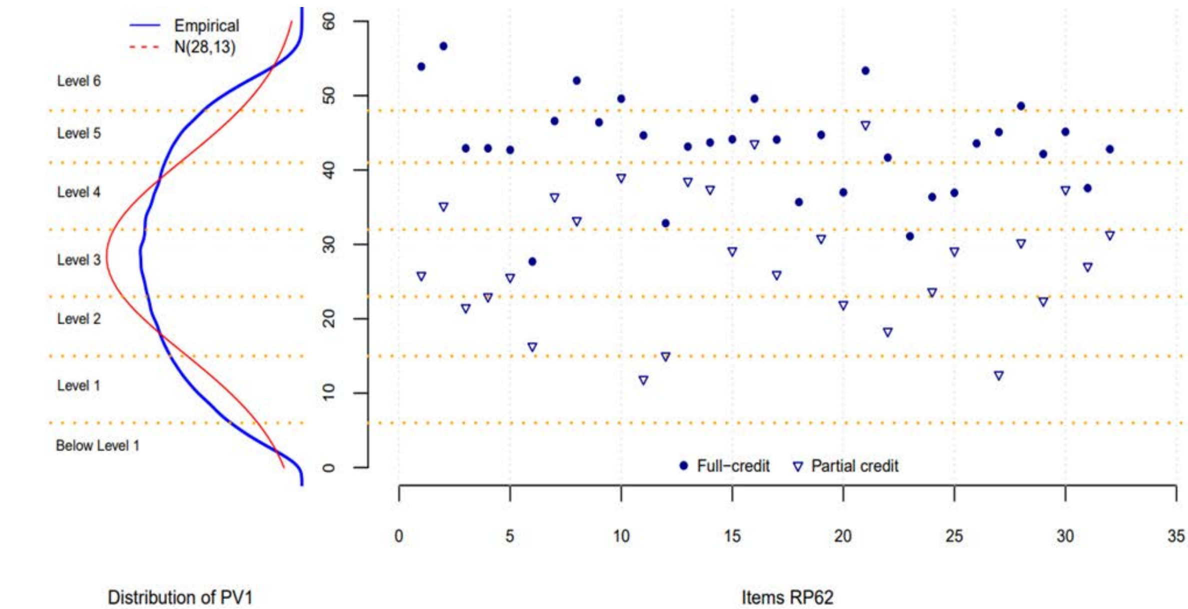
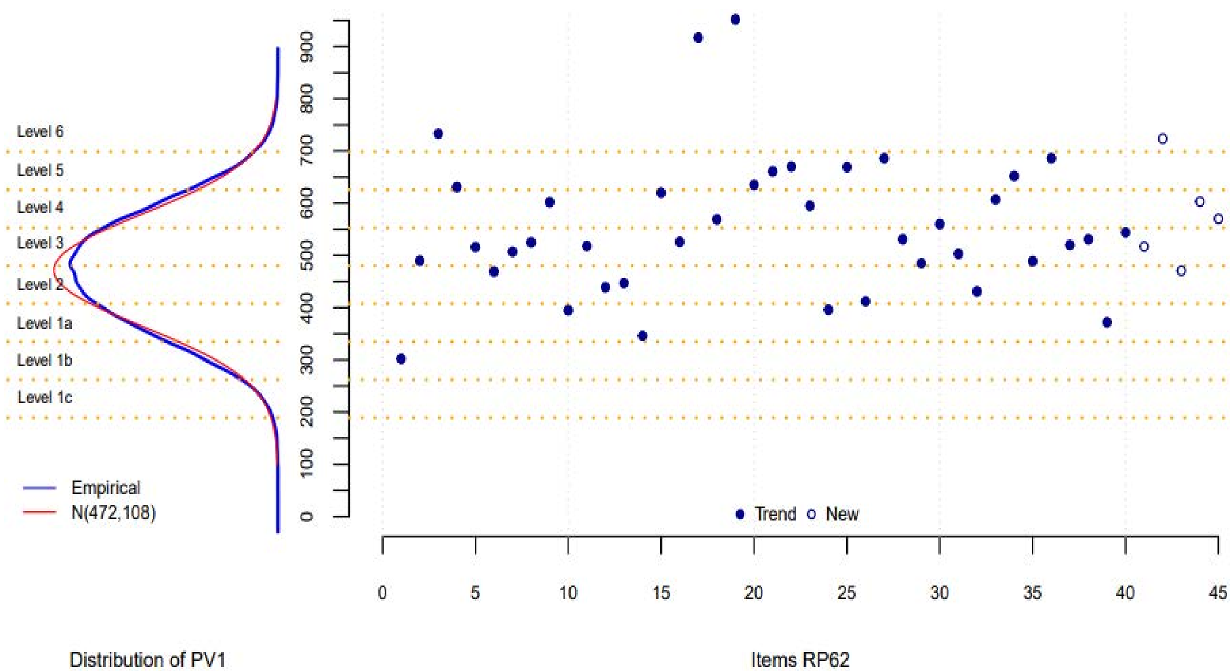


Figure 14.10. Distribution of the first plausible values and item RP62 values in financial literacy



Population modelling outcomes

The population modelling outcomes include the multivariate latent regressions models estimated for each country/economy and the plausible values (PVs) generated from them, which are included in the international and national databases. Because the latent part of the population model comes from the IRT scaling, the plausible values are generated on their underlying PISA IRT metric used when estimating IRT item and group parameters and then transformed to the PISA scale. For example, mathematics IRT scaled PVs are produced and then transformed to the PISA metric of mean 500 and standard deviation 100 across all participating OECD countries during the first mathematics assessment. Based on these PVs, then the overall and country/economy-level PISA scale reliability, average performance and students percentile by proficiency levels, and finally the correlations between domain scales were estimated. In the next sections, the methods used to transform the PVs from the IRT scale to the PISA reporting scales are described and the outcomes are reported.

Mathematics, reading, science and financial literacy scaling transformations

The mathematics, reading, and science PISA reporting scales were set when the domain became a major domain for the first time—in 2006 for science, 2009 for reading and 2012 in mathematics. This was done using a linear transformations of the senate weighted OECD participating countries/economies IRT scaled plausible values available at the time, so that the overall mean was 500 and the standard deviation 100, resulting in nearly all reported plausible values being between 200 and 800. The same approach was used for each new innovative domain and for the optional financial literacy domain.

However, because the IRT models used for scaling were updated in 2015, a bridge study was completed as part of the 2015 scaling analyses to establish new IRT to reported PISA scale parameters. This did not change the scales or the scores reported prior to 2015, but the new transformations have been applied since. Detailed descriptions of the bridge study and it results are provided in the PISA 2015 technical report OECD (2017[2]), Chapter 12)

Annex Table 14.A.9 provides the PISA IRT theta to reported PISA proficiency scale linear transformation A and B coefficients for the core and financial literacy domains. Given any IRT scaled theta (θ) value (e.g., item difficulty, item step parameters, or student PV or proficiency), the transformed value on the PISA scale is $A * \theta + B$.

Creative Thinking scaling transformation

For the creative thinking innovative domain developed for the PISA 2022 main survey, it was found that the use of a non-linear transformation provided a more appropriate reporting scale. This was because the particular challenges in creating such an innovative measure resulted in a relatively small pool that did not provide much information towards the lower end of the scale. To best support scale interpretations, the creative thinking item pool IRT test characteristic curve transformation of the theta plausible values PV_θ was applied to obtain the reported plausible values PV_{NC} = $\sum_{i \in V_p} T_i(PV_\theta)$, where V_P indicates the set of 32 creative thinking items and $T_i(PV_\theta)$ is the expected score on item *i* as a function of PV_θ and item *i*'s IRT parameters. In this way the reported creative thinking plausible values can be interpreted as the expected number correct on a hypothetical form made up of all the items in the creative thinking item pool, given the proficiency level that the plausible value represents.

Reliability of the PISA scales

As was done in prior PISA cycles, test reliability was estimated using the well known theoretical formula: 1 – (expected error variance/total variance). In practice, the expected error variance is the weighted average of the students' posterior variance, computed as the variance of the 10 plausible values, which is an expression of the posterior measurement error. The total variance was computed using a resampling approach (Efron, 1982[3]), using each country/economy set of resampling weights.

Annex Table 14.A.10 presents the test reliability descriptive statistics across countries/economies for the cognitive domains and the mathematics subscales. The reliabilities for each country/economy are presented in Annex Table 14.A.11. Overall, we observe that in average test reliability is high for the core and financial literacy domains (0.84 to 0.90) and a bit less for creative thinking (0.8), and that most countries/economies' reliability is close to the average. As expected, since the number of items is smaller than for the full mathematics instrument, the reliability of the mathematics subscales are much lower and more variable cross countries/economies.

Annex Table 14.A.12 shows the average transformed plausible values as well as the resampling-based standard errors for each country and domain.

Domain inter-correlations

Estimated correlations between the domains, based on the 10 reported plausible values and averaged across all countries and assessment modes, are presented in Annex Table 14.A.13 and Annex Table 14.A.14 for the core domains and creative thinking, and in Annex Table 14.A.15 for the financial literacy sample. The estimated correlations for each country are presented in Annex Table 14.A.16.

Mathematics subscales correlations

There were two sets of subscales reported for mathematics. The first set, measuring content domains, was composed of the following four subscales: space and shape (MCSS), quantity (MCQN), change and relationships (MCCR), and uncertainty and data (MCUD). The second set, based on the cognitive processes, comprised the following four subscales: employing mathematical concepts, facts, and procedures (MPEM), interpreting, applying, and evaluating mathematical outcomes (MPIN), formulating situations mathematically (MPFS), and reasoning (MPRE).

The correlations between reading, science and the mathematics content domain subscales are presented in Annex Table 14.A.17.Annex Table 14.A.18 shows the correlations between reading, science and the cognitive process domains.

Note that, as indicated in Chapter 11, because of the way in which these subscale plausible values were estimated, it is not appropriate to correlate the cognitive process subscales with the cognitive contents subscales, or any of the subscales with the overall mathematics proficiency.

Countries/economies average proficiency and percentages of students at each proficiency level

Figure 14.11, Figure 14.12. Figure 14.13, Figure 14.14 and Figure 14.15 show the average proficiency and percentages of students at each proficiency level across countries/economies for each domain.

Linking error

The estimation of the linking error between two PISA cycles was accomplished by considering the differences between the reported country means from the previous PISA cycles and new estimates of these country means based on the new PISA cycle item parameters. To estimate the linking error for trend comparisons between PISA 2022 and a previous PISA cycle down to 2006, the subset of countries that had participated in both cycles being compared was used. In the cases of trends to 2000 or 2006 or financial literacy, since the number of participating countries was relatively small, all countries were used.

The 2022 linking errors are reported in Annex Table 14.A.19. Using these values help evaluate the extent to which changes in a country/economy or subgroup's performance between PISA 2022 and a previous PISA cycle are significantly different.

Note that for each domain, the earliest cycle for which comparisons can be made between PISA 2022 and a previous PISA cycle is the cycle in which the domain first became a major domain. Thus, the comparison of mathematics scores between PISA 2022 and PISA 2000 is not possible, nor is the comparison of science scores between PISA 2022 and PISA 2000 or between PISA 2022 and PISA 2003. Detail on the methodology used to calculate the linking errors can be found in Chapter 11.

Table 14.1. Global Analysis of Item Dynamics and Student Proficiency in PISA 2022

Figure	Title
Figure 14.1	Frequency of invariant, variant, and dropped items for mathematics, by country/economy
Figure 14.2	Frequency of invariant, variant, and dropped items for reading, by country/economy
Figure 14.3	Frequency of invariant, variant, and dropped items for science, by country/economy
Figure 14.4	Frequency of invariant, variant, and dropped items for creative thinking competence, by country/economy
Figure 14.5	Frequency of invariant, variant, and dropped items for financial literacy, by country/economy
Figure 14.6	Distribution of the first plausible values and item RP62 values in mathematics
Figure 14.7	Distribution of the first plausible values and item RP62 values in readiing
Figure 14.8	Distribution of the first plausible values and item RP62 values in science
Figure 14.9	Distribution of the first plausible values and item RP62 values in creative thinking
Figure 14.10	Distribution of the first plausible values and item RP62 values in financial literacy
Figure 14.11	Percentage of students in each country/economy at each proficiency level for mathematics
Figure 14.12	Percentage of students in each country/economy at each proficiency level for reading
Figure 14.13	Percentage of students in each country/economy at each proficiency level for science
Figure 14.14	Percentage of students in each country/economy at each proficiency level for creative thinking
Figure 14.15	Percentage of students in each country/economy at each proficiency level for financial literacy

StatLink https://stat.link/4trwd2

References

Efron, B. (1982), “The jackknife, the bootstrap, and other resampling plans”*, CBMS-NSF Regional Conference Series in Applied Mathematics, Monograph 38*, https://doi.org/10.1137/1.9781611970319. [3]

OECD (2022), *PISA 2018 Technical Report*. [1]

OECD (2017), *PISA 2015 Technical Report*, OECD Publishing, Paris, https://www.oecd.org/en/about/programmes/pisa/pisa-data.html. [2]

Notes

1. The dropped items are: CMA112Q02, CR547Q07S, DF082Q01C, and DT520Q01C, DT560Q01C, DT560Q02C, DT450Q01C, DT450Q02C and DT450Q03C.

2. For the trend items classified as variant in a specific group (yellow), the 2018 parameters did not appropriately fit the 2022 data; thus, new unique parameters were estimated. For new items classified as variant in a specific group (yellow), unique parameters were needed due to the misfit of the common international parameters to the 2022 data.

Annex 14.A. IRT Scaling Outcomes and Population Modelling Analysis

Annex Table 14.A.1. Comprehensive Statistical Insights from PISA 2022: Item Dynamics, Proficiency Assessments, and Domain Interrelations

Tables	Title
Table 14.A.2	Proportion of invariant, variant, and dropped CBA items averaged across countries/economies, for each domain
Table 14.A.3	Proportion of invariant, variant, and dropped new PBA items averaged across countries/economies, for each domain
Table 14.A.4	Proficiency levels for mathematics and the classification of items and students
Table 14.A.5	Proficiency levels for reading and the classification of items and students
Table 14.A.6	Proficiency levels for science and the classification of items and students
Table 14.A.7	Proficiency levels for creative thinking and the classification of items and students
Table 14.A.8	Proficiency levels for financial literacy and the classification of items and students
Table 14.A.9	PISA IRT theta to reported PISA proficiency scale linear transformation coefficients
Table 14.A.10	Test reliability descriptive statistics across countries/economies for the cognitive domains and the mathematics subscales
Table 14.A.11	Countries/economies reliability values for the cognitive domains
Table 14.A.12	Average plausible values (PV) and resampling-based standard errors (SE) by country and domain.
Table 14.A13	Core domain inter-correlations for the main sample
Table 14.A.14	Creative Thinking inter-correlations with core domains for the main sample
Table 14.A.15	Domain inter-correlations for the financial literacy sample
Table 14.A.16	Domain inter-correlations by country/economy
Table 14.A.17	Mathematics content subscales inter-correlations
Table 14.A.18	Mathematics cognitive process subscales inter-correlations
Table 14.A.19	Linking error for score comparisons between PISA 2022 and previous PISA cycles

StatLink https://stat.link/y3l6ba

Annex Table 14.A.2. Proportion of invariant, variant, and dropped CBA items averaged across countries/economies, for each domain

	Mathematics		Reading		Science	Financial	Creative
	Trend	New	Fluency	MSAT	All	Literacy	Thinking
Total items	74	159	65	196	115	40	32
Total countries	68	68	68	68	68	19	55
Invariant	86.0%	92.6%	93.7%	76.4%	83.0%	84.1%	77.4%
Group-specific invariant	6.5%	-	3.5%	11.8%	11.8%	2.6%	-
Invariant total[1]	92.5%	92.6%	97.2%	88.2%	94.8%	86.7%	77.4%
Noninvariant	6.1%	7.0%	1.7%	10.6%	4.0%	11.0%	20.8%
Dropped	1.4%	0.4%	1.1%	1.2%	1.2%	2.3%	1.8%

Annex Table 14.A.3. Proportion of invariant, variant, and dropped new PBA items averaged across countries/economies, for each domain

	Mathematics	Reading		Science
		Fluency	Reading	
Total items	64	79	66	66
Total countries	3	3	3	3
Invariant	82.3%	90.7%	79.3%	80.3%
Noninvariant	16.1%	9.3%	17.2%	18.7%
Dropped	1.6%	0.0%	3.5%	1.0%

Annex Table 14.A.4. Proficiency levels for mathematics and the classification of items and students

Classification	Number of items		Percentage of items		Percentage of respondents	
	CBA	*New PBA*	*CBA*	*New PBA*	*CBA*	*New PBA*
Level 6	38	1	16%	2%	2%	
Level 5	34	4	15%	6%	5%	0%
Level 4	50	7	21%	11%	10%	0%
Level 3	48	10	21%	16%	16%	3%
Level 2	45	25	19%	39%	21%	10%
Level 1a	13	8	6%	13%	23%	26%
Level 1b	5	6	2%	9%	24%	33%
Level 1c		2		3%		20%
Below Level 1		1		2%		7%

Annex Table 14.A.5. Proficiency levels for reading and the classification of items and students

Classification	Number of items				Percentage of items				Percentage of respondents	
	RF CBA	*CBA*	*RF New PBA*	*New PBA*	*RF CBA*	*CBA*	*RF New PBA*	*New PBA*	*CBA*	*New PBA*
Level 6		7		4		4%		6%	1%	
Level 5		14		1		7%		2%	4%	0%
Level 4		25		10		13%		15%	11%	1%
Level 3		42		10		21%		15%	19%	5%
Level 2		59	1	15		30%	1%	23%	24%	18%
Level 1a	1	40	3	18	2%	20%	4%	27%	23%	37%

Level 1b	22	8	30	3	34%	4%	38%	5%	15%	30%
Level 1c	11	1	46	5	17%	1%	58%	8%	5%	8%
Below Level 1	31				48%				1%	1%

Annex Table 14.A.6. Proficiency levels for science and the classification of items and students

Classification	Number of items		Percentage of items		Percentage of respondents	
	CBA	New PBA	CBA	New PBA	CBA	New PBA
Level 6	3	3	3%	4%	1%	
Level 5	15	8	13%	9%	4%	0%
Level 4	31	23	27%	27%	11%	0%
Level 3	36	30	31%	35%	20%	4%
Level 2	22	17	19%	20%	25%	18%
Level 1a	7	3	6%	4%	24%	43%
Level 1b	1	1	1%	1%	13%	30%
Below 1b					2%	5%

Annex Table 14.A.7. Proficiency levels for creative thinking and the classification of items and students

Classification	Number of items		Percentage of items		Percentage of respondents
	Partial Credit*	Full Credit	Partial Credit*	Full Credit	
Level 6		7		22%	4%
Level 5	2	17	7%	53%	16%
Level 4	7	6	25%	19%	19%
Level 3	11	2	39%	6%	22%
Level 2	6		21%		20%
Level 1	2		7%		16%
Below Level 1					4%

Annex Table 14.A.8. Proficiency levels for financial literacy and the classification of items and students

Classification	Number of items	Percentage of items	Percentage of respondents
Level 5	12	27%	8%
Level 4	8	18%	17%
Level 3	14	31%	25%
Level 2	6	13%	24%
Level 1	4	9%	17%
Below 1	1	2%	9%

Annex Table 14.A.9. PISA IRT theta to reported PISA proficiency scale linear transformation coefficients

Domain	A	B
Mathematics	135.9030	514.1848
Reading	131.5532	437.9244
Science	168.3189	494.5360
Financial literacy	140.0807	490.7259

Creative thinking*	-	-

Note: * Not applicable because a non-linear test characteristic curve transformation was used.

Annex Table 14.A.10. Test reliability descriptive statistics across countries/economies for the cognitive domains and the mathematics subscales

MODE		Domains	Median	S.D.	Max	Min
CBA	Mathematics		0.90	0.03	0.93	0.81
	Content	Change and Relationships	0.85	0.05	0.91	0.66
		Quantity	0.87	0.04	0.91	0.75
		Space and Shape	0.80	0.08	0.87	0.57
		Uncertainty and Data	0.84	0.05	0.90	0.71
	Cog. Process	Employing Mathematical Concepts, Facts, and Procedures	0.87	0.04	0.91	0.75
		Formulating Situations Mathematically	0.83	0.08	0.90	0.57
		Interpreting, Applying, and Evaluating Mathematical Outcomes	0.86	0.04	0.90	0.74
		Reasoning	0.85	0.08	0.91	0.59
	Reading		0.86	0.03	0.91	0.77
	Science		0.87	0.03	0.92	0.79
	Financial literacy		0.90	0.02	0.92	0.85
	Creating Thinking		0.80	0.04	0.89	0.65
PBA	Reading		0.87	0.03	0.90	0.84
	Mathematics		0.87	0.01	0.89	0.85
	Science		0.84	0.03	0.87	0.81

Annex Table 14.A.11. Countries/economies reliability values for the cognitive domains

Mode	Country/Economy	Mathematics	Reading	Science	Financial Literacy	Creative Thinking
CBA	Albania	0.85	0.77	0.80		0.80
CBA	United Arab Emirates	0.90	0.86	0.85	0.89	0.77
CBA	Argentina	0.87	0.85	0.85		
CBA	Australia	0.92	0.85	0.87		0.76
CBA	Austria	0.93	0.90	0.91	0.91	
CBA	Belgium*	0.92	0.86	0.89	0.90	0.79
CBA	Bulgaria	0.90	0.87	0.86	0.89	0.82
CBA	Brazil	0.86	0.84	0.85	0.87	0.77
CBA	Brunei Darussalam	0.92	0.91	0.91		0.87
CBA	Canada*	0.89	0.82	0.83	0.90	0.69
CBA	Switzerland	0.92	0.90	0.91		
CBA	Chile	0.87	0.84	0.86		0.77
CBA	Colombia	0.87	0.85	0.86		0.81
CBA	Costa Rica	0.86	0.84	0.83	0.87	0.82
CBA	Czech Republic	0.92	0.87	0.89	0.90	0.79
CBA	Germany	0.92	0.88	0.90		0.82
CBA	Denmark	0.90	0.85	0.89	0.90	0.78
CBA	Dominican Republic	0.82	0.85	0.81		0.79
CBA	Spain	0.89	0.82	0.83	0.86	0.65
CBA	Estonia	0.90	0.84	0.86		0.77
CBA	Finland	0.91	0.86	0.87		0.81
CBA	France	0.92	0.87	0.88		0.80
CBA	United Kingdom	0.92	0.87	0.89		
CBA	Georgia	0.88	0.84	0.83		
CBA	Greece	0.89	0.84	0.86		0.81
CBA	Hong Kong (China)	0.92	0.85	0.86		0.77
CBA	Croatia	0.91	0.84	0.87		0.77

Mode	Country/Economy	Mathematics	Reading	Science	Financial Literacy	Creative Thinking
CBA	Hungary	0.92	0.88	0.90	0.90	0.84
CBA	Indonesia	0.85	0.87	0.85		0.81
CBA	Ireland	0.91	0.88	0.89		
CBA	Iceland	0.89	0.84	0.86		0.77
CBA	Israel	0.92	0.86	0.88		0.85
CBA	Italy	0.91	0.86	0.88	0.90	0.80
CBA	Jamaica	0.88	0.89	0.88		0.89
CBA	Jordan	0.81	0.81	0.82		0.79
CBA	Japan	0.92	0.86	0.89		
CBA	Kazakhstan	0.85	0.83	0.81		0.74
CBA	Korea	0.92	0.85	0.88		0.80
CBA	Kosovo	0.85	0.85	0.84		
CBA	Lithuania	0.91	0.85	0.89		0.81
CBA	Latvia	0.90	0.85	0.88		0.74
CBA	Macao (China)	0.91	0.84	0.88		0.80
CBA	Morocco	0.84	0.82	0.81		0.83
CBA	Republic of Moldova	0.89	0.88	0.86		0.81
CBA	Mexico	0.86	0.86	0.86		0.79
CBA	North Macedonia	0.88	0.84	0.84		0.85
CBA	Malta	0.91	0.87	0.88		0.85
CBA	Montenegro	0.89	0.86	0.86		
CBA	Mongolia	0.89	0.84	0.86		0.79
CBA	Malaysia	0.90	0.88	0.88	0.92	0.85
CBA	Netherlands	0.93	0.89	0.91	0.91	0.84
CBA	Norway	0.91	0.86	0.87	0.85	
CBA	New Zealand	0.92	0.88	0.89		0.83
CBA	Panama	0.87	0.88	0.88		0.85
CBA	Peru	0.87	0.84	0.86	0.89	0.80
CBA	Philippines	0.88	0.90	0.87		0.89
CBA	Poland	0.91	0.87	0.87	0.90	0.78
CBA	Portugal	0.91	0.85	0.88	0.87	0.78
CBA	Palestinian Authority	0.83	0.82	0.81		0.81
CBA	Qatar	0.91	0.87	0.88		0.84
CBA	Baku (Azerbaijan)	0.87	0.81	0.81		0.73
CBA	Cyprus	0.90	0.84	0.84		0.81
CBA	Ukrainian regions (18 of 27)	0.89	0.86	0.87		0.82
CBA	Romania	0.92	0.90	0.90		0.85
CBA	Saudi Arabia	0.83	0.81	0.79	0.86	0.76
CBA	Singapore	0.92	0.86	0.88		0.78
CBA	El Salvador	0.82	0.84	0.84		0.79
CBA	Serbia	0.90	0.86	0.87		0.79
CBA	Slovak Republic	0.92	0.87	0.89		0.85
CBA	Slovenia	0.91	0.87	0.90		0.82
CBA	Sweden	0.92	0.88	0.90		
CBA	Chinese Taipei	0.93	0.89	0.90		0.81
CBA	Thailand	0.88	0.86	0.87		0.85
CBA	Türkiye	0.92	0.88	0.90		
CBA	Uruguay	0.89	0.85	0.87		0.81
CBA	United States	0.92	0.90	0.92	0.91	
CBA	Uzbekistan	0.81	0.80	0.79		0.76
New PBA	Guatemala	0.88	0.90	0.87		
New PBA	Cambodia	0.85	0.84	0.81		
New PBA	Paraguay	0.89	0.89	0.87		
PBA	Viet Nam	0.87	0.84	0.81		

Note: Ukranian regions (18 out of 27) administered the assessment.
*Denotes a country/economy for which the financial literacy domain was not fully sampled across the population; it is not a nationally-representative sample.

Annex Table 14.A.12 Average plausible values (PV) and resampling-based standard errors (SE) by country and domain

Country	Reading		Mathematics		Science		Financial Literacy		Creative Thinking	
	Average PV	SE	Average PV	SE	Average PV	SE	Average PV	SE	Average PV	SE
International average	435.04	0.30	437.63	0.27	446.89	0.28	474.59	0.67	27.83	0.04
Albania	358.43	1.93	368.22	2.09	375.97	2.22			13.09	0.28
Argentina	400.74	2.57	377.53	2.25	406.19	2.49				
Australia	498.05	2.01	487.08	1.78	507.00	1.93			37.31	0.25
Austria	480.41	2.67	487.27	2.34	491.27	2.65	506.22	2.79		
Baku (Azerbaijan)	365.21	2.45	396.88	2.38	380.14	2.21			22.78	0.31
Belgium*	478.85	2.52	489.49	2.20	490.58	2.48	526.63	3.19	34.91	0.27
Brazil	410.36	2.09	378.69	1.58	403.00	1.93	415.51	2.29	23.32	0.29
Brunei Darussalam	429.23	1.16	442.09	0.93	445.86	1.32			23.74	0.19
Bulgaria	404.30	3.40	417.30	3.30	420.99	3.17	426.07	3.70	20.72	0.38
Cambodia	328.84	2.08	336.40	2.69	347.10	2.10				
Canada*	507.13	1.97	496.95	1.56	515.02	1.93	518.74	2.42	37.93	0.22
Chile	447.98	2.63	411.70	2.08	443.54	2.47			30.67	0.31
Chinese Taipei	515.17	3.25	547.09	3.78	537.38	3.31			32.62	0.39
Colombia	408.67	3.75	382.70	3.03	411.12	3.28			25.55	0.49
Costa Rica	415.23	2.66	384.58	1.89	410.99	2.42	418.23	3.10	27.48	0.32
Croatia	475.50	2.44	463.11	2.38	482.67	2.40			30.46	0.31
Cyprus	381.08	1.16	418.31	1.18	410.90	1.46			23.73	0.20
Czech Republic	488.60	2.25	487.00	2.09	497.74	2.30	506.61	2.22	32.64	0.29
Denmark	488.80	2.58	489.27	1.95	493.82	2.50	520.54	2.44	35.49	0.24
Dominican Republic	351.31	2.44	339.11	1.62	360.43	2.04			15.49	0.26
El Salvador	364.90	2.80	343.47	2.00	373.14	2.62			22.97	0.35
Estonia	511.03	2.36	509.95	1.98	525.81	2.07			35.85	0.27
Finland	490.22	2.26	484.14	1.86	510.96	2.50			35.82	0.30
France	473.85	3.07	473.94	2.49	487.23	2.73			32.43	0.31
Georgia	373.86	2.29	390.02	2.37	384.07	2.31				
Germany	479.79	3.61	474.83	3.06	492.43	3.48			32.53	0.40
Greece	438.44	2.83	430.15	2.34	440.79	2.77			27.00	0.33
Guatemala	374.12	2.44	344.20	2.21	372.96	2.23				
Hong Kong (China)	499.70	2.85	540.35	2.99	520.42	2.79			31.57	0.35
Hungary	472.97	2.83	472.78	2.51	485.89	2.71	492.41	3.11	30.94	0.33
Iceland	435.90	2.06	458.90	1.58	446.93	1.76			30.46	0.25
Indonesia	358.57	2.91	365.53	2.35	382.86	2.56			18.96	0.39
Ireland	516.01	2.33	491.65	2.02	503.85	2.26				
Israel	473.83	3.49	457.90	3.27	464.75	3.38			32.28	0.39
Italy	481.60	2.68	471.26	3.09	477.46	3.18	483.51	3.11	31.40	0.31
Jamaica	409.63	4.21	377.42	3.14	402.93	3.88			25.54	0.54
Japan	515.85	3.18	535.58	2.93	546.63	2.80				
Jordan	342.17	2.40	361.23	2.03	374.53	2.35			20.21	0.36
Kazakhstan	386.28	1.66	425.44	1.69	423.17	1.72			23.84	0.29
Korea	515.42	3.63	527.30	3.86	527.82	3.58			38.09	0.39
Kosovo	342.19	1.06	354.96	1.02	357.02	1.26				
Latvia	474.57	2.46	483.16	2.03	493.84	2.30			35.07	0.27

Country	Reading		Mathematics		Science		Financial Literacy		Creative Thinking	
	Average PV	SE	Average PV	SE	Average PV	SE	Average PV	SE	Average PV	SE
Lithuania	471.83	2.21	475.15	1.84	484.46	2.33			32.86	0.28
Macao (China)	510.41	1.35	551.92	1.10	543.10	1.11			31.62	0.20
Malaysia	388.09	2.75	408.69	2.40	416.31	2.35	405.75	2.94	25.11	0.38
Malta	445.30	1.90	466.02	1.58	465.59	1.70			31.32	0.24
Mexico	415.36	2.92	395.03	2.27	409.89	2.42			28.99	0.32
Mongolia	378.42	2.25	424.59	2.57	412.38	2.36			24.92	0.32
Montenegro	405.02	1.35	405.60	1.12	403.13	1.21				
Morocco	339.36	3.97	364.77	3.35	365.40	3.38			15.48	0.58
Netherlands	459.24	4.28	492.68	3.77	488.32	4.07	516.88	4.42	32.39	0.46
New Zealand	500.85	2.12	479.07	1.99	504.13	2.24			36.43	0.29
North Macedonia	358.52	0.81	388.58	0.87	379.88	0.93			19.11	0.23
Norway	476.52	2.54	468.45	2.06	478.23	2.37	488.73	2.63		
Palestinian Authority	349.16	2.03	365.75	1.84	368.82	2.10			18.46	0.34
Panama	391.95	3.41	356.57	2.84	387.77	3.54			23.22	0.34
Paraguay	373.16	2.44	337.54	2.16	368.33	2.06				
Peru	408.25	2.73	391.24	2.34	407.78	2.64	420.75	3.04	23.45	0.35
Philippines	346.55	3.40	354.72	2.58	356.17	3.11			14.20	0.51
Poland	488.71	2.74	488.96	2.27	499.16	2.55	505.84	2.69	34.44	0.28
Portugal	476.59	2.66	471.91	2.35	484.37	2.56	494.42	2.37	33.90	0.29
Qatar	419.30	1.45	414.11	1.14	432.40	1.48			27.66	0.24
Republic of Moldova	410.94	2.51	414.20	2.31	416.86	2.39			23.95	0.32
Romania	428.50	3.98	427.76	4.00	427.51	3.87			26.25	0.47
Saudi Arabia	382.55	1.99	388.78	1.76	390.39	1.96	412.47	2.58	23.32	0.31
Serbia	440.35	2.79	439.88	2.97	447.46	2.89			28.68	0.35
Singapore	542.55	1.87	574.66	1.23	561.43	1.33			40.96	0.17
Slovak Republic	446.86	3.10	463.99	2.89	462.27	3.03			29.22	0.40
Slovenia	468.54	1.64	484.53	1.24	499.96	1.45			29.99	0.23
Spain	474.31	1.65	473.14	1.50	484.53	1.60	486.08	2.70	32.75	0.22
Sweden	486.98	2.49	481.77	2.06	493.55	2.35				
Switzerland	483.33	2.26	507.99	2.14	502.52	2.19				
Thailand	378.66	2.82	393.95	2.68	409.26	2.78			20.93	0.37
Türkiye	456.08	1.85	453.15	1.59	475.94	1.93				
Ukrainian regions (18 of 27)	427.53	3.93	440.85	4.06	450.19	3.78			26.89	0.61
United Arab Emirates	417.35	1.34	431.11	0.95	431.98	1.31	441.12	1.58	28.43	0.16
United Kingdom	494.40	2.37	488.98	2.22	499.67	2.38				
United States	503.94	4.33	464.89	4.01	499.41	4.32	505.23	4.92		
Uruguay	430.36	2.41	408.71	2.02	435.38	2.48			28.64	0.35
Uzbekistan	335.50	2.00	363.94	2.02	354.86	2.01			14.50	0.25
Viet Nam	461.89	3.94	469.40	3.93	472.38	3.59				

Note: Ukranian regions (18 out of 27) administered the assessment.
*Denotes a country/economy for which the financial literacy domain was not fully sampled across the population; it is not a nationally-representative sample.

Annex Table 14.A.13. Core domain inter-correlations for the main sample

DOMAIN		Reading	Science
Mathematics	Average	**0.80**	**0.85**
	Average (CBA)	0.80	0.86
	Average (PBA)	0.81	0.83
	Range	0.65 ~ 0.89	0.75 ~ 0.92
Reading	Average		**0.79**

	Average (CBA)		0.79
	Average (PBA)		0.81
	Range		0.67~ 0.88

Annex Table 14.A.14. Creative Thinking inter-correlations with core domains for the main sample

DOMAIN		Mathematics	Reading	Science
Creative Thinking	Average	**0.68**	**0.68**	**0.67**
	Range	0.53 ~ 0.80	0.55 ~ 0.83	0.54 ~ 0.80

Annex Table 14.A.15. Domain inter-correlations for the financial literacy sample

DOMAIN		Mathematics	Reading
Financial Literacy	Average	**0.86**	**0.84**
	Range	0.80 ~ 0.90	0.79 ~ 0.88

Annex Table 14.A.16. Domain inter-correlations by country/economy

Country	Mathematics & Reading	Mathematics & Science	Mathematics & Financial literacy	Mathematics & Creative Thinking	Reading & Science	Reading & Financial literacy	Reading & Creative Thinking	Science & Creative Thinking
Albania	0.69	0.76		0.66	0.67		0.58	0.60
Argentina	0.75	0.81			0.75			
Australia	0.80	0.86		0.65	0.78		0.63	0.64
Austria	0.84	0.90	0.88		0.85	0.86		
Baku (Azerbaijan)	0.75	0.82		0.64	0.72		0.63	0.63
Belgium*	0.82	0.90	0.89	0.69	0.82	0.85	0.69	0.69
Brazil	0.80	0.84	0.84	0.69	0.78	0.82	0.70	0.68
Brunei Darussalam	0.88	0.92		0.80	0.87		0.81	0.80
Bulgaria	0.83	0.86	0.87	0.76	0.80	0.85	0.74	0.74
Cambodia	0.79	0.78			0.75			
Canada*	0.75	0.80	0.85	0.56	0.72	0.81	0.55	0.54
Chile	0.79	0.86		0.61	0.78		0.58	0.57
Chinese Taipei	0.84	0.90		0.68	0.82		0.67	0.67
Colombia	0.80	0.86		0.69	0.77		0.68	0.68
Costa Rica	0.79	0.83	0.87	0.71	0.78	0.83	0.69	0.66
Croatia	0.79	0.86		0.67	0.77		0.67	0.68
Cyprus	0.75	0.82		0.72	0.74		0.70	0.68
Czech Republic	0.81	0.88	0.87	0.68	0.80	0.83	0.67	0.68
Denmark	0.79	0.87	0.87	0.62	0.77	0.84	0.61	0.61
Dominican Republic	0.80	0.80		0.64	0.77		0.67	0.62
El Salvador	0.80	0.81		0.67	0.76		0.66	0.66
Estonia	0.77	0.86		0.62	0.74		0.58	0.62
Finland	0.79	0.87		0.68	0.77		0.71	0.70
France	0.84	0.88		0.71	0.82		0.72	0.70
Georgia	0.75	0.81			0.74			
Germany	0.85	0.90		0.76	0.86		0.76	0.76
Greece	0.78	0.83		0.69	0.77		0.65	0.68
Guatemala	0.84	0.87			0.88			
Hong Kong (China)	0.79	0.84		0.63	0.76		0.61	0.60

Country	Mathematics & Reading	Mathematics & Science	Mathematics & Financial literacy	Mathematics & Creative Thinking	Reading & Science	Reading & Financial literacy	Reading & Creative Thinking	Science & Creative Thinking
Hungary	0.84	0.91	0.89	0.76	0.84	0.85	0.74	0.74
Iceland	0.77	0.85		0.67	0.77		0.68	0.68
Indonesia	0.78	0.77		0.57	0.72		0.55	0.54
Ireland	0.81	0.88			0.84			
Israel	0.81	0.88		0.76	0.80		0.74	0.73
Italy	0.76	0.84	0.82	0.64	0.75	0.79	0.62	0.61
Jamaica	0.84	0.86		0.67	0.82		0.71	0.69
Japan	0.81	0.88			0.84			
Jordan	0.72	0.80		0.66	0.74		0.68	0.66
Kazakhstan	0.65	0.75		0.53	0.71		0.62	0.59
Korea	0.76	0.85		0.59	0.74		0.59	0.61
Kosovo	0.78	0.83			0.77			
Latvia	0.79	0.88		0.57	0.78		0.55	0.57
Lithuania	0.81	0.88		0.71	0.81		0.69	0.69
Macao (China)	0.75	0.87		0.66	0.78		0.64	0.66
Malaysia	0.79	0.87	0.90	0.75	0.82	0.88	0.79	0.78
Malta	0.78	0.87		0.73	0.80		0.73	0.72
Mexico	0.82	0.86		0.66	0.80		0.67	0.66
Mongolia	0.79	0.87		0.71	0.78		0.69	0.70
Montenegro	0.79	0.86			0.77			
Morocco	0.77	0.83		0.72	0.75		0.70	0.68
Netherlands	0.86	0.90	0.90	0.72	0.85	0.88	0.74	0.71
New Zealand	0.81	0.88		0.69	0.85		0.71	0.71
North Macedonia	0.80	0.84		0.75	0.76		0.72	0.74
Norway	0.78	0.86	0.80		0.80	0.82		
Palestinian Authority	0.76	0.81		0.71	0.72		0.67	0.67
Panama	0.82	0.86		0.64	0.79		0.66	0.65
Paraguay	0.84	0.87			0.86			
Peru	0.82	0.86	0.86	0.71	0.79	0.87	0.70	0.68
Philippines	0.89	0.87		0.80	0.85		0.83	0.77
Poland	0.81	0.87	0.87	0.70	0.80	0.84	0.68	0.68
Portugal	0.81	0.87	0.85	0.70	0.80	0.83	0.70	0.69
Qatar	0.81	0.86		0.72	0.79		0.70	0.70
Republic of Moldova	0.83	0.87		0.70	0.81		0.74	0.71
Romania	0.86	0.90		0.78	0.85		0.77	0.77
Saudi Arabia	0.75	0.78	0.80	0.66	0.73	0.82	0.67	0.65
Serbia	0.81	0.87		0.70	0.79		0.68	0.70
Singapore	0.82	0.89		0.67	0.81		0.66	0.66
Slovak Republic	0.83	0.89		0.74	0.81		0.72	0.73
Slovenia	0.77	0.89		0.60	0.77		0.59	0.58
Spain	0.76	0.82	0.86	0.59	0.75	0.79	0.59	0.58
Sweden	0.81	0.88			0.84			
Switzerland	0.83	0.89			0.86			
Thailand	0.79	0.83		0.68	0.77		0.67	0.68
Türkiye	0.82	0.90			0.83			
Ukrainian regions (18 of 27)	0.79	0.86		0.72	0.79		0.67	0.72
United Arab Emirates	0.81	0.85	0.87	0.71	0.79	0.84	0.71	0.69
United Kingdom	0.81	0.86			0.79			
United States	0.83	0.89	0.89		0.87	0.86		
Uruguay	0.80	0.87		0.72	0.79		0.69	0.71
Uzbekistan	0.72	0.78		0.67	0.70		0.63	0.63
Viet Nam	0.77	0.82			0.75			

Note: Ukranian regions (18 out of 27) administered the assessment.
*Denotes a country/economy for which the financial literacy domain was not fully sampled across the population; it is not a nationally-representative sample.

Annex Table 14.A.17. Mathematics content subscales inter-correlations

	MCCR1	MCQN2	MCSS3	MCUD4
Reading	0.71	0.72	0.63	0.71
Science	0.76	0.77	0.68	0.75
MCCR1		0.86	0.77	0.82
MCQN2			0.79	0.85
MCSS3				0.76

Annex Table 14.A.18. Mathematics cognitive process subscales inter-correlations

	MPEM1	MPFS2	MPIN3	MPRE4
Reading	0.72	0.66	0.73	0.69
Science	0.77	0.71	0.77	0.74
MPEM1		0.83	0.87	0.84
MPFS2			0.81	0.79
MPIN3				0.82

Annex Table 14.A.19. Linking error for score comparisons between PISA 2022 and previous PISA cycles

Comparison	Mathematics	Reading	Science	Financial literacy
PISA 2000 to 2022		6.67		
PISA 2003 to 2022	5.55	5.25		
PISA 2006 to 2022	4.09	8.56	3.68	
PISA 2009 to 2022	4.28	4.66	5.92	
PISA 2012 to 2022	3.58	6.01	5.20	4.05
PISA 2015 to 2022	2.74	3.63	1.38	3.47
PISA 2018 to 2022	2.24	1.47	1.61	2.20

15 Coding Design, Coding Process, and Reliability Studies

Introduction

The PISA 2022 assessment consisted of both constructed-response (CR) and multiple-choice items (MC). MC items could be simple multiple choice, with a single correct response selection, or complex multiple choice, with multiple correct response selections required. MC items had a predefined correct answer that could be computer coded. While a few CR items were designed to be coded by computer, most required a person to read the response and provide a code or score. These items are referred to as human-coded constructed response items.

This chapter describes the design, preparation, and processing of coding human-coded constructed-response (CR) items, and reports the reliability statistics and volume of responses that could be automatically coded for these items. A summary of all test items by domain, item format, and coding method is shown in Annex Table 15.A.2.

The CBA mathematics assessment was administered within each country/economy as both a linear test and a Multistage Adaptive Test (MSAT). The CBA reading assessment was also administered as an MSAT with three stages, while the science assessment was administered using a linear design. Countries participating in the CBA also had the option of administering a Financial Literacy assessment and a Creative Thinking assessment. One country chose to participate in the paper-based assessment (PBA), which has been administered since 2015, and three countries participated in the new paper-based assessment. More on the PISA 2022 test design is presented in Chapter 2 of this Technical Report.

Coding design

Coding designs for CBA, PBA, and the new PBA were developed to accommodate the various needs of countries/economies in terms of the number of languages assessed, sample size, and assessed domains (i.e., meaning whether Financial Literacy or the innovative domain were to be coded in the country/economy). In general, it was expected that coders would be able to code approximately 1 000 responses per day, over a two- to three-week period. The number of expected student responses per domain was based on the sample size completing the assessment in each assessed language in the core domains and in the optional domains of Financial Literacy and Creative Thinking.

Annex Table 15.A.3 shows the number of coders recommended by domain in the CBA coding designs based on the sample size. This design is exclusive by language of assessment. CBA participants were able to determine the appropriate design for their country/economy and language(s) with a coding estimation tool, which estimated the coding workload for each coder (duration of coding and the number of responses to be coded by each coder).

Annex Table 15.A.3 also includes an example of this estimated workload.

PBA and new PBA countries' sample sizes had little variation, and there were no additional domain options; therefore, all countries participating in these assessments were advised to recruit six coders for each domain. Annex Table 15.A.4 shows the estimated workload for six coders in each domain.

Designs for within-country and across-country scoring reliability

Reliable human coding is critical for ensuring the validity of assessment results within a country, as well as the comparability of assessment results across countries (Shin, von Davier and Yamamoto, 2019[1]). Throughout the chapter, we use the term *coding* to refer to the assignment of a numerical value to a student text response, which indicates the type of response provided by the student, and the term *scoring* to refer to the assignment of full credit, partial credit, or no credit, which is derived from the codes applied. Scoring reliability in PISA 2022 was evaluated and reported at both within- and across-country levels.

The purpose of monitoring and evaluating within-country scoring reliability is to ensure accurate scoring of student responses across coders in the same county-by-language group and identify any coding inconsistencies or problems in the scoring process throughout the process so that they can be promptly addressed and resolved. Within-country scoring reliability was evaluated by reviewing the codes assigned by two or more human coders on the same student responses in a process called multiple coding. *Multiple coding* refers to the coding of the same student response data by different independent coders, such that inter-rater agreement statistics can be calculated and evaluated.

It was also important to check the consistency of coders across countries and language groups. Accurate and consistent *scoring* (full credit, partial credit, no credit) within a country does not necessarily mean that coders from all countries and language groups are applying the coding rubric in the same manner. Coding bias may be introduced if, for example, one country codes a certain type of response differently than other countries (Shin, von Davier and Yamamoto, 2019[1]). Across-country scoring reliability was evaluated by checking the correctness of the codes assigned by two bilingual human coders on a set of English anchor responses in a process called anchor coding. *Anchor coding* refers to the coding of a set of common (across-country-by-language groups) responses in English for each item, for which the correct code for each response is already known by the PISA international contractor (but not provided to coders). Because countries coded the same anchor responses for each human-coded CR item, their coding results on the anchor responses could be compared to the anchor key and, thereby, to each other. For each human-coded CR item, a set of thirty anchor responses in CBA, and ten in PBA and New PBA, were distributed to the designated bilingual coders for coding.

In CBA, item responses were randomly selected from all student responses and gathered into coding sets for multiple coding. In the domains of Mathematics, Science, Financial Literacy, and Creative Thinking, one coding set was compiled, such that all coders contributed to the multiple-coding agreement for all items. In the domain of Reading, items were distributed among four coding sets, such that each coder only saw responses to half of the items and thus contributed only to the scoring reliability for the items in their assigned coding set. Each domain had two bilingual coders – always coders 01 and 03 – who additionally coded thirty anchor responses in English for each item in their coding set. The design for multiple coding for the CBA is shown in Figure 15.1.

For multiple coding in the paper-based designs, student test booklets are first sorted by booklet number. Because each test booklet contains responses from two administered domains (for example, Mathematics and Science were administered in booklets 1-6 in PBA), coding sets are first multiple coded by coders in one administered domain and then single coded by the coders in the other administered domain. A specified number of booklets (52 booklets of each booklet number in PBA and 90 of each number in the new PBA) are designated for multiple coding. These booklets are distributed equally among six coding sets and distributed to coders. In PBA, all coders code all coding sets, whereas a subset of coders code each coding set in new PBA. The PBA and new PBA coding designs are shown in Figure 15.2.

Figure 15.1. Organization of multiple coding for the CBA designs

Mathematics

	Number of Coded Responses	301 (bilingual)	302	303 (bilingual)	304	305	306	307	308	309	310	311	312	313	314	315	316
Coding Set	128/item	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Anchor Set	30/item	●		●													

Coder IDs

Reading

	Number of Coded Responses	201 (bilingual)	202	203 (bilingual)	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232
Coding Set 1	100/item	●	●					●	●	●			●	●			●	●			●	●			●	●			●	●			●
Coding Set 2		●	●			●	●			●		●		●		●		●		●		●		●		●		●		●		●	
Coding Set 3				●	●	●	●				●	●			●	●			●	●			●	●			●	●			●	●	
Coding Set 4				●	●			●	●		●		●		●		●		●		●		●		●		●		●		●		●
Anchor Set	30/item	●		●																													

Coder IDs

Science

	Number of Coded Responses	101 (bilingual)	102	103 (bilingual)	104	105	106	107	108	109	110	111	112
Coding Set	128/item	●	●	●	●	●	●	●	●	●	●	●	●
Anchor Set	30/item	●		●									

Coder IDs

Financial Literacy

	Number of Coded Responses	401 (bilingual)	402	403 (bilingual)	404	405	406	407	408	409	410	411	412
Coding Set	100/item	●	●	●	●	●	●	●	●	●	●	●	●
Anchor Set	30/item	●		●									

Coder IDs

Creative Thinking

	Number of Coded Responses	501 (bilingual)	502	503 (bilingual)	504	505	506	507	508	509	510	511	512	513	514	515	516	517	518	519	520	521	522	523	524
Coding Set	100/item	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●	●
Anchor Set	30/item	●		●																					

Figure 15.2. Organization of multiple coding for the PBA and New PBA standard coding design

PBA

Mathematics

		Number of Coded Responses	301 (bilingual)	302	303 (bilingual)	304	305	306
Booklets 7-12 52 of each booklet number (312 total)	Coding Set 13	120-138/item	●	●	●	●	●	●
	Coding Set 14		●	●	●	●	●	●
	Coding Set 15		●	●	●	●	●	●
	Coding Set 16		●	●	●	●	●	●
	Coding Set 17		●	●	●	●	●	●
	Coding Set 18		●	●	●	●	●	●
Booklet 92	Anchor Coding Set	10/item	●		●			

Reading

		Number of Coded Responses	201 (bilingual)	202	202 (bilingual)	204	205	206
Booklets 13-18 52 of each booklet number (312 total)	Coding Set 25	104/item	●	●	●	●	●	●
	Coding Set 26		●	●	●	●	●	●
	Coding Set 27		●	●	●	●	●	●
	Coding Set 28		●	●	●	●	●	●
	Coding Set 29		●	●	●	●	●	●
	Coding Set 30		●	●	●	●	●	●
Booklet 91	Anchor Coding Set	10/item	●		●			

Science

		Number of Coded Responses	101 (bilingual)	102	102 (bilingual)	104	105	106
Booklets 1-6 52 of each booklet number (312 total)	Coding Set 1	104/item	●	●	●	●	●	●
	Coding Set 2		●	●	●	●	●	●
	Coding Set 3		●	●	●	●	●	●
	Coding Set 4		●	●	●	●	●	●
	Coding Set 5		●	●	●	●	●	●
	Coding Set 6		●	●	●	●	●	●
Booklet 93	Anchor Coding Set	10/item	●		●			

New PBA

Mathematics

		Number of Coded Responses	301 (bilingual)	302	303 (bilingual)	304	305	306
Booklets 5-8 90 of each booklet number (360 total)	Coding Set 13	180/item	●	●	●		●	
	Coding Set 14			●	●	●		●
	Coding Set 15		●		●	●	●	
	Coding Set 16			●		●	●	●
	Coding Set 17		●		●		●	●
	Coding Set 18		●	●		●		●
Booklet 93	Anchor Coding Set	10/item	●		●			

Reading

		Number of Coded Responses	201 (bilingual)	202	202 (bilingual)	204	205	206
Booklets 1-4 90 of each booklet number (360 total)	Coding Set 1	180/item	●	●	●		●	
	Coding Set 2			●	●	●		●
	Coding Set 3		●		●	●	●	
	Coding Set 4			●		●	●	●
	Coding Set 5		●		●		●	●
	Coding Set 6		●	●		●		●
Booklet 92	Anchor Coding Set	10/item	●		●			

Science

		Number of Coded Responses	101 (bilingual)	102	102 (bilingual)	104	105	106
Booklets 9-12 90 of each booklet number (360 total)	Coding Set 25	180/item	●	●	●		●	
	Coding Set 26			●	●	●		●
	Coding Set 27		●		●	●	●	
	Coding Set 28			●		●	●	●
	Coding Set 29		●		●		●	●
	Coding Set 30		●	●		●		●
Booklet 91	Anchor Coding Set	10/item	●		●			

Coding preparation

Prior to the assessment, key activities were completed by National Centres to prepare for the process of coding CR items.

Recruitment of national coder teams

The first task of National Project Managers (NPMs) on the coding workflow was to assemble a national coder team. The size of the coding teams varied in each country, but the following criteria were used for selecting members of the team:

- All coders should have more than an upper secondary education qualification (i.e., high school degree); university graduates were preferred.
- All should have a good understanding of secondary education level studies in the relevant domains.
- All should be available for the duration of the coding period, which was expected to last two to three weeks.

- Due to normal attrition rates and unforeseen absences, it was strongly recommended that lead coders train a backup coder for their teams.
- Two coders for each domain must be bilingual in English and in the language(s) of the assessment.

After the national coding team was assembled, the next task was to identify a *lead coder* who was part of the coding team but also responsible for the following tasks:

- training coders within the country,
- organising all materials and distributing them to coders,
- monitoring the coding process,
- monitoring inter-rater reliability and taking action when the coding results were unacceptable or required further investigation,
- Producing reliability reports
- retraining or replacing coders if necessary, and
- consulting with the international experts if item-specific issues arose.

Additionally, the lead coder was required to be proficient in English, as international trainings and interactions with the PISA international contractors were in English only, and was encouraged to attend the international coder trainings. It was also assumed that the lead coder for the field trial would retain the role for the main survey. When this was not the case, it was the responsibility of the National Centre to ensure that the new lead coder received training equivalent to that provided at the international coder training prior to the main survey.

Coder training materials

Detailed coding guides were developed for all the new items in the domains of Mathematics, Financial Literacy, and Creative Thinking. These coding guides included coding rubrics for each item and example responses corresponding to each level (i.e., correct, partially correct, and incorrect) of the rubric. Coding rubrics for new items were revised for the main survey based on information learned from the field trial. Coding guides for trend domains were also prepared, but changes were limited to the correction of errors.

In addition to the coding guides, a separate workshop-materials file was either created for new domains or updated for trend domains. Unlike the coding guides which remain relatively static across cycles, the workshop-materials file can be updated. The workshop materials files contain additional example responses and annotations, which could be used to supplement the coder trainings. The additional example responses better illustrate the depth and breadth of the coding levels, and the lines between levels. Following the international trainings, final versions of all materials were prepared and released to participating countries/economies.

International coder trainings

Prior to the field trial, NPMs and lead coders were provided with a full item-by-item coder training for CBA, PBA, and new PBA participants in Athens, Greece in January 2020. The field trial training covered all items in all domains. Due to the one-year delay caused by the COVID-19 pandemic, a second international field trial training was held in January and February 2021. The second field trial training took place over several sessions and was conducted virtually. Additionally, the second field trial training covered only new material. That is, the sessions offered were for the new Mathematics items, all Creative Thinking items, and the four new Financial Literacy CR items.

Prior to the main survey, international coder trainings were held in January and February 2022, and were again conducted virtually for all domains. Full trainings were offered for all the new and all the trend Mathematics items, all the Creative Thinking items, and the new Financial Literacy items. Targeted trainings were offered for the trend domains (Science, Reading, and the trend items in Financial Literacy); that is, the international experts reviewed analysis results from the field trial and considered items that have been historically challenging to code, and targeted items for which a refresher training would be most beneficial. Participants were also given the opportunity to ask questions about trend items if they were not already on the list prepared by the experts. Participants were also provided with the recorded trainings that were prepared for PISA 2018, which cover the items in the trend domains, and could be used to supplement the targeted virtual trainings.

Full trainings were provided virtually in April and May 2022 for PBA and New PBA participants for all domains. During these trainings, the coding guides were presented and explained, and participants had the opportunity to ask questions to have the coding rubrics clarified. Participants also practiced coding on sample responses and discussed any ambiguous or problematic situations as a group. When the discussion revealed areas where rubrics could be improved, those changes were noted and eventually implemented in an updated version of the coding guide that was made available after the meeting. The workshop-materials files were also updated as needed following the international trainings.

To support the national teams during the coding process, a coding query service was offered, which allowed national teams to submit coding questions and receive responses from the relevant domain experts. National teams were also able to see questions submitted by other countries/economies pertaining to the coding of new items, along with the responses from the test developers. In the case of trend items, responses to queries from previous cycles were also provided. A summary report of coding issues was provided on a regular basis, and all related materials were stored on the PISA Portal for reference by national coding teams.

National coder training provided by the National Centres

Each National Centre was required to develop a training package and replicate as much as possible from the international training for their own coders. The training package consisted of an overview of the survey and their own training manuals based on the source manuals and materials provided by the PISA international contractors. Coding teams were asked to facilitate discussion about any items that were challenging to code. Past experience has shown that when coders discuss items among themselves and with their lead coder, many issues can be resolved, and more consistent coding can be achieved.

The National Centres were responsible for organising training and coding. The recommended approach was to train at the item level. Under this approach, coders were fully trained on the coding rules for one item, and then proceeded with coding all responses for that one item. Once the item was fully coded, training was provided for the next item (blocked by unit), and so on. The approach of coding item by item has been shown to improve reliability by helping coders to apply the scoring rubric more consistently.

For PBA and new PBA participants, coder training was also recommended at the item level; however, training could be given at the unit level. Once the training was complete on the items within a single unit, coding could take place across booklet for all the items within that one unit.

Coding procedures

Since PISA 2015, coding CBA item responses has been facilitated through use of the Open-Ended Coding System (OECS), which allows coders to view student responses, defer responses for further review, and code responses directly in the system interface. The OECS supported coding teams in their work to code the CBA responses while ensuring that the coding design was appropriately implemented. Especially important during the COVID-19 pandemic, the OECS afforded coders the ability to work remotely. Detailed information about the system was included in the OECS manual provided to countries/economies.

Computer-based responses were coded on an item-by-item basis. For each item, coders receive a set of responses to be coded. Each set includes 1) student responses to be multiple coded as part of the within-country reliability monitoring process, and 2) student responses to be single coded. If the coder is one of the national team's two bilingual coders they also received anchor responses in English will also be included for across-country reliability monitoring. Because the generation of inter-rater agreement statistics were continuously being updated by the OECS as coders code (see Formula 13.1 in the *Reliability Studies* section), no pause in coding is required in the CBA to manually calculate these statistics, allowing coders to work at their own pace through all assigned responses.

When a coder logs into the system and selects an item to code, responses that require human coding appear on screen. Buttons at the top of the screen allow the coder to scroll through responses. In general, multiple-coded responses are populated first and then single-coded responses; for bilingual coders, anchor responses appear ahead of all student responses. For each response, the OECS displays the item stem or question, the individual response, and the available codes for the item, as well as a checkbox to *defer* the response to the lead coder and a checkbox to indicate that the response has been *recoded* from the originally applied code to a new code for some reason. It is expected that coders will code most responses assigned to them and defer responses only in unusual circumstances. When deferring a response, coders were encouraged to note the reason for deferral into an associated comment box. Coders generally worked on one item at a time until all responses in that item set were coded. The process was repeated until responses for all items were coded. Detailed information about the system was provided in the OECS manual.

For the PBA and New PBA, the coding designs were supported by the Data Management Expert (DME) system, and reliability was monitored through the Open-Ended Reporting System (OERS), an additional software that worked in conjunction with the DME to evaluate and report reliability for CR items. The coding process for paper-based participants involved using the actual paper booklets, with sections of some booklets single-coded and some sections coded multiple times. When a response is single coded, coders mark directly in the booklets. When a response is coded multiple times, only the final coder codes directly in the booklet, while all others code on coding sheets; this allows coders to remain independent in their coding decisions and provides an accurate evaluation of scoring reliability. Detailed information about the system was provided in the OERS manual to PBA countries/economies.

Unlike coding in the CBA, the process of coding in PBA and New PBA does require a pause between coding different sets of responses (anchor responses, multiple-coded responses, and single-coded responses), resulting in three distinct coding phases. In the first phase of coding, bilingual coders code the anchor responses, enter the data into the project database using the DME and evaluate the across-country scoring reliability using the OERS. In the second phase, at least 100 student responses for each item are multiple coded. Single-coded responses are addressed in the final phase. All anchor- and multiple-coded response codes are entered into the project database using the DME and run the OERS reliability software for review. Any coding issues identified by the OERS are investigated and corrected before moving forward. The distributions of single codes are also reviewed in the OERS, as a quality check.

National Centres used the output reports generated by the OECS and OERS to monitor irregularities and deviations in the coding process. The OECS and OERS generate the following reports of scoring reliability: i) percentage of first-digit code agreement on multiple and anchor coded responses and ii) coding category distribution across coders. NPMs were instructed to investigate whether a systematic pattern of irregularities existed and if the observed pattern was attributable to a particular coder or item. In addition, NPMs were instructed not to carry out *coding resolution* (changing coding on individual responses to reach higher coding consistency). Instead, if systematic irregularities were identified, coders were to be retrained and all responses from a particular item or a particular coder were to be recoded, including those codes that showed agreement. Coding inconsistencies usually come from a misunderstanding of the general coding guidelines and/or a rubric for a particular item. Reliability studies conducted by the PISA contractors also made use of the OECS and OERS reports submitted by National Centres.

Reliability studies

Careful monitoring of scoring reliability plays an important role in data quality control. National Centres used the output reports generated by the OECS and OERS to monitor irregularities and deviations in the coding process for both items and individual coders. Through these processes of reliability monitoring, coding inconsistencies or problems within and across countries could be detected early in the coding process, and action could be taken quickly to address these concerns.

Within-country monitoring of scoring reliability

While coding was ongoing, score agreement and coding category distribution were the main indicators used by National Centres for monitoring coding.

- *Score agreement* refers to the proportion of scores (generally the first digit of assigned codes, denoting full, partial, and no credit) from one coder that exactly matched the scores of other coders on an identical set of multiple-coded responses for an item (including scores on partial credit item responses). Agreement can vary from 0 (0% agreement) to 1 (100% agreement). Each country/economy was expected to meet a scoring *standard* within-country and across-country proportion of at least 85% agreement on each item or coder in Mathematics, Reading, Science, and Financial Literacy; this standard was set to 70% agreement for Creative Thinking. Further, an average domain-level standard across all items in a domain of 92% was expected, except for Creative Thinking, which was also set to 70%. The design called for a minimum of one-hundred responses for each item to be multiple coded for the calculation of within-country score agreement; when fewer than 100 responses for an item in a particular country-by-language group were collected, as was the case of small samples, all responses were multiple coded. Additionally, ten (paper-based) or thirty (computer-based) English responses for each item were anchor coded for the calculation of across-country score agreement.
- *Coding category distribution* refers to the distributions of coding categories (such as “full credit”, “partial credit” and “no credit”) assigned by a coder to two sets of responses: a set of 100 responses for multiple coding and responses randomly allocated to the coder for single coding. Notwithstanding that negligible differences of coding categories among coders were tolerated, the coding category distributions between coders were expected to be statistically equivalent based on the standard chi-square distribution due to the random assignment of the single-coded responses.

During coding, the formula used to by the OECS to calculate ongoing interrater agreement was:

$$R_{ji} = \frac{G_{ji}\left(\frac{N-A}{N}\right)}{D_{ji}(C-1)} + \frac{A}{N} \qquad \text{Formula 15.1}$$

where R_{ji} is the calculated agreement rate for coder C_j for item i, N is the total number of responses for item i, A is the number of automatically coded responses for item i, C is number of coders for the item, G_{ji} is the number of agreed codes for coder C_j for item i (max = (C-1)), and D_{ji} is the number of multiple-coded responses for item i coded by coder j so far (at the end of coding, this will equal 100 in a standard sample). The OERS reports calculated agreement similarly, with the exception that no responses were automatically coded (so, A = 0, simplifying the equation).

Score agreement across countries/economies, languages, and items

Scoring reliability was again reviewed by the PISA contractor following the completion of coding to check for scoring consistency of human-coded CR items within and across countries participating in PISA 2022. For comparability among country-by-language groups and between multiple-coded student responses and anchor-coded English responses, the proportion of automatically coded responses were disregarded, and only the scoring reliability of human coders was considered. The reliability studies included 78 CBA countries/economies, resulting in 124 country-by-language groups. One PBA country, and three new PBA countries, each with one language group. In total there were 128 country-by-language groups across modes of assessment.

In a review of country-level data, quality and consistency of score agreement within and across country-by-language groups was evaluated. High score agreement is generally reflective of quality coding: that national and international coder trainings were well-implemented, coding guides were reflective of the student responses, such that scores could be consistently applied, and the scores applied on human-coded CR items are reliably accurate. All country-by-language groups were reviewed to see if the score agreement standard was met on all items and domains. Annex Web Table 15.A.5, (please refer to Annex Table 15.A.1 for access to this table) Annex Table 15.A.6, Annex Table 15.A.7 report the domain-level score agreement for all PISA 2022 participating countries and economies.

Overall, the majority of country-by-language groups administering the CBA met the domain-level within-country score agreement standard of 92% (or 70% in Creative Thinking):

- In Mathematics, 98.4% of country-by-language groups met the domain-level scoring standard; those below averaged 91.0% score agreement on multiple-coded responses.
- In Reading, 89.5% of country-by-language groups met the domain-level scoring standard; those below averaged 90.7% score agreement on multiple-coded responses.
- In Science, 73.4% of country-by-language groups met the domain-level scoring standard; those below averaged 90.7% score agreement on multiple-coded responses.
- In Financial Literacy, 86.7% of country-by-language groups met the domain-level scoring standard; those below averaged 91.4% score agreement on multiple-coded responses.
- In Creative Thinking, 97.0% of country-by-language groups met the domain-level scoring standard; those below averaged 69.6% score agreement on multiple-coded responses.

Note that in some cases, 100% score agreement was observed in certain country-by-language groups. This is more likely to occur when the number of responses being multiple coded is fewer than the recommended 100 student responses, usually due to a small sample size.

Quality in the coding of the English anchor responses is also important for ensuring that the coding guides have applied in the same way across countries/economies and language groups. Most country-by-language groups administering the CBA also met the relevant domain-level across-country score agreement standard of 85% (or 70% in Creative Thinking):

- In Mathematics, 92.7% of country-by-language groups met the domain-level scoring standard; those below averaged 86.0% score agreement on 30 anchor responses.
- In Reading, 87.1% of country-by-language groups met the domain-level scoring standard; those below averaged 86.0% score agreement on 30 anchor responses.
- In Science, 68.5% of country-by-language groups met the domain-level scoring standard; those below averaged 88.4% score agreement on 30 anchor responses.
- In Financial Literacy, 83.3% of country-by-language groups met the scoring standard; those below averaged 90.0% score agreement on 30 anchor responses.

- In Creative Thinking, 97.0% of country-by-language groups met the scoring standard; those below averaged 63.7% score agreement on 30 anchor responses.

Finally, all paper-based countries met the standard for across- and within-country score agreement all domains.

Annex Table 15.A.8 summarizes Annex Web Table 15.A.5 (please refer to Annex Table 15.A.1), Annex Table 15.A.6 and Annex Table 15.A.7 providing an overall breakdown of score agreement of items by domain.

Across most domains and modes of assessment, across-country score agreement tended to be slightly lower than the within-country agreement by domain in the majority of country-by-language groups. This may be expected because, compared to multiple-coding, there are fewer bilingual coders (only two from the coding team) and fewer anchor-coded responses contributing to the calculation of agreement. However, the difference between multiple-coding and anchor-coding agreement by domain is generally minimal across country-by-language groups. In the domains of Mathematics, Reading, Science, and Financial Literacy, there was about 1-3% difference between the within-country agreement and the across-country agreement at the domain level in country-by-language groups, with only a few exceptions. In Creative Thinking, the domain level difference in agreement was closer to 7%, but with a lower threshold for standard of agreement, there is more room for fluctuation in agreement statistics, so this can also be expected.

Coder-level score agreement

Coder quality was also reviewed, particularly the percentage of coders in a country-by-language group that did not meet the standard level of agreement on across several items. Annex Table 15.A.9 and Annex Table 15.A.10 summarize overall coder quality and the impact of coder quality by item. In general, the coding standard indicates that all coders should agree with their fellow coders at least 85% of the time on each item, except in Creative Thinking, in which 70% score agreement was considered acceptable. Annex Table 15.A.9 shows the percentage of coders who were unable to reach the 85% agreement threshold on 20% or more items assigned to them. In Mathematics, 2.5% of coders agreed with their fellow coders less than 85% of the time on at least 20% of new item responses selected for multiple coding, and 0.8% were below this standard for trend item responses; this was also true of 8.8% of coders in Reading, 3.1% of coders in Science, and 0.7% of coders in Financial Literacy. In Creative Thinking, 15.7% of coders agreed with their fellow coders less than 70% of the time on at least 20% of responses.

Because coder quality is reflected at the item level, the percentage of items in the domain over which two or more coders did not meet the standard level agreement on that item was also evaluated, and the results are presented in Annex Table 15.A.10. Because there are a varying number of items in each domain, this table expresses the percentage of cases across all country-by-language groups. In other terms, however, about half of the CBA country-by-language groups had one new Mathematics item for which two coders did not meet the established 85% score agreement, and a fraction of that had this issue with a trend Mathematics item. Most country-by-language groups would have had about two reading items for which at least two coders did not meet the scoring standard, and in science, one item. About a third of all groups administering financial literacy would have had two coders below the standard on one item, and in Creative Thinking, all groups would have had about two items for which two or more coders did not reach 70% score agreement. There were no items across the paper-based domains for which there were two or more coders below the standard of score agreement on an item. These results overall suggest that any significant coding issues that may have arisen during coding were resolved at the National Centres.

Item-level agreement

The scales on which the PISA statistical framework is built are only as good as the scores used to establish them, so the overall agreement on student responses was also reviewed at the item-level, taking into account the proportion of responses that could be automatically coded. Here, the interest is to determine the proportion of items in a country-by-language group that did not meet the standard level of agreement. Again, at the item level, the score agreement standard was set to 85% in all domains and modes of assessment, except for Creative Thinking, in which the standard was set to 70% score agreement. These standards were met for most items in each country-by-language group. Annex Table 15.A.11 shows the number of country-by-language groups that had either no items in a domain (n = 0) below the standard, between one and five items (1 ≤ n ≤ 5), or up to ten items (6 ≤ n ≤ 10) in a domain below the score agreement standard. In the paper administration, all countries met the scoring standard on all items in all three domains. In the computer administration, all country-by-language groups met the standard on all items in Science and Financial Literacy. In Mathematics, one country-by-language group had items that failed to meet the standard, in Reading, four groups, and in Creative Thinking, five country-by-language groups had 1-5 items below the standard, and two had 6-10 items below the standard.

Machine-supported coding system

During the 2022 cycle, the CBA coding teams were able to benefit from the use of a machine-supported coding system (MSCS). The MSCS operates effectively due to a high response regularity among collected student data. Consider that, although an item's response field is open-ended, there is a commonality among students' raw responses, meaning that the same or similar correct or incorrect responses can be expected regularly throughout coding (Yamamoto et al., 2017[2]; 2018[3]). High regularity in responses means that variability among all responses for an item is small, and a large proportion of identical responses can receive the same code when observed a second or third time. In such cases, human coding can be replaced by machine coding, greatly reducing the human coding burden and minimizing the error present in human-coded data, often associated with fatigue or carelessness.

Unlike commonly used automated scoring systems that generally involve algorithms, the MSCS relies entirely on text data that have already been human coded in past PISA cycles and during the field trial. These observed text responses and their associated verified codes from past administrations are stored in a *Coded Unique Response* (CUR) pool for each country-by-language group. In order for a text response to receive a verified code and be added to the CUR pool, the response must have appeared at least five times, and coders must have 100% agreement on the code to apply. The MSCS approach parallels automated scoring in the sense that a scoring model is first trained on existing historic data (2015 and 2018 PISA cycles and the 2022 field trial) and then applied to future data (2022 main survey). When raw student responses are received, and before they are distributed to human coders in the OECS, they are first checked to see if the MSCS can automatically apply a code. Raw responses fall into one of three categories: 1) nonresponse, 2) responses with verified coding in the CUR pool, and 3) infrequent or unseen responses that require human judgment. The MSCS can be applied to the first two categories. Human coding would only be required for unique, unseen responses (3). The MSCS is specific to each country-by-language group; responses that are identified for automatic coding are not shared among country-by-language groups. In brief, the MSCS identifies blank responses and the exact same responses that have been previously coded by humans and automatically applies the appropriate code, minimizing the need to score responses that have already been added to the database (Yamamoto et al., 2017[2]; 2018[3]; OECD, 2018[4]).

Reduction of human-coding burden as the result of the MSCS

Annex Table 15.A.12 and Annex Table 15.A.13 summarize the efficiency of the MSCS with the reduction of human-coding burden in the PISA 2022 field trial and main survey. The tables summarize the percentage of responses coded by the MSCS and by human coders across all items in four domains (mathematics, reading, science, and financial literacy) and across country/economy language groups using mean and median. Given that the distribution of proportions for each item per group can be skewed, medians are reported in addition to the mean values.

The first two columns under the "machine-coded" header, *CUR* and *nonresponse*, indicate the average and median percentage of responses across CBA items that were automatically coded by the MSCS as either a nonresponse or a verified response (correct – full and partial credit – and incorrect). The total of these values is also presented, which can be compared to the percentage of human-coded responses, noted in the first column. Note that without the MSCS, all of the responses to CR items would have had to be coded by humans, including nonresponses. On average, across items and country-by-language groups, the coding burden for human coders was reduced for the 2022 field trial from a low of approximately 14% on new Mathematics items to a high of 31% on trend Mathematics items. For the 2022 main survey, the coding burden was reduced by a low of approximately 7% in Creative Thinking, for which only nonresponses were coded by the MSCS, to a high of 35% (about 15% CUR and 20% nonresponse) on trend Mathematics items.

For both field trial and main survey, approximately 7% to 20% of the total responses (on average) across all domains were empty responses and were automatically coded by the system. On new items, where no historic data were available, the MSCS reduced coding burden for human coders by 12% to 14% in Mathematics and 7% to 15% in Creative Thinking. For new Mathematics items that received modification following the field trial, only empty responses were automatically coded during the main survey, which may explain why only 5% of new Mathematics responses were automatically coded through the CUR pool. Because of the format and generally graphical nature of the Creative Thinking domain, only empty responses were automatically coded by the MSCS in both the field trial and the main survey. Overall, a similar or slightly higher percentage of responses were coded in each of the core domains in the 2022 PISA cycle than in the previous cycle.

References

OECD (2018), *PISA 2018 Technical Report*, PISA, OECD Publishing, Paris, https://www.oecd.org/en/about/programmes/pisa/pisa-data.html. [4]

Shin, H., M. von Davier and K. Yamamoto (2019), “Investigating Rater Effects in International Large-Scale Assessments”, in Veldkamp, B. and C. Sluijter (eds.), *Theoretical and Practical Advances in Computer-based Educational Measurement: Methodology of Educational Measurement and Assessment*, Springer, Cham. [1]

Yamamoto, K. et al. (2018), “Development and implementation of a machine-supported coding system for constructed-response items in PISA”, *Psychological Test and Assessment Modeling*, Vol. 60/2, pp. 145-164. [3]

Yamamoto, K. et al. (2017), “Developing a machine-supported coding system for constructed-response items in PISA”*, ETS Research Report*, No. RR-17-47, Educational Testing Service, Princeton, NJ, https://doi.org/10.1002/ets2.121. [2]

Annex 15.A. Detailed Overview of the Coding Process

Annex Table 15.A.1. Chapter 15: Comprehensive Analysis of Coding Practices

Tables	Title
Table 15.A.2	Number of cognitive items by domain, item format, and coding method
Table 15.A.3	CBA coding number of coders by domain
Table 15.A.4	PBA and New PBA number of coders by domain
Web Table 15.A.5	Summary of within- and across-country (%) scoring agreement for CBA participants for reading, mathematics and science
Table 15.A.6	Summary of within- and across-country (%) agreement for Financial Literacy and Creative Thinking domains
Table 15.A.7	Summary of within- and across-country (%) scoring agreement for Paper-based countries
Table 15.A.8	Average item-level score agreement (across country-language groups) by domain
Table 15.A.9	Percentage of coders whose soring was below the standard inter-rater agreement on 20% or more of items, averaged across countries
Table 15.A.10	Percentage of items in a domain with at least two coders below the standard scoring agreement on the item (in the same country-by-language group)
Table 15.A.11	Number of country-language groups with score agreement below the domain standard
Table 15.A.12	Percentage of responses coded by the MSCS and by human coders across countries in the 2022 field trial
Table 15.A.13	Percentage of responses coded by the MSCS and by human coders across countries in the 2022 main survey

StatLink https://stat.link/p4cuhz

Annex Table 15.A.2. Number of cognitive items by domain, item format, and coding method

			Mathematics (New)	Mathematics (Trend)	Reading	Science	Financial Literacy	Creative Thinking
CBA	Human Coded	Constructed Response	19	16	64	32	16	34
	Computer Scored	Simple Multiple Choice	80	18	104	33	12	0
		Complex Multiple Choice	35	14	27	47	14	2
		Constructed Response	26	26	2	3	4	0
		Total	**160**	**74**	**197**	**115**	**46**	**36**
PBA	Human Coded	Constructed Response		38	51	32		
	Computer Scored	Simple Multiple Choice		18	27	29		
		Complex Multiple Choice		12	9	24		
		Constructed Response		3	0	0		
		Total		**71**	**87**	**85**		
New PBA	Human Coded	Constructed Response		40	37	9		
	Computer Scored	Simple Multiple Choice		16	24	34		
		Complex Multiple Choice		8	5	23		
		Constructed Response		0	0	0		
		Total		**64**	**66**	**66**		

Annex Table 15.A.3. CBA coding number of coders by domain

	Recommended Number of Coders by Number of Students Assessed				Example Workload*		
	< 4,500	4,501 – 8,000	8,001 – 13,000	> 13,000	Coders	Expected Coding Days	Responses per Coder
Mathematics	2 – 3	4 – 5	6 – 9	10 – 12	8	7.1	6,853
Reading	2 or 4	4 or 8	8 or 12	12 – 32	8	7.4	5,194
Science	2 – 3	4 – 5	6 – 9	10 – 12	8	6.3	6,107
Financial Literacy	2 – 3	4 – 5	6 – 9	10 – 12	8	4.8	4,691
Creative Thinking	2 – 3	4 – 5	6 – 9	10 – 24	8	8.9	5,974

Note: Example assumes a main sample size of 6 300 and a Financial Literacy sample size of 1 650.
Example assumes that coders in the core domains and Financial Literacy would be able to code approximately 1 000 responses per day, and coders in the Creative Thinking domain would be able to code approximately 700 responses per day.

Annex Table 15.A.4. PBA and New PBA number of coders by domain

		Example Workload		
		Coders	Expected Coding Days	Responses per Coder
PBA	Mathematics	6	14	10,418
	Reading	6	15	7,733
	Science	6	9	3,328
New PBA	Mathematics	6	17	11,667
	Reading	6	16	10,500
	Science	6	4	2,625

Note: Example assumes that coders would be able to code approximately 1 000 responses per day.

Annex Web Table 15.A.5. Summary of within- and across-country (%) scoring agreement for CBA participants for reading, mathematics and science

This table is available online. The link can be found under the Annex Table 15.A.1.

Annex Table 15.A.6. Summary of within- and across-country (%) agreement for Financial Literacy and Creative Thinking domains

			Within-country		Across-country	
		Country/Economy - Language	Financial Literacy	Creative Thinking	Financial Literacy	Creative Thinking
OECD		Australia - English		74.6%		88.7%
		Austria - German	93.0%		93.5%	
		Belgium - Dutch	96.6%	84.2%	96.6%	91.2%
		Belgium - French		76.8%		87.5%
		Belgium - German		76.8%		88.9%
		Canada - English	91.4%	76.2%	95.9%	91.0%
		Canada - French	92.6%	75.8%	94.6%	87.7%
		Chile - Spanish		76.7%		87.5%
		Colombia - Spanish		81.9%		86.7%
		Czech Republic - Czech	94.9%	89.7%	96.5%	91.7%
		Denmark - Danish	94.1%	86.8%	95.6%	90.6%

		Denmark - Faroese		98.1%		92.1%
		Estonia - Estonian		83.9%		96.9%
		Estonia - Russian		78.1%		86.6%
		Finland - Finnish		83.0%		92.9%
		Finland - Swedish		96.2%		93.9%
		France - French		84.1%		90.7%
		Germany - German		86.0%		88.7%
		Greece - Greek		80.6%		87.4%
		Hungary - Hungarian	92.5%	80.7%	95.5%	92.7%
		Iceland - Icelandic		80.3%		93.0%
		Israel - Arabic		83.2%		92.9%
		Israel - Hebrew		80.6%		93.0%
		Italy - German	95.6%	82.2%	93.4%	94.2%
		Italy - Italian	92.8%	91.1%	95.0%	96.4%
		Korea - Korean		85.7%		87.2%
		Latvia - Latvian		86.1%		85.8%
		Latvia - Russian		86.4%		88.9%
		Lithuania - Lithuanian		90.4%		95.6%
		Lithuania - Polish		89.0%		93.0%
		Lithuania - Russian		89.7%		95.0%
		Mexico - Spanish		80.7%		83.6%
		Netherlands - Dutch	92.0%	77.0%	92.2%	89.4%
		New Zealand - English		80.3%		91.0%
		Norway - Bokmål	95.4%		97.1%	
		Norway - Nynorsk	96.3%		97.3%	
		Poland - Polish	93.3%	79.5%	95.3%	91.0%
		Portugal - Portuguese	90.9%	81.4%	93.7%	85.4%
		Slovak Republic - Hungarian		98.1%		90.4%
		Slovak Republic - Slovak		87.3%		88.9%
		Slovenia - Slovenian		85.1%		89.2%
		Spain - Basque	95.0%	69.9%	87.4%	79.8%
		Spain - Catalan	92.6%	75.6%	93.3%	82.3%
		Spain - Galician	92.4%	72.5%	92.4%	81.0%
		Spain - Spanish	92.1%	69.0%	91.1%	83.6%
	*	Spain - Valencian	100.0%	82.8%	93.9%	83.6%
		United States - English	95.4%		97.7%	
Partners		Albania - Albanian		93.2%		83.0%
		Baku (Azerbaijan) - Azeri		76.6%		68.8%
		Baku (Azerbaijan) - Russian		77.3%		74.8%
		Brazil - Portuguese	99.3%	86.8%	98.8%	95.7%
		Brunei Darussalam - English		76.3%		87.8%
		Bulgaria - Bulgarian	91.3%	81.0%	96.4%	89.2%
		Chinese Taipei - Chinese		79.1%		86.0%
		Costa Rica - Spanish	93.1%	80.0%	94.3%	82.6%
		Croatia - Croatian		94.8%		85.8%
		Cyprus - English		87.6%		90.4%
		Cyprus - Greek		78.0%		87.8%
		Dominican Republic - Spanish		90.7%		62.7%
		El Salvador - Spanish		84.8%		77.2%
		Hong Kong (China) - Chinese		94.0%		95.1%
		Hong Kong (China) - English		97.2%		94.1%
		Indonesia - Indonesian		83.4%		84.7%
		Jamaica - English		69.8%		86.0%

		Jordan - Arabic		83.3%		83.4%
		Kazakhstan - Kazakh		83.1%		89.2%
		Kazakhstan - Russian		87.0%		89.7%
		Macao (China) - Chinese		93.2%		93.6%
		Macao (China) - English		91.8%		93.6%
	*	Macao (China) - Portuguese		100.0%		84.9%
		Malaysia - English	94.4%	76.2%	97.8%	86.6%
		Malaysia - Malay	92.9%	81.0%	98.0%	86.8%
		Malta - English		77.0%		87.6%
		Malta - Maltese		81.4%		88.1%
		Mongolia - Mongolian		81.8%		86.2%
		Morocco - Arabic		81.7%		88.6%
		Morocco - French		82.9%		87.4%
		North Macedonia - Macedonian		86.3%		89.3%
		Palestinian Authority - Arabic		95.9%		86.2%
	*	Palestinian Authority - English		98.5%		86.5%
	*	Panama - English		95.2%		77.8%
		Panama - Spanish		97.1%		59.5%
		Peru - Spanish	96.2%	87.5%	96.1%	91.4%
		Philippines - English		87.8%		90.9%
		Qatar - Arabic		93.2%		86.9%
		Qatar - English		100.0%		87.4%
		Republic of Moldova - Romanian		93.3%		100.0%
		Republic of Moldova - Russian		92.7%		99.7%
		Romania - Hungarian		79.4%		85.8%
		Romania - Romanian		81.0%		88.2%
		Saudi Arabia - Arabic	91.8%	83.7%	90.8%	88.5%
		Saudi Arabia - English	92.0%	97.2%	92.3%	90.0%
	*	Serbia - Hungarian		89.7%		90.7%
		Serbia - Serbian		85.7%		90.8%
		Singapore - English		86.7%		91.3%
		Thailand - Thai		92.0%		93.1%
	*	Ukraine - Russian		100.0%		87.7%
		Ukraine - Ukranian		82.0%		90.0%
		United Arab Emirates - Arabic	93.1%	79.1%	90.3%	79.5%
		United Arab Emirates - English	92.8%	78.3%	90.3%	81.0%
		Uruguay - Spanish		80.6%		92.5%
	*	Uzbekistan - Karakalpak		88.2%		84.7%
		Uzbekistan - Russian		91.8%		91.1%
		Uzbekistan - Uzbek		88.6%		87.0%

* Denotes a country-language group which assessed fewer than 200 students; therefore, there are fewer multiple coded responses contributing to the calculation of agreement in these groups.
Note: Originally assigned codes for Creative Thinking were rescored for some items during scaling; agreement in this table reflects the original human scoring.

Annex Table 15.A.7. Summary of within- and across-country (%) scoring agreement for Paper-based countries

		Within-country Agreement			Across-country Agreement		
	Country/Economy - Language	Mathematics	Reading	Science	Mathematics	Reading	Science
Partners	Guatemala - Spanish	99.9%	99.8%	99.1%	98.5%	97.5%	96.5%
	Cambodia - Khmer	99.5%	99.5%	99.1%	99.6%	99.2%	99.3%
	Paraguay - Spanish	99.3%	97.5%	97.4%	99.0%	97.3%	97.2%

	Viet Nam - Vietnamese	99.9%	99.3%	99.7%	98.2%	94.0%	96.6%

Annex Table 15.A.8. Average item-level score agreement (across country-language groups) by domain

		Mathematics (New)	Mathematics (Trend)	Reading	Science	Financial Literacy	Creative Thinking
CBA	Multiple-coded	95.4%	97.5%	95.4%	94.0%	93.9%	85.0%
	Anchor	93.8%	97.6%	94.8%	92.7%	94.4%	87.9%
PBA and New PBA	Multiple-coded		99.7%	99.0%	98.8%		
	Anchor		98.8%	97.0%	97.4%		

Annex Table 15.A.9. Percentage of coders whose soring was below the standard inter-rater agreement on 20% or more of items, averaged across countries

	Mathematics (New)	Mathematics (Trend)	Reading	Science	Financial Literacy	Creative Thinking
CBA	2.5%	0.8%	8.8%	3.1%	0.7%	15.7%
PBA and New PBA		0.0%	0.0%	0.0%		

Note The standard is set to 85% agreement in mathematics, science, reading, and financial literacy; in Creative Thinking, it is set to 70% agreement.

Annex Table 15.A.10. Percentage of items in a domain with at least two coders below the standard scoring agreement on the item (in the same country-by-language group)

	Mathematics (New)	Mathematics (Trend)	Reading	Science	Financial Literacy	Creative Thinking
CBA	3.0%	1.1%	4.2%	4.2%	2.1%	7.5%
PBA and N-PBA		0.0%	0.0%	0.0%		

Annex Table 15.A.11. Number of country-language groups with score agreement below the domain standard

	N Items below the Standard	Mathematics (New)	Mathematics (Trend)	Science	Reading	Financial Literacy	Creative Thinking
CBA	$N = 0$	123	123	124	120	30	93
	$1 \leq N \leq 5$	1	1	0	2	0	5
	$6 \leq N \leq 10$	0	0	0	2	0	2
PBA and New PBA	$N = 0$		4	4	4		
	$1 \leq N \leq 5$		0	0	0		
	$6 \leq N \leq 10$		0	0	0		

Note: The standard is set to 85% agreement in Mathematics, Science, Reading, and Financial Literacy and 70% in Creative Thinking.

Annex Table 15.A.12. Percentage of responses coded by the MSCS and by human coders across countries in the 2022 field trial

		Human Coded	Machine Coded		
			CUR	Nonresponse	Total
Mathematics (New)	Mean	85.81%	NA	14.19%	14.19%
	Median	93.85%	NA	6.15%	6.15%

Mathematics (Trend)	Mean	69.04%	11.94%	19.02%	30.96%
	Median	73.86%	1.59%	16.11%	26.14%
Reading	Mean	71.06%	10.70%	18.23%	28.94%
	Median	77.78%	0.00%	13.51%	22.22%
Science	Mean	78.44%	8.65%	12.90%	21.56%
	Median	81.48%	0.00%	8.44%	18.52%
Financial Literacy	Mean	84.42%	0.98%	14.60%	15.58%
	Median	87.09%	0.00%	12.27%	12.91%
Creative Thinking	Mean	85.23%	NA	14.77%	14.77%
	Median	89.91%	NA	10.09%	10.09%

Note: Mean values are the mean of the total percentage of responses coded by the MSCS within each domain across countries; median values are the median of those percentages across countries and, therefore, may not add up to 100%.
Note: CUR pool responses were not available for new items in Mathematics, Financial Literacy, or Creative Thinking.

Annex Table 15.A.13. Percentage of responses coded by the MSCS and by human coders across countries in the 2022 main survey

		Human Coded	Machine Coded		
			CUR	Nonresponse	Total
Mathematics (New)	Mean	82.85%	4.76%	12.39%	17.15%
	Median	91.30%	0.00%	6.70%	8.70%
Mathematics (Trend)	Mean	65.22%	14.88%	19.90%	34.78%
	Median	69.53%	4.48%	16.67%	30.47%
Reading	Mean	71.32%	10.68%	18.00%	28.68%
	Median	78.20%	0.43%	13.67%	21.80%
Science	Mean	75.49%	11.42%	13.09%	24.51%
	Median	77.89%	1.70%	8.78%	22.11%
Financial Literacy	Mean	84.77%	2.99%	12.24%	15.23%
	Median	86.19%	0.00%	10.44%	13.81%
Creative Thinking	Mean	92.36%	NA	7.64%	7.64%
	Median	94.37%	NA	5.63%	5.63%

Note: Mean values are the mean of the total percentage of responses coded by the MSCS within each domain across countries; median values are the median of those percentages across countries and, therefore, may not add up to 100%.
Note: CUR pool responses were not available for some new items in Mathematics that had changes following the field trial; CUR pool responses were not applied in Creative Thinking.

16 Data Adjudication

Introduction

Data adjudication is the process through which each national dataset is reviewed and a judgement about the appropriateness of the data for the main reporting goals is formed. The PISA Technical Standards (see Annex I) specify the way in which PISA must be implemented in each participating jurisdiction and adjudicated region. International contractors monitor the implementation in each of these and adjudicate on their adherence to the standards. This chapter describes the process used to adjudicate the PISA 2022 data for each of the adjudicated entities (i.e. the participating countries and economies – hereafter, "jurisdictions" – and the adjudicated regions) and gives the outcomes of data adjudication that are mainly based on the following aspects:

- the extent to which each adjudicated entity met PISA sampling standards
- the outcomes of the adaptation, translation, and verification process
- the outcomes of the PISA Quality Monitoring visits
- the quality and completeness of the submitted data, including concerns about the quality of the data that were identified during scaling and in preparation for reporting
- the outcomes of the international coding review.

Not all regions (i.e. subnational jurisdictions that report their results separately) opt to undergo the full adjudication that would allow their results to be compared statistically to all other participating economies and adjudicated regions. For example, the states of Australia are not adjudicated regions, whereas the Flemish Community of Belgium is an adjudicated region.

PISA 2022 Technical Standards

The areas covered in the PISA 2022 Technical Standards include several aspects connected to the implementation of PISA, including the definition and sampling of its target population on appropriate languages for testing, translation and adaptation of materials, school and student participation in the Field Trial and Main Survey, test administrations and handling of test materials, coding of responses, data management, privacy, and submissions, to cite a few key aspects. A comprehensive list of Technical Standards used for adjudication is available in Annex Table 16.A.1.

Implementing the standards – quality assurance

National Project Managers of participating jurisdictions are responsible for implementing the standards based on the international contractors' advice as contained in the various operational manuals and guidelines. Throughout the cycle of activities for each PISA survey, the international contractors carried out quality-assurance activities in two steps. The first step was to set up quality- assurance procedures using the operational manuals, as well as the agreement processes for national submissions on various aspects of the project. These processes gave the international contractor staff the opportunity to ensure that PISA implementation was planned in accordance with the PISA 2022 Technical Standards and to provide advice on taking rectifying action when required and before critical errors occurred. The second step was quality monitoring, which involved the systematic collection of data that monitored the implementation of the assessment in relation to the standards. For the data adjudication, information collected during both the quality-assurance and quality-monitoring activities was used to determine the level of compliance with the standards.

Information available for adjudication

The international contractors' quality monitoring of a participating jurisdiction's data collection is carried out from a range of perspectives during many stages of the PISA cycle. These perspectives include monitoring a participating jurisdiction's adherence to the deadlines, communication from the sampling contractor about each participating jurisdiction's sampling plan, information from the linguistic verification team, data from the PISA Quality Monitors, and information gathered from direct interviews at National Project Manager and Coder Training meetings. The information was combined together in the database so that:

- indications of non-compliance with the standards could be identified early on in order to enable rectifying measures
- the point at which the problem occurred could be easily identified
- information relating to said non-compliance could be cross-checked between different areas or sources.

Many of these data collection procedures refer to specific key documents, specified in the National Project Manager's Manual and the Sampling Manual in particular. These are procedures that the international contractors require for Field Trial and Main Survey preparation from each National Centre. The data adjudication process provides a motivation for collating and summarising the specific information relating to PISA Technical Standards collected in these documents, combined with information collected from specific quality monitoring procedures such as the PISA Quality Monitor visits and from information in the submitted data.

The quality monitoring information was collected from various quality monitoring instruments and procedures and covered the following main areas:

- international contractors' administration and management: information relating to administration processes, agreement of adaptation spreadsheets, submission of information.
- translation: information from linguistic verification of test items, questionnaire items, and the test administration script.
- sampling: information from the submitted data such as school and student response rates, exclusion rates and eligibility problems.
- school-level materials: information from the agreement of adaptations to test administration procedures and field operations.
- student materials: information from the pre- and post- Main Survey final optical checks of MS test booklets and background questionnaires.
- National Centre operations: School Coordinator, Test Administrator or School Associate trainings; information gathered through interviews conducted during meetings of National Project Managers or at other times.
- PISA Quality Monitors (PQMs): co-ordination of PISA Quality Monitor activities including recruitment; information gathered via the Data Collection Forms from PQMs and through their interactions with School Co-ordinators and Test Administrators.
- data cleaners: issues identified during the data cleaning checks and from data cleaners' reports.

- data processing: issues relating to the eligibility of students tested; issues identified in the coder query service and training of coders.
- data analysis: information from item level reports, from the Field Trial data, and from data cleaning steps, including consistency checks.
- questionnaire data: issues relating to the questionnaire data in the national questionnaire reports provided by the international contractor.
- Main Survey and Field Trial Reviews: information provided by the National Project Managers in the Field Trial and Main Survey Review Questionnaires.

Quality monitoring reports

There were two types of PISA quality monitoring reports: The Session Report Form containing data for each session in each school, and the Data Collection Form detailing the general observations across all schools visited by PQMs. The Session Report Form was completed by the Test Administrator after each test session and also contained data related to test administration. The data from this report were recorded by the National Centre and submitted as part of the national dataset to ETS, the PISA international contractor in charge of coordinating PISA implementation (Core A, see Chapter 1) where it was aggregated by the international project manager at the contractor. The PQM reports contained data related to test administration in selected schools, and the PISA quality monitoring data were collected independently of the National Project Manager.

Data adjudication process

Data adjudication is the process through which each national dataset is reviewed and a judgement about the appropriateness of the data for the main reporting goals is formed. The different steps in the data adjudication process ensure that the final judgement is transparent, based on evidence, and defensible.

The data adjudication process achieved this through the following steps:

- Step 1: International contractors collected quality-assurance and quality monitoring data throughout the survey administration period. The international project manager compiled this information into an adjudication database that was updated or amended as new information arose and provided an overview of the national implementation of PISA throughout the cycle.
- Step 2: The international project manager compiled individual reports for each jurisdiction that contained quality-assurance data for key areas of project implementation.
- Step 3: The international project director, together with the international contractor leads, identified data issues that were in need of adjudication. Where necessary, the relevant National Project Manager was contacted to provide additional information. After this stage, for each dataset, a summary report detailing whether and how the PISA Technical Standards had been met was drafted.
- Step 4: The PISA Adjudication Group, formed by representatives of the OECD, of international contractors, the Technical Advisory Group and the Sampling Referee, reviewed the summary reports to recommend adequate treatment of the data from each adjudicated entity in international PISA products (database and reports).
- Step 5: The recommendations of the PISA Adjudication Group were presented to the PISA Governing Board representatives and to the countries concerned.

Monitoring compliance to any single standard occurred through responses to one or more quality-assurance questions regarding test implementation and national procedures which may come from more than one area. For example, the session report data were used in conjunction with the PISA Quality Monitor reports, computer system tracking of timings, and information from the adaptation of national manuals to assess compliance with the PISA session timing standard (Standard 6.1, Annex I and Annex Table 16.A.1).

Information was collected in relation to these standards through a variety of instruments and information sources:

- through PISA Quality Monitor reports
- through the Field Trial and Main Survey reviews submitted by National Centres
- through information negotiated and stored on the communications portal for PISA 2022
- through a system database specific to the implementation of PISA tasks
- through the formal and informal exchanges between the international contractors and National Centres over matters such as sampling, translation and verification, specially requested analyses (such as non-response bias analysis)
- through a detailed post-hoc inspection of all Main Survey assessment materials
- through the data cleaning and data submission process.

For PISA 2022, an adjudication database was developed to capture, summarise, and store the most important information derived from these various sources. International contractor staff who led each area of work were responsible for identifying relevant information and entering it into the database. This means that at the time of data adjudication, relevant information was easily accessible for making recommendations about the appropriate and comparable use of data from each PISA adjudicated entity.

The adjudication database captured information related to the major phases of the data operation: field operations, sampling, questionnaires, and tests. Within each of these phases, the specific activities are identified, and linked directly to the corresponding standards.

Within each section of the database, specific comments are entered that describe the situation of concern, the source of the evidence about that situation, and the recommended action. Each entry is classified as serious, minor, or of no importance for adjudication. Typically, events classified as serious would warrant close expert scrutiny and possibly action affecting adjudication outcomes. Events classified as minor would typically not directly affect adjudication outcomes but will be reported back to National Centres to assist them in reviewing their national procedures.

The adjudication process for PISA 2022 had an increased challenge imposed by the COVID-19 global pandemic that caused school closures worldwide. The onset of the pandemic was in early 2020 – right as early testing countries were about to implement their Field Trial data collections for the – then named – PISA 2021 cycle.

School closures affected education systems differently, but overall, significantly disrupted survey operations, which led to a decision by the PISA Governing Board to postpone data collection for one year, thus renaming the cycle PISA 2022. As schools reopened and instruction was normalized throughout 2021 and 2022, data collection for PISA resumed, but nonetheless required a degree of reactivity and flexibility from education systems and international contractors. Indeed, as schools reopened at an uneven pace throughout jurisdictions and attempted to get back on track for the rest of the (sometimes expedited) school year, participation of said schools in PISA 2022 Main Survey data collection could not be taken for granted, neither could access from Test Administrators to students, or even high attendance of the latter.

These limitations compelled some changes to Standard 1.3 (assessment period), namely:

- Extension of the assessment period beyond 56 days where students remain within the PISA-eligible age range would be agreed to with the OECD's implicit approval.
- Extension of the assessment period that would not exceed the allowed 56 days but would result in assessed students who are outside of the PISA-eligible age range **by less than a week** would be agreed to with the OECD's implicit approval.

- Extension of the assessment period that would both exceed 56 days AND result in assessed students who are outside of the PISA-eligible age range will require further consultation with the contractors and the OECD before approval of such a deviation would be granted.

The changes were proposed by the international contractors, endorsed by the PISA Technical Advisory Group (TAG) on its December 2021 meeting, and implemented throughout PISA 2022 MS data collection. All participating jurisdictions managed to successfully conclude MS data collection using this added flexibility.

Data adjudication outcomes

It was expected that the data adjudication would result in a range of possible recommendations to the PISA Governing Board. Some possible, foreseen recommendations included:

- that the data be declared fit for use
- that the data be declared fit for use with explicit cautions advised regarding its representativity of the jurisdiction's student cohort, or international comparability of its results
- that some data be removed for a particular participating jurisdiction or adjudicated region, such as the removal of data for some open-ended items or the removal of data for some schools
- that rectifying action be performed by the National Project Manager, such as providing additional evidence to demonstrate that there was no non-response bias, or rescoring open-ended items
- that the data not be endorsed for use in certain types of analyses
- that the data not be endorsed for inclusion in the PISA 2022 database.

Throughout PISA 2022, the international contractors concentrated their quality control activities to ensure that the highest scientific standards were met. However, during data adjudication a wider definition of quality was used, especially when considering data that were at risk. In particular, the underlying criterion used in adjudication was fitness for use; that is, data were endorsed for use if they were deemed to be fit for meeting the major intended purposes of PISA.

General outcomes

It is important to recognise that PISA data adjudication is a late but not necessarily final step in the quality assurance process. By the time each participating jurisdiction was adjudicated at the adjudication group meeting in July 2023, the quality assurance and monitoring processes outlined earlier in this chapter, foreseen in the Technical Standards, and described throughout this report had been implemented. Data adjudication focused on residual issues that remained after these quality assurance processes had been carried out.

Overall, the Adjudication Group's review suggests good adherence of national implementations of PISA to the technical standards in spite of the challenging circumstances that affected not only PISA operations but schooling more generally during the COVID-19 pandemic. Thanks to the reactivity and flexibility of participating countries and international contractors, to carefully constructed instruments, to a test design that is aligned to the main reporting goals and is supported by adequate sample design, and to the use of appropriate statistical methods for scaling, population estimates are highly reliable and comparable across countries and time, and particularly with 2018 results.

Nevertheless, a number of deviations from standards were noted and their consequences for data quality were reviewed in depth. The following overall patterns of deviations from standards were identified:

- About one in five of all adjudicated entities had exclusion rates exceeding the limits set by the technical standards (Standard 1.7).

- Seven entities failed to meet the required school response rates, with three of them failing to meet the stricter level of 65% before replacement (Standard 1.11). This is in line with earlier cycles of PISA.
- There was a significant increase in the number of entities that failed to meet the required student response rates (Standard 1.12): 10 entities did not meet this standard.
- There were delays in data submission in a significant number of entities (Standard 19.1): 14 entities did not meet this standard, and 13 only partially met it. The Adjudication Group noted that delayed submissions may affect the quality of the international contractors' work; and if shorter reporting timelines are expected, it may no longer be possible to accommodate such delays.
- A large number of entities did not conduct the field trial as intended (Standard 3.1) or did not attend all meetings (Standard 23.1). While this may also be a consequence of the pandemic, the Adjudication Group noted that these violations may be particularly consequential for new participants and for less-experienced teams. The Group underlined the importance of attendance at coder training sessions for ensuring comparability of the data.

At the international level, these frequent deviations should guide future efforts of the PISA Governing Board, the OECD Secretariat and Contractors to review the corresponding standards, prevent future deviations from standards, or mitigate the consequences of such violations.

At the level of individual adjudicated countries, economies and regions, in most cases, these issues did not result in major threats to the validity of reports, and the data could be declared fit for use. Where school or student participation rates fell short of the standard and created a potential threat for non-response/non-participation bias, countries/economies were requested to submit non-response-bias analyses. The evidence produced by countries/economies (and in some cases, by the sampling contractor) was reviewed by the Adjudication Group.

The Adjudication Group reviewed and discussed major adjudication issues in June 2023. The major adjudication issues reviewed by the group fall under three broad categories: (1) exclusions and response rates, (2) invariance of item parameters, and (3) issues originating from the Chromebook online administration pilot.

Overview of exclusion and response rate issues

Exclusion rates

PISA Technical Standard 1.7 states that the target population - the population that sits PISA - covers at least 95% of the desired population, the one for which broader conclusions from the assessment are sought. This means that overall exclusions, at both school and student levels, cannot exceed 5%. Sixteen jurisdictions excluded students in excess of this threshold at varying degrees.

Data collection was severely disrupted by the onset of Russia's war of aggression on February 2022, meaning that only data for 18 out of Ukraine's 27 regions could be collected. Exclusions were computed with respect to the original sampling frame, covering the entire country. After February 2022, however, survey operations could not be completed successfully in the regions most affected by wartime disruptions adding to an exclusion rate of 36.1%. Results for the remaining regions were deemed fit for reporting, but comparisons with previous results should be made only with great caution, and with due consideration to the differences in target populations. The Adjudication Group recommended that the results are presented in such a way to alert readers to the difference in the target population between prior cycles and PISA 2022.

Exclusions in Denmark increased by a large margin, presenting a marked increase compared to previous cycles, at 11.6% in PISA 2022. The Adjudication Group noted that high levels of student exclusions may bias performance results upwards. In Denmark, a major cause behind the rise appears to be the increased share of students with diagnosed dyslexia, and the fact that more of these students are using electronic assistive devices to help them read on the screen, including during exams. The lack of such accommodation in PISA led schools to exclude many of these students, meaning that rates are likely to fall should said accommodations, especially those supporting dyslexic students, are allowed.

Rather elevated rates were also observed in the Netherlands and in Latvia, at 8.4% and 7.9% of schools respectively. On the former participant, the Dutch National Centre submitted non-response bias analysis was submitted, analysing differences in performance and in other characteristics between responding schools and the total population of schools, as well as differences between replacement schools and originally sampled, but non-responding schools. This supported the case that no large bias would result from non-response; furthermore, given the available evidence, there is no clear indication about the direction of any residual bias.

The Adjudication Group noted that exclusions exceeded the acceptable rate by a small margin in:

- Croatia (5.4%), Lithuania (6.5%), and in the United States (6.1%), which showed a marked increase in exclusions due to students with functional or intellectual disabilities, which are also bound to fall in the presence of increased accessibility in future cycles. The Adjudication Group invited the national centres to investigate the reasons for this increase in exclusion rates and take remedial action for future cycles. It is expected that exclusion rates will fall again in the future, as a result.
- Australia (6.9%), Canada (5.8%), Estonia (5.9%), New Zealand (5.8%), Norway (7.3%), Scotland (6.6%), Switzerland (5.8%), and Türkiye (5.6%) where the exclusion rates observed in 2022 remained relatively close to exclusion rates observed in 2018.
- Sweden, (7.4%), which showed a marked decrease from the high levels observed in 2018. The Adjudication Group noted that this might be the combined result of falling rates of refugee students and of the national centre's further effort to ensure uniform application of guidelines across schools.

School response rates

The PISA school response rate requirements are foreseen by Technical Standard 1.11, stating that all jurisdictions are to reach a weighted school response rate of 85%, which can be accomplished by administering PISA in replacement schools as needed. A comprehensive account of both jurisdictions and adjudicated regions' response rates either before and after replacement can be found in Annex Tables 13.A.4, 13.A.5, 13.A.6, and 13.A.7 respectively in Chapter 13 of this report.

Seven jurisdictions did not meet the 85% school participation threshold before replacements, with only two meeting the standard with the use of replacement schools. Nonetheless, the increase in participation of replacement schools was rather heterogeneous across jurisdictions. This lack of response, in particular at the school level, has the potential to induce bias in observed results, and thus further investigations in the form of a non-response bias analysis (NRBA) were conducted by affected jurisdictions supported by the international sampling contractor.

Such is the case of the Netherlands, where 66% of sampled schools responded, a share that increased to 90% upon replacement. A NRBA was submitted, analysing differences in performance and in other characteristics between responding schools and the total population of schools, as well as differences between replacement schools and originally sampled, but non-responding schools. This supported the case that no large bias would result from non-response; furthermore, given the available evidence, there is no clear indication about the direction of any residual bias.

Similarly, in Chinese Taipei, where the standard was nearly met before and after replacement (83% before replacement, 84% after), a thorough NRBA was produced, using school-level achievement data as auxiliary information, which provided convincing evidence that the potential bias is minimal after non-response adjustments are considered.

The effect of replacement schools was less pronounced in the United States, where 51% of schools responded before replacement, with the school response rate going up to 62% once replacement schools were invited to participate. Participation rates thus missed the standard by a significant margin, with particularly low participation rates among private schools (representing about 7% of the student population). A NRBA was submitted, indicating that, after replacement schools and non-response adjustments are considered, a number of characteristics (not including direct measures of school performance) are balanced across respondents and non-respondents. Based on the available information, it was not possible for the Adjudication Group to exclude the possibility of bias, nor to determine its most likely direction.

Four other jurisdictions: Canada (81% before replacement, 86% after), Hong Kong (China; 60% before replacement, 80% after), New Zealand (61% before replacement, 72% after), and the United Kingdom (excl. Scotland; 66% before replacement, 80% after) have also submitted NRBAs supporting the case that any bias resulting from school non-participation is most likely be negligible, and are discussed in further detail below, as student participation was also a concern.

Student response rates

Technical Standard 1.12 states that student response rates must be in excess of 80% across responding schools. Students are not replaced in PISA, and thus only those sampled and present at the testing sessions are to sit the test. Student response rates for all jurisdictions and adjudication entities can be found at Tables 13.A.8 and 13.A.9 in Chapter 13 of this report.

Ten jurisdictions did not meet this standard. Albeit some observed rates were close to the 80% threshold, a downwards trend in student participation is of particular concern. Checking for this potential bias is particularly significant, as students' absence from school for the PISA 2022 cycle comes in the aftermath of a global pandemic with severe economic consequences, which might affect some students more than others.

In Malta the student response rate observed in PISA 2022 (79%), fell short of the standard by a small margin, but decreased significantly (from 86%) compared to PISA 2018. A thorough non-response bias analysis (NRBA) was produced, using student-level academic track variable as auxiliary information, along with demographic characteristics. Because students were tracked in the previous academic years based on their grades in mathematics, track information can be expected to correlate strongly with performance on the PISA test. The NRBA provides convincing evidence that the potential bias is minimal after non-response adjustments are taken into account.

Similarly, in the United Kingdom (excl. Scotland), student response rates decreased to 75% from 83% with respect to PISA 2018. School response rates also fell short of the target as discussed above. An informative NRBA was submitted, using external achievement data at student level as auxiliary information, along with demographic characteristics; the analysis was limited to England as the largest subnational entity within the UK, and thus covered over 90% of the intended sample. The analysis provided evidence to suggest a small residual upwards bias, after non-response adjustments are considered, driven entirely by student non-response while school non-participation did not result in significant bias. On the PISA scale, considering that the standard deviation in Scotland (in 2018) was about 95 score points in reading and mathematics, this could translate in a bias of approximately 9 or 10 points.

On the other hand, in Scotland, student response rates missed the standard by one percentage point but were otherwise similar to response rates in PISA 2018 (81%). A thorough non-response bias analysis was submitted, using several external achievement variables at student level as auxiliary information, along with demographic characteristics. The analysis provided evidence to suggest a residual upwards bias of about 0.1 standard deviations, after non-response adjustments are taken into account. On the PISA scale, considering that the standard deviation in Scotland (in 2018) was about 95 score points in reading and mathematics, this could translate in an estimated bias of approximately 9 or 10 points. Given the similarity of response rates between 2018 and 2022, it cannot be excluded that a similar bias might be present in 2018 as well, and in many PISA 2022 participants whose response rates were similarly close to the target. For this reason, data were deemed to be comparable to previous cycles.

Decreases in student participation were more severe for other jurisdictions. In the challenging circumstances surrounding schooling in Panama in 2022 (teacher strikes, road blockades, and student absenteeism), student response rates decreased from 90% in PISA 2018 to 77% in PISA 2022. However, no NRBA was submitted; the PISA national centre explained that non-response was potentially related to the agitated school climate the students found themselves when returning to their schools after the strikes. A limited NRBA was prepared by the international sampling contractor, to compare respondent characteristics (both before and after nonresponse adjustment) to characteristics of the full eligible sample of students. This analysis suggested that (before non-response adjustments were taken into account), non-response was related to students' grade level, and to special needs status. Based on the available information, it is not possible to exclude the possibility of bias; considering the analyses on student non-response conducted in other countries, the residual bias after non-response adjustments are taken into account is likely to correspond to an upward bias.

In Canada, response rates decreased to 77% in PISA 2022 from 84% observed in PISA 2018. A thorough NRBA was submitted, with analyses conducted separately for each Canadian province, using students' academic achievement data as auxiliary information. School response rates also fell short of the target, driven by low participation rates in two provinces (Québec and Alberta). For these provinces, non-response bias was also examined at the school level. The analyses clearly indicate that school nonresponse has not led to any appreciable bias, but student nonresponse has given rise to a small upwards bias.

A similar decrease in participation was observed in Ireland, where student response rates decreased to 77% in PISA 2022 from 86% with respect to PISA 2018. A thorough NRBA was submitted, using external achievement data at student level as auxiliary information. The analysis provided evidence to suggest a residual upwards bias of about 0.1 standard deviations, after non-response adjustments are taken into account. On the PISA scale, considering that the standard deviation in Ireland ranged (in 2018) from 78 score points in mathematics to 91 score points in reading, this could translate in an estimated bias of approximately 8 or 9 points. The Adjudication Group also noted that the bias associated with trend and cross-country comparisons might be smaller, if past data or data for other countries are biased in the same direction.

Australia also observed a decline in student response rates, from 85% in PISA 2018 to 76% in PISA 2022. A technically sound NRBA was submitted; however, the strength of the evidence was limited by the fact that no external student-level achievement variables could be used in the analysis. Based on the available evidence, and on the experience of other countries participating in PISA, the Adjudication Group considered that while non-response adjustments likely limited the severity of non-response biases, a small residual upward bias could not be excluded.

Hong Kong (China) had a similar decrease from 85% in PISA 2018 to 75% in PISA 2022. School response rates also fell short of the standard (as they did in 2018). At the school level, the fact that a raw, but direct measure of school performance is used to assign schools to sampling strata (and therefore, differential non-response across strata is unlikely to cause bias), limits the risk of bias due to non-response. A NRBA was submitted; however, the strength of the evidence was limited by the fact that no external student-level achievement variables could be used in the analysis (only student grade information, already used in non-response adjustments, was available). The proxies for school and student achievement (school size and student grade) that were used in the analyses showed no or very limited relationship with participation rates. Nevertheless, based on the available evidence, and on the experience of other countries participating in PISA, the Adjudication Group considered that while non-response adjustments likely limited the severity of non-response biases, a small residual upward bias could not be excluded.

New Zealand also experienced a decline in student participation and did not meet this standard in PISA 2022. Indeed, student response rates decreased from 83% in PISA 2018 to 72% in PISA 2022. School response rates also fell short of the target as shown above. A thorough and detailed NRBA was submitted, using external achievement data at student level, but also information on chronic absenteeism along with demographic characteristics to further support comparisons. The analysis provided evidence to suggest a residual upwards bias of about 0.1 standard deviations, after non-response adjustments are considered, driven entirely by student non-response with no discernible bias due to school non-response. The analysis also suggested that chronically absent students are over-represented among non-respondents in PISA. On the PISA scale, considering that the standard deviation in New Zealand ranged (in 2018) from 93 score points in mathematics to 106 score points in reading, this could translate in an estimated bias of approximately 10 points. The Adjudication Group also noted that the bias associated with trend and cross-country comparisons might be smaller, if past data or data for other countries are biased in the same direction.

A new participant in PISA 2022, Jamaica also observed student participant rates well below the standard, at 66%. A simple NRBA was submitted by the National Centre, analysing student response rates by school characteristics: this showed in particular lower response rates in schools located in rural areas. A limited NRBA was also prepared by the international sampling contractor, to compare respondent characteristics (both before and after nonresponse adjustment) to characteristics of the full eligible sample of students. This suggested that non-response was also related to students' grade level and gender (both variables are used in non-response adjustments). Based on the available information, it is not possible to exclude the possibility of bias; considering the analyses on student non-response conducted in other countries, the residual bias after non-response adjustments are taken into account is likely to correspond to an upward bias.

The Adjudication Group noted that a number of issues encountered during the Main Survey data collection could have been prevented, had Jamaica been able to do a full Field Trial, which was not possible due to COVID-related disruptions to schooling in 2021. In particular, enrolment information available to the national centre for school-level sampling often turned out to be imprecise; and low student participation rates could have been anticipated, had a regular Field Trial been conducted. As a result of inaccurate sampling frames and low student response rates, the achieved sample size for the Main Survey was well below target, and sampling errors for Jamaica are larger than desired. In spite of the violations of sampling standards, the Adjudication Group considered the data of sufficient quality for reporting if reports are annotated with appropriate notes of caution.

Invariance of item parameters

Albeit not a formal PISA Technical Standard for the 2022 cycle, the share of non-invariant items (i.e. presenting differential item functioning) was considered for adjudication, as measurement invariance underpins the international comparability of results, and thus, one of the main goals of PISA itself. During its 2021 December meeting, the PISA Technical Advisory Group (TAG) provided guidance on this matter[1] and set the share of two thirds of invariant items for each significant language group within a jurisdiction as a threshold for adjudication on whether there is sufficient alignment with the international PISA scale for results to be deemed comparable. Viet Nam did not meet this psychometric threshold.

In Viet Nam, mathematics and science scores were considered fit for reporting, given that for the vast majority of items, student responses were in line with the expectations derived from the experience of other countries participating in PISA. In reading however, a strong linkage to the international PISA scale could not be established as 40% of items in reading (35 of 87) were assigned unique (group-specific) parameters. The Adjudication Group noted that this lack of fit in reading might also reflect differences in construct coverage of the PISA paper-based instrument used in Viet Nam and noted that this instrument will no longer be available in PISA 2025 and further cycles. Furthermore, in addition to item invariance, the Adjudication Group also noted that the response patterns in all subjects deviated significantly from those observed in Viet Nam in earlier cycles; for this reason, the Adjudication Group recommended breaking the trend for Viet Nam, and avoiding comparisons of scale scores to those reported in past cycles.

Chromebook pilot administration issues

In PISA 2022, Iceland, and Norway participated in a pilot administration of online data collection using Chromebooks. Schools in both countries, especially those tested at the first or last days of testing windows in both countries, were affected by server outages during testing. This outage caused slowness and unresponsiveness for some students taking the cognitive assessment, thus resulting in inferior test conditions.

While the PISA Consortium solved this problem during the testing period, 579 students in Iceland (17.2% of the final student sample, unweighted) and 584 students in Norway (8.8%) were assessed on Chromebooks before the problem was solved. According to Iceland, test administrators reported the issue having affected at most 13% of the unweighted final sample (438 students).Data analyses for both countries indicated noticeable differences in the overall response time and overall response rate between students that took the test on days affected by technical problems and those tested on other days, but no noticeable differences were observed in the fit statistics, proficiency estimates or performance between the two testing sessions (first and second hour). In December 2022, the TAG reviewed these results and supported the proposal to keep the entire data for both countries, including that of days when Chromebook issues were reported, also considering that in PISA students are not penalised for having non-reached items at the end of each session. The Adjudication Group, in June 2023, confirmed that overall, the data, including those of students who took the test in these circumstances, were considered to be fit for reporting. The group noted that while it is not possible to exclude that the issue affected students' engagement and motivation to give their best effort, and therefore may have resulted in a small negative impact, their responses did show good fit with the model, and were not remarkably different from the performance of students in other schools.

Notes

1. Namely, the summary record of the PISA TAG meeting reads as follows: *For each major language of assessment within a participating country/economy, over two-thirds of items per domain are expected to be invariant from the international item parameters for the Field Test and the Main Survey. Cases with less than two-thirds of common items will undergo further review of their items and response data.*

Annex 16.A. Data adjudication additional items

Annex Table 16.A.1. PISA 2022 Technical Standards considered in data adjudication

Area	Standard
Target population and sampling	**Standard 1.2**: Unless otherwise agreed upon only PISA-Eligible students participate in the test.
	Standard 1.3: Unless otherwise agreed upon, the testing period: • is no longer than eight consecutive weeks in duration for computer-based testing participants, • is no longer than six consecutive weeks in duration for paper-based testing participants, • does not coincide with the first six weeks of the academic year, and • begins exactly three years from the beginning of the testing period in the previous PISA cycle NOTE: TAG approved deviations to the testing period when necessary due to covid-19. a) Extension of assessment period beyond 8 weeks, students still in PISA-eligible age range b) Extension of assessment period does not exceed 8 weeks but students assessed may be outside of PISA-eligible age range by less than one week c) OECD and contractors pre-approved extension beyond 8 weeks that resulted in students outside the PISA-eligible age range being assessed.
	Standard 1.7: The PISA Defined Target Population covers 95% or more of the PISA Desired Target Population. That is, school-level exclusions and within-school exclusions combined do not exceed 5%.
	Standard 1.8: The student sample size for the computer-based mode is a minimum of 6300 assessed students, and 2100 for additional adjudicated entities, or the entire PISA Defined Target Population where the PISA Defined Target Population is below 6300 and 2100 respectively. The student sample size of assessed students for the paper-based mode is a minimum of 5250.
	Standard 1.9: The school sample size needs to result in a minimum of 150 participating schools, and 50 participating schools for additional adjudicated entities, or all schools that have students in the PISA Defined Target Population where the number of schools with students in the PISA Defined Target Population is below 150 and 50 respectively. Countries not having at least 150 schools, but which have more students than the required minimum student sample size, can be permitted, if agreed upon, to take a smaller sample of schools while still ensuring enough sampled PISA students overall.
	Standard 1.10: The minimum acceptable sample size in each school is 25 students per school (all students in the case of school with fewer than 25 eligible students enrolled).
	Standard 1.11: The final weighted school response rate is at least 85% of sampled eligible and non-excluded schools. If a response rate is below 85% then an acceptable response rate can still be achieved through agreed upon use of replacement schools.
	Standard 1.12: The final weighted student response rate is at least 80% of all sampled students across responding schools.
	Standard 1.13: The final weighted teacher response rate is at least 75% of all sampled teachers across responding schools.
	Standard 1.14: The final weighted sampling unit response rate for any optional cognitive assessment is at least 80% of all sampled students across responding schools.
	Standard 1.16: Unless otherwise agreed upon, the international contractors will draw the school sample for the Main Survey.
	Standard 1.17: Unless otherwise agreed upon, the National Centre will use the sampling contractor's software to draw the student sample, using the list of eligible students provided for each school.
	Other sampling issues such as undercoverage, poor student listing etc.
Language of Testing	**Standard 2.1**: The PISA test is administered to a student in a language of instruction provided by the sampled school to that sampled student in the major domain (Mathematics) of the test. If the language of instruction in the major domain is not well defined across the set of sampled students then, if agreed upon, a choice of language can be provided, with the decision being made at the student, school, or National Centre level. Agreement with the international contractor will be subject to the principle that the language options provided should be languages that are common in the community and are common languages of instruction in schools in that adjudicated entity. If the language of instruction differs across domains then, if agreed upon, students may be tested using assessment instruments in more than one language on the condition that the test language of each domain matches the language of instruction for that domain. Information obtained from the Field Trial will be used to gauge the suitability of using assessment instruments with more than one language in the Main Survey. In all cases the choice of test language(s) in the assessment instruments is made prior to the administration of the test.
Field Trial Participation	**Standard 3.1**: PISA participants participating in the PISA2021 Main Survey will have successfully implemented the Field Trial. Unless otherwise agreed upon:

Area	Standard
	A Field Trial should occur in an assessment language if that language group represents more than 5% of the target population. For the largest language group among the target population, the Field Trial student sample should be a minimum of 200students per item. For all other assessment languages that apply to at least 5% of the target population, the Field Trial student sample should be a minimum of 100students per item. For additional adjudicated entities, where the assessment language applies to at least 5% of the target population in the entity, the Field Trial student sample should be a minimum of 100students per item.
Adaptation of tests, questionnaires and manuals	**Standard 4.1**: The majority of test items used in previous cycles will be administered unchanged from their previous administration, unless amendments have been made to source versions, or outright errors have been identified in the national versions.
	Standard 4.2: All assessment instruments are equivalent to the source versions. Agreed upon adaptations to the local context are made if needed.
	Standard 4.3: National versions of questionnaire items used in previous cycles will be administered unchanged from their previous administration, unless amendments have been made to source versions, outright errors have been identified in the national versions, or a change in the national context calls for an adjustment.
	Standard 4.4: The questionnaire instruments are equivalent to the source versions. Agreed upon adaptations to the local context are made if as needed.
	Standard 4.5: School-level materials are equivalent to the source versions. Agreed upon adaptations to the local context are made as needed.
Translation of assessment instruments, questionnaires, and manuals	**Standard 5.1**: The following documents are translated into the assessment language in order to be linguistically equivalent to the international source versions. • All administered assessment instruments • All administered questionnaires • The Test Administrator script from the Test Administrator (or School Associate) Manual • The Coding Guides
	Standard 5.2: Unless otherwise agreed upon, school-level materials are translated/adapted into the assessment language to make them functionally equivalent to the international source versions.
Testing of national software versions	**Standard 6.1**: The international contractors must test all national software versions prior to their release to ensure that they were assembled correctly and have no technical problems.
	Standard 6.2: Once released, countries must test the national software versions following testing plans to ensure the correct implementation of national adaptations and extensions, display of national languages, and proper functioning on computers typically found in schools in each country. Testing results must be submitted to the international contractors so that any errors can be promptly resolved.
Test administration	**Standard 8.1**: All test sessions follow international procedures as specified in the PISA operations manuals, particularly the procedures that relate to: • test session timing, • maintaining test conditions, • responding to students' questions, • student tracking, and • assigning assessment materials.
	Standard 8.2: The relationship between Test Administrators and participating students must not compromise the credibility of the test session. In particular, the Test Administrator should not be the reading, mathematics, or science instructor of any student in the assessment sessions he or she will administer for PISA.
	Standard 8.3: National Centres must not offer rewards or incentives that are related to student achievement in the PISA test to students, teachers, or schools.
Training Support	**Standard 9.1**: Qualified contractor staff will conduct trainer training sessions with NPMs or designees on PISA materials and procedures to prepare them to train PISA test administrators.
	Standard 9.2: NPMs or designees shall use the comprehensive training package developed by the contractors and provided on the PISA Portal to train PISA test administrators.
	Standard 9.3: All test administrator training sessions should be scripted to ensure consistency of presentations across training sessions and across countries. Failure to do so could cause errors in data collection and make results less comparable.
	Standard 9.4: In-person and/or web based test administrator trainings should be conducted by the NPMs or designees, unless a suitable alternative is agreed upon.
	Standard 9.5: PQMs need to successfully complete self-training materials, attend webinars to review and enhance the self-training, and attend the test administrator training, unless otherwise agreed upon.
National Options	**Standard 10.1**: Only national options that are agreed upon between the National Centre and the international contractors are implemented.
	Standard 10.2: Any national option instruments that are not part of the core components of PISA are administered after all the test and questionnaire instruments of the core component of PISA have been administered to students that are part of the international PISA sample.

Area	Standard
Security of the material	**Standard 11.1**: PISA materials designated as secure are kept confidential at all times. Secure materials include all test materials, data, and draft materials. In particular: • no-one other than approved project staff and participating students during the test session is able to access and view the test materials, • no-one other than approved project staff will have access to secure PISA data and embargoed material, and • formal confidentiality arrangements will be in place for all approved project staff.
	Standard 11.2: Participating schools, students and/or teachers should only receive general information about the test prior to the test session, rather than formal content-specific training. In particular, it is inappropriate to offer formal training sessions to participating students, in order to cover skills or knowledge from PISA test items, with the intention to raise PISA scores.
Quality Monitoring	**Standard 12.1**: PISA Main Survey test administration is monitored using site visits by trained independent quality monitors.
	Standard 12.2: Fifteen site visits to observe test administration sessions are conducted in each PISA participating country/economy, and five site visits in each adjudicated region.
	Standard 12.3: Test administration sessions that are the subject of a site visit are selected by the international contractors to be representative of a variety of schools in a country/economy.
Printing of Materials	**Standard 13.1**: All paper-based student assessment material will be centrally assembled by the international contractors and must be printed using the final printready file and agreed upon paper and print quality. New countries/entities must submit a printed copy of all Field Trial instruments (booklets and questionnaires) for approval of the printing quality for the Main Survey. The same printing standard must be used for both the Field Trial and the Main Survey.
	Standard 13.2: The cover page of all national PISA test paper-based materials used for students and schools must contain all titles and approved logos in a standard format provided in the international version.
	Standard 13.3: The layout and pagination of all test paper-based material is the same as in the source versions, unless otherwise agreed upon.
	Standard 13.4: The layout and formatting of the paper-based questionnaire material is equivalent to the source versions, with the exception of changes made necessary by national adaptations.
Response Coding	**Standard 14.1**: The coding scheme described in the coding guides is implemented according to instructions from the international contractors' item developers.
	Standard 14.3: Both the single and multiple coding procedures must be implemented as specified in the PISA operations manuals (see Note 14.1). These procedures are implemented in all software that countries will be required to use.
	Standard 14.4: Coders are recruited and trained following agreed procedures.
	Standard 14.2: Representatives from each National Centre attend the international PISA coder training session for both the Field Trial and the Main Survey.
Data Submission	**Standard 15.1**: Each PISA participant submits its data in a single complete database, unless otherwise agreed upon.
	Standard 15.2: All data collected for PISA will be imported into a national database using the Data Management Expert (DME) data integration software provided by the international contractors following specifications in the corresponding operational manuals and international/national record layouts (codebooks). Data are submitted in the DME format.
	Standard 15.3: Data for all instruments are submitted. This includes the assessment data, questionnaires data, and tracking data as described in the PISA operations manuals.
	Standard 15.4: Unless agreed upon, all data are submitted without recoding any of the original response variables.
	Standard 15.5: Each PISA participating country's database is submitted with full documentation as specified in the PISA operations manuals.
Communication with the International Contractors	**Standard 16.2**: The National Centre ensures that qualified staff are available to respond to requests by the international contractors during all stages of the project. The qualified staff: • Are authorized to respond to queries, • Are able to communicate in English, • Acknowledge receipt of queries within one working day, • Respond to queries from international contractors within five working days, or, if processing the query takes longer, give an indication of the amount of time required to respond to the query.
Schedule for submission of materials	**Standard 18.1**: An agreed upon Translation Plan will be negotiated between each National Centre and the international contractors.
	Standard 18.2: The following items are submitted to the international contractors in accordance with agreed timelines: • the Translation Plan • a print sample of booklets prior to final printing, for new countries/entities using the paper-based instruments (where this is required, see Standard 13.1), • results from the national checking of adapted computer-based assessment materials and questionnaires, • adaptations to school-level materials, • sampling forms (see Standard 1), • demographic tables, • completed Field Trial and Main Survey Review Forms, and • documents related to PISA Quality Monitors: nomination information, Test Administrator training schedules, translated school-level materials, school contact information, test dates, and

Area	Standard
	• other documents as specified by the PISA operations manuals
	Standard 18.3: Questionnaire materials are submitted for linguistic verification only after all adaptations have been agreed upon.
	Standard 18.4: All adaptations to those elements of the school-level materials that are required to be functionally equivalent to the source as specified in Standard 5.2, need to be agreed upon.
Management of data	**Standard 19.1**: The timeline for submission of national databases to the international contractors is within eight weeks of the last day of testing for the Field Trial and within eight weeks of the last day of testing for the Main Survey, unless otherwise agreed upon.
	Standard 19.2: National Centres execute data checking procedures as specified in the PISA operations manuals before submitting the database.
	Standard 19.3: National Centres make a data manager available upon submission of the database. The data manager: • is authorized to respond to international contractor data queries, • is available for a three-month period immediately after the database is submitted unless otherwise agreed upon, • is able to communicate in English, • is able to respond to international contractor queries within three working days, and • is able to resolve data discrepancies.
	Standard 19.5: To enable the PISA participant to submit a single dataset, all instruments for all additional adjudicated entities will contain the same variables as the primary adjudicated entity of the PISA participant.
Archiving of Materials	**Standard 19.4**: A complete set of PISA paper-based instruments as administered and including any national options, is forwarded to the international contractors on or before the first day of testing. The submission includes the following: • hard copies of instruments, • electronic PDF copies of instruments
	Standard 20.2: The National Project Manager must submit one copy of each of the following translated and adapted Main Survey materials to the international contractors: • electronic versions (Word and/or PDF) of all administered Test Instruments, including international and national options • electronic versions (Word and/or PDF) of all administered Questionnaires, including international and national options (paper-based countries only); • electronic versions of the school-level materials; and • electronic versions of the Coding Guides.
Data Suppression for Privacy Rights	**Standard 21.4**: Each National Centre must facilitate requests from participants to exercise their data rights. • Data access requests will be possible using the raw data from the assessment. No scaled data will be provided in breach of the PISA data embargo. • Data erasure requests will be possible for a limited period before submission to the Contractors. This is to be decided by each National Centre, with two options, up to the submission of ST12 or to upload of student data files to the OECS. • Each National Centre will retain and update a log of completed data requests for data erasure, to facilitate quality control processes. This information must be submitted to the PISA contractors in a timely manner to comply with the requests and for the purpose of data management and sampling processes.
Meeting Attendance	**Standard 23.1**: Representatives from each National Centre are required to attend all PISA international meetings including National Project Manager meetings, coder training, and any separate within-school sampling training, and data management training, as necessary. Up to 6 international meetings are planned per cycle.
	Standard 23.2: Representatives from each National Centre who attend international meetings must be able to work and communicate in English.
Invariant Item Parameters	*TAG Memo Dec 2021*: For each major language of assessment within a participating country/economy, over two-thirds of items per domain are expected to be invariant from the international item parameters for the Field Test and the Main Survey.

Note: Albeit not a Technical Standard *per se*, a significant proportion of common items are essential for the linking of PISA national versions among jurisdictions and thus central for the comparability of assessment results. The PISA Technical Advisory Group (TAG) has fixed a threshold for invariant item parameters that will be incorporated into PISA technical documents in future cycles.
Source: PISA 2022 Technical Standards (Annex I)

17 Proficiency Scale Construction for the Core Domains

Introduction

This chapter discusses the methodology used to develop the PISA Mathematics reporting scales. These describe levels of proficiency in the domain and presents the outcomes of the development process for mathematics literacy, the major domain in the PISA 2022 assessment.

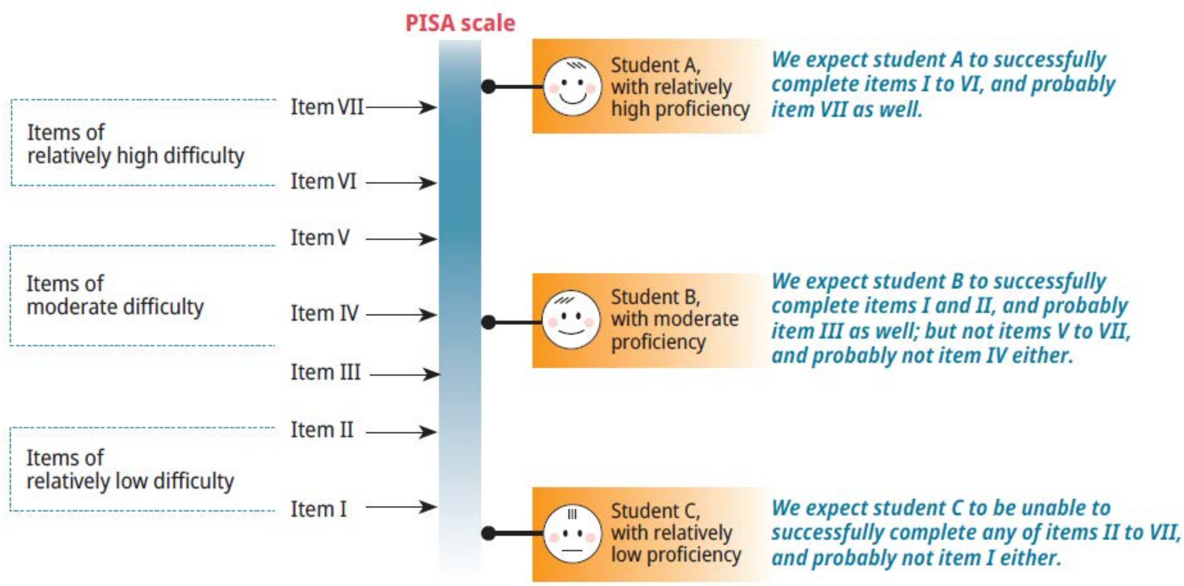
The reporting scales are called "proficiency scales" rather than "performance scales" because they describe what students *typically* know and can do at given levels of proficiency, rather than how individuals who were tested *actually* performed on a single test administration. This emphasis reflects the primary goal of PISA, which is to report general population-level results rather than the results for individual students. PISA uses samples of students and items to make estimates about populations. A sample of 15-year-old students is selected to represent all 15-year-olds in a country/economy and a sample of test items from a large pool is administered to each student. Results are then analysed using statistical models that estimate the likely proficiency of the population, based on this sampling.

The PISA test design makes it necessary to use techniques of modern item response modelling to both, estimate the ability of all students taking the PISA assessment and the statistical characteristics of all PISA items. These techniques are described in Chapter 11 [*Scaling PISA Data*].

The PISA data are collected using a rotated matrix test design in which students take different but overlapping sets of items. The mathematical model employed to analyse the PISA data is implemented through test analysis software that uses iterative procedures to simultaneously estimate the distribution of students along the proficiency dimension assessed by the test, as well as a mathematical function that describes the association of student proficiency and the likelihood of a correct response for each item on the test. The result of these procedures is a set of item parameters that represents, among other things, locations of the items on a proficiency continuum reflecting the domain being assessed. On that continuum, it is possible to estimate the distribution of groups of students, and thereby the average (location) and range (variability) of their skills and knowledge in this domain. This continuum represents the overall PISA scale in the relevant test domain, such as reading, mathematics, or science.

PISA assesses students and uses the outcomes of that assessment to produce estimates of students' proficiency in relation to the skills and knowledge being assessed in each domain. The skills and knowledge of interest, as well as the kinds of tasks that represent those abilities, are described in the PISA frameworks (OECD, 2023[1]). For each domain, one or more scales are defined, each ranging from very low levels of proficiency to very high levels. Students whose ability estimate places them at a certain point on a PISA proficiency scale would be more likely to be able to successfully complete tasks at or below that point. Those students would be increasingly *more likely* to complete tasks located at progressively lower points on the scale, and increasingly *less likely* to complete tasks located at progressively higher points on the scale. Figure 17.1 depicts a simplified hypothetical proficiency scale, ranging from relatively low levels of proficiency at the bottom of the figure, to relatively high levels towards the top. Seven items of varying difficulty are placed along the scale, as are three students of varying ability. The relationship between the students and items at various levels is described in the figure.

Figure 17.1. Simplified relationship between items and students on a proficiency scale



In addition to defining the numerical range of the proficiency scale, it is also possible to define the scale by describing the competencies typical of students at particular points along the scale. The distribution of students along this proficiency scale is estimated, and locations of students can be derived from this distribution and their responses on the test. Those location estimates are then aggregated in various ways to generate and report useful information about the proficiency levels of 15-year-old students within and among participating countries/economies.

The development of a method for describing proficiency in PISA reading, mathematical and scientific literacy occurred in the lead-up to the reporting of outcomes of PISA 2000 and was revised in the lead-up to each of the subsequent surveys. The same basic methodology has again been used to develop proficiency descriptions for the core domains of PISA 2022, even though, like in the PISA 2015 and PISA 2018 cycles, a more general statistical model to describe the items was used in the scaling procedure compared to PISA cycles before 2015.

The proficiency descriptions that had been developed for the science domain in PISA 2015, for the reading domain in PISA 2018, and for financial literacy in 2012 were used again to report the results of PISA 2022. The proficiency descriptors for creative thinking, the innovative domain for PISA 2022, are entirely new and these are described in the next chapter of this Technical Report.

Reporting for mathematics, the major domain in PISA 2022, was linked back to the 2012 proficiency scale and was based on the detailed proficiency level descriptions developed in 2012, the last PISA cycle in which mathematics was the major domain. These proficiency level descriptors were revised based on PISA 2022 data in order to incorporate the new aspects of the mathematics framework and the performance of the new items, including the reasoning and interactive items.

The mathematics expert group (MEG) worked with the Core A contractor (ETS) to revise the sets of described proficiency scales and subscales for PISA mathematics. Similarly, the Creative Thinking Expert

Group (CTEG) worked with Core B3 contractor (ACT) to develop the described proficiency scale for that domain. More detail on the development of the Creative Thinking scale is given in Chapter 18.

Development of the PISA scales

The development of described proficiency scales for PISA has been carried out through a process that typically involves several tasks conducted by the expert groups and the item development team. The process of developing the described scales involved several iterations as the data were collected and analysed during PISA 2022. It should be noted that, as each PISA cycle builds upon the work implemented in previous cycles, the same tasks are not completed for every domain in every administration. The following description of the development process focuses on the development of described proficiency scales for mathematics.

Classification of items

As part of new item development for mathematics, test developers classified all items based on the specifications provided in the framework. Item classifications for the trend mathematics items were also revised to reflect the PISA 2022 assessment framework. All trend classifications were reviewed by the MEG and revised as needed.

Defining the overall proficiency scale

Using Main Survey data with preliminary student weights, the mathematics expert group met over several days and reviewed representative items, particularly those that were classified as representing the new reasoning process scale or having an interactive component (e.g. a spreadsheet or data simulator) and discussing key characteristics that differentiated performance along the proficiency scale. Following this meeting, the descriptors for each level in the overall proficiency scale were refined and finalised.

Identifying possible subscales

For each major domain assessed in PISA, reporting includes an overall proficiency scale based on the combined results for all items within that domain. In addition, the assessment framework may support subscales based on the various dimensions of the framework. Where subscales are included, they must arise clearly from the domain framework, be meaningful and potentially useful for feedback and reporting purposes and be defensible with respect to their measurement properties. Thus, the first stage in the process involves having the experts articulate possible reporting subscales based on the most recent framework.

In the case of mathematics, a single mathematical scale was developed for PISA 2000. With the additional data available in PISA 2003, when mathematics was the major test domain, subscales based on the four overarching subdomains – *space and shape, change and relationships, quantity* and *uncertainty* – were reported. In PISA 2006 and PISA 2009, when mathematics was again a minor domain, only a single scale was reported. For PISA 2012, the expert group carried out a comprehensive revision of the framework at the specific behest of the PISA Governing Board that indicated an interest in seeing mathematical process dimensions used as the primary basis for reporting in mathematics. As well as considering ways in which this could be done, the mathematics expert group also had to consider how the addition of the optional computer-based assessment component included in PISA 2012 could be incorporated into the reporting for the cycle. The outcome of these considerations was, first, a decision that the computer-based items would be used to expand the same mathematical literacy dimension that was expressed through the paper-based items. Second, the expert group recommended that three process-based subscales should be reported. These included: *formulating situations mathematically* (or "formulate"), *employing mathematical concepts, facts, and procedures* (or "employ"), and *interpreting, applying and evaluating mathematical outcomes* (or "interpret"). In addition, for continuity with the PISA 2003 reporting scales, the content-based scales including *space and shape, change and relationships, quantity*, and *uncertainty and data* (formerly just "uncertainty"), were also reported. For PISA 2015 and 2018, where mathematics was once more the minor domain, only a single scale representing overall proficiency in the mathematics domain was reported.

For the PISA 2022 cycle, the MEG decided that additional proficiency scales should be reported for the four mathematical processes (i.e. *mathematical reasoning; formulate; employ; and interpret*). Since the last three processes were part of the domain in previous cycles, proficiency scales already existed and just needed to be updated based on the new items classified to each process. However, mathematical reasoning was "new" this cycle as a separate process scale, so that a proficiency scale needed to be fully developed. As part of their work updating the mathematics framework, the MEG developed a range of actions that students would be expected to perform for each of the mathematical processes. These actions represented a hierarchy of "demands" that the items make of the students in order to solve a problem and were designed to span the proficiency scale. These lists of actions proved useful during the item development phase and when updating/writing the proficiency scales.

Scales in the minor domains

For science, the subscales selected for inclusion in the PISA 2006 database were the three competency-based subscales based on the scientific dimensions documented in the framework: *explaining phenomena scientifically, identifying scientific issues* and *using scientific evidence*. The 2015 expert group recommended reporting again on the three scientific competencies, as they were defined in the updated framework: *explain phenomena scientifically, evaluate and design scientific enquiry,* and *interpret data and evidence scientifically*. In addition, the expert group recommended that two knowledge subscales be reported: *content knowledg*e and *procedural/epistemic knowledge*. Procedural and epistemic knowledge were combined into a single reporting subscale due to a limited number of epistemic items in some of the administered forms. Finally, for continuity with previous reporting scales, three systems – *physical, living* and *Earth and space* – were recommended as a third reporting scale. For PISA 2018 and PISA 2022, only a single scale representing overall proficiency in the science domain is reported.

For reading, which was the major domain in PISA 2018, work on identifying possible subscales began with a review of the subscales used in PISA 2009, when reading was also a major domain. In PISA 2009, volume I of the *PISA 2009 Results* included an overall reading scale and descriptions of subscales that described the types of reading tasks or "cognitive aspects": access and retrieve, integrate and interpret and reflect and evaluate and subscales based on the form of reading material: continuous texts and non-continuous texts (OECD, 2010[2]). For digital reading, a separate, single scale was developed based on the digital reading assessment items administered in 19 countries/economies in PISA 2009, as an international option (OECD, 2011[3]). In PISA 2012, when reading was a minor domain, a single print reading scale was reported, along with a single digital reading scale. For PISA 2018, the reading expert group decided the former distinction of "cognitive aspects" should be updated to "cognitive processes". This terminology better connects the PISA 2018 assessment framework with the literature on reading psychology and better reflects the actual skills and proficiencies assessed. The subscales that correspond to the ways students interact and process text were updated to the following: locate information, understand, and evaluate and reflect. The former subscales that were based on the form of reading material are not included in PISA 2018. Instead, scales are included corresponding to using a single unit of text or multiple units of texts for answering the questions. For PISA 2022, only a single scale representing overall proficiency in reading was reported.

For creative thinking, the innovative domain in PISA 2022, a proficiency description on a single overall reporting scale was developed and is described in the next Chapter. The optional assessment of financial literacy used the same proficiency description from PISA 2015 and PISA 2018.

Defining the proficiency levels

The proficiency levels for each of the PISA domains were defined when each was first introduced as a major domain. The goal of that process was to decide how to divide up the proficiency continuum into levels that might be more interpretable. And, having defined those levels, decisions needed to be made about how to decide on the level to which a particular student should be assigned.

The relationship between the observed responses and student proficiency and item characteristics is probabilistic. That is, there is some probability that a particular student can correctly solve a particular item and each item can be differentially responsive to the proficiency being measured.

One of the basic tenets of the measurement of human skills or proficiencies is this: if a student’s proficiency level exceeds the item’s demands, the probability that the student can successfully complete that item is relatively high, and if the student’s proficiency is lower than that required by the item, the probability of success for that student on that item is relatively low. The rate of change of the probability of success across the range of proficiency for each item is also affected by the sensitivity of the item to student proficiency.

This leads to the question as to the precise criterion that should be used to locate a student on the same scale as that on which the items are located. How can we assign a location that represents student proficiency in meaningful ways? When placing a student at a particular point on the scale, what probability of success should we deem sufficient in relation to items located at the same point on the scale? If a student were given a test comprising a large number of items, each with the same item characteristics, what proportion of those items would we expect the student to successfully complete? Or, thinking of it in another way, if a large number of students of equal ability were given a single test item with a specified item characteristic, about how many of those students would we expect to successfully complete the item?

The answers to these questions depend on assumptions about how items differ in their characteristics or how items function, as well as on what level of probability is deemed a *sufficient probability of success*. In order to define and report PISA outcomes in a consistent manner, an approach is needed to define performance levels and to associate students with those levels. The same basic methodology has again been used to develop proficiency descriptions for PISA 2022.

Defining proficiency levels for PISA progressed in two broad phases. The first, which came after the development of the described scales, was based on a substantive analysis of PISA items in relation to the aspects that underpin each assessment domain. This produces descriptions of increasing proficiency that reflect observations of student performance and a detailed analysis of the cognitive demands of PISA assessment items. The second phase involves decisions about where to set cut-off points for levels and how to associate students with each level in order to lay out how a *sufficient probability of success* plays out in these levels. This is both a technical and a very practical matter of interpreting what it means to be at a level and has significant consequences for reporting national and international results.

Several principles were considered in developing and establishing a useful meaning of being at a level, and therefore for determining an approach to locating cut-off points between levels and associating students with them. For the levels to provide useful information to the PISA assessment stakeholders, it is important to develop a common understanding of what performance at each of those levels means.

First, it is important to understand that the skills measured in each PISA domain fall along a continuum: There are no natural breaking points to mark borderlines between stages along said continuum. Dividing the continuum into levels, though useful for communication about students' development, is essentially arbitrary. Like the definition of units on, for example, a scale of length, there is no fundamental difference between 1 metre and 1.5 metres – it is a matter of degree. It is useful, however, to define stages, or levels along the continua, because they enable us to communicate about the proficiency of students in terms other than continuous numbers. This is a rather common concept, an approach we all know from categorising clothing and portions by size (i.e. small, medium, large, extra-large, etc.).

The approach adopted since PISA 2000 was that it would only be useful to regard students as having attained a particular level if this would mean that we can have certain expectations about what these students are capable of, in general, when they are said to be at that level. It was thus decided that this expectation would have to mean, at a minimum, that students at a particular level would be more likely than not to successfully complete tasks at that level. By implication, it must be expected that they would succeed on at least half of the items on a test composed of items uniformly spread across that level. This definition of being "at a level" is useful in helping to interpret the proficiency of students at different points across the proficiency range defined at each level.

For example, the expectation is that students located at the bottom border of a level would complete at least 50% of items correctly on a test set at the level, while students at the middle and top of each level would be expected to achieve a higher success rate. At the top border of a level would be the students who would be likely to solve a high proportion of the tasks at that level. But, being at the top border of that level, they would also be at the bottom border of the next highest level where, according to the reasoning here, they should have at least a 50% likelihood of solving any tasks defined to be at that higher level.

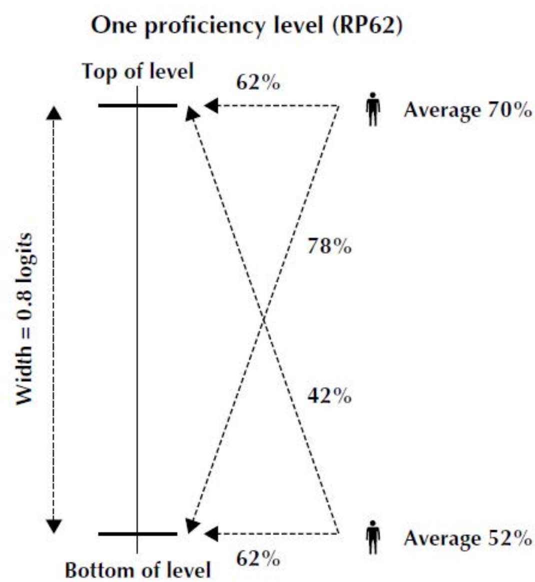
Furthermore, the meaning of being at a level for a given scale should be more or less consistent for each level and, indeed, also for scales from the different domains. In other words, to the extent possible within the substantively based definition and description of levels, cut-off points should create levels of more or less constant range. Some small variation may be appropriate, but for interpretation and definition of cut-off points and levels to be consistent, the levels have to be about equally broad within each scale. The exception would be the highest and lowest proficiency levels, which are unbounded.

Thus, a consistent approach should be taken to defining levels for the different scales. Their range may not be exactly the same for the proficiency scales in different assessment domains, but the same kind of interpretation should be possible for each scale that is developed. This approach links the following three variables:

- the expected success of a student at a particular level on a test containing items at that level (proposed to be set at a minimum that is near 50% for the student at the bottom of the level and greater for students who are higher in the level)
- the width of the levels in that scale (determined largely by substantive considerations of the cognitive demands of items at the level and data related to student performance on the items)
- the probability that a student in the middle of a level would correctly answer an item of average difficulty for that level (in fact, the probability that a student at any particular level would get an item at the same level correct), sometimes referred to as the "RP value" for the scale, where "RP" indicates "response probability".

Figure 17.2 summarises the relationship among these three mathematically linked variables under a particular scenario. The vertical line represents a segment of the proficiency scale, with marks delineating the "top of level" and "bottom of level" for any level one might want to consider, with a width of 0.8 logits between the boundaries of the level (noting that this width can vary somewhat for different assessment domains). The RP62 indicates that students will be located on the scale at a point that gives them a 62% chance of getting a typical item at that same level correct. The student represented near the top of the level shown has a 62% chance of getting an item correct that is located at the top of the level, and similarly the student represented at the bottom of the level has the same chance of correctly answering a question at the bottom of the level. A student at the bottom of the level will have an average score of about 52% correct on a set of items spread uniformly across the level. Of course, that student will have a higher likelihood (62%) of getting an item at the bottom of the level correct, and a lower likelihood (about 42%) of getting an item at the top of the level correct. A student at the top of the level will have an average score of about 70% correct on a set of items spread uniformly across the level. That student will have a higher likelihood (about 78%) of getting a typical item at the bottom of the level correct and a lower likelihood (62%) of getting an item at the top of the level correct.

Figure 17.2. Calculating the RP values used to define PISA proficiency levels



In PISA we have implemented the following solution: Start with the range of described abilities for each bounded level in each scale (the desired band breadth); then determine the highest possible RP value that will be common across domains potentially having bands of slightly differing breadth that would give effect to the broad interpretation of the meaning of being at a level (an expectation of correctly responding to a minimum of 50% of the items in a test comprising items spread uniformly across that level). The value RP = 0.62 is a probability value that satisfied the logistic equations for typical items in that level through which the scaling model is defined, subject to the two constraints mentioned earlier (a width per level of about 0.8 logits and the expectation that a student would get at least half of the items correct on a hypothetical test composed of items spread evenly across the level). In fact, RP=0.62 satisfied the requirements for any scales having band widths up to about 0.97 logits.

The highest and lowest levels are unbounded. For a certain high point on the scale and below a certain low point, the proficiency descriptions could, arguably, cease to be applicable. At the high end of the scale, this is not such a problem since extremely proficient students could reasonably be assumed to be capable of at least the achievements described for the highest level. At the other end of the scale, however, the same argument does not hold. A lower limit therefore needs to be determined for the lowest described level, below which no meaningful description of proficiency is possible. It was proposed that the floor of the lowest described level be set so that it was the same range as the other described levels. Student performance below this level is lower than that which PISA can reliably assess and, more importantly, describe.

Reporting PISA results for Mathematics

In this section, the ways in which levels of mathematics are defined, described and reported will be discussed. This will be illustrated using a subset of released new mathematics items from PISA 2022.

Building an item map for mathematics

The data from the PISA mathematics assessment were analysed to estimate a set of item characteristics for the 234 items included in the Main Survey. During the process of item development, each item was classified to reflect the content area and mathematical process it required. Following data analysis, the items were associated with their difficulty. Annex Table 17.A.1 shows an item map, which includes information for a set of new mathematics units released after the PISA 2022 Main Survey. Each row in Annex Table 17.A.1 represents a level on the mathematics proficiency scale. The selected items have been ordered according to their difficulty, with the most difficult at the top, and the least difficult at the bottom of the table. The difficulty estimate for each item expressed in the reporting scale is given in the rightmost column. For items with a partial-credit response category, the item is listed twice to show the level for a full-credit response and the level for a partial-credit response. Partial-credit responses are listed in italicised font. Note that four new mathematics units were also released after the Field Trial, but those units are not included here because there were no estimates of item difficulty.

Defining levels of mathematical literacy

The reporting approach used by the OECD has been defined in previous PISA cycles and is based on the definition of a number of levels of proficiency. Descriptions were developed to characterise typical student performance at each level. The levels were used to summarise the performance of students, to compare performances across subgroups of students, and to compare average performances among groups of students, in particular among the students from different participating countries/economies. A similar approach has been used here to analyse and report PISA 2022 assessment outcomes for mathematics.

Since the PISA 2000 assessment, results have been reported on a scale with a mean of 500 and a standard deviation of 100. The metric has been set using the participating OECD countries at the time when the subject was the major domain for the first time. In PISA 2012, the last time mathematics was the major domain, the scale consisted of Levels 1 through 6. Starting with the PISA for Development (PISA-D) assessment, Level 1 on the mathematics scale was split into Levels 1c, 1b, and 1a, with Level 1a corresponding to what had previously just been Level 1. This was done to further describe what students at the lower levels of proficiency can do. The level definitions on the PISA mathematical literacy scale are given in Annex Table 17.A.2.

Information about the items in each level is used to develop summary descriptions of the kinds of mathematical skills and abilities associated with different levels of proficiency. These summary descriptions can then be used to encapsulate typical mathematical proficiency of students associated with each level. As a set, they describe a progression in mathematical ability.

For PISA 2022, there was already a set of proficiency level descriptors upon which to build. The new items that were developed for PISA 2022 were considered in relation to the existing level descriptions. Annex Table 17.A.3 presents the updated description for the overall mathematical literacy scale Annex Table 17.A.4 Annex Table 17.A.5 and Annex Table 17.A.6 present updated descriptions for each process (i.e. Formulate, Employ, and Interpret, respectively) that was part of the mathematical problem-solving model also used in PISA 2012.

Annex Table 17.A.7 presents a description of the mathematical reasoning process, which for PISA 2022 was treated as separate process.

Annex Table 17.A.8, Annex Table 17.A.9, Annex Table 17.A.10 and Annex Table 17.A.11 present updated descriptions for each content that was part of the mathematics assessment in PISA 2022.

Cutpoints defining proficiency levels for Reading, Science and Financial Literacy in PISA 2022

Annex Table 17.A.12, Annex Table 17.A.13 and Annex Table 17.A.14 present the cut points used to assign items and students to a proficiency level for the minor domains of reading and science, as well as for the financial literacy domain. As with the mathematics cut points, values in the table are the lower bound for the corresponding level. For example, in the reading scale, Level 6 begins with 698.32. Level 5 begins with 625.61 and ends just below 698.32, where Level 6 begins. Below Level 1c are those with values lower than 189.33. In other words, those reaching a level are those with a score or difficulty at or above the given cut point. This same interpretation applies to all proficiency scales used in PISA.

References

OECD (2023), *PISA 2021 Creative Thinking Framework*, PISA, OECD Publishing, Paris, https://www.oecd.org/pisa/publications/PISA-2021-Creative-Thinking-Framework.pdf. [5]

OECD (2023), *PISA 2021 Financial Literacy Framework*, PISA, OECD Publishing, Paris, https://www.oecd.org/pisa/sitedocument/PISA-2021-Financial-Literacy-Framework.pdf. [6]

OECD (2023), *PISA 2021 Mathematics Framework*, PISA, OECD Publishing, Paris, https://www.oecd.org/pisa/sitedocument/PISA-2021-mathematics-framework.pdf. [4]

OECD (2023), *PISA 2022 Assessment and Analytical Framework*, PISA, OECD Publishing, Paris, https://doi.org/10.1787/dfe0bf9c-en. [1]

OECD (2011), *PISA 2009 Results: Students On Line: Digital Technologies and Performance (Volume VI)*, PISA, OECD Publishing, Paris, https://doi.org/10.1787/9789264112995-en. [3]

OECD (2010), *PISA 2009 Results: What Students Know and Can Do: Student Performance in Reading, Mathematics and Science (Volume I)*, PISA, OECD Publishing, Paris, https://doi.org/10.1787/9789264091450-en. [2]

Annex 17.A. PISA Mathematics reporting scales.

Annex Table 17.A.1. A map for released mathematics items

Level	Cut point	Item	Item Difficulty
6	669.30	Forested Area (CMA161Q03) – Full Credit Forested Area (CMA161Q04) Points (CMA156Q01) – Full Credit	840 739 672
5	606.99	Forested Area (CMA161Q02) *Points (CMA156Q01) – Partial Credit* Forested Area (CMA161Q01) – Full Credit Triangular Pattern (CMA150Q03) – Full Credit *Forested Area (CMA161Q03) – Partial Credit*	647 *642* 636 620 *617*
4	544.68	*Forested Area (CMA161Q01) – Partial Credit* *Triangular Pattern (CMA150Q03) – Partial Credit*	*575* *545*
3	482.38	Solar System (CMA123Q01) – Full Credit (*Partial Credit*)	514 (*503*)
2	420.07	Triangular Pattern (CMA150Q02) Solar System (CMA123Q02)	448 430
1a	357.77	Triangular Pattern (CMA150Q01)	411
1b	295.47	There were no released items at this level	
1c	233.17	There were no released items at this level	

Annex Table 17.A.2. Mathematical literacy performance band definitions on the PISA scale

Level	Score points on the PISA Scale
6	At or above 669.30
5	At or above 606.99 but less than 669.30
4	At or above 544.68 but less than 606.99
3	At or above 482.38 but less than 544.68
2	At or above 420.07 but less than 482.38
1a	At or above 357.77 but less than 420.07
1b	At or above 295.47 but less than 357.77
1c	At or above 233.17 but less than 295.47

Annex Table 17.A.3. Summary descriptions of the proficiency levels on the Mathematical Literacy scale

Level	What students can typically do
6	At Level 6, students can work through abstract problems and demonstrate creativity and flexible thinking to develop solutions. For example, they can recognise when a procedure that is not specified in a task can be applied in a non-standard context or when demonstrating a deeper understanding of a mathematical concept is necessary as part of a justification. They can link different information sources and representations, including effectively using simulations or spreadsheets as part of their solution. Students at this level are capable of critical thinking and have a mastery of symbolic and formal mathematical operations and relationships that they use to clearly

Level	What students can typically do
	communicate their reasoning. They can reflect on the appropriateness of their actions with respect to their solution and the original situation.
5	At Level 5, students can develop and work with models for complex situations, identifying or imposing constraints, and specifying assumptions. They can apply systematic, well-planned problem-solving strategies for dealing with more challenging tasks, such as deciding how to develop an experiment, designing an optimal procedure, or working with more complex visualisations that are not given in the task. Students demonstrate an increased ability to solve problems whose solutions often require incorporating mathematical knowledge that is not explicitly stated in the task. Students at this level reflect on their work and consider mathematical results with respect to the real-world context.
4	At Level 4, students can work effectively with explicit models for complex concrete situations, sometimes involving two variables, as well as demonstrate an ability to work with undefined models that they derive using a more sophisticated computational-thinking approach. Students at this level begin to engage with aspects of critical thinking, such as evaluating the reasonableness of a result by making qualitative judgements when computations are not possible from the given information. They can select and integrate different representations of information, including symbolic or graphical, linking them directly to aspects of real-world situations. At this level, students can also construct and communicate explanations and arguments based on their interpretations, reasoning, and methodology.
3	At Level 3, students can devise solution strategies, including strategies that require sequential decision-making or flexibility in understanding of familiar concepts. At this level, students begin using computational-thinking skills to develop their solution strategy. They are able to solve tasks that require performing several different but routine calculations that are not all clearly defined in the problem statement. They can use spatial visualisation as part of a solution strategy or determine how to use a simulation to gather data appropriate for the task. Students at this level can interpret and use representations based on different information sources and reason directly from them, including conditional decision-making using a two-way table. They typically show some ability to handle percentages, fractions and decimal numbers, and to work with proportional relationships.
2	At Level 2, students can recognise situations where they need to design simple strategies to solve problems, including running straightforward simulations involving one variable as part of their solution strategy. They can extract relevant information from one or more sources that use slightly more complex modes of representation, such as two-way tables, charts, or two-dimensional representations of three-dimensional objects. Students at this level demonstrate a basic understanding of functional relationships and can solve problems involving simple ratios. They are capable of making literal interpretations of results.
1a	At Level 1a, students can answer questions involving simple contexts where all information needed is present, and the questions are clearly defined. Information may be presented in a variety of simple formats and students may need to work with two sources simultaneously to extract relevant information. They are able to carry out simple, routine procedures according to direct instructions in explicit situations, which may sometimes require multiple iterations of a routine procedure to solve a problem. They can perform actions that are obvious or that require very minimal synthesis of information, but in all instances the actions follow clearly from the given stimuli. Students at this level can employ basic algorithms, formulae, procedures, or conventions to solve problems that most often involve whole numbers.
1b	At Level 1b, students can respond to questions involving easy to understand contexts where all information needed is clearly given in a simple representation (i.e. tabular or graphic) and, as necessary, recognise when some information is extraneous and can be ignored with respect to the specific question being asked. They are able to perform simple calculations with whole numbers, which follow from clearly prescribed instructions, defined in short, syntactically simple text.
1c	At Level 1c, students can respond to questions involving easy to understand contexts where all relevant information is clearly given in a simple, familiar format (for example, a small table or picture) and defined in a very short, syntactically simple text. They are able to follow a clear instruction describing a single step or operation.

Annex Table 17.A.4. Summary descriptions of the proficiency levels on the Formulating situations mathematically scale

Level	What students can typically do
6	Students at Level 6 can typically apply a wide variety of mathematical content knowledge to transform and represent information from a broad variety of contexts into a mathematical form amenable to analysis. At this level, students can formulate and solve complex real-world problems involving significant modelling steps and extended calculations, such as applying their geometric knowledge to irregular shapes, inferring relevant parameters of a large data set, or analysing an experiment to recognise the mathematical relationship between objects. Students at level 6 are able to identify the relationship between the key components of a problem and to develop algebraic formulations that accurately represent them.
5	At level 5, students show an ability to use their understanding across a range of mathematical areas to transform information or data from a problem context into mathematical form, sometimes involving two or more variables. They are able to recognise a situation where statistical counting techniques can be applied or formulate inequalities based on given conditions. Students are able to manipulate relatively large data sets by determining appropriate mathematical operations to perform using a spreadsheet tool. They are able to analyse more complex geometric figures, for example, by recognising the relationship between the properties of a compound figure and the properties of individual shapes that comprise the compound figure. Students at this level can formulate a process to solve a problem where some of the information used is given as a range instead of a single value or when information is not given explicitly in the task.

Level	What students can typically do
4	At Level 4, students are able to solve complex problems in a variety of contexts that may require designing a sequence of steps to reach the solution. They also recognise when a single process, repeated iteratively, can lead to the solution. Students are able to run simulations to identify the underlying relationship between two or more variables. They can determine probabilities from data presented in two-way tables. Students at this level can also formulate linear algebraic expressions of relatively simple contexts involving one constraint, recognise an application of a known procedure from a data table and use that procedure to determine missing values, or formulate a method to compare information, such as the prices of several sale items. They can work with more complex geometric models of practical situations which contain all the relevant information needed for formulating the solution.
3	At Level 3, students can identify and extract information from a variety of sources, including text, geometric models, tables, and diagrams, where all necessary information is provided. They can identify basic mathematical concepts relevant for the model or identify how to transform information given in a diagram to data that can be input into a simulation. Students at this level are able to solve problems by recognising situations in which quantities are related proportionally or by performing a computation using a percentage in real-life contexts such as medical testing or ticket sales. They are able to solve simple multi-step problems where the sequence of steps needs to be determined, and each step requires translating some of the given information into a form that can be operated on mathematically.
2	At this level, students can understand clearly formulated instructions and information about simple processes and tasks in order to express them in a mathematical form. They can determine a rule used in a simple pattern, and then use that rule to extend the pattern to the next term. They are able to use information presented in tables or diagrams to identify or build a simple model of a practical situation. For example, they can revise a given formula to determine the number of seats in any row of a theatre. Students at this level are able to translate descriptions of situations to be operated on mathematically that first require identifying information relevant to the particular task. At this level, students begin to formulate situations involving non-integer quantities, provided all necessary information is given in the task.
1a	At this level, students can recognise an explicit model of a contextual situation from a list or translate a short verbal description so that it can be operated on using basic mathematical tools. Students at this level are able to work with simple models involving one operation and at most two variables. For example, they can select the appropriate model that represents the total number of items that can be produced based on a production rate. Students at this level are capable of formulating situations that involve whole numbers and where all relevant information is given.
1b	*There were no items to describe this level on the scale.*
1c	*There were no items to describe this level on the scale.*

Annex Table 17.A.5. Summary descriptions of the proficiency levels on the Employing mathematical concepts, facts, and procedures scale

Level	What students can typically do
6	Students at Level 6 are typically able to employ a strong repertoire of knowledge and procedural skills in a wide range of mathematical areas. They can solve problems involving several stages or a problem that does not have a well-defined solution method, such as computing the area of an irregularly shaped figure. They demonstrate an understanding of statistical data, and can apply that understanding, for example, to determine the probability of different events. Students at this level can observe regularities in information and use that to determine algorithms to apply to a situation. At Level 6, students' work is consistently precise and reflects a strong ability to work with different data formats and representations.
5	Students at Level 5 can employ a broader range of knowledge and skills to solve problems. They can sensibly link information in graphical and diagrammatic formats to textual information. Students can reason proportionally to find a unit rate or understand and apply the meaning of a concept to extract relevant information from a table to solve a problem. At this level, they can devise a strategy to extrapolate from a sample or to determine which of two savings options would be better in a situation involving variously priced items. Students demonstrate the ability to solve problems that require converting between units or working with constraints and can provide mathematical or conceptual arguments to support their results. They also demonstrate proficiency working with percentages and ratios.
4	At Level 4, students show an understanding of the context and can recognise efficient strategies for solving problems. For example, they can typically identify relevant data and information from contextual material and use it to perform such tasks as, calculating distances from a map, analysing a model based on percentages, or comparing the results from two different formulae to compute the same measure. They are able to determine how a rating system was used to support a claim or evaluate several construction designs to rank order them based on a given criterion. At this level, students can estimate values from a graph and use them to solve a problem or analyse statements relating quantities expressed in different numerical formats. They demonstrate an ability to work with ratios or problems that require a series of steps be performed in a specific order.
3	Students at Level 3 demonstrate more flexibility in devising and implementing solution strategies for problems that can be solved in a variety of ways. They are able to solve problems where the information given in the task must first be analysed to determine which of a given set of processes should be implemented, such as determining a fine for exceeding a speed limit based on different driving speeds or a model for computing charges for water-usage. At this level, students are able to use the basic properties of angles to solve a geometric problem or are able to translate between graphical and tabular representations of the same data. Students show an ability to approximate a final solution from interim results or to recognise how a given constraint affects the conclusion. They can work with percentages, fractions, decimal numbers, proportional relationships, and simple non-linear contexts.
2	Students at Level 2 show an ability to work with given models in flexible ways, such as identifying the relevant information to input or manipulating information to make it amenable to use in the model (including models with multiple inputs or tasks that require using a

Level	What students can typically do
	calculator tool specific to the context). They are also able to determine the input when given the output. Students can apply familiar geometric concepts to analyse a spatial pattern. At this level, students show an understanding of place value in decimal numbers and can use that understanding to compare numbers presented in a familiar context. They can apply a known procedure that first requires understanding a data table to extract the necessary information. Students are able to solve simple problems using proportional reasoning and work with ratios.
1a	Students at Level 1a can solve well-defined problems that require minimal decisions. For example, they can make direct inferences from textual information that points to an obvious strategy to solve a given problem, particularly where the mathematical procedures are one- or two-step arithmetic operations with whole numbers or require application of a familiar procedure. Students are able to extract information presented in a variety of formats, such as advertisements, simple pie charts, diagrams, or tables, which contain all the needed information to solve a problem. At this level, students can compute simple percentages, recognise when quantities are related proportionally, find the total area of a standard region, or determine a cost saving.
1b	At Level 1b, students can employ straightforward, one-step procedures that are clearly defined in the task, and where all information is presented in simple tabular format. For example, they are able to determine the winner of a tournament given the criterion for winning or locate information in a table based on a set of conditions.
1c	*There were no items to describe this level on the scale.*

Annex Table 17.A.6. Summary descriptions of the proficiency levels on the Interpreting, applying and evaluating mathematical outcomes scale

Level	What students can typically do
6	At Level 6, students are able to link multiple complex mathematical representations in an analytical way to identify and extract data and information that enables conceptual and contextual questions to be answered. Students at this level demonstrate creativity in order to evaluate claims or interpret solutions to problems that require greater insight to solve, such as using a simulation to determine a design that satisfies several conditions. They are able to interpret data sets with multiple variables that typically require having to perform two or more operations before being able to evaluate a set of given claims related to the data set. Students can recognise different possible subdivisions of an irregular shape based on interpreting a list of geometric properties of the irregular shape. At this level, students can readily interpret or evaluate percentages, frequency distributions, and statistical measures, such as means and medians, in a variety of contexts.
5	At Level 5, students demonstrate the ability to interpret complex situations that require analyses of the underlying mathematics and can apply their understanding of mathematical concepts to real-world situations to make judgements on the reasonableness of claims or results. For example, students can explain why a possible mathematical model does not fit the real-world context. They can interpret experimental results and devise a method for comparing and ranking the results based on a given criterion. At this level, students can evaluate statistical statements based on means or product ratings presented in multiple formats, or they can manipulate a data set so that the presentation facilitates interpretation of the provided information.
4	At Level 4, students are able to interpret and evaluate situations or outcomes that typically involve satisfying multiple conditions, in a range of real-world contexts. They are able to interpret simple statistical or probabilistic statements from data presented in tables or charts in such contexts as fitness levels or genetics. Students at this level are able to interpret experimental results to infer a relationship between two variables in order to evaluate a claim or explain how the computational result of an experiment relates to a given set of specifications. They can determine if a solution is compatible with a particular context or recognise how different adjustments to an algorithm affect the results. At this level, students also are able to approach problems where their interpretation of the given information or model can influence the solution strategy they choose for the task.
3	Students at Level 3 show an ability to reflect on an outcome, or the process used to reach an outcome, in more complex contexts. For example, they can interpret an algebraic model of a design plan to determine what quantity a variable in the model represents or manipulate a set of data using a spreadsheet tool to analyse claims related to energy usage or changes in population data. Students are able to use simulation results to determine a relationship between two contextual variables or explain if a conjecture about a simple algorithm is true. Students demonstrate spatial reasoning by translating between two- and three-dimensional representations of solids or by understanding how properties of geometric figures are related. At this level, students can analyse relatively unfamiliar data presentations to support their conclusions or interpret solutions of non-integer values or ratios with respect to real-world contexts.
2	At Level 2, students can link conceptual and contextual elements of the problem to the mathematics in order to solve problems in a variety of real-world contexts where the information is presented clearly. Students are able to evaluate outcomes, often without having to perform calculations, such as determining the angle measures of an object based on interpreting a description of its properties. They can interpret context-specific language into simple mathematical relationships, sometimes involving one or two constraints, or understand how relationships presented in graphical formats relate to the context, such as a graph of distance versus time. At this level, students can run simulations and interpret the results with respect to the conditions of the task involving one variable.
1a	At Level 1a, students are able to locate and utilise information in order to make sense of the context. They can interpret information that requires relating two simple data sources, such as tables. For example, they can relate information in one table showing how points are awarded to another table of match outcomes to solve a problem in a familiar context or to understand how data from one source is represented in another source. Students at this level can also recognise when some of the given information can be ignored with respect to the specific task.

Level	What students can typically do
1b	At Level 1b, students are able to interpret contextual information presented in one of a variety of formats, such as two-way tables or work schedules. They demonstrate an ability to process the information given basic constraints imposed by the task, such as determining which rule from a table to apply or when to plan an event.
1c	Students at Level 1c can interpret information from real-world contexts presented in simple diagrams or tables and then use that information to solve well-defined problems involving a single operation with whole numbers or straightforward comparisons.

Annex Table 17.A.7. Summary descriptions of the proficiency levels on the Mathematical reasoning scale

Level	What students can typically do
6	At Level 6, students use deductive and inductive reasoning to devise strategies to solve real-world problems that require inference and creativity to recognise the mathematical nature of the task. Tasks at this level are often presented abstractly and require reasoning to recognise how the context-specific language can be transformed into known mathematical concepts or procedures, which underlies making the mathematical context suitable for analysis. Students can solve problems that require visualising a nonstandard geometric model not explicitly shown or described in the task or that require a solid understanding of known algorithms. For example, they can transform given information to construct a visual model to represent a situation or they can use the definition of a procedure for computing a statistical measure to justify if a mathematical result is possible without having numerical values to manipulate. At this level, they use reasoning to critique the limits of a model, such as identifying if a model can or cannot be used in a particular situation, which is necessary for being able to interpret/evaluate the mathematical outcome in context. Students also use reasoning to construct mathematical arguments based on logic and contradictions, such as justifying if a conclusion can be made from a given data set or developing a counterexample in response to a claim.
5	At Level 5, students can recognise structure in problem situations that can be solved using an algorithmic approach. Students use computational thinking to design an optimal procedure, such as programming a sequence of commands, and then reflect on the solution to determine if it meets the given constraints. They can analyse situations and recognise how a known procedure or set of procedures can be applied as a way to justify, for example, if an object can fit into a particular space or if a plan for a geometric design is possible. At this level, they can determine how to develop an experiment and run simulations to collect data necessary for evaluating a context. Students can identify a counterexample or analyse a rule used in a pattern as a way to support a mathematical argument. Students also use reasoning to develop solution strategies by identifying which elements of a model vary and which are invariant.
4	At Level 4, students demonstrate reasoning ability by reflecting on solutions to explain mathematical concepts in real-world contexts. They can evaluate the reasonableness of a claim and provide mathematical justifications to either support or refute the claim, such as recognising how to apply a common procedure in a novel context or determining how to interpret data or information presented in articles, tables, or phone apps. At this level, students can use their understanding of arithmetic and algebraic properties to analyse how manipulating the variables in a model or the steps in a procedure will help explain the real-world results, or they can develop a model to derive a relationship between the variables used in an equation. Students can identify more complex geometric relationships from images of shapes or descriptions of their properties. They are able to reason inductively from sample results to inform decision making or reason about the likelihood of various outcomes related to a probability context.
3	At Level 3, students can apply reasoning by utilising definitions and making judgements necessary for transforming conceptual and contextual situations into mathematical problems. Students at this level can evaluate a claim based on devising simple strategies to connect the underlying mathematics with the context. They are able to solve problems that require making minimal assumptions, such as recognising the relative size of a region from a diagram or comparing graphs of population data. Students can reason about properties in a description of a geometric model to determine a simple algebraic relationship. At this level, they can also apply reasoning to solve problems involving familiar concepts presented in nonstandard ways, such as race results or statistical measures represented graphically on a coordinate plane.
2	At Level 2, students are able to use reasoning to infer relationships between conceptual and contextual elements in a problem or to devise a straight-forward strategy for evaluating a claim. For example, they can order objects by recognising how the size of various objects relates to distance traveled or how to use given assumptions to compare two rate plans with varying prices. Students at this level can also use spatial reasoning, when provided with a model or diagram, to recognise an alternate representation of an image or to analyse simple geometric properties of the model.
1a	At Level 1a, students use reasoning to draw conclusions based on their understanding of simple mathematical concepts, such as evaluating the likelihood of an outcome in a familiar probability context.
1b	*There were no items to describe this level on the scale.*
1c	*There were no items to describe this level on the scale.*

Annex Table 17.A.8. Summary descriptions of the proficiency levels on the mathematical content subscale: Change and relationships

Level	What students can typically do

6	At Level 6, students use significant insight, abstract reasoning and argumentation skills and technical knowledge and conventions to solve problems involving relationships among variables and to generalise mathematical solutions to complex real-world problems. They are able to create and use an algebraic model of a functional relationship incorporating multiple quantities. They apply deep geometrical insight to work with complex patterns. And they are typically able to use complex proportional reasoning, and complex calculations with percentage to explore quantitative relationships and change.
5	At Level 5, students solve problems by using algebraic and other formal mathematical models, including in scientific contexts. They are typically able to use complex and multi-step problem-solving skills, and to reflect on and communicate reasoning and arguments, for example in evaluating and using a formula to predict the quantitative effect of change in one variable on another. They are able to use complex proportional reasoning, for example to work with rates, and they are generally able to work competently with formulae and with expressions including inequalities.
4	Students at Level 4 are typically able to understand and work with multiple representations, including algebraic models of real-world situations. They can reason about simple functional relationships between variables, going beyond individual data points to identifying simple underlying patterns. They typically employ some flexibility in interpretation and reasoning about functional relationships (for example in exploring distance-time-speed relationships) and are able to modify a functional model or graph to fit a specified change to the situation; and they are able to communicate the resulting explanations and arguments.
3	At Level 3, students can typically solve problems that involve working with information from two related representations (text, graph, table, formulae), requiring some interpretation, and using reasoning in familiar contexts. They show some ability to communicate their arguments. Students at this level can typically make a straight-forward modification to a given functional model to fit a new situation; and they use a range of calculation procedures to solve problems, including ordering data, time difference calculations, substitution of values into a formula, or linear interpolation.
2	Students at Level 2 are typically able to locate relevant information on a relationship from data provided in a table or graph and make direct comparisons, for example to match given graphs to a specified change process. They can reason about the basic meaning of simple relationships expressed in text or numeric form by linking text with a single representation of a relationship (graph, table, simple formula), and can correctly substitute numbers into simple formulae, sometimes expressed in words. At this level, student can use interpretation and reasoning skills in a straight-forward context involving linked quantities.
1a	Students at Level 1a are typically able to evaluate single given statements about a relationship expressed clearly and directly in a formula, table, or graph. Their ability to reason about relationships, and change in those relationships, is limited to simple expressions and to those located in familiar situations, such as contexts involving unit rates. They may apply simple calculations needed to solve problems related to clearly expressed relationships.
1b	*There were no items in the PISA 2022 Mathematics assessment to describe this level on the scale.*
1c	*There were no items in the PISA 2022 Mathematics assessment to describe this level on the scale.*

Annex Table 17.A.9. Summary descriptions of the proficiency levels on the mathematical content subscale: Quantity

Level	What students can typically do
6	At Level 6 and above, students conceptualise and work with models of complex quantitative processes and relationships; devise strategies for solving problems; formulate conclusions, arguments and precise explanations; interpret and understand complex information, and link multiple complex information sources; interpret graphical information and apply reasoning to identify, model and apply a numeric pattern. They are able to analyse and evaluate interpretive statements based on data provided; work with formal and symbolic expressions; plan and implement sequential calculations in complex and unfamiliar contexts, including working with large numbers, for example to perform a sequence of currency conversions, entering values correctly and rounding results. Students at this level work accurately with decimal fractions; they use advanced reasoning concerning proportions, geometric representations of quantities, combinatorics and integer number relationships; and they interpret and understand formal expressions of relationships among numbers, including in a scientific context.
5	At Level 5, students are able to formulate comparison models and compare outcomes to determine best price; interpret complex information about real-world situations (including graphs, drawings and complex tables, for example two graphs using different scales); they are able to generate data for two variables and evaluate propositions about the relationship between them. Students are able to communicate reasoning and argument; recognise the significance of numbers to draw inferences; provide a written argument evaluating a proposition based on data provided. They can make an estimation using daily life knowledge; calculate relative and/or absolute change; calculate an average; calculate relative and/or absolute difference, including percentage difference, given raw difference data; and they can convert units (for example calculations involving areas in different units).
4	At Level 4, students are typically able to interpret complex instructions and situations; relate text-based numerical information to a graphic representation; identify and use quantitative information from multiple sources; deduce system rules from unfamiliar representations; formulate a simple numeric model; set up comparison models; and explain their results. They are typically able to carry out accurate and more complex or repeated calculations, such as adding 13 given times in hour/minute format; carry out time calculations using given data on distance and speed of a journey; perform simple division of large multiples in context; carry out calculations involving a sequence of steps and accurately apply a given numeric algorithm involving a number of steps. Students at this level can perform calculations involving proportional reasoning, divisibility or percentages in simple models of complex situations.

Level	What students can typically do
3	At Level 3, students typically use basic problem-solving processes, including devising a simple strategy to test scenarios, understand and work with given constraints, use trial and error, and use simple reasoning in familiar contexts. At this level students typically can interpret a text description of a sequential calculation process, and correctly implement the process; identify and extract data presented directly in textual explanations of unfamiliar data; interpret text and diagrams describing a simple pattern; perform calculations including working with large numbers, calculations with speed and time, conversion of units (for example from an annual rate to a daily rate). They understand place value involving mixed 2- and 3-decimal values and including working with prices; and are typically able to order a small series of (4) decimal values; calculate percentages of up to 3-digit numbers; and apply calculation rules given in natural language.
2	At Level 2, students can typically interpret simple tables to identify and extract relevant quantitative information; interpret a simple quantitative model (such as a proportional relationship) and apply it using basic arithmetic calculations. They are able to identify the links between relevant textual information and tabular data to solve word problems; interpret and apply simple models involving quantitative relationships; identify the simple calculation required to solve a straight-forward problem; carry out simple calculations involving the basic arithmetic operations, as well as ordering 2- and 3-digit whole numbers and decimal numbers with one or two decimal places, and calculate percentages.
1a	At Level 1a, students are typically able to solve basic problems in which relevant information is explicitly presented, and the situation is straightforward and limited in scope. They are able to handle situations where the required computational activity is obvious and the mathematical task is basic, such as performing one or two simple arithmetic operations with whole numbers or percentages. Students at this level can manipulate quantitative information to make it amenable to computational analysis, such as determining the total number of points earned by teams given a record of their wins and losses.
1b	At Level 1b, students can solve straight-forward problems that require single arithmetic operations with whole numbers or retrieving numerical information from a table or chart. For example, students can total the columns of a simple table and compare the results, or they can read and interpret a simple table of monetary amounts or a work schedule to satisfy a situation with a single constraint.
1c	*There were no items in the PISA 2022 Mathematics assessment to describe this level on the scale.*

Annex Table 17.A.10. Summary descriptions of the proficiency levels on the mathematical content subscale: Space and shape

Level	What students can typically do
6	At Level 6, students are able to solve complex problems involving multiple representations or calculations; identify, extract, and link relevant information, for example by extracting relevant dimensions from a diagram or map and using scale to calculate an area or distance; they use spatial reasoning, significant insight and reflection, for example by interpreting text and related contextual material to formulate a useful geometric model and applying it taking into account contextual constraints; they are able to recall and apply relevant procedural knowledge from their mathematical knowledge base such as in circle geometry, trigonometry, Pythagoras's rule, or area and volume formulae to solve problems; and they are typically able to generalise results and findings, communicate solutions and provide justifications and argumentation.
5	At Level 5, students are typically able to solve problems that require appropriate assumptions to be made, or that involve reasoning from assumptions provided and taking into account explicitly stated constraints, for example in exploring and analysing the layout of a room and the furniture it contains. They solve problems using theorems or procedural knowledge such as symmetry properties, or similar triangle properties or formulas including those for calculating area, perimeter or volume of familiar shapes; they use well-developed spatial reasoning, argument and insight to infer relevant conclusions and to interpret and link different representations, for example to identify a direction or location on a map from textual information.
4	Students at Level 4 typically solve problems by using basic mathematical knowledge such as angle and side-length relationships in triangles, and doing so in a way that involves multistep, visual and spatial reasoning, and argumentation in unfamiliar contexts; they are able to link and integrate different representations, for example to analyse the structure of a three dimensional object based on two different perspectives of it; and typically they can compare objects using geometric properties.
3	At Level 3, students are able to solve problems that involve elementary visual and spatial reasoning in familiar contexts, such as calculating a distance or a direction from a map or a GPS device; they are typically able to link different representations of familiar objects or to appreciate properties of objects under some simple specified transformation; and at this level students can devise simple strategies and apply basic properties of triangles and circles, and can use appropriate supporting calculation techniques such as scale conversions needed to analyse distances on a map.
2	At Level 2, students are typically able to solve problems involving a single familiar geometric representation (for example, a diagram or other graphic) by comprehending and drawing conclusions in relation to clearly presented basic geometric properties and associated constraints. They can also evaluate and compare spatial characteristics of familiar objects in a situation where given constraints apply (such as comparing the height or circumference of two cylinders having the same surface area; or deciding whether a given shape can be dissected to produce another specified shape).

Level	What students can typically do
1a	Students at Level 1a can typically recognise and solve simple problems in a familiar context using pictures or drawings of familiar geometric objects and applying basic spatial skills such as recognising elementary symmetry properties, or comparing lengths or angle sizes, or using procedures such as dissection of shapes.
1b	*There were no items in the PISA 2022 Mathematics assessment to describe this level on the scale.*
1c	*There were no items in the PISA 2022 Mathematics assessment to describe this level on the scale.*

Annex Table 17.A.11. Summary descriptions of the proficiency levels on the mathematical content subscale: Uncertainty and data

Level	What students can typically do
6	At Level 6, students are able to interpret, evaluate and critically reflect on a range of complex statistical or probabilistic data, information and situations to analyse problems. Students at this level bring insight and sustained reasoning across several problem elements; they understand the connections between data and the situations they represent and are able to make use of those connections to explore problem situations fully; they bring appropriate calculation techniques to bear to explore data or to solve probability problems; and they can produce and communicate conclusions, reasoning and explanations.
5	At Level 5, students are typically able to interpret and analyse a range of statistical or probabilistic data, information and situations to solve problems in complex contexts that require linking of different problem components. They can use proportional reasoning effectively to link sample data to the population they represent, can appropriately interpret data series over time and are systematic in their use and exploration of data. Students at this level can use statistical and probabilistic concepts and knowledge to reflect, draw inferences and produce and communicate results.
4	Students at Level 4 are typically able to activate and employ a range of data representations and statistical or probabilistic processes to interpret data, information and situations to solve problems. They can work effectively with constraints, such as statistical conditions that might apply in a sampling experiment, and they can interpret and actively translate between two related data representations (such as a graph and a data table). Students at this level can perform statistical and probabilistic reasoning to make contextual conclusions.
3	At Level 3, students are typically able to interpret and work with data and statistical information from a single representation that may include multiple data sources, such as a graph representing several variables, or from two simple related data representations such as a simple data table and graph. They are able to work with and interpret descriptive statistical, probabilistic concepts and conventions in contexts such as coin tossing or lotteries and make conclusions from data, such as calculating or using simple measures of centre and spread. Students at this level can perform basic statistical and probabilistic reasoning in simple contexts.
2	Students at Level 2 are typically able to identify, extract and comprehend statistical data presented in a simple and familiar form such as a simple table, a bar graph or pie chart; they can identify, understand and use basic descriptive statistical and probabilistic concepts in familiar contexts, such as tossing coins or rolling dice. At this level students can interpret data in simple representations, and apply suitable calculation procedures that connect given data to the problem context represented.
1a	At Level 1a, students can typically read and extract data from charts or two-way tables, and recognise how these data relate to the context. Students at this level can also use basic concepts of randomness to identify misconceptions in familiar experimental contexts, such as flipping a coin.
1b	Students at Level 1b, can typically read information presented in a well-labelled table to locate and extract specific data values while ignoring distracting information.
1c	*There were no items in the PISA 2022 Mathematics assessment to describe this level on the scale.*

Annex Table 17.A.12. Cutpoints for the Reading Scale

Cut point	Level Name
698.32	Level 6
625.61	Level 5
552.89	Level 4
480.18	Level 3
407.47	Level 2
334.75	Level 1a
262.04	Level 1b
189.33	Level 1c

Annex Table 17.A.13. Cutpoints for the Science Literacy Scale

Cut point	Level Name
707.93	Level 6
633.33	Level 5
558.73	Level 4
484.14	Level 3
409.54	Level 2
334.94	Level 1a
260.54	Level 1b[1]

Note: 1. Level 1b bandwidth is slightly narrower than others.

Annex Table 17.A.14.Cut points for the Financial Literacy Scale

Cut point	Level Name
624.63	Level 5
549.86	Level 4
475.10	Level 3
400.33	Level 2
325.57	Level 1

18 PISA 2022 Innovative Domain Test Design and Test Development

Introduction

This chapter describes the assessment design framework for the PISA 2022 innovative domain of creative thinking as well as the processes used by the PISA Core B contractor, ACT, the PISA Secretariat, and the international test development team to develop the creative thinking assessment for the PISA 2022 cycle.

Activities undertaken in the context of the innovative domain test design and development included the following:

- The creation of a Creative Thinking Expert Group (CTEG) to guide the assessment framework, test design and test development;
- The development of a creative thinking assessment framework;
- The assessment design and development;
- A series of small-scale validation studies;
- Field trial activities; and
- The main survey administration.

The role of the Creative Thinking Expert Group (CTEG) in the framework and item development

As the contractor for the creative thinking instrument development, Core B was responsible for working with the creative thinking expert group (CTEG) and the PISA Secretariat. Work focused on understanding the CTEG and PISA Secretariat's vision for the creative thinking assessment framework as well as the range and types of items to be developed for the test and questionnaire instruments. The PISA Secretariat and CTEG members began work on the framework in September 2017 finalised the framework in September 2022. Core B's work with the PISA Secretariat and CTEG began in February 2018 and focused on the following tasks:

- describing the kinds of items needed to assess the skills and abilities in each domain as defined in the framework;
- reviewing and understanding the proposed assessment design in order to define the number and types of items that were needed for each of the domains;
- defining the testing functionalities that would be desirable to develop for measuring the construct and that would be feasible to implement in the context of the PISA 2022 administration.

Work with the CTEG continued beyond the initial meeting in February 2018 through the entire phase of instrument development and during data analysis. CTEG members played an important role in reviewing and providing feedback on the assessment tasks as they were developed, providing input into the analysis of the data from multiple small scale validation exercises and the field trial(s), approving the set of items for the main survey administration, and working with instrument development and data analysis staff to develop the described scales and performance level descriptors used for reporting the PISA 2022 creative thinking results.

PISA 2022 creative thinking assessment framework

The PISA Secretariat, together with guidance from the CTEG, developed the PISA 2022 creative thinking assessment framework. The PISA 2022 creative thinking assessment focused on the creative thinking processes that can be reasonably expected from 15-year-old students around the world. It does not aim to single out exceptionally creative individuals but rather to describe the extent to which students can think creatively when searching for and expressing ideas, and to describe how this capacity is related to teaching approaches, school activities and other features of education systems.

The main objective of PISA is to provide internationally comparable data on students' competencies that have clear implications for education policies and pedagogies. In the context of the PISA 2022 assessment, the creative thinking processes in question therefore need to be malleable through education; the different enablers of these thinking processes in the classroom context need to be clearly identified and related to performance in the assessment; the content domains covered in the assessment need to be closely related to subjects taught in common compulsory schooling; and the test tasks should resemble real activities in which students engage, both inside and outside of their classroom, so that the test has some predictive validity of creative achievement and progress in school and beyond.

While closely related to the broader construct of creativity, the PISA 2022 assessment focuses on creative thinking understood as the cognitive processes that are required to engage in creative work. Creative thinking was considered a more appropriate construct to assess in the context of PISA as it is a malleable individual capacity that can be developed through practice, and it refers more to specific cognitive processes than to the subjective quality of an output.

PISA defines creative thinking as:

> *The competence to engage productively in the generation, evaluation, and improvement of ideas, that can result in original and effective solutions, advances in knowledge, and impactful expressions of imagination. (OECD, 2023[1])*

The PISA definition builds on definitions of creativity and creative thinking found in the literature, following a comprehensive review, and it was developed with the guidance of a wider interdisciplinary group of experts in the field (the CTEG). The definition is aligned with the cognitive processes and outcomes associated with "little-c" creativity – in other words, it reflects the types of creative thinking that 15-year-old students around the world can reasonably demonstrate in everyday contexts. It emphasises that students need to learn to engage productively in generating ideas, reflecting upon ideas by valuing their relevance and novelty, and iterating upon ideas before reaching a satisfactory outcome. This definition of creative thinking applies to learning contexts that require imagination and the expression of one's inner world, such as creative writing or the arts, as well as contexts in which generating ideas is functional to the investigation of problems or phenomena.

The competency model

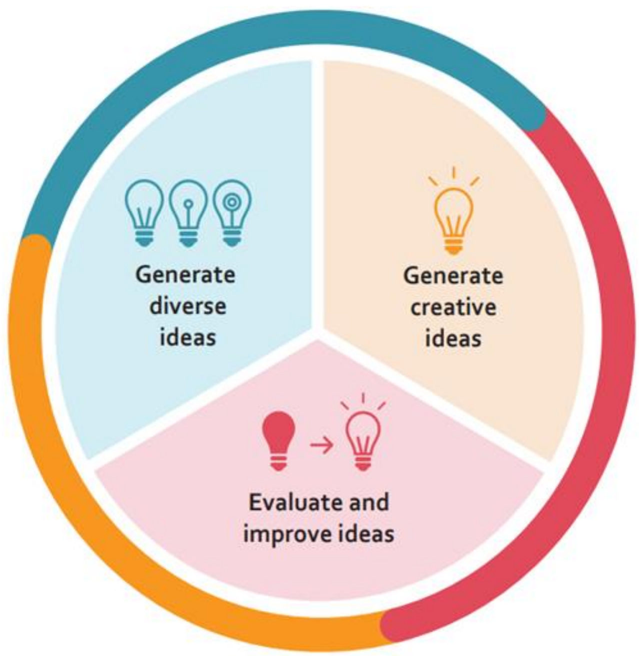
Three cognitive facets support creative thinking and constitute the competency model for the PISA 2022 creative thinking assessment (see Figure 18.1). These three facets are:

- generate diverse ideas;

- generate creative ideas; and
- evaluate and improve ideas.

These three facets reflect the PISA definition of creative thinking and incorporate both divergent cognitive processes (the ability to generate diverse ideas and to generate creative ideas) and convergent cognitive processes (the ability to evaluate other people's ideas and identify improvements to those ideas). "Ideas" in the context of the PISA assessment can take many forms, and the test units provide a meaningful context and sufficiently open tasks in which students can demonstrate their capacity to produce different ideas and think outside of the box.

Figure 18.1. The PISA 2022 competency model for creative thinking



Generate diverse ideas

Typically, attempts to measure creative thinking have focused on the number of ideas that individuals are able to generate – often referred to as ideational fluency. Going one step further is ideational flexibility, or the capacity to generate ideas that are different to each other. When it comes to measuring the quality of ideas that an individual generates, some researchers have argued that fundamentally different ideas should be weighted more than similar ideas (Guilford, 1956[2]). The facet 'generate diverse ideas' of the competency model encompasses these notions and refers to a student's capacity to think flexibly by generating multiple distinct ideas. Test items for this facet present students with a stimulus and ask them to generate two or three appropriate ideas in response that are as different as possible from one another.

Generate creative ideas

The literature generally agrees that creative ideas and outputs are defined as being both novel and useful (Plucker, Beghetto and Dow, 2004[3]). Expecting 15-year-olds around the world to generate ideas that are completely unique or novel is clearly neither a feasible nor appropriate approach for the PISA assessment. Instead, originality represents a useful concept as a proxy for measuring the novelty of ideas. Defined by Guildford (1950[4]) as "statistical infrequency", originality encompasses the qualities of newness, remoteness, novelty or unusualness, and generally refers to deviance from patterns that are observed within the population at hand. In the PISA assessment context, originality is therefore a relative measure established with respect to the responses of other students who complete the same task.

The facet ‘generate creative ideas’ focuses on a student’s capacity to generate appropriate and original ideas. This dual criterion ensures the measurement of creative ideas – ideas that are both original *and* of use – rather than ideas that make random associations that are original yet not meaningful. Test items for this facet present students with a stimulus and ask them to develop one original idea in response.

Evaluate and improve ideas

Evaluative cognitive processes help to identify and remediate deficiencies in initial ideas as well as ensure that ideas or solutions are appropriate, adequate, efficient and effective (Cropley, 2006[5]). They often lead to further iterations of idea generation or the reshaping of initial ideas to improve a creative outcome. Evaluation and iteration are thus at the heart of the creative thinking process. The facet ‘evaluate and improve ideas’ focuses on a student’s capacity to evaluate limitations in ideas and improve their originality. To reduce problems of dependency across items in the test, students are not asked to iterate upon their own ideas but rather to modify a provided “idea”. Test items for this facet thus present students with a given scenario and idea and ask them to suggest an original improvement in response, defined as a change that preserves the essence of the initial idea but that adds or incorporates original elements.

Task contexts: domains of creative thinking

The literature suggests that the larger the number of domains included in an assessment of creative thinking, the better the coverage of the construct given that creative thinking draws on both domain-general and domain-specific resources. The choice of which domains to include in the PISA test was thus a central design question. Given the age and diversity of PISA test takers (15 years-old in over 60 countries), and the fact that domain knowledge is an important enabler of creative thinking, the domain contexts included in the assessment needed to be familiar and accessible to most students around the world, be relevant to schooling, reflect realistic manifestations of creative thinking that 15-year-olds could achieve in a constrained test context, and represent a sufficiently diverse coverage of different types of “everyday” creative thinking as reflected in the literature. Further practical constraints, including the available testing time (a maximum of one hour for the creative thinking test) and testing technology, also informed design choices.

Taking these main constraints into account, the PISA test of creative thinking includes tasks situated within four distinct domain contexts:

- written expression;
- visual expression;
- social problem solving; and
- scientific problem solving.

In the PISA test, the written and visual expression domains involve communicating one’s imagination to others, and creative work in these domains tends to be characterised by originality, aesthetics, imagination, and affective intent and impact. In contrast, the social and scientific problem-solving domains involve investigating and solving open problems. They draw on a more functional employment of creative thinking that is a means to a better end, and creative work in these domains is characterised by ideas or solutions that are original, innovative, effective and efficient.

The inclusion of tasks situated in several domain contexts will allow the PISA 2022 creative thinking test to provide information about students’ strengths and weaknesses in creative thinking across countries. The test items were distributed across the three facets and four domain contexts to allow for a range of opportunities for students to engage and express creative thinking. Annex Table 18.A.2 sets out the distribution of items across facets and domains for the field trial(s) and the main survey administration.

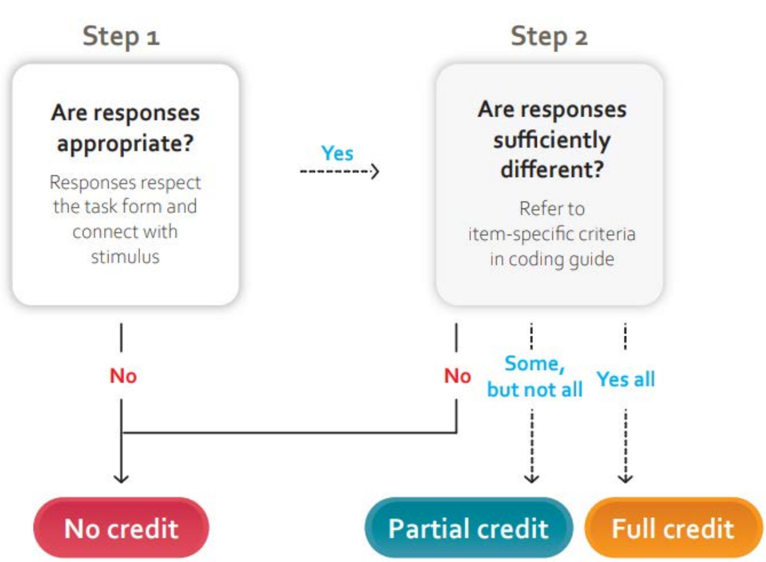
Coding and scoring processes

Every task in the PISA creative thinking test is open-ended and scoring student responses relies on human judgement following detailed scoring rubrics and well-defined coding procedures. All items corresponding to the same facet of the competency model (i.e. ‘generate diverse ideas’, ‘generate creative ideas’ and ‘evaluate and improve ideas’) apply the same general coding procedure. However, as the form of response varies by domain and task (e.g. a title, a solution, a design, etc.), so do the item-specific criteria for evaluating whether an idea is different or original. ACT developed detailed coding guides to describe the item-specific criteria for each item and provide annotated example responses to help human coders score consistently.

Scoring of ‘generate diverse ideas’ items

All items corresponding to the ‘generate diverse ideas’ facet of the competency model require students to provide two or three responses. The general coding procedure for these items involves two steps, as summarised in Figure 18.2. First, coders must determine whether responses are appropriate**.** Appropriate in the context of the creative thinking assessment means that students’ responses respect the required form and connect (explicitly or implicitly) to the task stimulus. Second, coders must determine whether responses are sufficiently different from one another based on item-specific criteria described in the coding guide**.**

Figure 18.2. General coding process for ‘generate diverse ideas’ items



The item-specific criteria are as objective and inclusive as possible of the range of different potential responses. For example, for a written expression item, sufficiently different ideas must use words that convey a different meaning (i.e. are not synonyms). For items in the problem-solving domains, the coding guides list pre-defined response categories to help coders distinguish between similar and different ideas. The coding guides provide detailed example responses and explanations for how to code each example.

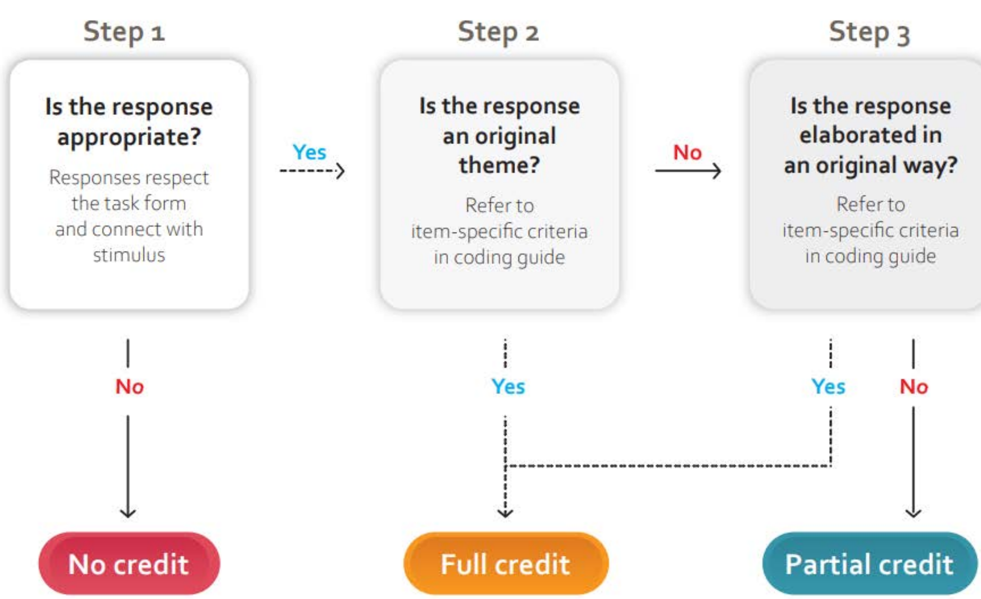
Full credit is assigned where all the responses required in the task are both appropriate and different from each other. Partial credit is assigned in tasks requiring students to provide three responses and where two or three responses are appropriate, but only two are different from each other. No credit is assigned in all other cases.

Scoring of 'generate creative ideas' items

All items corresponding to the facet 'generate creative ideas' of the competency model require a single response. The general coding procedure for these items involves two or three steps depending on the content of the response. First, as with all items, coders must determine whether the response is appropriate. Then, coders must determine whether the response is original by considering two criteria (see Figure 18.3).

An original idea is defined as a relatively uncommon idea with respect to the entire pool of student responses. The coding guide identifies conventional themes for each item according to the patterns of genuine student responses revealed in the validation studies. If a response does not correspond to a conventional theme as described in the coding guide, it is directly coded as original; however, if an idea does correspond to a conventional theme, then coders must determine whether it is original based on its elaboration. The coding guide provides item-specific explanations and examples of original ways to elaborate on conventional themes. For example, a student might add an unexpected twist to a story idea that otherwise centres on a conventional theme.

Figure 18.3. General coding process for 'generate creative ideas' and 'evaluate and improve' items



This twofold originality criteria ensures that the scoring model takes into account both the general idea and the details of a response. While this approach does not single out the most original responses in the entire response pool, it does ensure that the coding process is less susceptible to culturally-sensitive grading styles that favour middle points or extremes and it provides some mitigation against potential cultural bias in the identification of conventional themes across countries.

Full credit is assigned where the response is both appropriate and original. Partial credit is assigned where the response is appropriate only, and no credit is assigned in all other cases.

Scoring of 'evaluate and improve ideas' items

All items corresponding to the facet 'evaluate and improve ideas' of the competency model require a single response and generally ask students to adapt a given idea in an original way rather than coming up with an idea from scratch. The general coding procedure for these items involves the same steps as those for the 'generate creative ideas' items. However, appropriate responses for these items must be both relevant and constitute an improvement. The threshold for achieving the appropriateness criteria for these items is thus somewhat strengthened with respect to items measuring the other two facets, as responses must explicitly connect to the task stimulus and attempt to address its deficiencies. The coding guide provides item-specific criteria, examples and explanations to help orient coders. For responses considered appropriate, coders must then establish the originality of the improvement by considering the same two originality criteria as for 'generate creative ideas' items.

Full credit is assigned where the response is both appropriate and an original improvement. Partial credit is assigned where the response is appropriate only, and no credit is assigned in all other cases.

PISA 2022 innovative domain test assembly design

According to the PISA assessment design, about 28% of the sample of PISA students were administered the creative thinking assessment. Students who took the creative thinking assessment spent one hour on creative thinking test items with the remaining hour of testing time assigned to one of the other core domains (mathematics, reading or scientific literacy).

The creative thinking items were organised into test units. The units vary in terms of the facets that are measured (i.e. generate diverse ideas, generate creative ideas, and evaluate and improve ideas), the domain context (i.e. written expression, visual expression, social problem solving, or scientific problem solving) and the duration of the unit (guidelines of between 5 and 15 minutes). Some units are composed of a single item and some units have multiple items. Dependencies between items within units was minimised.

The creative thinking units were then organised into five, mutually exclusive 30-minute blocks or clusters. The clusters were rotated according to the integrated design presented in Chapter 3 of this Technical Report.

Constructed-response tasks accounted for 92% of the items in the creative thinking assessment. The tasks typically call for a written response, ranging from a few words (e.g. cartoon caption or scientific hypothesis) to a short text (e.g. creative ending to a story or explanation of a design idea). Some constructed-response items call for a visual design response (e.g. designing a poster combining a set of given shapes and stamps) that is supported by a simple drawing editor tool. The assessment also included 2 items that were part of an interactive simulation-based task and two (possible) multiple-choice items where students are given the option to select a previously suggested idea or to generate a new idea.

PISA 2022 innovative domain assessment design and development

Test development for the PISA 2022 creative thinking assessment cycle began in early-2018 and focused on the development of items for a computer-based assessment. Through a process that included both CTEG contributions, as well as country submission and country review, Core B along with the PISA Secretariat selected an initial set of unit and item scenarios. Core B test developers then further developed the unit and item scenarios. The PISA Secretariat reviewed all unit scenarios and items early in the review process, prior to country reviews, to ensure the items fulfilled the goals of the assessment framework.

The developed units were submitted for translatability review at the same time that they were released for country review. Linguists representing different language groups provided feedback on potential translation, adaptation and cultural issues arising from the initial wording of items. Experts at cApStAn and the translation referee for the 2022 cycle alerted test developers to both general wording patterns and specific item wording that are known to be problematic for some translations and suggested alternatives. This allowed test developers to make wording revisions at an early stage, in some cases simply using the alternatives provided and in others working with cApStAn to explore other possibilities.

To ensure that the creative thinking assessment items were understood the same way across linguistic and cultural groups, participating countries also engaged in several cycles of review of the test material to help identify items that may be likely to suffer from cross-cultural bias. This enabled problematic cultural and linguistic characteristics to be identified during the early stages of the assessment development process. Countries had two weeks to perform reviews and submit feedback on all draft stimuli and items.

Preparation of the French source version for all of the test units provided another opportunity to identify issues with the English source version related to content and expression. The development of the two source versions helped to identify instances where wording may prove problematic for translation and could be modified to simplify translation into other languages, and specified where translation notes would be needed to ensure the required accuracy in translating items to other languages.

Cognitive laboratories

Experienced testing professionals were engaged to conduct cognitive laboratory exercises with students in Australia, Singapore and the United States. A total of 66 students across the three countries participated in these cognitive labs in the period between August and September 2018 on set of eight prototype units across the four domain contexts. In the format of concurrent and retrospective thinking-out-loud exercises, students around the age of the PISA population were asked to explain their thought processes while completing the test items and to point out any difficulties or misunderstandings in the instructions or stimulus material. After students completed all of the tasks within each unit, they were asked to answer a series of probing questions about their experience working through the tasks including specific questions on the comprehensibility of the task prompt and perceived difficulty of the task. Students also went through each task a second time to verbalise any thoughts they had had when working through the task. The cognitive laboratories helped to evaluate whether students could understand what they were asked to do during the test, whether students perceived the tasks as engaging, excessively demanding or frustrating, and whether they needed more clarifications to be added to the task prompts.

The analysis of the information collected during these sessions, as well as from video recordings, identified opportunities for the revision and optimisation of items as well as to correct several identified bugs in the testing platform (ACT, 2018[6]). Insights from the cognitive laboratories included:

- **Refining the number of required responses in ‘generate diverse ideas’ items.** In the prototype units tested in the cognitive labs, students could enter as many responses as they wished on these items. In general, students created up to three responses for these items with relative ease but expended considerable effort to move beyond three responses. Moreover, their fourth or fifth responses were rarely their most creative ones. As a result, in successive revisions of the test material, the test development team decided to ask students for up to three responses only and emphasise in the task prompt that students should aim to provide responses that are as different as possible from each other.
- **Choosing the number of required responses in time-intensive items.** Some of the prototype items required a significantly greater investment of time, elaboration and careful execution than others. It was evident from students’ feedback and actual responses to some tasks that asking them to iterate upon or produce more the one response could easily generate fatigue; it was thus

decided that in these types of time-intensive tasks, students should be asked to generate no more than one response.

- **Providing guidance about time required on the task.** In the cognitive labs, students could spend as much time as they wished on each task, although many expressed the need to have some kind of guidance on timing. Different solutions for providing guidance on time usage were considered; the test development team decided to provide an indication of the maximum amount of time that students should spend on each task in the respective task prompt to help students manage their time.
- **Clarifying task expectations and instructions.** Some students requested further clarification on certain terms used in the tasks prompts (e.g. “original”). The prompts were subsequently revised to reduce subjective interpretations of such terms as much as possible. For example, a clarification was added to explain that “original” refers to a solution that other students might not have thought of (clearly associating originality to statistical frequency).
- **Selecting the right images as task stimuli.** Several tasks use a visual stimulus and ask students to engage in idea association in order to generate a response inspired by the image. Some images used in the prototype units evoked associations that were strongly culturally-mediated (for example, some students thought of the Beatles’ song when they saw the image of a yellow submarine). While cultural influences upon student responses cannot be completely eliminated, the development team revised any images that were clearly susceptible to inspiring culture-specific associations.
- **Defining features of the drawing tool.** Most students rapidly understood how to use the drawing tool provided in the platform. However, in some cases students clearly lost precious time trying to complete specific actions that were not immediately intuitive (e.g. deleting an object.) These issues have been addressed by including a tutorial on the use of the drawing tool. The test development team evaluated the potential advantages and disadvantages of including additional features in the drawing tool and decided to keep a relatively simple tool with limited graphical instruments to limit any potential unfair advantage to those students who are more proficient in doing graphical work on a computer. The analysis of responses from the cognitive labs confirmed that it is possible to generate highly creative outputs using only a limited version of the drawing tool.

Six of the eight prototype units were further developed after the cognitive labs for inclusion in subsequent validation studies, while two units were abandoned at this stage due to unsatisfactory performance in the cognitive labs. A further set of units were also developed in accordance with the insights from the cognitive laboratories.

Small scale validation exercises

Further small-scale validation exercises were conducted in parallel to the overall test development process, in an iterative manner, to observe how the then-current test materials functioned under similar test conditions to the field trial and main survey. The purpose of these validation studies was severalfold:

- to provide evidence on the performance of the creative thinking assessment in PISA-like classroom settings;
- to collect sample student responses in multiple countries to inform the development of the coding and scoring guides;
- to assess the inter-rater reliability of human coded items (i.e. the agreement between raters);
- to gain insights into the difficulty of the items;
- to determine the extent to which a creative thinking score or sub-scores could be obtained from the creative thinking assessment; and

- to gain preliminary insights on the essential coder training materials and processes needed for human coders.

A total of 703 15-year-old students from Singapore (n=206), Australia (n=234) and Canada (n=263) participated in the first validation study between October to November 2018. Samples were recruited through the PISA National Project Managers and coordinated with the PISA Secretariat. The validation study instrument included 12 fully functional prototype units delivered in 3 test forms, with 4 units per form. Each form contained one unit per domain.

The coding of the units was carried out according to the preliminary coding guides developed by ACT. Student responses were scored by a team of professional scorers at ACT. As a group, the team reviewed and assigned scores to 5% of the available responses for each task, which enabled scorers to build a common understanding of the coding procedures. Each response was then coded independently by two scorers. Any questions or issues that arose during the scoring of the data were referred to the Scoring Supervisor and the Assessment Design team at ACT.

An analysis of the genuine student data indicated items that did not perform as intended and informed evidence-based improvements to the test material, as well as development of and improvements to coder training material such as the coding guide (ACT, 2019[7]). The validation study also helped to refine the methodology followed for scoring students' responses – in particular, it informed the introduction of a double criteria for coding the originality of responses taking into consideration both the originality of the theme of a students' response and the originality of their approach – and provided genuine responses for the international coder workshops.

A total of 202 15-year-old students from the Republic of South Africa participated in the second validation study from February to March 2019. ACT and the PISA Secretariat partnered with the Care for Education in Republic of South Africa, with support from the LEGO Foundation, to carry out the validation study. It included 16 units, delivered in two test forms. This validation study was delivered on paper to simplify administration procedures (only a limited time was available for the recruitment of schools and it was not possible to condition participation to the availability of computer equipment). Each test form contained the same units, but the order of the units presented to students varied to mitigate potential order effects on performance.

In the second validation study, student responses were scored by a team of non-professional scorers at Care for Education that were trained by the scoring team at ACT following a standard training process and with the support of the international coding guides. Similarly to the first validation study, each response was coded independently by two scorers.

The second validation study provided valuable insights into the success of revisions to the units throughout the test development process. In general, the performance of the creative thinking item pool in the second validation study improved upon the performance of the items included in the first validation study, particularly in terms of inter-rater reliability.

Field trial

The field trial for creative thinking was initially scheduled for 2020; however, this timeline was disrupted by the COVID-19 global pandemic meaning only a limited field trial (LFT) was carried out, with findings further investigated during a second administration of the field trial in 2021. The LFT conducted in 2020 with 11 countries provided preliminary evidence in support of:

1. the psychometric quality of the PISA 2022 creative thinking assessment units in terms of their validity, reliability, and comparability across participating countries;
2. the ability to construct a creative thinking scale and, possibly, subscales;

3. the inclusion of all the creative thinking units and forms in Field Trial 2021; and
4. further enrichment of the coder training materials utilised in coder training for the full field trial in 2021 and the main survey administration in 2022 (ACT, 2020[8]).

In 2021, a further field trial was conducted with 44 countries to provide additional evidence of the validity and reliability of the creative thinking assessment.

Field trial coder training

Among the total 38 items administered, two items were machine-scored (the simulation-based items) and the remaining 36 items were human-scored items. For the human-scored items, all coding processes were performed by each country's coders. The ACT team provided international coder training and supported the national coding teams through a standard PISA query service.

Limited field trial (2020)

The coding guide for the PISA 2022 creative thinking assessment was developed by test developers and performance scoring experts at ACT, with the support of the PISA Secretariat. Coder training procedures and materials were informed by the cognitive labs and validation studies and included examples of genuine student responses.

The English master version of the coding guide was released in a draft form prior to the in-person PISA International Coder Training meeting in January 2020. The training objectives included developing a foundational understanding of the creative thinking construct and an in-depth understanding of the coding processes so that attending representatives would be prepared to train coders in their countries using the provided materials. Test developers and performance scoring experts from ACT, with the support of the PISA Secretariat, facilitated discussions at that meeting. The coding guide used in the limited field trial was finalised based on these discussions. The updated English version of the coding guide and the French source version were subsequently released to countries in February 2020 prior to the beginning of the limited field trial data collection period.

Field trial (2021)

The International Coder Training meeting for creative thinking ahead of the full field trial was held virtually over 5 days due to the COVID-19 pandemic in February 2021. Performance scoring experts from ACT developed online coding training modules and facilitated an interactive coder training workshop, held with representatives from the participating countries in the 2021 field trial prior to coding. To facilitate the online coder training, ACT's team developed comprehensive exemplar sets consisting primarily of authentic student responses that were selected and intended to demonstrate a typical response for each credit level and theme assignment (i.e. codes 00, 11, 12, 13, 21, 22, 23, etc., with code 29 used to designate an unlisted theme). Discussion was also dedicated to reinforcing understanding and consensus about the coding rules for each item to better ensure consistency of coding within and between countries.

Facilitators reviewed the layout of the coding guide, general coding principles, common problems, and guidelines for applying special codes. Workshop materials were optimised based on feedback from the LFT coder training, LFT coder queries and translation referee updates to the earlier version of the coding guide. Attendees were required to code the workshop materials (i.e. the exemplar sets) "live" during the interactive workshop; where there were disagreements about the coding for an item, those were discussed in detail so that all attendees understood, and would be able to follow, the intent of the coding guides. In some instances, disagreements – particularly those highlighting possible cultural bias – led to modifications of the coding guide and/or workshop materials.

Preparation of the field trial data collection instruments

The process for creating the field trial national student delivery system (SDS) began with the assembly and testing of the master SDS, followed by the process for assembling national versions of the field trial SDS. After all components of the national materials were locked, including the questionnaires and cognitive instruments, the student delivery system was assembled and tested first by Core 2. Countries were then asked to check their SDS and identify any remaining content or layout issues. Once countries signed off on their national SDS, their final systems were released for the field trial. The PISA 2022 creative thinking assessment was only administered on computers.

Field trial coding procedures

The field trial design required that two independent coders review and code each student's responses at a credit level of either 0,1 (i.e. no credit or credit), or 0, 1, or 2 (i.e.no credit, partial credit or full credit), thus generating inter-rater reliability at the credit level. In addition, two selected English-fluent bilingual coders from each country reviewed and coded 30 pre-designated anchor responses to verify coder reliability across countries. These anchor responses were selected from earlier validation studies conducted in Australia, Canada, Colombia, Singapore and South Africa, and represented a range of responses at all credit levels (ACT, 2019[7]). Inter-rater reliability (IRR) on the anchor responses across all items and coder pairs was high (0.71). The average quadratic Kappa was also high (0.79).

For the items measuring either the 'generate creative ideas' or 'evaluate and improve ideas' facets, coders were required to use a second digit to indicate the primary theme of each response that earned either partial or full credit. Partial credit responses could only be coded using values of 1-3 as their second digit (i.e. codes 11, 12 or 13), to represent correspondence with the initial conventional themes designated in the coding guide based on an analysis of available student responses in the validation studies; however, responses that received full credit could use up to 9 different values for the second digit (i.e. codes 21 through 29), with the ninth value representing all themes not associated with themes 1-8. The resulting data informed distinctions between "conventionality" and "unconventionality" of themes across a diverse international student cohort.

Field trial coder queries procedures

As was the case during previous cycles, Core A set up and maintained a coder query service for the 2020 and 2021 field trials. Countries were encouraged to send coder queries to the service so that a common adjudication process was consistently applied to all coder questions about constructed-response items. Core B test developers and performance scoring experts from ACT reviewed and responded to coder queries that were specific to the creative thinking test.

In addition to responses to new queries, Core B curated a selection of queries to include in the Coder Query Log containing accumulated responses from previous cycles of PISA. This helped foster consistent coding of creative thinking items. The query log was regularly updated and posted for National Centres on the PISA portal as new queries were received and processed.

National item review post-Field Trial

The item feedback process began in August 2021 and concluded in October 2021 and was conducted in two phases. Phase 1 occurred before countries received their field trial data and Phase 2 after receipt of their data. This two-phase process was implemented to allow for the most efficient correction of any remaining errors in item content or layout given the extremely short turnaround period between the field trial and main survey.

Phase 1 allowed countries to report any linguistic or layout issues that were noted during the field trial, including errors to the coding guides. All requests were reviewed by Core B. Following the release of the field trial data, countries received their Phase 2 updated item feedback forms that included flags for any items that had been identified as not fitting the international trend parameters. Flagged items were then reviewed by national teams. As was the case in Phase 1, countries were asked to provide comments about these specific items in instances where they could identify serious errors. Requests for corrections were reviewed by Core B and, where approved, implemented.

Field Trial outcomes

The 2021 Field Trial data analyses addressed the issue of construct and score validity and reliability, within and across countries, in addition to differential item functioning. Following the field trial data collection, the items were analysed for inter-rater reliability on anchor responses, inter-rater reliability on all responses, average Quadratic Kappa, item category response functions, item quality, and item omit and not-reached rates. Items that exceeded the omit and not-reached rates were identified and investigated; in some cases, this could be attributed to technical issues with some items during the administration of the test, and cluster placement was also considered to be a contributing factor.

Other analyses of the data included item difficulty, item discrimination, item response time, position effects, IRT scaling, item model fit, IRT parameters and student theta estimates, the evaluation of sub-scores on domain and facet levels, and differential item functioning (DIF) via the item-total score curves from different country-by-language groups. Any flagged items for DIF were further reviewed in terms of their sample size, contents, translations and coding guides (i.e. verified translation vs non-verified translation of coding guides), student responses (indications of misunderstanding), performance in alternative languages for that country, performance on similar items in assessment for that country/language, performance on the other items in that unit, additional item flags for that item, LFT data vs. FT data, and planned optimisations for that item (e.g., theme changes, coding optimisations or cluster placement).

Due to the operational timeline in PISA, it was not possible to include new items in the creative thinking test after this phase and no substantial modifications were made to existing test items, i.e. poorly performing items were removed from the test item pool to ensure a proper coverage of the construct in the main survey. Following the field trial analyses, one unit consisting of two items was removed (see Annex Table 18.A.2).

In summary, the findings from the field trial analysis supported:

1. the psychometric quality of the PISA 2022 creative thinking assessment units in terms of their validity, reliability, and comparability across participating countries;
2. the ability to construct a creative thinking scale; and
3. the inclusion of 20 of the 21 creative thinking units administered in the field trial for administration in the 2022 main survey.

The field trial(s) also generated insights for the further enrichment of the coder training materials, including the coding guide, prior to the 2022 main survey. Substantial work was undertaken including reviewing large amounts of genuine student responses, conducting an additional frequency analysis of response themes, and identifying instructions that caused coding issues by being absent, too vague or too restrictive. This resulted in substantial modifications of the coding guide, including updates to the designation of conventional and unconventional themes, the refinement of theme descriptions, the increased representation of exemplar responses, and edits to the item-specific instructions to facilitate effective and consistent coding.

PISA 2022 main survey

The PISA 2022 main survey was conducted between March and December 2022. The majority of countries completed the main survey data collection by August 2022. In preparation for the main survey, countries reviewed items based on their performance in the field trial and were asked to identify any serious errors still in need of correction. The Core B contractors worked with countries to resolve any remaining issues and prepare the national instruments for the main survey.

Item review and selection

The PISA 2022 field trial provided evidence in support of the psychometric quality of the PISA 2022 creative thinking assessment units in terms of validity, reliability, and comparability across participating countries. Maintaining the same range of contexts from the field trial to the main survey provided good continuity and kept a consistent representation of skills and domains. Clusters were created following the final item selection and balanced based on the coverage of cognitive processes, the discrimination and difficulty of the items, and the total number of units and items. The duration of each unit was between 5 and 15 minutes. The units were organised into five mutually exclusive 30-minute blocks or clusters, and the clusters were rotated according to the integrated design presented in Chapter 3 of this Technical Report. The assessment aimed to achieve a good balance between units that situate creative thinking within the two thematic content areas (creative expression, and knowledge creation and problem solving) and the four domains.

The CTEG reviewed the field trial data and outcomes, the approach to item selection, the content and balance of the proposed main survey clusters, and signed off on the selection.

Main survey coder training

The main survey International Coder Training for creative thinking was held in February 2022. Analysis of student responses and coder queries during the field trial administration helped performance scoring experts from ACT improve upon the online coding training modules and other coder training and workshop materials. Additional sample responses were included in the coding guide to better illustrate different types of student responses. Workshop materials were also enhanced to include additional authentic student responses that better illustrated the boundaries between full credit, partial credit (where appropriate) and no credit.

The main survey coder training process was similar to that ahead of the 2021 field trial in that self-guided online training modules were completed before full-group discussions. The training objectives again included developing a foundational understanding of the construct and an in-depth understanding of the coding processes so that attending representatives would be prepared to train coders in their countries using the provided materials. Facilitators again reviewed the layout of the coding guide, general coding principles, common problems, guidelines for applying special codes, and workshop materials for each item. Following the international coder training, additional and final revisions were made to the coding guide in response to discussions that took place at the meeting.

Preparation of data collection instruments

The process for creating the main survey national student delivery system (SDS) followed the approach used during the field trial, beginning with the assembly and testing of the master SDS followed by the process for assembling national versions of the main survey SDS. After all components of national materials were locked, including the questionnaires and cognitive instruments, the student delivery system was assembled and tested first by Core 2. Countries were then asked to check their SDS and identify any remaining content or layout issues. Once countries signed off on their national SDS, their final systems were released for the main study. The PISA 2022 creative thinking assessment was only administered on computers.

Main survey coder queries

The coder query service was again used in the main survey as it was in the field trial to assist countries in clarifying any uncertainty around the coding process or students' responses. Queries were reviewed and responses were provided by domain-specific teams including test developers and coding experts. Core B test developers and performance scoring experts from ACT reviewed and responded to queries specific to the creative thinking test. Relevant queries were included in the Coder Query Log, a resource maintained by Core A and accessible by all participant NPMs in the PISA Portal.

Data adjudication and approach to scaling the data for reporting

In June 2023, Core A presented the Technical Advisory Group (TAG) with the PISA 2022 creative thinking data and preliminary psychometric analyses for data adjudication. Following the initial feedback of the TAG on the scalability of the data given the relatively low inter-item correlations and the creation of plausible values, the PISA Secretariat conducted further analyses of the creative thinking data including modifying some of the scoring rules with the goal of increasing the validity of inferences drawn from the creative thinking data, and improving the scalability and comparability across countries.

Following a thorough review of the data, the following changes were implemented:

- **Four items were dropped from the scaling.** The four items identified for exclusion were drawn from two units (one visual expression, and one scientific problem solving) and were all in the same test cluster. These four items showed poor discrimination and high omit rates, likely due to their position within the cluster.
- **The scoring rules for 14 items were modified.** All 'generate creative ideas' and 'evaluate and improve' items were reviewed following the main survey in terms of the distribution of double-digit codes across countries. The scoring process for these items required coders to use a second digit to indicate the primary theme of each response, and those coded using values of 1-3 as their second digit (i.e. 11, 21, 12, 22, 31 or 32) represented correspondence with the initial conventional themes designated in the coding guide. The double-digit codes were intended to serve as a mechanism through which to review the distribution of codes across countries and adjust the themes designated as conventional following the field trial and main survey. The number of conventional themes were modified for 14 of the 18 items corresponding to 'generate creative ideas' and 'evaluate and improve ideas' based on the results of the main survey to improve the validity of the scoring rules for these items and to align the scoring with the framework (i.e. originality as statistical infrequency, with respect to the responses of other students who completed the same task).
- **Responses submitted in fewer than 15 seconds were invalidated (i.e. converted to missing responses).** For most items in the creative thinking test, students must generate a written or visual artefact in response to a written or visual stimulus (i.e. task prompt with instructions and material for inspiration). The construct of creative thinking also aims to measure the cognitive processes associated with idea generation, evaluation and improvement, which are considered to be slow and thoughtful processes rather than reflective of opportunistic or rapid processes. For most items in the test, responses submitted within 15 seconds of viewing the item cannot be considered reflective of creative thinking processes. A review of the timing data for the items also showed a clear bimodal distribution of response submission, with one peak prior to 15 seconds and another peak a significant time afterwards. This modification was applied to all items, with the exception of

three: in two cases, students were able to select a response to a previous question akin to a multiple-choice mechanism; and in the other item, students were asked to generate a very short written artefact. In these three cases, it was judged that students could submit a response that reflected creative thinking processes within 15 seconds and thus no minimum response time was imposed.

In October 2023, the PISA Secretariat, Core A and the TAG reconvened for the data adjudication of the creative thinking data following the further analyses conducted by the PISA Secretariat and to finalise the reporting approach. The TAG recommended to report the creative thinking data according to a non-linear transformation of the "theta" scale, using the test-characteristic curve for a hypothetical test using the final pool of 32 creative thinking items and based on international item parameters. The advantages of this approach include:

- Reporting student performance according to a bounded scale (between 0-60, reflecting the maximum sum-score of all items) that is the same for all countries. This solution maintains the possibility to report performance on a scale, but signals a clear difference to the PISA scales used for the other domains and the broader "grain" size of the creative thinking scale signals its relative lower reliability compared to the other PISA scales (a 1 point change in the creative thinking scale reflects about 10% of a standard deviation).
- Scores can be easily interpreted in terms of the number of items correct on this specific test (rather than a more general reflection of students' creative thinking ability applied to other performance tests), drawing attention to the actual test content and the framework that guided its development and facilitating the interpretation of the relatively high frequency of low scores in this test (i.e. students scored 0 on the test, rather than not having any creative thinking skill).
- Test scores differ more where the test has more information about students.
- The international database still includes 10 "plausible scores" per student.

Performance level descriptors

Following the data adjudication process and the finalisation of the scale for reporting the creative thinking data, the PISA Secretariat, in collaboration with Core B and the CTEG, defined performance level descriptors. Performance on the creative thinking scale was split into 6 performance levels.

Level 1

At level 1, students can generate very simple visual designs using isolated shapes or existing visual elements, and in some cases very short written artefacts (e.g. a few words), that require them to engage their imagination. In general, students at this level rely on obvious themes or idea associations as the basis for their response and struggle to generate more than one appropriate idea even for open and simple imagination tasks. These students typically generate simple visual or written artefacts with few details that reflect a minimal level of engagement with the task.

Level 2

At level 2, students can generate appropriate ideas for simple visual and written expression tasks as well as those that focus on solving familiar, everyday social problems. With respect to students at level 1, students in level 2 can develop simple written ideas in the form of longer captions or short dialogues. Students at level 2 typically suggest ideas that rely on obvious idea associations for expressive tasks or that refer to existing solutions for problems in social problem-solving tasks. Students can generate more than one appropriate idea for some written expression and social problem-solving tasks, but these ideas are not qualitatively different to one another.

Level 3

At level 3, students can generate one or several appropriate ideas for simple to moderately complex expressive and problem-solving tasks, including extended written ideas that require them to engage and express their imagination and coherently build upon others' ideas. Students at level 3 still typically suggest ideas that rely on obvious idea associations or common themes with respect to their peers, but they begin to demonstrate the ability to recognise and generate original solutions for familiar, everyday problems with a social focus. They may suggest solution ideas that not many other students think of or add an innovative or different twist to more conventional solution ideas.

Level 4

At level 4, students can productively engage in idea generation across a range of expressive and problem-solving tasks. Students at level 4 can also generate original and diverse ideas for simple tasks in more familiar domain contexts. With respect to students at level 3, students at this level can generate an appropriate idea for most types of idea generation task, including more complex or unfamiliar problem-solving tasks and tasks in a scientific context. They can also build on others' ideas for solutions in social and scientific contexts, although they tend to provide an obvious or common iteration with respect to their peers. Students at level 4 can generate their own original ideas in written expression tasks and sometimes when iterating on others' ideas. They can express their imagination in unexpected ways, making unconventional idea associations between elements of the stimulus and their written artefact, or they can add atypical details to elaborate creatively on more common ideas. Students at this level can often suggest two or three qualitatively different ideas in open written expression and social problem contexts but are less successful in more complex or constrained social and scientific problem contexts.

Level 5

At level 5, students can productively engage in creative idea generation, generating both original and diverse ideas for a range of expressive and problem-solving tasks. Students at level 5 can think of qualitatively different ways to express their imagination and to address familiar social and scientific problems. They can make several different idea associations, considering different interpretations and perspectives on the same issue or stimulus. For both simple and more abstract written expression tasks, they can use their imagination to create original written artefacts that make unconventional associations between ideas or that add atypical details to elaborate creatively on common themes. With respect to students at level 4, students can create original visual artefacts that combine elements in an unusual or unexpected way for open visual design tasks. Students at this level can also generate unconventional solution ideas that integrate innovative approaches in familiar social, and sometimes scientific, problem contexts. This includes when tasked to iterate on and improve an existing solution idea in more open, familiar problem contexts.

Level 6

At level 6, students can productively engage in creative idea generation, generating both original and diverse ideas for a wide range of expressive and problem-solving tasks including those in more complex, abstract and unfamiliar contexts. With respect to students at level 5, students at this level can identify weaknesses in existing solutions to social or scientific problems, including those that are in less familiar contexts, and build on this understanding to suggest original and innovative ways to improve solutions. They can also generate several appropriate solution ideas for complex social and scientific problems that require more specific knowledge of the domain context and that have a more restricted solution space. For expressive tasks, students at level 6 can create and improve more abstract visual designs, combining visual elements and representations in unexpected ways and conveying an original interpretation or iteration of an existing representation.

Cutpoints defining the proficiency levels for creative thinking

Annex Table 18.A.3 presents the cut points used to assign items and students to a proficiency level for the creative thinking assessment. As with the other PISA domains (see Chapter 17), values in the table are the lower bound for the corresponding level. For example, Level 6 begins with 48.00. Level 5 begins with 41.00 and ends just below 48.00 (i.e. 47.99), where Level 6 begins. Below Level 1 are those with values lower than 6.00. In other words, those reaching a level are those with a score or difficulty at or above the given cut point. This same interpretation applies to all proficiency scales used in PISA. Annex Table 18.A.4 presents a mapping of the released items to the different levels of the proficiency scales.

References

ACT (2020), *PISA 2022 Creative Thinking Limited Field Trial Research Report*, ACT, Iowa City, IA. [8]

ACT (2019), *PISA 2021 Creative Thinking Validation Study Research Report*, ACT, Iowa City, IA. [7]

ACT (2018), *PISA 2021 Creative Thinking Cognitive Lab Research Report*, ACT, Iowa City, IA. [6]

Cropley, A. (2006), “In Praise of Convergent Thinking”, *Creativity Research Journal*, Vol. 18/3, pp. 391-404. [5]

Guilford, J. (1956), “The structure of intellect”, *Psychological Bulletin*, Vol. 53/4, pp. 267-293, https://doi.org/10.1037/h0040755. [2]

Guilford, J. (1950), “Creativity”, *American Psychologist*, Vol. 5/9, pp. 444-454, https://doi.org/10.1037/h0063487. [4]

OECD (2023), *PISA 2022 Assessment and Analytical Framework*, PISA, OECD Publishing, Paris, https://doi.org/10.1787/dfe0bf9c-en. [1]

Plucker, J., R. Beghetto and G. Dow (2004), “Why isn’t creativity more important to educational psychologists? Potentials, pitfalls, and future directions in creativity research”, *Educational Psychologist*, Vol. 39/2, pp. 83-96. [3]

Annex 18.A. Development and Validation of the Creative Thinking Assessment in PISA 2022

Annex Table 18.A.1. Overview of Creative Thinking Test Metrics and Distribution in PISA 2022

Tables	Title
Table 18.A.2	Distribution of items across facets and domains for the PISA 2022 creative thinking test
Table 18.A.3	Cutpoints for the Creative Thinking Scale
Table 18.A.4	A map for released creative thinking items

Annex Table 18.A.2. Distribution of items across facets and domains for the PISA 2022 creative thinking test

Domain	Facet					
	Field trial			Main survey		
	Generate diverse ideas	Generate creative ideas	Evaluate and improve ideas	Generate diverse ideas	Generate creative ideas	Evaluate and improve ideas
Written expression	4	6	2	4	6	2
Visual expression	2	2	4	1	1	2
Social problem solving	4	3	3	4	3	3
Scientific problem solving	4	1	3	3	1	2
Total	14	12	12	12	11	9

Annex Table 18.A.3. Cutpoints for the Creative Thinking Scale

Cutpoint	Level name
48.00	Level 6
41.00	Level 5
32.00	Level 4
23.00	Level 3
15.00	Level 2
6.00	Level 1

Annex Table 18.A.4. A map for released creative thinking items

Level	Cutpoint	Item	Item difficulty
Level 6	48.00	Science Fair Poster (DT200Q02C2) – Full credit	56.66
		Science Fair Poster (DT200Q01C2) – Full credit	53.91
		Library Accessibility (DT500Q02C2) – Full credit	53.35
		Save the River (DT690Q02C2) – Full credit	49.59
Level 5	41.00	*Library Accessibility (DT500Q02C2) – Partial credit*	*46.73*
		Save the River (DT690Q01C)	46.41
		Carpooling (DT630Q01C2) – Full credit	45.14
		Illustration Titles (DT300Q01C2) – Full credit	44.65

		Save the Bees (DT400Q02C2) – Full credit	43.69
		Space Comic (DT240Q01C2) – Full credit	42.92
Level 4	32.00	*Carpooling (DT630Q01C2) – Partial credit*	*39.40*
		2983 (DT370Q01C2) – Full credit	37.56
		Library Accessibility (DT500Q01C) – Full credit	37.00
		Save the River (DT690Q02C2) – Partial credit	*36.63*
		Save the Bees (DT400Q02C2) – Partial credit	*36.14*
Level 3	23.00	Robot Story (DT570Q01)	31.09
		2983 (DT370Q01C2) – Partial credit	*27.18*
Level 2	15.00	*Library Accessibility (DT500Q01) – Partial credit*	*19.02*
		Space Comic (DT240Q01C2) – Partial credit	*18.50*
Level 1	6.00	*Science Fair Poster (DT200Q02C2) – Partial credit*	*14.59*
		Science Fair Poster (DT200Q01C2) – Partial credit	*11.80*
		Illustration Titles (DT300Q01C2) – Partial credit	*6.74*

19 Scaling procedures and construct validation of context questionnaire data

Introduction

The PISA 2022 Context Questionnaires are based on the questionnaire framework (OECD, 2023[1]) described in Chapter 5 of this technical report. Many questionnaire items were designed to be combined in some way in order to represent latent constructs that cannot be observed directly (e.g., a student's mathematics self-efficacy; sense of belonging; or economic, social, and cultural status). To construct meaningful indices, transformations or scaling procedures were applied to these items.

In the following sections, these indices are referred to as *derived variables* (DVs). This chapter describes the DVs based on one or more items that were constructed and validated for all questionnaires across the respondent groups – students, parents, schools, and teachers – administered in PISA 2022.

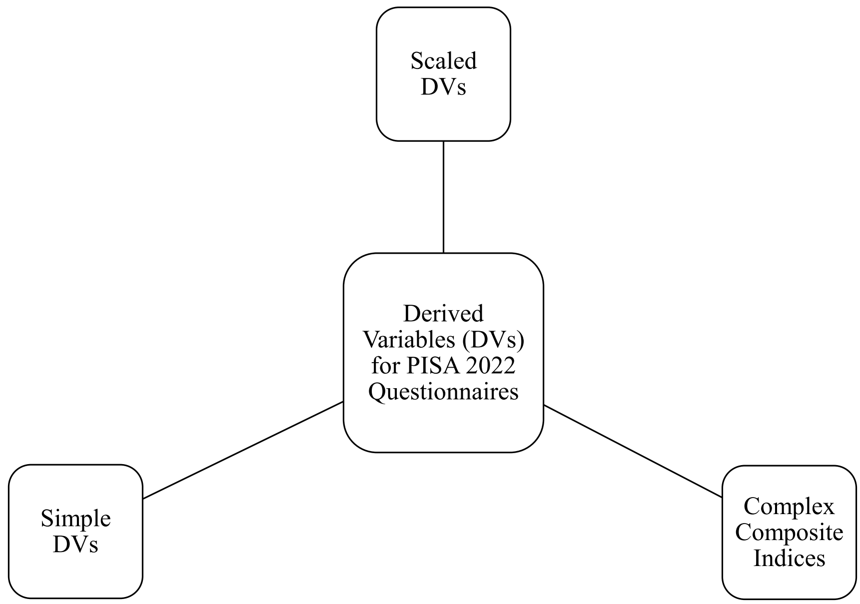
As in the previous PISA surveys, three different kinds of DVs can be distinguished (see Figure 19.1):

- *simple questionnaire indices* constructed through the arithmetical transformation or recoding of one or more items;
- *scaled indices* based on item response theory (IRT) scaling; and
- *complex composite indices* based on a combination of two or more indices.

As described in Chapter 5, the PISA 2022 Context Questionnaires included a broad scope of contextual factors assessed with different questionnaire instruments. While the student and school questionnaires were mandatory in all countries/economies, many countries/economies also administered an optional questionnaire for the parents of the participating students. In addition, countries/economies could choose to administer the optional Financial Literacy Questionnaire, the Information and Communication Technology (ICT) Familiarity Questionnaire, and the Well-Being Questionnaire to students. Moreover, several countries/economies also chose to administer the optional Teacher Questionnaire, which included questionnaires for mathematics teachers and general teachers.

This chapter describes the methodology used for the scaled DVs and also presents an overview of all the simple and scaled DVs for each questionnaire.

Figure 19.1. Types of derived variables for questionnaires in PISA 2022



Within-construct matrix sampling

Previous PISA cycles have used different strategies for collecting data on relevant contextual variables via the Student Questionnaire. For example, PISA 2012 used a three-form booklet design through which each student was administered items for some but not all of the constructs in the questionnaire. The main benefit of this design was that it allowed for the collection of data on approximately 33% more contextual items at the population level without overburdening students with a single-booklet design. However, a disadvantage of this design was the introduction of systematic missing data for students at the construct level, preventing researchers conducting secondary analyses to fully study the relationships between all possible sets of constructs, since no student was administered items for all of the constructs. PISA 2015 and PISA 2018 used a single-form booklet design through which all students were administered the same items. The main benefit of this design was that it allowed for the creation of a database without systematic missing data at the construct or item level, enabling researchers conducting secondary analyses to study the relationships between all possible sets of items and constructs. However, a disadvantage of this design was that it only allowed for the administration of a smaller set of items for each construct compared to the design used in PISA 2012, leading to a large number of relatively short 3-item scales with somewhat limited representation of the broad underlying construct.

PISA 2022 used a new within-construct matrix sampling design that combined the advantages of the multi-form and single-form booklet designs. This design was studied extensively using data from previous PISA cycles as well as the Field Trial data before it was implemented in the Main Survey (Bertling and Weeks, 2018[2]; 2020[3]; Bertling et al., 2020[4]). Specifically, with this new design, every student was administered a random subset of five items for each construct. This design ensured that each item was administered to approximately the same number of students in each country/economy as well as the overall sample. It also allowed each construct to be assessed in larger breadth, kept individual students' burden comparable to previous cycles, and substantially reduced the reading load for students by displaying only five items on each screen.

This within-construct matrix design was only used for the IRT based scales in the Student Questionnaire, as the primary reporting objective for these scales was at the construct level instead of the item level. Also, this design was not used for any scales pertaining to the economic, social, and cultural status index. In addition, it was not used for any of the optional questionnaires administered to students (i.e., Financial Literacy Questionnaire, ICT Familiarity Questionnaire, Well-Being Questionnaire) or questionnaires administered to adult respondents (i.e., Parent Questionnaire, School Questionnaire, Teacher Questionnaire) due to the smaller sample sizes for these questionnaires. Table 19.A.2 provides a list of the 32 scales in the Student Questionnaire that were administered using the within-construct matrix sampling design.

Scaling methodology and reporting of scores

Scaling methodology

As in previous cycles of PISA, some of the DVs were constructed using IRT. More specifically, the two-parameter logistic model (2PLM) (Birnbaum, 1968[5]) was used to scale items with only two response categories (i.e., dichotomous items), while the generalised partial credit model (GPCM) (Muraki, 1992[6]), was used to scale items with more than two response categories (i.e., polytomous items).[1] A detailed explanation of each model is in the following sections. The software mdltm (version 1.965) (Shin et al., 2017[7]; von Davier, 2015[8]) was used for the scaling.

In the initial scaling, item parameters were estimated using data from all individuals with available data from all participating countries/economies. Each country/economy was included in the analysis using a senate weight (SENWT). The senate weight is a linear transformation of the student full sampling weight (W_FSTUWT) such that the sum of SENWT for all cases within a country/economy add up to a constant of 5 000. Due to missing responses within each country/economy, the sum of the SENWT of the cases used in the calibration of each scale varied on a scale-by-scale basis.

For countries/economies with more than one language group, a language group was treated as an independent group in the scaling process if the group's sample size was over 150 and the sum of the weights was over 300. The groups used in the scaling are called *country-by-language groups* since they are defined by both country/economy and language group. For simplicity, the country-by-language groups are also called *groups* in the remainder of this chapter. Note that if the sample size for an entire country/economy was 150 or less, data from the country/economy were not included in the estimation of the item parameters, and the country/economy was assigned international item parameters (explained below) that had been estimated with data from the other countries/economies.

Several of the scales had items with negative valence. These are items for which a higher response category signified a lower level of the construct being measured, and vice versa. The responses to these items were reverse-coded prior to scaling. For all items, including the reverse-coded items, the responses were recoded so that the response corresponding to the lowest level of the construct was coded as 0. Any missing response data, whether it was because an item was not administered or a student did not respond to an item, were ignored and were not included in the analysis.

The two parameter logistic model (2PLM)

The 2PLM, a generalisation of the Rasch model (Rasch, 1960[9]), assumes that the probability of a response x to be positive (coded as 1 in this case) by individual v to item i depends on the difference between the respondent v's trait level θ and the location of the item β. In addition, the 2PLM postulates that for every item, the association between this difference and the response probability depends on an additional item discrimination parameter α. The equation for the response probability for an item under the 2PLM is presented in Formula 19.1:

Formula 19.1

$$P(x_{iv} = 1|\theta_v, \beta_i, \alpha_i) = \frac{\exp\left(D\alpha_i(\theta_v - \beta_i)\right)}{1 + \exp\left(\mathrm{D}\alpha_i(\theta_v - \beta_i)\right)}$$

The item location parameter β can be regarded as the item's general location on the latent continuum of the construct being measured. Items with a higher β parameter require a higher latent trait for a positive response to be selected.

The item discrimination parameter α, which was scaled by a constant D = 1.7 starting in PISA 2015 when the 2PLM was used instead of the Rasch model, characterises how quickly the probability of responding positively to an item approaches 1 with an increase in the trait level θ. In other words, α describes how well a certain item relates to the latent trait θ and, therefore, discriminates between individuals with different trait levels. To solve the indeterminacy of the IRT scale, the average of the item discrimination parameters α across all the items in the scale was constrained to 1. A special case of the 2PLM is when α = 1 for all items, in which case the model is equivalent to the Rasch model.

The generalised partial credit model (GPCM)

The GPCM (Muraki, 1992[6]) is a mathematical model for the probability that an individual will select a certain response category for an item with more than two response categories. Note that the GPCM is a generalisation of the 2PLM and that it reduces to the 2PLM when applied to items with only two response categories. For an item *i* with *m* + 1 ordered categories, the probability of an individual selecting a certain response category *k* (0, 1, 2,…, *m*) under the GPCM and adopting the same notation employed above can be written as:

Formula 19.2

$$P(x_i = k|\theta_v, \beta_i, \alpha_i, d_i) = \frac{\exp\{\sum_{r=0}^{k} D\alpha_i\,(\theta_v - \beta_i + d_{ir})\}}{\sum_{u=0}^{m_i}\exp\{\sum_{r=0}^{u} D\alpha_i\,(\theta_v - \beta_i + d_{ir})\}}$$

As with the 2PLM, the overall item location parameter β can be regarded as the item's general location on the latent continuum of the construct being measured. Items with a higher β parameter require a higher latent trait for a higher response category to be selected. *d* is the step parameter (of which there are *m* for an item with *m* + 1 categories, with the step parameters for each item summing to 0) which represents the deviation of the category intersection δ from the general location β.

The category intersection δ is the intersection between two neighbouring category characteristic curves, in other words, the point on the latent continuum θ at which a higher response category is more likely to be selected (e.g., when the individual is more likely to select "disagree" than "strongly disagree"). Note that β and *d* can be used to calculate the category intersection δ using Formula 19.3.

Formula 19.3

$$\delta_k = \beta - d_k$$

The discrimination parameter α, which was scaled by a constant D = 1.7 starting in PISA 2015, signifies the slope of the category characteristic curves. In other words, it indicates how well selecting a certain response category discriminates between individuals on the latent continuum θ. To solve the indeterminacy of the IRT scale, the average of the item discrimination parameters α across all the items in the scale was constrained to 1. A special case of the GPCM is when α = 1 for all items, in which case the model is equivalent to the partial credit model (PCM) (Masters, 1982[10]).

Figure 19.2 displays the category characteristic curves of a four-category item (e.g., a Likert-type item with response categories “strongly disagree”, “disagree”, “agree”, and “strongly agree”), with the three item parameters used in the GPCM (i.e., α, β and *d*) represented in the figure. For comparison, Figure 19.3 displays the category characteristic curves of an item for which only the α parameter has been increased while the β and *d* parameters were kept the same as in Figure 19.2.

Figure 19.2. Category characteristic curves for a four-category item under the generalised partial credit model (GPCM)

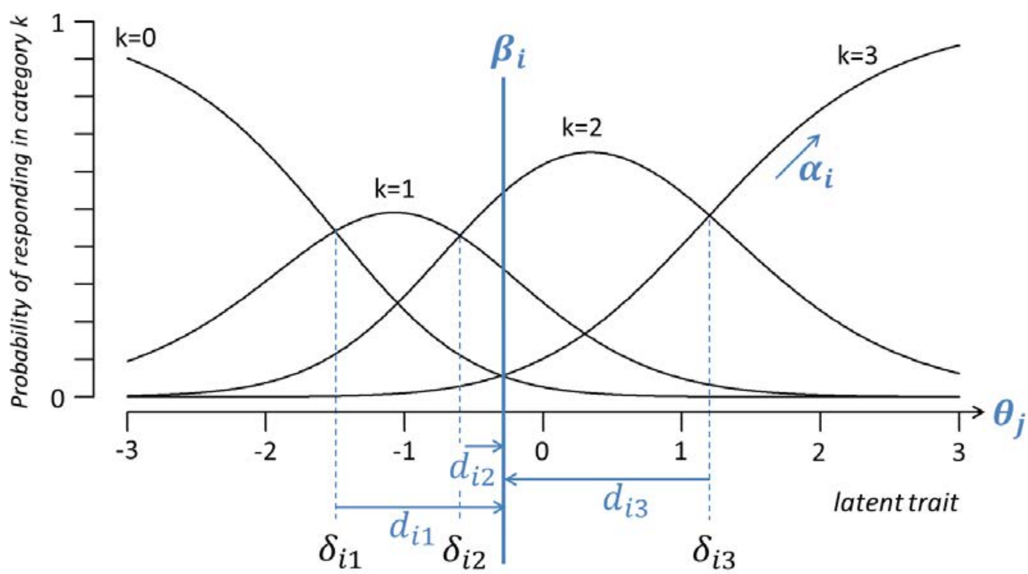
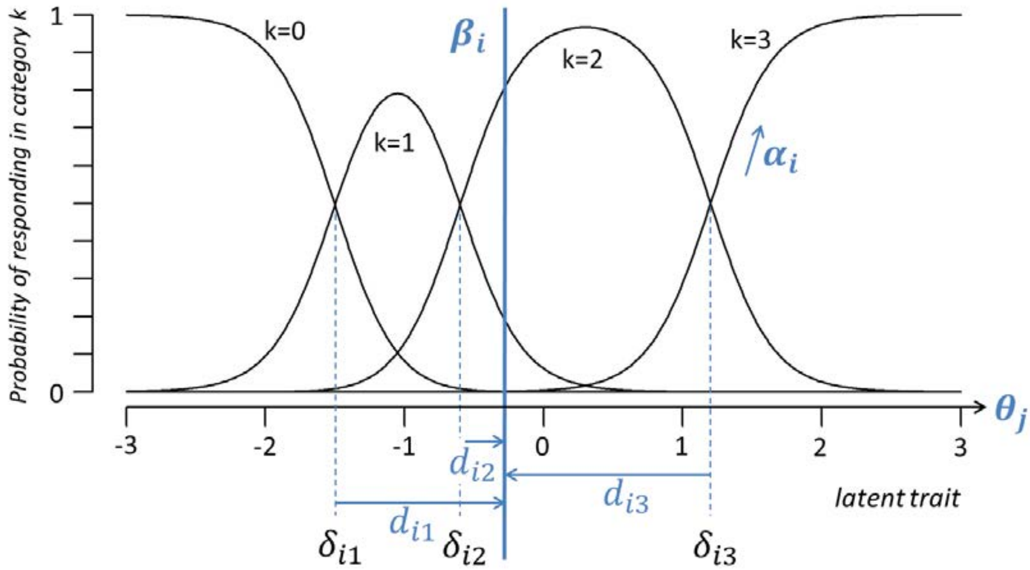


Figure 19.3. Illustration of how an increase in the slope parameter α affects the category characteristic curves of the model above



Special handling of trend scales

For the trend scales, the scaling process began by fixing the item parameters of the trend items to the parameters that had been estimated for each group in the previous cycle, a procedure called fixed parameter linking. *Also, in line with the models that were used in the past cycles, the trend scales linked to PISA 2018 were scaled using the 2PLM and GPCM, while the trend scales linked to PISA 2012 were scaled using the Rasch model and PCM.* This was done so that the scale scores from the current cycle would be comparable to the scale scores from the previous cycle. To compute trends, a scale needed to have at least three trend items, but some trend scales consisted of both trend items and new items. In this case, the item parameters for the trend items were fixed at the beginning of the scaling process, but the item parameters for the new items were estimated using the PISA 2022 data. Note that all the items in the trend scales were also evaluated for the goodness-of-fit of the trend parameters, a process described below. Please see Table 19.A.3 for a full list of trend scales in PISA 2022.

Releasing item parameters

PISA 2022 adopted and further refined the approach for evaluating the invariance of latent constructs across groups using multiple-group concurrent calibration with partial invariance constraints, a method which was first introduced in PISA 2015.

As explained above, in the initial scaling, item parameters were estimated using data from all individuals with available data from all countries/economies. The item parameters that were estimated in this initial scaling process are called *international parameters* since they were estimated using responses from most or all participating countries/economies. After the initial scaling, the fit of the international parameters for each item was evaluated for each group using the root mean square deviance (RMSD). The $RMSD_g$ for group g is defined as:

Formula 19.4

$$RMSD_g = \sqrt{\int \left[p_g^{obs}(\theta) - p_g^{exp}(\theta)\right]^2 f_g(\theta) d\theta}$$

quantifying the difference between the observed item characteristic curve (ICC) for the group based on the pseudo counts from the E-step of the Expectation-Maximisation (EM) algorithm ($\mathrm{P}_{\mathrm{obs,gk}}(\theta)$) with the model-based Item Characteristic Curve ICC ($\mathrm{P}_{\mathrm{exp,gk}}(\theta)$) (Shin et al., 2017[7]). RMSD values range from 0 to 1, with values close to 0 indicating good item fit, meaning that the model-based item parameters fit the data for the group well. Note that the RMSD statistic is sensitive to group-specific deviations of both the item location parameter β and the item discrimination parameter α.

When the RMSD for an item*group exceeded a pre-defined cut-off of 0.25, it was considered that the model-based item parameters did not fit the group's data well, and unique item parameters (also called *group-specific item parameters*) were estimated for the group using data only from that group. However, if more than one group had similar response patterns for an item, data from those groups were pooled together and the same unique parameters were estimated for those groups. This process is called *releasing item parameters*. Item parameters were released until all item*groups had an RMSD value under 0.25.

In PISA 2015 and PISA 2018, an RMSD value of 0.3 was used as the cut-off criterion for releasing item parameters for the context questionnaires. However, an analysis of the scaling results from the PISA 2018 Student Questionnaire suggested that this threshold may have over-emphasised international comparability of the model over group-level model-data fit, as very few item*groups received unique parameters. For PISA 2022, the RMSD threshold for releasing item parameters for the context questionnaires was lowered to 0.25, as it was found that this new threshold could improve the group-level model-data fit without weakening the comparability of the model across groups (as measured by the percent of item*groups with unique parameters, number of groups with international parameters for three or more items in a scale, and the rank order correlation of the scale scores from the models with and without unique parameters).

The final distribution of the RMSD values across groups for each scale item after the final scaling is documented in Annex E. Note that the figures do not include RMSD values for item*groups with an unweighted sample size of 150 or less, as the sample size was too small to estimate stable group-specific ICCs.

Scale scores

The scaling process described above produced weighted likelihood estimates (WLE) (Warm, 1989[11]) for each individual. These WLE scores were subsequently standardised through the process described in the following sections. Note that if an individual had fewer than three valid responses for a scale, a WLE score was not produced for the individual and his/her scale score was replaced with "99" in the SPSS file and ".M" in the SAS file.

New scales

For the new scales, the original WLE scores were transformed into a reporting metric to have a mean of 0 and a standard deviation of 1 across the OECD countries, using senate weights for all cases with available data. The transformation was achieved by applying Formula 19.5:

Formula 19.5

$$\theta'_v = \frac{\theta_v - \bar{\theta}_{OECD}}{\sigma_{\theta(OECD)}}$$

where θ'_v is the scale score on the reporting metric, θ_v is the original WLE, $\bar{\theta}_{OECD}$ is the mean of the original WLEs across the OECD countries, and $\sigma_{\theta\,OECD}$ is the standard deviation of the original WLEs across the OECD countries. The transformation constants that were used to transform the original WLEs into the reporting scale are displayed in Table 19.A.4.

For the new scales, an average scale score of 0 is expected when calculated across all OECD countries using the senate weights. A negative scale score does not imply that a student responded negatively to the items in the scale. Rather, it means that the student is below the OECD average.

Trend scales

For the trend scales, to ensure the comparability of the scale scores from the current cycle to the scale scores from the previous cycle, the original WLEs of PISA 2022 were transformed using the same transformation constants of the original WLEs from the cycle to which the current cycle was linked. Table 19.A.5 presents the transformation constants of the original WLEs in PISA 2018 for the trend scales linked to PISA 2018, while Table 19.A.6 presents the transformation constants used in PISA 2012 for the trend scales linked to PISA 2012.

Criteria for suppressing scale scores

The scale scores of individuals or groups were suppressed under the following conditions.

Low internal consistency

Cronbach’s alpha coefficient was used to check the internal consistency of each scale for each group. This coefficient ranges from 0 to 1, with a higher value indicating higher internal consistency. A group needed to have a Cronbach’s alpha of at least 0.60 for a scale in order for the group’s scale scores to be reported. Scale scores were suppressed for countries/economies in which one or more language groups had a Cronbach’s alpha under 0.60 for the scale.

Few items with international parameters

For each scale, a group needed to have at least three items with international parameters in order for the scale scores of the group to be considered comparable to the scale scores of the other groups. Scale scores were suppressed for countries/economies in which one or more language groups had less than three items with international parameters for the scale. The scale scores for the individuals in these countries/economies were replaced with “97” in the SPSS file and “N” in the SAS file.

Lack of trend items with international parameters

For the trend scales, a group needed to have at least three trend items with international parameters in order for the PISA 2022 scales scores for the group to be considered comparable to the scale scores of the previous cycle to which the current cycle was linked. Scale scores were suppressed for countries/economies in which one or more language groups had less than three trend items with international parameters for the scale. The scale scores for the individuals in these countries/economies were replaced with “97” in the SPSS file and “N” in the SAS file.

Student Questionnaire derived variables

There were 86 variables derived from the Student Questionnaire, including 43 simple DVs, 42 IRT scaled DVs, and one complex composite index. The DVs are shown in Table 19.A.7 and will be described in the following sections; the first section covers all simple DVs, the second section covers those that are based on IRT scaling, and the last section covers the complex composite index. The simple and scaled DVs are organised first by framework module (please see Chapter 5 for a description of modules) and then alphabetical order within modules.

Simple questionnaire indices

Basic demographics (Module 1)

Student's age (AGE)

The age of a student (AGE) was calculated as the difference between the year and month of the testing and the year and month of a student's birth, which was obtained from school records from the student sampling data and validated by comparing to the students' responses in the questionnaire. Data on students' age were obtained from both the questionnaire (ST003) and the student tracking forms. The formula for computing *AGE* was:

Formula 19.6

$$AGE = (100 + Ty - Sy) + (Tm - Sm)/12$$

where *Ty* and *Sy* are the year of the test and the year of the students' birth, respectively, in two-digit format (for example "06" or "92"), and *Tm* and *Sm* are the month of the test and month of the students' birth, respectively. The result is rounded to two decimal places.

Grade compared to modal grade in country (GRADE)

The relative grade index (GRADE) was computed to capture between-country/economy variation. It indicates whether students are in the country/economy's modal grade (value of 0), or the number of grades below or above the modal grade in the country. The information about the students' grade level was obtained from school records from the student sampling data and validated by comparing the students' responses in the Student Questionnaire (ST001).

Gender (ST004D01T)

The gender of a student which was obtained from school records from the student sampling data and validated by comparing to the student's responses in the questionnaire (ST004).

Economic, social and cultural status (Module 2)

Mother's level of education (MISCED)

Student responses to questions ST005 and ST006 regarding their mothers' education were used to derive the mother's level of education (MISCED) index, where education level ranged from "1" less than ISCED level 1 to "10" ISCED level 8, as noted in Table 19.A.8.[2]

Father’s level of education (FISCED)

Student responses to questions ST007 and ST008 regarding their fathers’ education were used to derive the father’s level of education (FISCED) index, where education level ranged from “1” less than ISCED level 1 to “10” ISCED level 8, as noted in Table 19.A.8 above.

Highest level of education of parents (HISCED)

Students’ responses to questions ST005, ST006, ST007, and ST008 regarding their mothers’ and fathers’ education were used to derive the index of highest education level of parents (HISCED). The index is equal to the highest ISCED level of either parent.

Highest education of parents in years (PAREDINT)

The index of the highest education of parents in years, PAREDINT, was based on the median cumulative years of education associated with completion of the highest level of parental education (HISCED). Cumulative years of education values used in PISA 2018 were assigned to each ISCED level (see Table 19.A.8.).

Mother’s occupational code (OCOD1)

Students’ responses to the fill-in question ST014 about their mothers’ occupation were human-coded based on the International Standard Classification of Occupations (ISCO)-08 classification system, resulting in the mother’s occupational code (4-digit ISCO; ILO, 2007) index, OCOD1. These 4-digit codes range from 0000 to 9705. Codes 0000 to 9629 are occupations from the ISCO-08 classification system. Codes 9701-9705 were used to classify responses that fell outside of the ISCO-08 classification system. Specifically, the code 9701 indicates “stay-at-home parent”, 9702 indicates “student”, and 9703 indicates “social beneficiary (e.g., unemployed, retired, sick)”. Lastly, “I don’t know” responses were coded 9704 and vague responses (e.g., a good job, a well-paid job) were coded 9705.

Father’s occupational code (OCOD2)

Students’ responses to the fill-in question ST015 about their fathers’ occupation were human-coded based on the ISCO-08 classification system, resulting in the father’s occupational code (4-digit ISCO) index, OCOD2. These 4-digit ISCO-08 codes range from 0000 to 9705. Codes 0000 to 9629 are occupations from the ISCO-08 classification. Codes 9701-9705 were used to classify responses that fell outside of the ISCO-08 classification. Specifically, the code 9701 indicates “stay-at-home parent”, 9702 indicates “student”, and 9703 indicates “social beneficiary (e.g., unemployed, retired, sick)”. Lastly, “I don’t know” responses were coded 9704 and vague responses (e.g., a good job, a well-paid job) were coded 9705.

Mother’s occupational status (BMMJ1)

The mother’s occupational status index, BMMJ1, was derived from the OCOD1 index and international socio-economic index of occupational status (ISEI) (Ganzeboom and Treiman, 2003[12]) scores. The 4-digit ISCO-08 occupation codes in OCOD1 were mapped onto ISEI ratings.

Father’s occupational status (BFMJ2)

The father’s occupational status index, BFMJ2, was derived from the OCOD2 index and international socio-economic index of occupational status (ISEI) scores. The 4-digit ISCO-08 occupation codes in OCOD2 were mapped onto ISEI occupational status scores.

Highest parental occupational status (HISEI)

This highest parental occupational status index (HISEI) was based on the 4-digit ISCO-08 occupational codes that were human coded from students' responses to questions ST014 and ST015 about their mother and father's occupations, respectively. The index was equal to the higher of the mother's (BMMJ1) and father's (BFMJ2) ISEI scores.

Educational pathways and post-secondary aspirations (Module 3)

Duration in early childhood education and care (DURECEC)

Questions ST125 and ST126 measure the starting age in ISCED 1 and ISCED 0. The indicator DURECEC is built as the difference of ST126 and ST125 plus the value of "2" to indicate the number of years a student spent in early childhood education and care.

Study programme level and orientation (ISCEDP)

PISA collects data on study programmes available to 15-year-old students in each country/economy. This information is obtained through the student tracking form and the Student Questionnaire (ST002). In the final database, all national programmes are included in a separate DV (PROGN) where the first six digits represent the National Centre code, and the last two digits are the nationally specific programme code. All study programmes were classified using the International Standard Classification of Education (ISCED 2011).

The study programme level and orientation index (ISCEDP) is a three-digit index that describes whether students were at the lower or upper secondary level and (ISCED 2 or ISCED 3) and whether their programmes were general or vocational and sufficient for level completion with direct access to tertiary or post-secondary non-tertiary education. ISCEDP values and labels can be found in Table 19.A.9.

Grade repetition (REPEAT)

Students' answers on question ST127 of whether and, if yes, how often they have ever repeated a grade at ISCED levels 1, 2, and 3 were combined into the index REPEAT. Each item included three response options ("No, never", "Yes, once", "Yes, twice or more"). REPEAT took the value of "0" if the student never repeated a grade (student did not select options 2 or 3 for any of the three items) and the value of "1" if the student repeated a grade at least once (student selected options 2 or 3 for at least one of the three items). The index was assigned a missing value if none of the three response options were selected in any levels.

Missing school (MISSSC)

Students' answers on question ST260 of whether and, if yes, how often they have ever missed school for more than three months in a row at ISCED levels 1, 2, and 3 were combined into the index MISSSC. Each item included three response options ("No, never", "Yes, once", "Yes, twice or more"). MISSSC took the value of "1" if the student selected options 2 or 3 for at least one of the three items, and the value "0" elsewise. The index was assigned a missing value if none of the three response options were selected in any levels.

Skipping classes or days of school (SKIPPING)

Students' responses to whether, in the two weeks prior to the PISA test, they had skipped classes (ST062Q02TA) or days of school (ST062Q01TA) at least once were used to derive an indicator of student truancy. Both questions have four response options ("Never", "One or two times", "Three or four times", "Five or more times"). The indicator takes a value of 0 if students reported that they had not skipped any class or day of school in the two weeks before the PISA test, and a value of 1 if students reported that they had skipped classes or days of school at least once in the same period.

Arriving late for school (TARDYSD)

Students responded to a question about whether and how frequently they had arrived late for school during the two weeks prior to the PISA test (ST062Q03TA). TARDYSD takes a value of “0” for on-time students if students reported that they had not arrived late for school, a value of “1” for occasional late arrivals if students report they arrived late for school one or two times, and “2” for frequent late arrivals if students reported they had arrived late for school three or more times.

Highest expected educational level (EXPECEDU)

Students’ responses which of a list of possible educational levels they expect to complete in question ST327 were transformed into the index of “Highest Expected Educational Level”. This DV has been newly created for 2022. Values on the index can range from “Less than ISCED level 2” to “ISCED level 8”. Scores are assigned as shown in Table 19.A.10.

Expected occupation (OCOD3) and Expected occupation status (BSMJ)

Students’ responses to the fill-in question ST329 about what kind of job they expect to have when they are about 30 years old were human-coded based the ISCO-08 classification system, resulting in the index “Expected Occupation (OCOD3)”. These ISCO codes were then mapped to the international socio-economic index of occupational status (ISEI) (Ganzeboom and Treiman, 2003[12]) in variable BSMJ. Higher scores on this variable indicate higher levels of a student’s expected occupational status.

Clear idea about future job (SISCO)

The students who had a clear idea about their future job index (SISCO) was based on the human-coded open-ended expected occupation index, OCOD3, which was derived from question ST329. Students who had no clear idea about their future jobs were considered those who indicated “I do not know” or gave a vague answer such as “a good job”, “a quiet job”, “a well-paid job”, “an office job” in response to question ST329. In the OCOD3 index, “I don’t know” responses were coded 9704 and vague responses were coded 9705. Examples of invalid responses include students who did not answer the question or gave an answer, such as a smiley face. Specifically, a value of “0” is assigned on the index if OCOD3 values are 9704 or 9705, and a value of “1” is assigned if OCOD3 values are 0000 to 9703.

Migration and language exposure (Module 4)

Based on students’ responses to question ST019 (“In what country were you and your parents born?”), five indices are created as outlined below.

Student’s country of birth (COBN_S)

This index has the value “1” if the student selected the country of test (“Country A”) in question ST019AQ01T, and “0” otherwise.

Student mother’s country of birth (COBN_M)

This index has the value “1” if the student selected the country of test (“Country A”) in question ST019BQ01T, and “0” otherwise.

Student father's country of birth (COBN_F)

This index has the value "1" if the student selected the country of test ("Country A") in question ST019CQ01T, and "0" otherwise.

Index on immigrant background (IMMIG)

The index on immigrant background (IMMIG) is calculated from the three variables above (COBN_S, COBN_M, COBN_F), and has the categories as listed below. Students with missing responses for either the student or for both parents were given missing values for this variable.

- Native students (those students who had at least one parent born in the country/economy);
- Second-generation students (those born in the country/economy of assessment but whose parent[s] were born in another country/economy);
- First-generation students (those students born outside the country/economy of assessment and whose parents were also born in another country/economy).

Language spoken at home (LANGN)

Students also indicated what language they usually spoke at home, and the database includes a variable (LANGN) containing country/economy-specific code for each language.

Subject-specific beliefs, attitudes, feelings and behaviours (Module 7)

Relative motivation to do well in mathematics compared to other core subjects (MATHMOT)

This simple index captures whether students indicate being more motivated to do well in mathematics than in Test Language and Science class. If students endorsed question ST268Q07JA ("I want to do well in my mathematics class.") stronger than both items ST268Q08JA ("I want to do well in my <test language> class.") and ST268Q09JA ("I want to do well in my <science> class."), they received a "1" on this index, otherwise "0". Please note that this index captures students' relative motivation for math rather than their absolute motivation for mathematics. The latter is captured by the original response to the item.

Perception of mathematics as easier than other core subjects (MATHEASE)

This simple index captures whether students indicate they perceive mathematics as easier compared to the Test Language and Science. If students endorsed question ST268Q014A ("Mathematics is easy for me.") stronger than both items ST268Q05JA ("<Test language> is easy for me.") and ST268Q06JA ("<Science> is easy for me"), they received a "1" on this index, otherwise "0". Please note that this index captures students' relative easiness of math rather than their absolute easiness rating for mathematics. The latter is captured by the original response to the item.

Preference of mathematics over other core subjects (MATHPREF)

This simple index captures whether students indicate they preferred mathematics over Test Language and Science. If students endorsed question ST268Q01JA ("Mathematics is one of my favourite subjects") stronger than both items ST268Q02JA ("<Test language> is one of my favourite subjects.") and ST268Q03JA ("<Science> is one of my favourite subjects."), they received a "1" on this index, otherwise "0". Please note that this index captures students' relative preference of math rather than their absolute preference for mathematics. The latter is captured by the original response to the item.

Out-of-school experiences (Module 10)

Exercising or practising a sport before or after school (EXERPRAC)

Students' answers on how many days during a typical school week they exercised or practised a sport before going to school and/or after leaving school in questions ST294 and ST295 were scaled into the index of "Exercise or practise a sport before or after school". Each item included six response options ("0 days", "1 day", "2 days", "3 days", "4 days", "5 or more days"). Values on this index range from 0 (no exercise or sports) to 10 (10 or more times exercise or sport a per week).

Studying for school or homework before or after school (STUDYHMW)

Students' answers on how many days during a typical school week they studied for school or homework before going to school and/or after leaving school in questions ST294 and ST295 were scaled into the index of "Study for school or homework before or after school". Each item included six response options ("0 days", "1 day", "2 days", "3 days", "4 days", "5 or more days"). Values on this index range from 0 (no studying) to 10 (10 or more times of studying per week).

Working for pay before or after school (WORKPAY)

Students' answers on how many days during a typical school week they worked for pay before going to school and/or after leaving school in questions ST294 and ST295 were scaled into the index of "Work for pay before or after school". Each item included six response options ("0 days", "1 day", "2 days", "3 days", "4 days", "5 or more days"). Values on this index range from 0 (no work for pay) to 10 (10 or more times of working for pay per week).

Working in household or taking care of family members (WORKHOME)

Students' answers on how many days during a typical school week they worked in the household or took care of a family member before going to school and/or after leaving school in questions ST294 and ST295 were scaled into the index of "Work in household or take care of family members". Each item included six response options ("0 days", "1 day", "2 days", "3 days", "4 days", "5 or more days"). Values on this index range from 0 (no work in household or care of family members) to 10 (10 or more times of working in household or caring for family members per week).

Derived variables based on IRT scaling

The Student Questionnaire provided data for 42 DVs based on IRT scaling. The Cronbach's alpha for each scale and group are presented in Table 19.A.11., the number of items with international parameters for each scale and group are presented in Table 19.A.12, the number of trend items with international parameters for each trend scale and group are presented in Table 19.A.13, the countries/economies for which the scale scores were suppressed for each scale are presented in Table 19.A.14, and the groups that did not administer each scale are presented in Table 19.A.15.

Economic, social and cultural status (Module 2)

Home possessions (HOMEPOS)

In the HOMEPOS scale (which included questions ST250, ST251, ST253, ST254, ST255, and ST256), students indicated whether their household possessed certain items (e.g., "A room of your own", "Educational software or apps") or how many of an item their household possessed (e.g., "Rooms with a <flush toilet>", "Cars, vans, or trucks"). This scale included 31 items, including four country/economy-specific items (ST250Q06JA, ST250Q07JA, ST251Q08JA, and ST251Q09JA) that were seen as local measures of family wealth within the country/economy's context.[3] In addition, students answered how many books (ST255) and digital devices with screens (ST253) were in their home. Note that all groups received unique item parameters for the country/economy-specific items (i.e., no international parameters were estimated for these items) and that for some items, the response categories were collapsed to align with the response categories used in previous cycles. Table 19.A.16 shows the item wording and item parameters for the items in this scale, while Table 19.A.17 shows how the response categories for each item were recoded prior to scaling.

ICT resources (ICTRES)

Students reported on the availability of 11 Information and Communications Technologies (ICT) resources in their home (e.g., "A computer (laptop, desktop, or tablet) that you can use for school work", "Internet access (e.g., Wi-fi) (excluding through smartphones)") in questions ST250 (which had two response categories), ST253 (which had eight response categories), and ST254 (which had four substantive response categories and an additional response category "I don't know." which was recoded as missing prior to scaling). These items were scaled into the index of "ICT resources". Table 19.A.18 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded and how the response categories were recoded prior to scaling.

Educational pathways and post-secondary aspirations (Module 3)

Information seeking regarding future career (INFOSEEK)

Students' ratings of whether they had undertaken a range of possible activities to find out about future study or types of work (e.g., "I did an internship.", "I researched the internet for information about careers.") in question ST330 were scaled into the index of "Information seeking regarding future career". Note that this scale used a within-construct matrix sampling design. Each of the 11 items included in this scale had three response options ("Yes, once", "Yes, two or more times", "No"). Table 19.A.19 shows the item wording and item parameters for the items in this scale.[4] It also shows how the response categories were recoded prior to scaling.

School culture and climate (Module 6)

Being bullied (BULLIED)

Students' ratings of how often they had a range of experiences at school that are indicative of being bullied during the past 12 months (e.g., "Other students left me out of things on purpose.", "Other students made fun of me.") in question ST038 were scaled into the index of "Being bullied". Note that this scale was linked to the BEINGBULLIED scale in PISA 2018. Each of the nine items included in this scale had four response options ("Never or almost never", "A few times a year", "A few times a month", "Once a week or more"). Table 19.A.20 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Feeling safe (FEELSAFE)

Students' ratings of their agreement with four statements about their perceived safety (e.g., "I feel safe on my way to school.", "I feel safe in my classrooms at school.") in question ST265 were scaled into the index of "Feeling safe". Each of the four items included in this scale had four response options ("Strongly agree", "Agree", "Disagree", "Strongly disagree"). Table 19.A.21 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Mathematics teacher support (TEACHSUP)

Students' frequency ratings of how often a range of situations occurred in their mathematics lessons (e.g., "The teacher shows an interest in every student's learning.", "The teacher gives extra help when students need it.") in question ST270 were scaled into the index of "Mathematics teacher support". Note that this scale was linked to the TEACHSUP scale in PISA 2012 and was scaled using the PCM, in line with the model used in PISA 2012. Each of the four items included in this scale had four response options ("Every lesson", "Most lessons", "Some lessons", "Never or almost never"). Table 19.A.22 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling and which items are trend items.

Quality of student-teacher relationships (RELATST)

Students' ratings of their agreement with the eight statements (e.g., "The teachers at my school are respectful towards me.", "When my teachers ask how I am doing, they are really interested in my answer.") in question ST267 were scaled into the index of "Quality of student-teacher relationships". Note that this scale used a within-construct matrix sampling design. Each of the eight items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.23 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

School safety risks (SCHRISK)

Students' answers of whether a range of events indicative of safety risks at school occurred during the past four weeks (e.g., "Our school was vandalised.", "I witnessed a fight on school property in which someone got hurt.") in question ST266 were scaled into the index of "School safety risks". Each of the five items included in this scale had two response options ("Yes", "No"). Table 19.A.24 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Sense of belonging (BELONG)

Students' ratings of their agreement with six statements (e.g., "I feel like I belong at school.", "I feel lonely at school.") in question ST034 were scaled into the index of "Sense of belonging". Note that this scale used a within-construct matrix sampling design and that it was linked to the BELONG scale in PISA 2018. Each of the six items included in this scale had four response options ("Strongly agree", "Agree", "Disagree", "Strongly disagree"). Table 19.A.25 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling and which items are trend items.

Subject-specific beliefs, attitudes, feelings, and behaviours (Module 7)

Growth mindset (GROSAGR)

Students' ratings of their agreement with a range of statements indicative of their mindset (e.g., "Your intelligence is something about you that you cannot change very much.", "Some people are just not good at mathematics, no matter how hard they study.") in question ST263 were scaled into the index of "Growth mindset". Each of the four items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.26 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Mathematics anxiety (ANXMAT)

Students' ratings of their agreement with statements about a range of attitudes towards mathematics (e.g., "I often worry that it will be difficult for me in mathematics classes.", "I feel anxious about failing in mathematics.") in question ST292 were scaled into the index of "Mathematics anxiety". Note that this scale was linked to the ANXMAT scale in PISA 2012 and was scaled using the PCM, in line with the model used in PISA 2012. Also, it used a within-construct matrix sampling design. Each of the six items included in this scale had four response options ("Strongly agree", "Agree", "Disagree", "Strongly disagree"). Table 19.A.27 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling and which items are trend items.

Mathematics self-efficacy: Formal and applied mathematics (MATHEFF)

Students' ratings of how confident they felt about having to do a range of formal and applied mathematics tasks (e.g., "Calculating how much more expensive a computer would be after adding tax", "Solving an equation like 2(x+3) = (x+3)(x-3)") in question ST290 were scaled into the index of "Mathematics self-efficacy: Formal and applied mathematics". Note that this scale was linked to the MATHEFF scale in PISA 2012 and was scaled using the PCM, in line with the model used in PISA 2012. Also, it used a within-construct matrix sampling design. Each of the nine items included in this scale had four response options ("Not at all confident", "Not very confident", "Confident", "Very confident"). Note that in PISA 2012, the response options were presented to the students ordered from "Very confident" to "Not at all confident" possibly eliciting different response patterns related to the format of the question, and not necessarily related the construct. Because of this, caution should be exercised when comparing scale scores across these two cycles. Table 19.A.28 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Mathematics self-efficacy: Mathematical reasoning and 21st century mathematics (MATHEF21)

Students' ratings of how confident they felt about having to do a range of mathematical reasoning and 21st century mathematics tasks (e.g., "Extracting mathematical information from diagrams, graphs, or simulations", "Using the concept of statistical variation to make a decision") in question ST291 were scaled into the index of "Mathematics self-efficacy: Mathematical reasoning and 21st century mathematics". Note that this scale used a within-construct matrix sampling design. Each of the 10 items included in this scale had four response options ("Not at all confident", "Not very confident", "Confident", "Very confident"). Table 19.28 shows the item wording and item parameters for the items in this scale.

Proactive mathematics study behaviour (MATHPERS)

Students' frequency ratings of how often they engaged in behaviours indicative of effort and persistence in mathematics (e.g., "I actively participated in group discussions during mathematics class.", "I put effort into my assignments for mathematics class.") in question ST293 were scaled into the index of "Proactive mathematics study behaviour". Note that this scale used a within-construct matrix sampling design. Each of the eight items included in this scale had five response options ("Never or almost never", "Less than half of the time", "About half of the time", "More than half of the time", "All or almost all of the time"). Table 19.A.30 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Subjective familiarity with mathematics concepts (FAMCON)

Students' ratings of how familiar they were with different mathematical concepts representative of different levels of mathematical skill or understanding (e.g., "Divisor", "Exponential function", "3-dimensional geometry") in question ST289 were scaled into the index of "Subjective familiarity with mathematics concepts". Note that this scale was linked to the FAMCON scale in PISA 2012 and was scaled using the PCM, in line with the model used in PISA 2012. Also, it used a within-construct matrix sampling design. Each of the 10 items included in this scale had five response options ("Never heard of it", "Heard of it once or twice", "Heard of it a few times", "Heard of it often", "Know it well, understand the concept"). Table 19.A.31 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

General social and emotional characteristics (Module 8)

All of the scales in this module used a within-construct matrix sampling design and included both positively and negatively valenced items. This allowed us to check the consistency of responses since we would expect those agreeing with the items with positive valence to disagree with items with negative valence, and vice versa. To this effect, some students were identified as extreme straightliners. These were students that were administered items with positive and negative valence and responded to all five items selecting the same extreme response category, "Strongly disagree" or "Strongly agree". Students that were identified as extreme straightliners were removed from the analysis.

Assertiveness (ASSERAGR)

Students' ratings of their agreement with statements about a range of behaviours indicative of assertiveness (e.g., "I take initiative when working with my classmates.", "I find it hard to influence people.") in question ST305 were scaled into the index of "Assertiveness". Note that this scale used a within-construct matrix sampling design. Each of the 10 items included in this scale had five response options ("Strongly disagree", "Disagree", "Neither agree nor disagree", "Agree", "Strongly agree"). Table 19.A.32 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling. Table 19.A.33 shows the percent of students in each country/economy that did not receive a scale score for ASSERAGR due to extreme straightlining or, for comparison, for not having enough responses (i.e., less than three responses for the scale). In both cases, the scale scores were replaced with "99" in the SPSS file and ".M" in the SAS file.

Cooperation (COOPAGR)

Students' ratings of their agreement with statements about a range of behaviours indicative of cooperation (e.g., "I work well with other people.", "I get annoyed when I have to compromise with others.") in question ST343 were scaled into the index of "Cooperation". Note that this scale used a within-construct matrix sampling design. Each of the 10 items included in this scale had five response options ("Strongly disagree", "Disagree", "Neither agree nor disagree", "Agree", "Strongly agree"). Table 19.A.34 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling. Table 19.A.35 shows the percent of students in each country/economy that did not receive a scale score for COOPAGR due to extreme straightlining or, for comparison, for not having enough responses (i.e., less than three responses for the scale). In both cases, the scale scores were replaced with "99" in the SPSS file and ".M" in the SAS file.

Curiosity (CURIOAGR)

Students' ratings of their agreement with statements about a range of behaviours indicative of curiosity (e.g., "I like to know how things work.", "I am more curious than most people I know.") in question ST301 were scaled into the index of "Curiosity". Note that this scale used a within-construct matrix sampling design. Each of the 10 items included in this scale had five response options ("Strongly disagree", "Disagree", "Neither agree nor disagree", "Agree", "Strongly agree"). Table 19.A.36 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling. Table 19.A.37 shows the percent of students in each country/economy that did not receive a scale score for CURIOAGR due to extreme straightlining or, for comparison, for not having enough responses (i.e., less than three responses for the scale). In both cases, the scale scores were replaced with “99” in the SPSS file and “.M” in the SAS file.

Emotional control (EMOCOAGR)

Students’ ratings of their agreement with statements about a range of behaviours indicative of emotional control (e.g., “I keep my emotions under control.”, “I get mad easily.”) in question ST313 were scaled into the index of “Emotional control”. Note that this scale used a within-construct matrix sampling design. Each of the 10 items included in this scale had five response options (“Strongly disagree”, “Disagree”, “Neither agree nor disagree”, “Agree”, “Strongly agree”). Table 19.A.38 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling. Table 19.A.39 shows the percent of students in each country/economy that did not receive a scale score for EMOCOAGR due to extreme straightlining or, for comparison, for not having enough responses (i.e., less than three responses for the scale). In both cases, the scale scores were replaced with “99” in the SPSS file and “.M” in the SAS file.

Empathy (EMPATAGR)

Students’ ratings of their agreement with statements about a range of behaviours indicative of empathy (e.g., “I predict the needs of others.”, “It is difficult for me to sense what others think.”) in question ST311 were scaled into the index of “Empathy”. Note that this scale used a within-construct matrix sampling design. Each of the 10 items included in this scale had five response options (“Strongly disagree”, “Disagree”, “Neither agree nor disagree”, “Agree”, “Strongly agree”). Table 19.A.40 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling. Table 19.A.41 shows the percent of students in each country/economy that did not receive a scale score for EMPATAGR due to extreme straightlining or, for comparison, for not having enough responses (i.e., less than three responses for the scale). In both cases, the scale scores were replaced with “99” in the SPSS file and “.M” in the SAS file.

Perseverance (PERSEVAGR)

Students’ ratings of their agreement with statements about a range of behaviours indicative of perseverance (e.g., “I keep working on a task until it is finished.”, “I give up after making mistakes.”) in question ST307 were scaled into the index of “Perseverance”. Note that this scale used a within-construct matrix sampling design. Each of the 10 items included in this scale had five response options (“Strongly disagree”, “Disagree”, “Neither agree nor disagree”, “Agree”, “Strongly agree”). Table 19.A.42 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling. Table 19.A.43 shows the percent of students in each country/economy that did not receive a scale score for PERSEVAGR due to extreme straightlining or, for comparison, for not having enough responses (i.e., less than three responses for the scale). In both cases, the scale scores were replaced with “99” in the SPSS file and “.M” in the SAS file.

Stress resistance (STRESAGR)

Students’ ratings of their agreement with statements about a range of behaviours indicative of stress resistance (e.g., “I remain calm under stress.”, “I get nervous easily.”) in question ST345 were scaled into the index of “Stress resistance”. Note that this scale used a within-construct matrix sampling design. Each of the 10 items included in this scale had five response options (“Strongly disagree”, “Disagree”, “Neither agree nor disagree”, “Agree”, “Strongly agree”). Table 19.A.44 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Table 19.A.45 shows the percent of students in each country/economy that did not receive a scale score for STRESAGR due to extreme straightlining or, for comparison, for not having enough responses (i.e., less than three responses for the scale). In both cases, the scale scores were replaced with “99” in the SPSS file and “.M” in the SAS file.

Exposure to mathematics content (Module 15)

Exposure to formal and applied mathematics tasks (EXPOFA)

Students’ frequency ratings of how often they had encountered a range of formal and applied mathematics tasks during their time at school (e.g., “Calculating how much more expensive a computer would be after adding tax.”, “Solving an equation like 2(x+3) = (x+3)(x-3)”) in question ST275 were scaled into the index “Exposure to formal and applied mathematics tasks”. Note that this scale used a within-construct matrix sampling design. Each of the nine items included in this scale had four response options (“Frequently”, “Sometimes”, “Rarely”, “Never”). Table 19.A.46 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Exposure to mathematical reasoning and 21st century mathematics tasks (EXPO21ST)

Students’ frequency ratings of how often they had encountered a range of different types of mathematics tasks related to mathematical reasoning and 21st century mathematics tasks during their time at school (e.g., “Extracting mathematical information from diagrams, graphs, or simulations”, “Using the concept of statistical variation to make a decision”) in question ST276 were scaled into the index “Exposure to mathematical reasoning and 21st century mathematics tasks”. Note that this scale used a within-construct matrix sampling design. Each of the 10 items included in this scale had four response options (“Frequently”, “Sometimes”, “Rarely”, “Never”). Table 19.A.47 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Mathematics teacher behaviour (Module 16)

Cognitive activation in mathematics: Foster reasoning (COGACRCO)

Students’ frequency ratings of how often their mathematics teacher showed a range of behaviours indicative of fostering mathematics reasoning during the ongoing school year (e.g., “The teacher asked us to explain our reasoning when solving a mathematics problem.”, “The teacher asked us to defend our answer to a mathematics problem.”) in question ST285 were scaled into the index of “Cognitive activation in mathematics: Foster reasoning”. Note that this scale used a within-construct matrix sampling design. Each of the nine items included in this scale had five response options (“Never or almost never”, “Less than half of the lessons”, “About half of the lessons”, “More than half of the lessons”, “Every lesson or almost every lesson”). Table 19.A.48 shows the item wording and item parameters for the items in this scale.

Cognitive activation in mathematics: Encourage mathematical thinking (COGACMCO)

Students’ frequency ratings of how often their mathematics teacher showed a range of behaviours indicative of encouraging mathematical thinking during the ongoing school year (e.g., “The teacher encouraged us to “think mathematically”.”, “The teacher asked us how different topics are connected to a bigger mathematical idea.”) in question ST283 were scaled into the index of “Cognitive activation in mathematics: Encourage mathematical thinking”. Note that this scale used a within-construct matrix sampling design. Each of the nine items included in this scale had five response options (“Never or almost never”, “Less than half of the lessons”, “About half of the lessons”, “More than half of the lessons”, “Every lesson or almost every lesson”). Table 19.A.49 shows the item wording and item parameters for the items in this scale.

Disciplinary climate in mathematics (DISCLIM)

Students’ frequency ratings of how often a range of situations occurred in their mathematics lessons (e.g., “Students do not listen to what the teacher said.”, “Students get distracted by using <digital resources> (e.g., smartphones, websites, apps).”) in question ST273 were scaled into the index of “Disciplinary climate in mathematics”. Note that this scale was linked to the DISCLIMA scale in PISA 2012 and was scaled using the PCM, in line with the model used in PISA 2012. Also, this scale used a within-construct matrix sampling design. Each of the seven items included in this scale had four response options (“Every lesson”, “Most lessons”, “Some lessons”, “Never or almost never”). Table 19.A.50 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Parental/guardian involvement and support (Module 19)

Family support (FAMSUP)

Students’ ratings of how often their parents or someone else in their family engaged in a range of behaviours indicative of family support (e.g., “Discuss how well you are doing at school”, “Spend time just talking with you”) in question ST300 were scaled into the index of “Family support”. Note that this scale used a within-construct matrix sampling design. Each of the 10 items included in this scale had five response options (“Never or almost never”, “About once or twice a year”, “About once or twice a month”, “About once or twice a week”, “Every day or almost every day”). Table 19.A.51 shows the item wording and item parameters for the items in this scale.

Creative thinking (Module 20)

Creative peers and family environment (CREATFAM)

Students’ ratings of their agreement with statements about the degree to which creative thinking is fostered and supported by their peers and family environment (e.g., “My friends are open to new ideas.”, “At home, I am encouraged to use my imagination.”) in question ST336 were scaled into the index of “Creative peers and family environment”. Note that this scale used a within-construct matrix sampling design. Each of the six items included in this scale had four response options (“Strongly disagree”, “Disagree”, “Agree”, “Strongly agree”). Table 19.A.52 shows the item wording and item parameters for the items in this scale.

Creative school and class environment (CREATSCH)

Students’ ratings of their agreement with statements about the degree to which creative thinking is fostered and supported in their school and class environment (e.g., “My teachers value students’ creativity.”, “At school, I am given a chance to express my ideas.”) in question ST335 were scaled into the index of “Creative school and class environment”. Note that this scale used a within-construct matrix sampling design. Each of the six items included in this scale had four response options (“Strongly disagree”, “Disagree”, “Agree”, “Strongly agree”). Table 19.A.53 shows the item wording and item parameters for the items in this scale.

Creative thinking self-efficacy (CREATEFF)

Students’ ratings of how confident they felt about having to do a range of tasks reflective of creative thinking skills (e.g., “Coming up with creative ideas for school projects”, “Inventing new things”) in question ST334 were scaled into the index of “Creative thinking self-efficacy”. Note that this scale used a within-construct matrix sampling design. Each of the 10 items included in this scale had four response options (“Not at all confident”, “Not very confident”, “Confident”, “Very confident”). Table 19.A.54 shows the item wording and item parameters for the items in this scale.

Creativity and openness to intellect (CREATOP)

Students’ ratings of their agreement with statements regarding their own views on their creativity and openness to intellect (e.g., “Doing something creative satisfies me.”, “I like games that challenge my creativity.”) in question ST340 were scaled into the index of “Creativity and openness to intellect”. Note that this scale used a within-construct matrix sampling design. Each of the 10 items included in this scale had four response options (“Strongly disagree”, “Disagree”, “Agree”, “Strongly agree”). Table 19.A.55 shows the item wording and item parameters for the items in this scale.

Imagination and adventurousness (IMAGINE)

Students’ ratings of their agreement with statements regarding their own views on their imagination and adventurousness (e.g., “I have difficulty using my imagination.”, “Coming up with new ideas is satisfying to me.”) in question ST342 were scaled into the index of “Imagination and adventurousness”. Note that this scale used a within-construct matrix sampling design. Each of the seven items included in this scale had four response options (“Strongly disagree”, “Disagree”, “Agree”, “Strongly agree”). Table 19.A.56 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Openness to art and reflection (OPENART)

Students’ ratings of their agreement with statements regarding their own views on their openness to art and reflection (e.g., “I enjoy creating art.”, “I reflect on movies I watch.”) in question ST341 were scaled into the index of “Openness to art and reflection”. Each of the five items included in this scale had four response options (“Strongly disagree”, “Disagree”, “Agree”, “Strongly agree”). Table 19.A.57 shows the item wording and item parameters for the items in this scale.

Participation in creative activities at school (CREATAS)

Students’ ratings of how often they participated in creative activities that were available in their school (e.g., “Art classes/activities (e.g., painting, drawing)”, “Debate club”) in question ST337 were scaled into the index of “Participation in creative activities at school”. Note that the activities sampled in this question are the same as the activities in the “outside of school” version of this question (CREATOOS – ST338). Each of the eight items included in this scale had five substantive response options (“Never or almost never”, “About once or twice a year”, “About once or twice a month”, “About once or twice a week”, “Every day or almost every day”) and an additional response option “Not available at school” which was recoded as missing prior to scaling. Table 19.A.58 shows the item wording and item parameters for the items in this scale. It also indicates how the response categories were recoded prior to scaling.

Participation in creative activities outside of school (CREATOOS)

Students’ ratings of how often they participated in creative activities outside of school (e.g., “Art classes/activities (e.g., painting, drawing)”, “Debate club”) in question ST338 were scaled into the index of “Participation in creative activities outside of school”. Note that the activities sampled in this question are the same as the activities in the “at school” version of this question (CREATAS – ST337). Each of the eight items included in this scale had five substantive response options (“Never or almost never”, “About once or twice a year”, “About once or twice a month”, “About once or twice a week”, “Every day or almost every day”) and an additional response option “Not available” which was recoded as missing prior to scaling.

Table 19.A.59 shows the item wording and item parameters for the items in this scale. It also indicates how the response categories were recoded prior to scaling.

Global crises (Module 21)

Note that the questions in this module were skipped for students who reported that their school had not been closed for more than a week due to COVID-19 in question ST347.

Family support for self-directed learning (FAMSUPSL)

Students' frequency ratings of how often someone in their family provided specific kinds of learning support (e.g., "Help me create a learning schedule"; "Help me access learning materials online") while the school building was closed due to COVID-19 in question ST353 were scaled into the index of "Family support for self-directed learning". Note that this scale used a within-construct matrix sampling design. Each of the eight items included in this scale had four response options ("Never", "A few times", "About once or twice a week", "Every day or almost every day"). Table 19.A.60 shows the item wording and item parameters for the items in this scale.

Feelings about learning at home (FEELLAH)

Students' ratings of their agreement with statements about how they felt about learning at home (e.g., "I enjoyed learning by myself.", "My teachers were well prepared to provide instruction remotely.") while the school building was closed due to COVID-19 in question ST354 were scaled into the index of "Feelings about learning at home". Note that this scale used a within-construct matrix sampling design. Each of the six items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.61 shows the item wording and item parameters for the items in this scale.

Problems with self-directed learning (PROBSELF)

Students' frequency ratings of how often they had various problems completing their school work (e.g., "Problems with Internet access", "Problems with understanding my school assignments") while their school building was closed due to COVID-19 in question ST352 were scaled into the index of "Problems with self-directed learning". Note that this scale used a within-construct matrix sampling design. Each of the eight items included in this scale had four response options ("Never", "A few times", "About once or twice a week", "Every day or almost every day"). Table 19.A.62 shows the item wording and item parameters for the items in this scale.

Self-directed learning self-efficacy (SDLEFF)

Students' ratings of how confident they felt about having to do a range of self-directed learning tasks (e.g., "Finding learning resources online on my own", "Completing school work independently") should their school building close again in the future in question ST355 were scaled into the index of "Self-directed learning self-efficacy". Note that this scale used a within-construct matrix sampling design. Each of the eight items included in this scale had four response options ("Not at all confident", "Not very confident", "Confident", "Very confident"). Table 19.A.63 shows the item wording and item parameters for the items in this scale.

School actions to sustain learning (SCHSUST)

Students' frequency ratings of how often someone from their school completed an activity to sustain their learning (e.g., "Sent me learning materials to study on my own", "Checked in with me to ensure that I was completing my assignments") while their school building was closed due to COVID-19 in question ST348 were scaled into the index of "School actions to sustain learning". Note that this scale used a within-construct matrix sampling design. Each of the eight items included in this scale had four response options ("Never", "A few times", "About once or twice a week", "Every day or almost every day"). Table 19.A.64 shows the item wording and item parameters for the items in this scale.

Types of learning resources used while school was closed (LEARRES)

Students' frequency ratings of how often they used specific learning resources (e.g., "Paper textbooks, workbooks, or worksheets", "Recorded lessons or other digital material provided by teachers from my school") while the school building was closed due to COVID-19 in question ST351 were scaled into the index of "Types of learning resources used while school was closed". Note that this scale used a within-construct matrix sampling design. Each of the eight items included in this scale had four response options ("Never", "A few times", "About once or twice a week", "Every day or almost every day"). Table 19.A.65 shows the item wording and item parameters for the items in this scale.

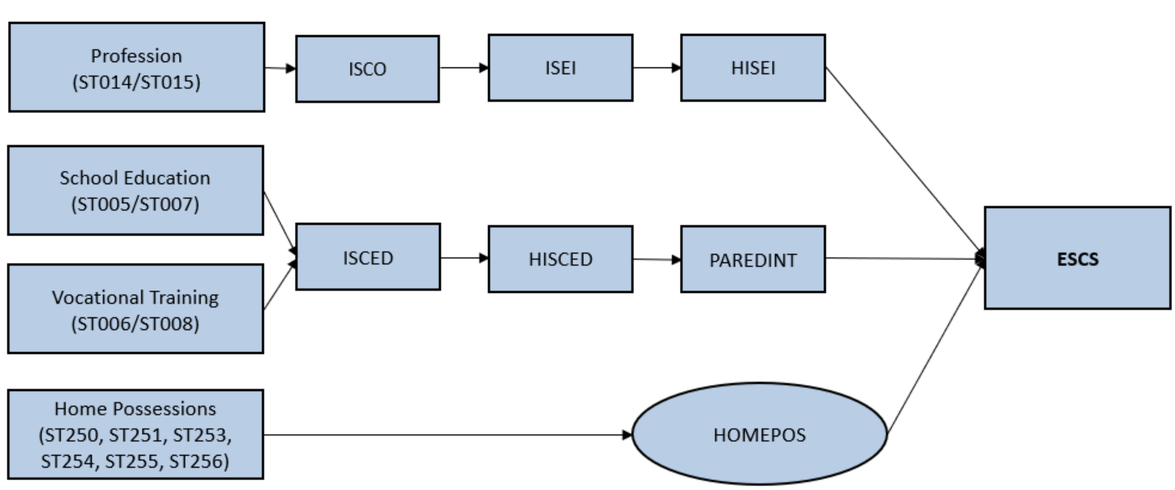
Complex composite index – Index of economic, social and cultural status (ESCS)

There was only one complex composite index derived from the Student Questionnaire – the index of economic, social and cultural status (ESCS).

Components of ESCS

The ESCS score was based on three indicators: highest parental occupation status (HISEI), highest education of parents in years (PAREDINT), and home possessions (HOMEPOS). The rationale for using these three components, which are consistent with the components used in previous PISA cycles, is that socio-economic status is most commonly theoretically conceptualized based on "the big 3" (occupational status, education, and income) (Cowan et al., 2012[13]). As no direct income measure is available in the PISA data, the existence of household items has been used as a proxy for family income. Figure 19.4 provides a schematic representation of ESCS and its components.

Figure 19.4. Computation of ESCS in PISA 2022



HISEI. For more information on HISEI, refer to the explanation on HISEI in the simple indices section above.

PAREDINT. For more information on PAREDINT, refer to the explanation on PAREDINT in the simple indices section above.

HOMEPOS. For more information on HOMEPOS, refer to the explanation on HOMEPOS in the IRT scale section above.

Computation of ESCS

The ESCS scores were computed using the same methodology used in PISA 2018 (Avvisati, 2020[14]; OECD, 2020[15]). For students with missing data on one out of the three components, the missing component was imputed using a regression equation which was created for each country/economy using data from students without any missing components. For each student with a missing component, this regression equation was used to predict the missing component with the two non-missing components and a random value was added to the predicted value to reflect the error of the regression model.[5] If a student had missing data on more than one component, the ESCS score was not computed for the student, and the student's ESCS score was replaced with "99" in the SPSS file and ".M" in the SAS file.

After the imputation process, each of the three components (including the imputed values) was standardised to have a mean of 0 and a standard deviation of 1 across the OECD countries, with each OECD country weighted approximately equally using senate weights.[6] The OECD means and standard deviations that were used to standardise each component of ESCS are displayed in Table 19.A.66.

Subsequently, the arithmetic mean of the three standardised components was calculated to create a preliminary ESCS score for each student. Lastly, the preliminary ESCS scores were standardised again to have a mean of 0 and a standard deviation of 1 across the OECD countries (again with each country weighted approximately equally using senate weights[7]), producing the final ESCS score for each student. The OECD mean and standard deviation that were used to transform the preliminary ESCS scores into the final ESCS scores are displayed in Table 19.A.66.

ESCS trend scores

In contrast to the other trend scales in the context questionnaires (for which the scale scores for the current cycle were made to be comparable to the scale scores from a previous cycle), the scores for each component of ESCS and the composite ESCS scores are not comparable to the scores from previous cycles. Instead, each of the component scores for ESCS and the composite ESCS scores for PISA 2012, PISA 2015, and PISA 2018 were recomputed to be comparable to the respective scores for PISA 2022. This was done by recoding the scores for each component of ESCS for PISA 2012, PISA 2015, and PISA 2018 using the coding scheme used in PISA 2022, then recomputing the composite ESCS score for these previous cycles using the ESCS computation methodology used in PISA 2022. More details are provided below.

HISEI. Until PISA 2009, ISCO-88 was used to code parental occupation. However, since PISA 2012, parental occupation has been coded using ISCO-08, the most recent version of ISCO. In PISA 2018, the coding scheme for ISEI was updated so that an ISEI value of 17 was attributed to ISCO codes 9701 ("stay-at-home parent"), 9702 ("student"), and 9703 ("social beneficiary"), equivalent to the ISEI value for ISCO code 9000 ("elementary occupations"). This coding scheme was also used in PISA 2022.

To make the HISEI scores for PISA 2012 and PISA 2015 comparable to the HISEI scores for PISA 2018 and PISA 2022, new HISEI scores were created for each student that participated in PISA 2012 and PISA 2015 using the coding scheme used in PISA 2018 and PISA 2022. These new HISEI scores were used in the computation of the trend ESCS scores.

PAREDINT. For some countries/economies, the mapping of ISCED levels to years of education was updated in 2009, 2015, and 2018, taking into account changes in the countries/economies' educational systems. In PISA 2022, PAREDINT was updated again to map each ISCED level (based on ISCED-11) to the PISA 2018 cumulative years of education values, as presented in Table 19.A.8.

To make the PAREDINT scores for PISA 2012, PISA 2015, and PISA 2018 comparable to the PAREDINT scores for PISA 2022, new PAREDINT scores were created for each student that participated in the previous cycles the mapping presented in Table 19.A.8. These new PAREDINT scores were used in the computation of the trend ESCS scores.

HOMEPOS. Indicators of HOMEPOS have been dropped or added in all PISA cycles, taking into account the social, technical, and economic changes in the participating countries/economies. Moreover, the method for estimating HOMEPOS changed in PISA 2009, PISA 2012, and PISA 2015.

To make the HOMEPOS scores comparable across cycles, prior to scaling, the response categories for some items in PISA 2012, PISA 2015, and PISA 2018 were collapsed to align with the response categories used in PISA 2022 (as presented in Table 19.A.17). Then, the HOMEPOS WLEs for each student that participated in the past three cycles were re-estimated by fixing the item parameters to the parameters that were estimated for each group in PISA 2022 (either the international parameters or the group's unique parameters, depending on whether the item parameters were released for the group in PISA 2022). For items that were not administered in PISA 2022, new international parameters were estimated by pooling data across all cycles and groups in which the item had been administered, then the item parameters were released until all groups in all cycles had an RMSD under 0.25. As an exception, unique parameters were estimated for all country/economy-specific items for all groups and all cycles. These newly estimated HOMEPOS WLEs were used in the computation of the trend ESCS scores.

ESCS. Prior to PISA 2018, the ESCS scores were computed using a principal component analysis (PCA), although there were differences across cycles regarding which countries/economies were included in the PCA and how the scores were standardised. In PISA 2018, the ESCS scores were computed as the arithmetic mean of the three components, with each of the component scores and the composite ESCS score standardised to have a mean of 0 and standard deviation of 1 across the OECD countries (with each country weighted approximately equally using senate weights[8]). As noted above, this methodology was also used in PISA 2022.

To make the ESCS scores for PISA 2012, PISA 2015, and PISA 2018 comparable to the ESCS scores for PISA 2022, new ESCS scores were computed for the previous cycles using the methodology used in PISA 2022. Specifically, for students with missing data on one out of the three components, the missing component was imputed using a regression equation which was created for each country/economy in each cycle using data from students without any missing components. For each student with a missing component, this regression equation was used to predict the missing component with the two non-missing components and a random value was added to the predicted value to reflect the error of the regression model.[9] If a student had missing data on more than one component, the ESCS score was not computed for the student, and the student's ESCS score was replaced with "99" in the SPSS file and ".M" in the SAS file.

After the imputation process, each of the three components (including the imputed values) was standardised using the OECD mean and standard deviation of the respective component in PISA 2022 (presented in Table 19.A.66 above). Next, the arithmetic mean of the three standardised components was calculated to create a preliminary ESCS score for each student. Lastly, the preliminary ESCS scores were standardised again using the OECD mean and standard deviation of the preliminary ESCS scores in PISA 2022 (also presented in Table 19.A.66). This process ensured that the trend ESCS scores produced for PISA 2012, PISA 2015, and PISA 2018 were directly comparable to the ESCS scores produced for PISA 2022.

Financial Literacy Questionnaire derived variables

The Financial Literacy Questionnaire is an international option that countries/economies could choose to implement. It was administered to students after they had completed the Student Questionnaire. It addresses familiarity of students related to financial literacy and their confidence about financial matters. There were 10 variables derived from this questionnaire, including one simple DV and nine IRT scaled DVs. An overview of all DVs in this questionnaire is shown in Table 19.A.67 and each are described in the following sections.

Simple questionnaire indices

Familiarity with concepts of finance (FCFMLRTY)

Students' ratings of how familiar they were with various financial topics in question FL164 were used to derive an indicator of familiarity with concepts of finance. There were three response options ("Never heard of it", "Heard of it, but I don't recall the meaning", "Learnt about it, and I know what I means"). For each item, a value of "1" was assigned to "Learnt about it, and I know what I means" responses, and all other responses were assigned a value of "0". This index was constructed as the sum of values across all 16 items. Values range from "0" to "16".

Derived variables based on IRT scaling

The Financial Literacy Questionnaire provided data for nine DVs based on IRT scaling. The Cronbach's alpha for each scale and group are presented in Table 19.A.68, the number of items with international parameters for each scale and group are presented in Table 19.A.69, the number of trend items with international parameters for each trend scale and group are presented in Table 19.A.70, and the groups that did not administer each scale are presented in Table 19.A.71 (in this case, the scale scores for the individuals in the group were replaced with "99" in the SPSS file and ".M" in the SAS file). Note that there were no countries/economies for which the scale scores were suppressed for the scales in the Financial Literacy Questionnaire.

Financial education in school lessons (FLSCHOOL)

Students' frequency ratings of how often they encountered financial tasks and activities in school lessons (e.g., "Describing the purpose and uses of money", "Exploring ways of planning to pay an expense") in question FL166 were scaled into the index of "Financial education in school lessons". Note that this scale was linked to the FLSCHOOL scale in PISA 2018. Each of the six items included in this scale had three response options ("Never", "Sometimes", "Often"). Table 19.A.72 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Financial education in school lessons – Multiple subjects (FLMULTSB)

Students' responses to questions about where they encountered lessons about financial topics (e.g., "During your mathematics class", "During classes about economics or business") in question FL174 were scaled into the index of "Financial education in school lessons – Multiple Subjects (FLMULTSB)". Each of the seven items included in this scale had two substantive response options ("Yes", "No") and two additional response options ("I don't know.", "I don't have this class.") which were recoded as missing prior to scaling. Table 19.A.73 shows the item wording and item parameters for the items in this scale. It also indicates how the response categories were recoded prior to scaling.

Parental involvement in matters of financial literacy (FLFAMILY)

Students' frequency ratings of how often they discuss various financial issues with their parents (e.g., "Your spending decisions", "Shopping online") in question FL167 were scaled into the index of "Parental involvement in matters of financial literacy". Note that this scale was linked to the FLFAMILY scale in PISA 2018. Each of the seven items included in this scale had four response options ("Never or hardly ever", "Once or twice a month", "Once or twice a week", "Almost every day"). Table 19.A.74 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Access to money and financial projects – Sources of money (ACCESSFP)

Students' frequency ratings about how often their money came from different sources (e.g., "An allowance or pocket money for doing chores at home", "Working in a family business") in question FL170 were scaled into the index of "Access to money and financial projects – Sources of Money (ACCESSFB)". Each of the seven items included in this scale had five response options ("Never or almost never", "About once or twice a year", "About once or twice a month", "About once or twice a week", "Every day or almost every day"). Table 19.A.75 shows the item wording and item parameters for the items in this scale.

Confidence about financial matters (FLCONFIN)

Students' ratings of their confidence with various financial matters (e.g., "Understanding bank statements", "Keeping track of my account balance") in question FL162 were scaled into the index of "Confidence about financial matters". Note that this scale was linked to the FLCONFIN scale in PISA 2018. Each of the six items included in this scale had four response options ("Not at all confident", "Not very confident", "Confident", "Very confident"). Table 19.A.76 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Confidence about financial matters using digital devices (FLCONICT)

Students' ratings of their confidence in doing various financial tasks with electronic devices (e.g., "Transferring money", "Paying with a mobile device (e.g., mobile phone or tablet) instead of using cash") in question FL163 were scaled into the index of "Confidence about financial matters using digital devices". Note that this scale was linked to the FLCONICT scale in PISA 2018. Each of the five items included in this scale had four response options ("Not at all confident", "Not very confident", "Confident", "Very confident"). Table 19.A.77 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Access to money and financial products – Financial activities (ACCESSFA)

Students' frequency ratings of how often they completed different financial activities (e.g., "Checked how much money you have", "Saved money at home") in question FL171 were scaled into the index of "Access to money and financial products – Financial activities (ACCESSFA)". Each of the 11 items included in this scale had five response options ("Never or almost never", "About once or twice a year", "About once or twice a month", "About once or twice a week", "Every day or almost every day"). Table 19.A.78 shows the item wording and item parameters for the items in this scale.

Attitudes towards and confidence about financial matters (ATTCONFM)

Students' rating of their agreement with different statements about their attitudes towards and confidence about financial matters (e.g., "I enjoy talking about money matters.", "I know how to manage my money.") in question FL169 were scaled into the index of "Attitudes towards and confidence about financial matters (ATTCONFM)". Each of the seven items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.79 shows the item wording and item parameters for the items in this scale.

Friends' influence on financial matters (FRINFLFM)

Students' ratings of their agreement with various statements about their friends' influence on finance decisions (e.g., "My friends have a strong influence on my spending decisions.", "Sometimes I spend more than I would like when I am with my friends.") in question FL172 were scaled into the index of "Friends' influence on financial matters". Each of the four items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.80 shows the item wording and item parameters for the items in this scale.

ICT Familiarity Questionnaire derived variables

The ICT Familiarity Questionnaire is an international option that countries/economies could choose to implement. It was administered to students after they had completed the Student Questionnaire. There were 15 variables derived from this questionnaire, including three simple DVs and 12 IRT scaled DVs. All of the IRT scaled DVs were new for PISA 2022, as the ICT framework had been revised for this cycle. An overview of all DVs in this questionnaire is shown in Table 19.A.81 and each are described in the following sections.

Simple questionnaire indices

Availability and usage of ICT at school (ICTAVSCH)

The availability of ICT at school was gathered from IC170 where students' frequency ratings of how often they use various digital resources at school (e.g., "Desktop or laptop computer", "Smartphone") was used for the index of "ICT availability at school". Each of the seven items in this question included six response options ("Never or almost never", "About once or twice a month", "About once or twice a week", "Every day or almost every day", "Several times a day", "This resource is not available to me at school"). The index was calculated as the number of all seven items that were marked with a value other than "This resource is not available to me at school"), thus ranging from 0-7. Items 2-4 were included in various previous versions of the ICT Questionnaire.

Availability and usage of ICT outside of school (ICTAVHOM)

The availability of ICT outside of school was gathered from IC171 where students' frequency ratings of how often they use various digital resources outside of school (e.g., "Desktop or laptop computer", "Smartphone") was used for the index of "ICT use outside of school". Each of the six items in this question included six response options ("Never or almost never", "About once or twice a month", "About once or twice a week", "Every day or almost every day", "Several times a day", "This resource is not available to me outside of school"). For each of the six items, a score of "0" was assigned when students choose the "This resource is not available to me outside of school" response options and all other responses were coded "1". The index was calculated as the sum of "0" and "1" designations across the six items that were marked with a value other than "This resource is not available to me at school", thus ranging from 0-6. Items 2-4 were included in various previous versions of the ICT Questionnaire.

Distress from online content and cyberbullying (ICTDISTR)

Students' ratings of how upset they were when various situation occurred online (e.g., "Encountering content online that was inappropriate for my age", "Receiving unkind, vulgar or offending messages, comments or videos”) in question IC181 were scaled into the index of “Distress from online content and cyberbullying”. Each item included five response options (“This did not happen to me”, “Not at all upset”, “A little upset”, “Quite upset”, “Very upset”). Values in this index range from 0-16, with “This did not happen to me” recoded as a missing variable, “Not at all upset” coded “1”, “A little upset” coded “2”, “Quite upset” coded “3”, and “Very upset” coded “4”. Values across all items were summed.

Derived variables based on IRT scaling

The ICT Familiarity Questionnaire provided data for 12 DVs based on IRT scaling. The Cronbach’s alpha for each scale and group are presented in Table 19.A.82, the number of items with international parameters for each scale and group are presented in Table 19.A.83, the countries/economies for which the scale scores were suppressed for each scale are presented in Table 19.A.84 (in this case, the scale scores for the individuals in the country/economy were replaced with “97” in the SPSS file and “.N” in the SAS file), and the groups that did not administer each scale are presented in Table 19.A.85 (in this case, the scale scores for the individuals in the group were replaced with “99” in the SPSS file and “.M” in the SAS file).

ICT availability at school (ICTSCH)

Students’ frequency ratings of how often they use various digital resources at school (e.g., “Desktop or laptop computer”, “Smartphone (i.e., mobile phone with internet access)”) in question IC170 were scaled into the index of “ICT availability at school”. Each of the seven items included in this scale had six response options (“Never or almost never”, “About once or twice a month”, “About once or twice a week”, “Every day or almost every day”, “Several times a day”, “This resource is not available to me at school”). “This resource is not available to me at school” was recoded as 0, while the five other response options were recoded as 1 prior to scaling. Table 19.A.86 shows the item wording and item parameters for the items in this scale.

ICT availability outside school (ICTHOME)

Students’ frequency ratings of how often they use various digital resources outside of school (e.g., “Desktop or laptop computer”, “Smartphone (i.e., mobile phone with internet access)”) in question IC171 were scaled into the index of “ICT availability outside school”. Each of the six items included in this scale had six response options (“Never or almost never”, “About once or twice a month”, “About once or twice a week”, “Every day or almost every day”, “Several times a day”, “This resource is not available to me outside of school”). “This resource is not available to me outside of school” was recoded as 0, while the five other response options were recoded as 1 prior to scaling. Table 19.A.87 shows the item wording and item parameters for the items in this scale.

Quality of access to ICT (ICTQUAL)

Students’ ratings of their agreement with various statements about ICT resources at their school (e.g., “There are enough digital devices with access to the Internet at my school.”, “The school’s Internet speed is sufficient.”) in question IC172 were scaled into the index of “Quality of access to ICT”. Each of the nine items included in this scale had four response options (“Strongly disagree”, “Disagree”, “Agree”, “Strongly agree”). Table 19.A.88 shows the item wording and item parameters for the items in this scale.

Subject-related ICT use during lessons (ICTSUBJ)

Students’ frequency ratings of how often digital resources are used in various subject lessons (e.g., “Mathematics”, “Science”) in question IC173 were scaled into the index of “Subject-related ICT use during lessons”. Each of the four items included in this scale had five substantive response options (“Never or almost never, “In less than half of the lessons”, “In about half of the lessons”, “In more than half of the lessons”, “In every or almost every lesson”) and an additional response option “I do not have this subject” which was recoded as missing prior to scaling. Table 19.A.89 shows the item wording and item parameters for the items in this scale.

Use of ICT in enquiry-based learning activities (ICTENQ)

Students' frequency ratings of how often they use digital resources for various school-related activities (e.g., "Create a multi-media presentation with pictures, sound or video", "Track the progress of your own work or projects") in question IC174 were scaled into the index of "Use of ICT in enquiry-based learning activities". Each of the 10 items included in this scale had five response options ("Never or almost never", "About once or twice a year", "About once or twice a month", "About once or twice a week", "Every day or almost every day"). Table 19.A.90 shows the item wording and item parameters for the items in this scale.

Support or feedback via ICT (ICTFEED)

Students' frequency ratings of how often they use digital resources in various activities related to support or feedback (e.g., "Read or listen to feedback sent by my teachers regarding my work and academic results", "Read or listen to feedback sent by other students on my work") in question IC175 were scaled into the index of "Support or feedback via ICT". Each of the four items included in this scale had five response options ("Never or almost never", "About once or twice a year", "About once or twice a month", "About once or twice a week", "Every day or almost every day"). Table 19.A.91 shows the item wording and item parameters for the items in this scale.

Use of ICT for school activities outside of the classroom (ICTOUT)

Students' frequency ratings of how often they use digital resources for various school-related activities outside of the classroom (e.g., "See my grades or results from specific assignments (e.g., homework or tests)", "Communicate with my teacher") in question IC176 were scaled into the index of "Use of ICT for school activities outside of the classroom". Each of the eight items included in this scale had five response options ("Never or almost never", "About once or twice a year", "About once or twice a month", "About once or twice a week", "Every day or almost every day"). Table 19.A.92 shows the item wording and item parameters for the items in this scale.

Frequency of ICT activity – Weekday (ICTWKDY)

Students' frequency ratings of how often they did various leisure activities using ICT during a typical week day (e.g., "Play video-games (using my smartphone, a gaming console or an online platform or apps)", "Look for practical information online (e.g., find a place, book a train ticket, buy a product)") in question IC177 were scaled into the index of "Frequency of ICT activity – Weekday". Each of the seven items included in this scale had six response options ("No time at all, "Less than 1 hour a day", "Between 1 and 3 hours a day", "More than 3 hours and up to 5 hours a day", "More than 5 hours and up to 7 hours a day", "More than 7 hours a day"). Table 19.A.93 shows the item wording and item parameters for the items in this scale.

Frequency of ICT activity – Weekend (ICTWKEND)

Students' frequency ratings of how often they did various leisure activities using ICT during a typical weekend day (e.g., "Play video-games (using my smartphone, a gaming console or an online platform or apps)", "Look for practical information online (e.g., find a place, book a train ticket, buy a product)") in question IC178 were scaled into the index of "Frequency of ICT activity – Weekend". Each of the seven items included in this scale had six response options ("No time at all, "Less than 1 hour a day", "Between 1 and 3 hours a day", "More than 3 hours and up to 5 hours a day", "More than 5 hours and up to 7 hours a day", "More than 7 hours a day"). Table 19.A.94 shows the item wording and item parameters for the items in this scale.

Views of regulated ICT use in school (ICTREG)

Students' ratings of their agreement with various statements about regulation of ICT use at school (e.g., "Students should not be allowed to bring mobile phones to class.", "The school should set up filters to prevent students from playing games online.") in question IC179 were scaled into the index of "Views of regulated ICT use in school". Each of the six items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.95 shows the item wording and item parameters for the items in this scale.

Students' practices regarding online information (ICTINFO)

Students' ratings of their agreement with various statements about their practices regarding online information (e.g., "When searching for information online I compare different sources.", "I discuss the accuracy of online information with friends or other students.") in question IC180 were scaled into the index of "Students' practices regarding online information". Each of the six items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.96 shows the item wording and item parameters for the items in this scale.

Self-efficacy in digital competencies (ICTEFFIC)

Students' ratings of how well they can do various tasks using digital resources (e.g., "Search for and find relevant information online", "Write or edit text for a school assignment") in question IC183 were scaled into the index of "Self-efficacy in digital competencies". Each of the 14 items included in this scale had four substantive response options ("I cannot do this", "I struggle to do this on my own", "I can do with a bit of effort", "I can easily do this") and an additional response option "I don't know what this is" which was recoded as missing prior to scaling. Table 19.A.97 shows the item wording and item parameters for the items in this scale.

Well-Being Questionnaire derived variables

The Well-Being Questionnaire is an international option that countries/economies could choose to implement. It was administered to students after they had completed the Student Questionnaire. It addresses the well-being of students. There were seven variables derived from this questionnaire, including one simple DV and six IRT scaled DVs. An overview of all DVs in this questionnaire is shown in Table 19.A.98 and each are described in the following sections.

Simple questionnaire indices

Body mass index (STUBMI)

The only simple DV from the Well-Being Questionnaire is STUBMI, indicating the student's body mass index (STUBMI). It is based on two questions, WB151 and WB152, which asked about the weight and the height of the student, respectively, in the units of measurement that are more common in the respective country/economy. The index is constructed as it was in PISA 2018. Specifically, the index was constructed as the weight (transformed to kilograms) divided by the square of the body height (transformed to metres).

Derived variables based on IRT scaling

The Well-Being Questionnaire provided data for six DVs based on IRT scaling. The Cronbach's alpha for each scale and group are presented in Table 19.A.99, the number of items with international parameters for each scale and group are presented in Table 19.A.100, the number of trend items with international parameters for each trend scale and group are presented in Table 19.A.101, and the groups that did not administer each scale are presented in Table 19.A.102 (in this case, the scale scores for the individuals in the group were replaced with "99" in the SPSS file and ".M" in the SAS file). Note that there were no countries/economies for which the scale scores were suppressed for the scales in the Well-Being Questionnaire.

Body image (BODYIMA)

Students' ratings of their agreement with statements about their body image (e.g., "I like my look just the way it is.", "I like my body.") in question WB153 were scaled into the index of "Body image". Note that this scale was linked to the BODYIMA scale in PISA 2018. Each of the five items included in this scale had four substantive response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree") and an additional response option "I don't have an opinion" which was recoded as missing prior to scaling. Table 19.A.103 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Social connection to parents (SOCONPA)

Students' ratings of how often their parents engage in various activities (e.g., "Show that they care", "Encourage me to make my own decisions") in question WB163 were scaled into the index of "Social connection to parents". Note that this scale was linked to the SOCONPA scale in PISA 2018. Each of the six items included in this scale had three response options ("Almost never", "Sometimes", "Almost always"). Table 19.A.104 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Students' life satisfaction across domains (LIFESAT)

Students' ratings of their satisfaction with different areas of their lives (e.g., "Your health", "The neighbourhood you live in") in question WB155 were scaled into the index of "Students' life satisfaction across domains". Each of the 10 items included in this scale had four response options ("Not at all satisfied", "Not satisfied", "Satisfied", "Totally satisfied"). Table 19.A.105 shows the item wording and item parameters for the items in this scale.

Psychosomatic symptoms (PSYCHSYM)

Students' ratings of how often they experienced different psychosomatic symptoms (e.g., "Headache", "Stomach pain") in question WB154 were scaled into the index of "Psychosomatic symptoms". Each of the nine items included in this scale had five response options ("Rarely or never", "About every month", "About every week", "More than once a week", "About every day"). Table 19.A.106 shows the item wording and item parameters for the items in this scale.

Social connections: Ease of communication about worries and concerns (SOCCON)

Students' ratings of how easy it is to communicate about their worries and concerns with different people (e.g., "Your father", "Your brother(s)") in question WB162 were scaled into the index of "Social connections: Ease of communication about worries and concerns". Each of the nine items included in this scale had four substantive response options ("Very difficult", "Difficult", "Easy", "Very Easy") and an additional response option “I don’t have or see this person” which was recoded as missing prior to scaling. Table 19.A.107 shows the item wording and item parameters for the items in this scale.

Experienced well-being – Previous day (EXPWB)

Students’ responses regarding their experienced well-being in the previous day (e.g., “Were you treated with respect all day yesterday?”, “Did you smile or laugh a lot yesterday?”) in question WB178 were scaled into the index of “Experienced well-being – Previous day”. Each of the six items included in this scale had two response options (“Yes”, “No”). Note that prior to scaling, all responses were recoded as missing if the student did not respond “yes” to WB178Q07JA (“Was yesterday a typical day?”). Table 19.A.108 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Parent Questionnaire derived variables

The Parent Questionnaire is an international option that countries/economies could choose to implement. It was administered to the parents of students participating in the PISA assessment. There were 12 variables derived from this questionnaire, including one simple DV and 11 IRT scaled DVs. An overview of all DVs in this questionnaire is shown in Table 19.A.109 and each are described in the following sections.

Simple questionnaire indices

Parents’ expectations in child’s future educational career (PAREXPT)

Parents’ responses to a list of possible educational levels they expected their children to complete in question PA183 were transformed into the index of “Parents’ expectations in child’s future educational career”. The categories were specified using country-specific terms that were understood by the respondents. Each qualification was mapped to the ISCED classification of educational levels [see *ISCED 2011 Operational Manual: Guidelines for Classifying National Educational Programmes and Related Qualifications* (OECD/Eurostat/UNESCO Institute for Statistics, 2015[16])]. Values on the index ranged from “Less than ISCED level 2” to “ISCED level 8” and scores were assigned as noted in Table 19.A.110.

Derived variables based on IRT scaling

The Parent Questionnaire provided data for 11 DVs based on IRT scaling. The Cronbach’s alpha for each scale and group are presented in Table 19.A.111, the number of items with international parameters for each scale and group are presented in Table 19.A.112, the number of trend items with international parameters for each trend scale and group are presented in Table 19.A.113, the countries/economies for which the scale scores were suppressed for each scale are presented in Table 19.A.114 (in this case, the scale scores for the individuals in the country/economy were replaced with “97” in the SPSS file and “.N” in the SAS file), and the groups that did not administer each scale are presented in Table 19.A.115 (in this case, the scale scores for the individuals in the group were replaced with “99” in the SPSS file and “.M” in the SAS file).

Current parental/guardian support (CURSUPP)

Parents’ frequency ratings of how often they or someone else in their home provides education-related support (e.g., “Discuss how well my child is doing at school,” “Talk to my child about any problems he/she may have at school”) in question PA003 were scaled into the index of “Current parental/guardian support”. Note that this scale was linked to the CURSUPP scale in PISA 2018. Each of the 14 items included in this scale had five response options (“Never or hardly ever”, “Once or twice a year”, “Once or twice a month”, "Once or twice a week", "Every day or almost every day"). Table 19.A.116 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Parent attitudes toward mathematics (PQMIMP)

Parents' ratings of their agreement with statements about the importance of mathematical knowledge (e.g., "Most jobs today require some mathematics knowledge and skills.", "It is an advantage in the job market to have good mathematics knowledge and skills.") in question PA196 were scaled into the index of "Parent attitudes toward mathematics". Note that this scale was linked to the PQMIMP scale in PISA 2012 and was scaled using the PCM, in line with the model used in PISA 2012. Each of the four items included in this scale had four response options ("Strongly agree", "Agree", "Disagree", "Strongly disagree"). Table 19.A.117 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items and which items were reverse-coded prior to scaling.

Mathematics career (PQMCAR)

Parents' responses to questions about mathematics-related careers (e.g., "Does anybody in your family (including you) work in a <mathematics-related career>?", "Does your child show an interest in working in a <mathematics-related career>?") in question PA197 were scaled into the index of "Mathematics career". Note that this scale was linked to the PQMCAR scale in PISA 2012 and was scaled using the PCM, in line with the model used in PISA 2012. Each of the five items included in this scale had two response options ("Yes", "No"). Table 19.A.118 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items and which items were reverse-coded prior to scaling.

Parental involvement (PARINVOL)

Parents' responses to questions about their involvement in their child's schooling in the past year (e.g., "Discussed my child's progress with a teacher on my own initiative", "Attended a scheduled meeting or conferences for parents") in question PA008 were scaled into the index of "Parental involvement". Each of the 10 items included in this scale had two substantive response options ("Yes", "No") and an additional response option "Not supported by school" which was recoded as missing prior to scaling. Table 19.A.119 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

School quality (PQSCHOOL)

Parents' ratings of their agreement with statements about school quality (e.g., "Most of my child's school teachers seem competent and dedicated.", "Standards of achievement are high in my child's school.") in question PA007 were scaled into the index of "School quality". Note that this scale was linked to the PQSCHOOL scale in PISA 2018. Each of the seven items included in this scale had four response options ("Strongly agree", "Agree", "Disagree", "Strongly disagree"). Table 19.A.120 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling and which items are trend items.

School policies for parental involvement (PASCHPOL)

Parents' ratings of their agreement with statements about school policies for parental involvement (e.g., "My child's school provides effective communication between the school and families.", "My child's school involves parents in the school's decision-making process.") in question PA007 were scaled into the index of "School policies for parental involvement". Note that this scale was linked to the PASCHPOL scale in PISA 2018. Each of the six items included in this scale had four response options ("Strongly agree", "Agree", "Disagree", "Strongly disagree"). Table 19.A.121 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling and which items are trend items.

Parents' attitudes towards immigrants (ATTIMMP)

Parents' ratings of their agreement with statements about immigrants (e.g., "Immigrant children should have the same opportunities for education that other children in the country have.", "Immigrants who live in a country for several years should have the opportunity to vote in elections.") in question PA167 were scaled into the index of "Parents' attitudes towards immigrants". Note that this scale was linked to the ATTIMMP scale in PISA 2018. Each of the four items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.122 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Creative home environment (CREATHME)

Parents' ratings of their agreement with statements about creativity in the home environment (e.g., "In our family, we encourage participating in extra-curricular activities that require creativity.", "At home, we try to fix things that are broken.") in question PA185 were scaled into the index of "Creative home environment". Each of the nine items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.123 shows the item wording and item parameters for the items in this scale.

Participation in creative activities outside of school (CREATACT)

Parents' ratings of how often their child participated in creative activities outside of school (e.g., "Art classes/activities (e.g., painting, drawing)", "Debate club") in question PA186 were scaled into the index of "Participation in creative activities outside of school". Each of the eight items included in this scale had five substantive response options ("Never or almost never", "About once or twice a year", "About once or twice a month", "About once or twice a week", "Every day or almost every day") and an additional response option "Not available" which was recoded as missing prior to scaling. Table 19.A.124 shows the item wording and item parameters for the items in this scale.

Creativity and openness to intellect (CREATOPN)

Parents' ratings of their agreement with statements regarding their views on their own creativity and openness to intellect (e.g., "I am very creative.", "I enjoy projects that require creative solutions.") in question PA188 were scaled into the index of "Creativity and openness to intellect". Each of the nine items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree," "Strongly agree"). Table 19.A.125 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Openness to creativity: Other's report (CREATOR)

Parents' ratings of their agreement with statements regarding their views about their child's creativity (e.g., "My child is very creative.", "My child enjoys projects that require creative solutions.") in question PA189 were scaled into the index of "Openness to creativity: Other's report". Each of the eight items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree," "Strongly agree"). Table 19.A.126 shows the item wording and item parameters for the items in this scale.

School Questionnaire derived variables

The School Questionnaire consisted mainly of questions used in previous cycles. There were 57 variables derived from this questionnaire, including 32 simple DVs and 25 IRT scaled DVs. An overview of all DVs in this questionnaire is shown in Table 19.A.127 and each are described in the following sections. The simple and scaled DVs are organised first by framework module (please see Chapter 5 for a description of modules) and then alphabetical order within modules.

Simple questionnaire indices

Out-of-school experiences (Module 10)

Creative extra-curricular activities (CREACTIV)

School principals were asked in SC053 to report what extra-curricular activities their schools offered to 15-year-old students. The two response categories were “Yes” and “No” for the 10 items. The index of creative extra-curricular activities at school (CREACTIV) was computed as the total number of the following 3 activities that occurred at school: i) band, orchestra or choir (SC053Q01TA); ii) school play or school musical (SC053Q02TA); and iii) art club or art activities (SC053Q09TA). The index ranges from 0 to 3. Additionally, a separate DV (SC053D11TA) combines all the customizations across countries to SC053C11TA (please see Annex D).

Mathematics extension courses offered at school (MATHEXC)

School principals were asked in SC181 to report what additional mathematics lessons are offered at their school. The two response categories were “Yes” and “No”. The index of mathematics extension course offered at school (MATHEXC) was computed from SC181 by assigning schools to one of three different categories based on the type of additional mathematics lessons offered at the school. Schools that responded “Yes” to offering additional mathematics courses without differentiation based on prior achievement (SC181Q03JA) and “No” to offering enrichment (SC181Q01JA) and remedial (SC181Q02JA) mathematics classes were assigned a ‘1’. Schools that responded “Yes” to offering either enrichment mathematics lessons or remedial mathematics lessons were assigned a ‘2’. Schools that responded “Yes” to offering both enrichment and remedial mathematics classes were assigned a ‘3’.

Mathematics-related extra-curricular activities at school (MACTIV)

School principals were asked in SC053 to report what mathematics-related extra-curricular activities their schools offered to 15-year-old students. The two response categories were “Yes” and “No”. The index of mathematics-related extra-curricular activities at school (MACTIV) was computed as follows. First the question SC181 was assigned the value of ‘1’ if “Yes” was selected for “Enrichment” (SC181Q01JA), “Remedial” (SC181Q02JA), or “Without differentiation depending on the prior achievement level of the students” (SC181Q03JA). SC181 was assigned the value of ‘2’ if “Yes” was selected for both “Enrichment” and “Remedial”. Second, each of three items about a mathematics club (SC053Q05NA), mathematics competitions (SC053Q06NA), or club with a focus on computers (SC053Q08TA) was assigned the value of ‘1’ if a school selected “Yes” to these activities. If a school did not offer one of these three activities (i.e., selected “No”), the corresponding variable received the value of ‘0’. Third, these recoded variables were summed up to result in a range of 0 to 5 for MACTIV. For example, if the purpose of additional lessons was both “Enrichment” and “Remedial” and the school offered a mathematics club, but not mathematics competitions or a club with a focus on computers, the value of MACTIV was coded as “3”.

School type and infrastructure (Module 11)

Availability of computers (RATCMP1)

School principals were asked in SC004 to report the number of digital devices available for 15-year-old students at their school. The index of availability of computers (RATCMP1) is the ratio of the number of desktop or laptop computers available for these students for educational purposes (SC004Q02TA) to the total number of students in the modal grade for 15-year-olds at their school (SC004Q01TA).

Availability of tablet devices (RATTAB)

School principals were asked in SC004 to report the number of tablet devices or e-book readers available for 15-year-old students at their school for educational purposes (SC004Q08JA). The index of availability of tablet devices (RATTAB) is the ratio of the number of tablet devices available for these students for educational purposes to the total number of students in the modal grade for 15-year-olds at their school (SC004Q01TA).

Computers connected to the Internet (RATCMP2)

School principals were asked in SC004 to report the number of desktop or laptop computers at their school that are connected to the Internet. The index of computers connected to the Internet (RATCMP2) is the ratio of the number of desktop or laptop computers available for 15-year-olds for educational purposes (SC004Q02TA) to the number of these computers that are connected to the Internet (SC004Q03TA).

Proportion of personnel for pedagogical support (PROPSUPP)

Principals were asked in SC168 to report the number of personnel for pedagogical support currently working in their school (SC168Q01JA). The proportion of personnel for pedagogical support (PROPSUPP) was calculated by dividing the number of these personnel by the total number of non-teaching staff at the school (TOTSTAFF).

Proportion of school administrative personnel (PROADMIN)

Principals were asked in SC168 to report the number of school administrative personnel currently working in their school (SC168Q02JA). The proportion of school administrative personnel (PROADMIN) was calculated by dividing the number of these personnel by the total number of non-teaching staff at the school (TOTSTAFF).

Proportion of school management personnel (PROMGMT)

Principals were asked in SC168 to report the number of school management personnel currently working in their school (SC168Q03JA). The proportion of school management personnel (PROMGMT) was calculated by dividing the number of these personnel by the total number of non-teaching staff at the school (TOTSTAFF).

Proportion of other non-teaching staff (PROOSTAF)

Principals were asked in SC168 to report the number of other non-teaching staff (i.e., not personnel for pedagogical support, school administrative personnel, or school management personnel) currently working in their school (SC168Q04JA). The proportion of other non-teaching staff (PROOSTAF) was calculated by dividing the number of these personnel by the total number of non-teaching staff at the school (TOTSTAFF).

School size (SCHSIZE)

The index of school size (SCHSIZE) contains the total enrolment at school. It is based on the enrolment data provided by the school principal in SC002, summing the number of girls (SC002Q02TA) and boys (SC002Q01TA) at a school.

School type (SCHLTYPE)

Schools are classified as either public or private according to whether a private entity or a public agency has the ultimate power for decision making concerning its affairs. As in previous PISA surveys, the index of school type (SCHLTYPE) was constructed by recoding SC013 and SC016. SC013 asks whether the school is public or private, and SC016 asks about the source and proportion of resources (government, resources from students/parents, benefactors/donations, or other). SCHLYTPE has the following three categories:

1. Private independent (if SC013Q01TA=2 and SC016Q01TA < 50), or (SC013Q01TA=2 and SUM(SC016Q02TA, SC016Q03TA, SC016Q04TA)>=50).
2. Private Government-dependent (if SC013Q01TA=2 and SC016Q01TA >=50), or
3. Public (if SC013Q01TA=1).

Since PISA 2018, PRIVATESCH was created from sampling information in order to improve the public/private indicators. If SC013 is missing, PRIVATESCH is used to create SCHLTYPE.

Similar to 2018, IRL had special treatment for this designation – based solely on the STRATUM sampling variable.

Student-teacher ratio (STRATIO)

The student-teacher ratio (STRATIO) was obtained by dividing the number of enrolled male and female students (SCHSIZE) provided by the principal in SC002 (SC002Q01TA, SC002Q02TA) by the total number of full-time and part-time teachers (TOTAT) provided by the principal in SC018 (SC018Q01TA01, SC018Q01TA02).

Student-mathematics teacher ratio (SMRATIO)

The student-mathematics teacher ratio (SMRATIO) was obtained by dividing the number of enrolled male and female students (SCHSIZE) provided by the principal in SC002 (SC002Q01TA, SC002Q02TA) by the total number of full-time and part-time mathematics teachers (TOTMATH) provided by the principal in SC182 (SC182Q01WA01, SC182Q01WA02).

Total number of mathematics teachers at school (TOTMATH)

Principals were asked in SC182 to report the number of full-time and part-time mathematics teachers at their school (SC182Q01WA01, SC182Q01WA02) and provide additional information on how many of the staff was full-time and part-time employed mathematics teachers qualified at different ISCED levels. The total number of mathematics teachers at the school (TOTMATH) was computed as the sum of full-time and part-time mathematics teachers.

Total number of non-teaching staff at school (TOTSTAFF)

Principals were asked in SC168 to report the number of non-teaching staff currently working in their school. The total number of non-teaching staff at the school (TOTSTAFF) is a sum of the numbers of personnel for pedagogical support (SC168Q01JA), school administrative personnel (SC168Q02JA), school management personnel (SC168Q03JA), and other non-teaching staff (SC168Q04JA).

Total number of all teachers at school (TOTAT)

Principals were asked in SC018 to report the total number of full-time and part-time teachers at their school (SC018Q01TA01, SC018Q01TA02) and provide additional information on how many of the staff was full-time and part-time employed teachers qualified at different ISCED levels.

Selection and enrolment (Module 12)

School selectivity (SCHSEL)

Principals were asked in SC012 about admittance policies at their school, including student academic performance and recommendation by feeder schools. The three response categories for this question were "Never", "Sometimes", and "Always". An index of academic school selectivity (SCHSEL) was computed by assigning schools to one of three categories based on how often two factors, namely "Student's record of academic performance" (SC012Q01TA) and "Recommendation of feeder schools" (SC012Q02TA), were considered when admitting students to the school as follows:

1. the two factors (student's record of academic performance and recommendation of feeder schools) were never considered (if SC012Q01TA=1 and SC012Q02TA=1),
2. at least one of the factors was considered sometimes but neither always (if SC012Q01TA=2 or SC012Q02TA=2, and if SC012Q01TA<3 and SC012Q02TA<3), and
3. at least one of the factors was considered always (if SC012Q01TA=3 or SC012Q02TA=3).

School autonomy (Module 13)

School responsibility for curriculum (SRESPCUR)

Principals were asked in SC202 about who had the main responsibility for various decisions or activities at their school. The six response categories for this question were "Principal", "Teachers or members of <school management team>", "<School governing board>", "<Local or municipal authority>", "<Regional or state authority>", and "<National or federal authority>". An index of the relative level of responsibility of school staff in deciding issues related to curriculum and assessment (RESPCUR) was computed from the school principals' reports regarding who had the main responsibility for 4 items in SC202. The index was calculated on the basis of the ratio of responses for "Principal", "Teachers or members of <school management team>", or "<School governing board>" on the one hand to responses for "<Local or municipal authority>", "<Regional or state authority>", or "<National or federal authority>" on the other hand. In the first step, a count for school responsibility was calculated by counting the number of "Principal", "Teachers or members of <school management team>", and "<School governing board>" responses. In the second step, a count for non-school responsibility was calculated by counting the number of "<Local or municipal authority>", "<Regional or state authority>", and "<National or federal authority>". In the third step, the school responsibility count was divided by the non-school responsibility count. To avoid dividing by "0", "1" was added to both the numerator and denominator; when the ratio of school responsibility to non-school responsibility was 4:0, an index value of 4 was assigned. Higher values indicated relatively higher levels of school responsibility in deciding issues related to curriculum and assessment.

School responsibility for resources (SRESPRES)

Principals were asked in SC202 about who had the main responsibility for various decisions or activities at their school. The six response categories for this question were "Principal", "Teachers or members of <school management team>", "<School governing board>", "<Local or municipal authority>", "<Regional or state authority>", and "<National or federal authority>". An index of the relative level of responsibility of school staff in deciding issues related to allocating resources (RESPRES) was computed from the school principals' reports regarding who had the main responsibility for 6 items in SC202. The index was calculated on the basis of the ratio of responses for "Principal", "Teachers or members of <school management team>", or "<School governing board>" on the one hand to responses for "<Local or municipal authority>", "<Regional or state authority>", or "<National or federal authority>" on the other hand. In the first step, a count for school responsibility was calculated by counting the number of "Principal", "Teachers or members of <school management team>", and "<School governing board>" responses. In the second step, a count for non-school responsibility was calculated by counting the number of "<Local or municipal authority>", "<Regional or state authority>", and "<National or federal authority>". In the third step, the school responsibility count was divided by the non-school responsibility count. To avoid dividing by "0", "1" was added to both the numerator and denominator; when the ratio of school responsibility to non-school responsibility was 6:0, an index value of 6 was assigned. Higher values on the scale indicated relatively higher levels of school responsibility in this area.

Organisation of student learning at school (Module 14)

Ability grouping for mathematics classes (ABGMATH)

School principals were asked in SC187 to report the extent to which their mathematics classes catered to students with different abilities. The three response categories were "For all classes", "For some classes", and "Not for any classes". An index of ability grouping between mathematics classes (ABGMATH) was derived from the first two items (SC187Q01WA, SC187Q02WA) by assigning schools to three categories: (1) schools with no ability grouping for any classes, (2) schools with one of these forms of ability grouping between some classes and (3) schools with one of these forms of ability grouping for all classes.

Class size (CLSIZE)

Principals were asked in SC003 about the average size of the test language classes in their school. The nine response categories were "15 students or fewer", "16-20 students", "21-25 students", "26-30 students", "31-35 students", "36-40 students", "41-45 students", "46-50 students", and "More than 50 students". The average class size (CLSIZE) was derived from the midpoint of each response category, resulting in a value of 13 for the lowest category, and a value of 53 for the highest.

Math class size (MCLSIZE)

Principals were asked in SC176 about the average class size of mathematics classes in their school. The nine response categories were "15 students or fewer", "16-20 students", "21-25 students", "26-30 students", "31-35 students", "36-40 students", "41-45 students", "46-50 students", and "More than 50 students". The average math class size (TBD) was derived from the midpoint of each response category, resulting in a value of 13 for the lowest category, and a value of 53 for the highest.

Teacher qualification, training, and professional development (Module 17)

Proportion of all teachers fully certified (PROATCE)

Principals were asked in SC018 to report the number of full-time and part-time teachers fully certified by the appropriate authority (SC018Q02TA01, SC018Q02TA02). The proportion of fully certified teachers (PROATCE) was computed by dividing the number of fully certified teachers by the total number of teachers (TOTAT).

Proportion of all teachers with at least ISCED level 6 bachelor qualification (PROPAT6)

Principals were asked in SC018 to report the number of full-time and part-time teachers with an ISCED level 6 (Bachelor's or equivalent level) qualification (SC018Q08JA01, SC018Q08JA02). The proportion of teachers with *at least* an ISCED 6 Bachelor qualification (PROPAT6) was calculated by dividing the number full-time and part-time teachers with an ISCED level 6 (Bachelor's or equivalent level) qualification (SC018Q08JA01, SC018Q08JA02), ISCED level 7 (Master's or equivalent level) qualification (SC018Q09JA01, SC018Q09JA02), and ISCED level 8 (Doctoral or equivalent level) qualification (SC018Q10JA01, SC018Q10JA02) by the total number of teachers (TOTAT).

Proportion of all teachers with at least ISCED level 7 master qualification (PROPAT7)

Principals were asked in SC018 to report the number of full-time and part-time teachers with an ISCED level 7 (Master's or equivalent level) qualification (SC018Q09JA01, SC018Q09JA02). The proportion of teachers with *at least* an ISCED 7 Master qualification (PROPAT7) was calculated by dividing the number of full-time and part-time teachers with an ISCED level 7 (Master's or equivalent level) qualification (SC018Q09JA01, SC018Q09JA02) and ISCED level 8 (Doctoral or equivalent level) qualification (SC018Q10JA01, SC018Q10JA02) by the total number of teachers (TOTAT).

Proportion of all teachers with ISCED level 8 doctoral qualification (PROPAT8)

Principals were asked in SC018 to report the number of full-time and part-time teachers with an ISCED level 8 (Doctoral or equivalent level) qualification (SC018Q10JA01, SC018Q10JA02). The proportion of teachers with an ISCED 8 Doctoral qualification (PROPAT8) was calculated by dividing the number of these teachers by the total number of teachers (TOTAT).

Proportion of mathematics teachers at school (PROPMATH)

The proportion of mathematics teachers (PROPMATH) was computed as the total number of full-time and part-time mathematics teachers at their school (TOTMATH) provided by the principal in SC182 (SC182Q01WA01, SC182Q01WA02), divided by the total number of teachers at their school (TOTAT) provided by the principal in SC018 (SC018Q01TA01, SC018Q01TA02).

Global crises (Module 21)

School closure support from education authorities (SCSUPRTED)

School administrators' responses to three items in SC222 comprise the index on school closure support from education authorities. School administrators who indicated that their school was closed for one or more school days because of COVID-19 were asked to rate the extent that they felt their school was supported by educational authorities during the time that their school building was closed to students because of COVID-19. They reported this information by selecting one of four response options: "Not at all"; "Very little"; "To some extent"; "A lot". Respondents may interpret support broadly to include any kind of assistance (i.e., financial support, volunteer support, etc). If school administrators chose "A lot" to any of the three items, they received a value of "2" on the index. If school administrators chose "Not at all" to all three items, they received a value of "0" on the index. All other responses received a value of "1". The values of the index range from 0-2. This variable was skipped for respondents who reported that their schools had not been closed for COVID-19 on question SC213.

School closure support from other sources (SCSUPRT)

School administrators' responses to two items in SC222 comprise the index on school closure support from other sources. School administrators who indicated that their school was closed for one or more school days because of COVID-19 were asked to rate the extent that they felt their school was supported by students' parents or guardians and by private donors during the time that their school building was closed to students because of COVID-19. They reported this information by selecting one of four response options: "Not at all"; "Very little"; "To some extent"; "A lot". Respondents may interpret support broadly to include any kind of assistance (i.e., financial support, volunteer support, etc). If school administrators chose "A lot" to either of the two items, they received a value of "2" on the index. If school administrators chose "Not at all" to both items, they received a value of "0" on the index. All other responses received a value of "1". The values of the index will range from 0-2. This variable was skipped for respondents who reported that their schools had not been closed for COVID-19 on question SC213.

Derived variables based on IRT scaling

The School Questionnaire provided data for 25 DVs based on IRT scaling. The Cronbach's alpha for each scale and group are presented in Table 19.A.128, the number of items with international parameters for each scale and group are presented in Table 19.A.129, the number of trend items with international parameters for each trend scale and group are presented in Table 19.A.130, the countries/economies for which the scale scores were suppressed for each scale are presented in Table 19.A.131, and the groups that did not administer each scale are presented in Table 19.A.132.

School culture and climate (Module 6)

Negative school climate (NEGSCLIM)

Principals were asked in SC172 about the extent of problem behaviours that contribute to a negative school climate in their school (e.g., "Profanity", "Vandalism"). The four response categories for the six items in the scale were "Not at all", "Small extent", "Moderate extent", and "Large extent". Higher scale score values indicate that problem behaviours contribute to a negative school climate to a greater extent, while lower scale score values indicate that problem behaviours impact school climate to a lesser extent. Table 19.A.133 shows the item wording and item parameters for the items in this scale.

School diversity and multi-cultural views (DMCVIEWS)

Principals were asked in SC173 about the school staff's efforts to promote a diversity-oriented culture and climate during the last academic year (e.g., "They encouraged students of different backgrounds to resolve disagreements by finding common ground.", "They taught students how to respond to discrimination."). The five response categories for the six items in the scale were "Never or almost never", "About once or twice a year", "About once or twice a month", "About once or twice a week", and "Every day or almost every day". Higher scale score values indicate that diversity-related views were encouraged in the school with greater frequency, while lower scale score values indicate that diversity-related views were encouraged with lesser frequency. Table 19.A.134 shows the item wording and item parameters for the items in this scale.

Student-related factors affecting school climate (STUBEHA)

Principals were asked in SC061 about the extent to which student learning is hindered by student behaviours (e.g., "Student truancy", "Student use of alcohol or illegal drugs"). Note that this scale was linked to the STUBEHA scale in PISA 2018. The four response categories for the six items in the scale were "Not at all", "Very little", "To some extent", and "A lot". Higher scale score values indicate that students' learning is hindered to a greater extent by negative student behaviours, while lower scale score values indicate that students' learning is hindered by negative student behaviours to a lesser extent. Table 19.A.135 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Teacher-related factors affecting school climate (TEACHBEHA)

Principals were asked in SC061 about the extent to which student learning is hindered by teacher behaviours (e.g., “Teacher absenteeism”, “Staff resisting change”). The four response categories for the five items in the scale were “Not at all”, “Very little”, “To some extent”, and “A lot”. Higher scale score values indicate that students’ learning is hindered to a greater extent by negative teacher behaviours, while lower scale score values indicate that students’ learning is hindered by negative teacher behaviours to a lesser extent. Table 19.A.136 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Out-of-school experiences (Module 10)

Extra-curricular activities offered (ALLACTIV)

School principals were asked in SC053 to report what extra-curricular activities their schools offered to 15-year-old students (e.g., “School play or school musical”, “Mathematics club”). The two response categories for the 10 items in the scale were “Yes” and “No”. Higher scale score values indicate that more extra-curricular activities were offered by the school, while lower scale score values indicate that fewer extra-curricular activities were offered. Table 19.A.137 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

School type and infrastructure (Module 11)

Shortage of educational material (EDUSHORT)

Principals were asked in SC017 about the extent to which instruction is hindered by a shortage of educational materials in their school (e.g., “A lack of educational material (e.g., textbooks, IT equipment, library or laboratory material)”, “Inadequate or poor quality educational material (e.g. textbooks, IT equipment, library or laboratory material)”). Note that this scale was linked to the EDUSHORT scale in PISA 2018. The four response categories for the four items in the scale were “Not at all”, “Very little”, “To some extent”, and “A lot”. Higher scale score values indicate that the school is impacted by a shortage of educational materials to a greater extent, while lower scale score values indicate that the school is impacted to a lesser extent by a shortage of educational materials. Table 19.A.138 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Shortage of educational staff (STAFFSHORT)

Principals were asked in SC017 about the extent to which instruction is hindered by a shortage of educational staff in their school (e.g., “A lack of teaching staff”, “Inadequate or poorly qualified assisting staff”). Note that this scale was linked to the STAFFSHORT scale in PISA 2018. The four response categories for the four items in the scale were “Not at all”, “Very little”, “To some extent”, and “A lot”. Higher scale score values indicate that the school is impacted by a shortage of educational staff to a greater extent, while lower scale score values indicate that the school is impacted to a lesser extent by a shortage of educational staff. Table 19.A.139 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

School autonomy (Module 13)

Educational leadership (EDULEAD)

Principals were asked in SC201 about how often they or other members of their school management team engaged in activities or behaviours related to educational leadership during the past 12 months (e.g., “Collaborating with teachers to solve classroom discipline problems”, “Providing parents or guardians with information on the school and student performance”). The five response categories for the seven items in the scale were “Never or almost never”, “About once or twice a year”, “About once or twice a month”, “About once or twice a week”, and “Every day or almost every day”. Higher scale score values indicate higher frequencies of engagement by the principal and school management team in educational leadership activities, while lower scale values indicate lower frequencies of engagement by the principal and school management team in educational leadership activities. Table 19.A.140 shows the item wording and item parameters for the items in this scale.

Instructional leadership (INSTLEAD)

Principals were asked in SC201 about how often they or other members of their school management team engaged in activities or behaviours related to teaching or instructional leadership during the last 12 months (e.g., “Providing feedback to teachers based on observations of instruction in the classroom”, “Taking actions to ensure that teachers feel responsible for their students' learning outcomes”). The five response categories for the five items in the scale were “Never or almost never”, “About once or twice a year”, “About once or twice a month”, “About once or twice a week”, and “Every day or almost every day”. Higher scale score values indicate higher frequencies of engagement by the principal and school management team in instructional leadership activities, while lower scale score values indicate lower frequencies of engagement by the principal and school management team in instructional leadership activities. Table 19.A.141 shows the item wording and item parameters for the items in this scale.

School autonomy (SCHAUTO)

Principals were asked in SC202 about who had the main responsibility for various decisions or activities at their school (e.g., “Appointing or hiring teachers”, “Determining teachers’ salary increases”). The six response categories for the 12 items in the scale were “Principal”, “Teachers or members of <school management team>”, “<School governing board>”, “<Local or municipal authority>”, “<Regional or state authority>”, and “<National or federal authority>”. Higher scale score values indicate that the principal, teachers or members of the school management team, and the school governing board had a greater level of autonomy in decision-making activities at their school. Lower scale score values indicate that these groups had less autonomy. Table 19.A.142 shows the item wording and item parameters for the items in this scale. It also indicates how the response categories were recoded prior to scaling.

Teacher participation (TCHPART)

Principals were asked in SC202 about who had the main responsibility for various decisions or activities at their school (e.g., “Formulating the school budget”, “Choosing which learning materials are used”). The six response categories for the 12 items in the scale were “Principal”, “Teachers or members of <school management team>”, “<School governing board>”, “<Local or municipal authority>”, “<Regional or state authority>”, and “<National or federal authority>”. Higher scale score values indicate that the teachers or members of the school management team participated to a greater extent in decision-making activities at their school. Lower scale score values indicate that they participated to a lesser extent. Table 19.A.143 shows the item wording and item parameters for the items in this scale. It also indicates how the response categories were recoded prior to scaling.

Organisation of student learning at school (Module 14)

Digital device policies at school (DIGDVPOL)

School principals were asked in SC190 to indicate whether their school had various policies regarding digital device use (e.g., “Teachers establish rules for when students may use digital devices during lessons.", "The school has a specific programme to prepare students for responsible internet behaviour."). The two response categories for the nine items in the scale were "Yes" and "No". Higher scale score values indicate that digital device policies are enforced at the school to a greater extent, while lower scale score values indicate that such policies are enforced to a lesser extent. Table 19.A.144 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Teacher qualification, training, and professional development (Module 17)

Mathematics teacher training (MTTRAIN)

School principals were asked in SC184 to indicate the areas in which professional development was offered to mathematics teachers in their school (e.g., "Mathematics content", "Mathematics curriculum"). The two response categories for the seven items in the scale were "Yes" and "No". Higher scale score values indicate that more opportunities for professional development are offered to mathematics teachers in the school, while lower scale score values indicate that fewer professional development opportunities are offered to them. Table 19.A.145 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Assessment, evaluation, and accountability (Module 18)

Feedback to teachers (TEAFDBK)

Principals were asked in SC193 how much impact teacher evaluations had on various matters (e.g., "A change in salary", "A change in the likelihood of career advancement"). The four response categories for the seven items in the scale were "No impact", "Small impact", "Moderate impact", and "Large impact". Higher scale score values indicate greater impact of teacher evaluations or feedback, while lower scale score values indicate lesser impact. Table 19.A.146 shows the item wording and item parameters for the items in this scale.

Use of standardised tests (STDTEST)

Principals were asked in SC035 to indicate whether standardised tests were used for various purposes (e.g., "To guide students' learning", "To group students for instructional purposes"). The two response categories for the 11 items in the scale were "Yes" and "No". Higher scale score values indicate that standardised tests are used for accountability purposes to a greater extent, while lower scale score values indicate that these tests are used to a lesser extent. Table 19.A.147 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Use of teacher-developed tests (TDTEST)

Principals were asked in SC035 to indicate whether teacher-developed tests were used for various purposes (e.g., "To guide students' learning", "To group students for instructional purposes"). The two response categories for the 11 items in the scale were "Yes" and "No". Higher scale score values indicate that teacher-developed tests are used for accountability purposes to a greater extent, while lower scale score values indicate that these tests are used to a lesser extent. Table 19.A.148 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Parental/guardian involvement and support (Module 19)

School encouragement of parent or guardian involvement (ENCOURPG)

Principals were asked in SC192 about how often their school staff engaged parents or guardians in various aspects of students' educational environment during the last academic year (e.g., "Invited parents or guardians to volunteer for school activities", "Initiated communications with parents or guardians about their child's progress"). The four response categories for the six items in the scale were "Never or almost never", "A few times a year", "A few times a month", and "Once a week or more". Higher scale score values indicate more frequent efforts by the school staff to engage parents or guardians in becoming involved at the school, while lower scale score values indicate less frequent engagement efforts. Table 19.A.149 shows the item wording and item parameters for the items in this scale.

Creative thinking (Module 20)

Beliefs about creativity (BCREATSC)

Principals were asked in SC204 to indicate their level of agreement with statements regarding their beliefs about creativity (e.g., "Creativity can be trained.", "There are many different ways to be creative."). The four response categories for the four items in the scale were "Strongly disagree", "Disagree", "Agree", and "Strongly agree". Higher scale score values indicate that principals endorse, to a greater extent, beliefs about the malleability of creativity and an expansive view of what it means to be creative. Lower scale score values indicate that principals endorse these beliefs to a lesser extent. Table 19.A.150 shows the item wording and item parameters for the items in this scale.

Creative school activities offered (ACTCRESC)

Principals were asked in SC207 to indicate how often creative activities are offered in their school (e.g., "Creative writing classes/activities", "Debate <club>"). The five substantive response categories for the eight items in the scale were "Never or almost never", "About once or twice a year", "About once or twice a month", "About once or twice a week", and "Every day or almost every day". There was an additional response category, "Not available at our school", which was recoded as missing prior to scaling. Higher scale score values indicate a greater frequency of creative activities being offered in school, while lower scale score values indicate creative activities are offered on a less frequent basis. Table 19.A.151 shows the item wording and item parameters for the items in this scale. It also indicates how the response categories were recoded prior to scaling.

Creative school environment (CREENVSC)

Principals were asked in SC205 to indicate their level of agreement with statements regarding the encouragement of creative thinking by teachers and through activities at the school (e.g., "Teachers in our school value students' creativity.", "Class activities in our school help students think about new ways to solve complex tasks."). The four response categories for the six items in the scale were "Strongly disagree", "Disagree", "Agree", and "Strongly agree". Higher scale score values indicate more agreement with the overall view that students' creativity is encouraged in the school, while lower scale score values indicate less agreement with this view. Table 19.A.152 shows the item wording and item parameters for the items in this scale.

Openness culture/climate (OPENCUL)

Principals were asked in SC208 to indicate the extent to which they agree or disagree with statements regarding their students' orientation towards openness and creativity (e.g., "Most students at my school are creative.", "Most students at my school enjoy learning new things."). The four response categories for the nine items in the scale were "Strongly disagree", "Disagree", "Agree", and "Strongly agree". Higher scale score values indicate that students have a greater orientation towards openness and creativity, while lower scale score values indicate they have less orientation towards openness and creativity. Table 19.A.153 shows the item wording and item parameters for the items in this scale.

Global crises (Module 21)

Note that the questions in this module were skipped for respondents who reported that their school had not been closed for one or more school days because of COVID-19 in question SC213.

Problems with schools' capacity to provide remote instruction (PROBSCRI)

School administrators were asked in SC216 to what extent specific challenges hindered their school's capacity to provide remote instruction during the time when the school building was closed to students because of COVID-19 (e.g., "Lack of access to <digital devices> among students", "Lack of learning management systems or school learning platforms (e.g., [Blackboard®], [Edmodo®], [Moodle®], [Google® Classroom™])"). The four response categories for the five items in the scale were "Not at all", "Very little", "To some extent", and "A lot". Higher scale score values indicate a greater level of problems with the schools' capacity to provide remote instruction while the school building was closed to students during the COVID-19 pandemic, while lower scale score values indicate fewer problems with the capacity to provide remote instruction. Table 19.A.154 shows the item wording and item parameters for the items in this scale.

School preparation for remote instruction – Before pandemic (SCPREPBP)

School administrators were asked in SC223 whether their school had taken specific actions to prepare for remote instruction (e.g., "Adapting existing curriculum plans for remote instruction (e.g., modifying course requirements, sequence of lessons, grading policies)", "Ensuring that students have access to <digital devices> for remote instruction"). The three response categories for the 10 items in the scale were "Yes, as a standard practice before the COVID-19 pandemic", "Yes, in response to the COVID-19 pandemic", and "No". Prior to scaling, the first response was coded as 1, while the two latter responses were coded as 0. Higher scale score values indicate a higher level of school preparation for remote instruction before the pandemic, while lower scale score values indicate less preparation for remote instruction before the pandemic. Table 19.A.155 shows the item wording and item parameters for the items in this scale.

School preparation for remote instruction – In response to pandemic (SCPREPAP)

School administrators were asked in SC223 whether their school had taken specific actions to prepare for remote instruction (e.g., "Adapting existing curriculum plans for remote instruction (e.g., modifying course requirements, sequence of lessons, grading policies)", "Ensuring that students have access to <digital devices> for remote instruction"). The three response categories for the 10 items in the scale were "Yes, as a standard practice before the COVID-19 pandemic", "Yes, in response to the COVID-19 pandemic", and "No". Prior to scaling, the first two responses were coded as 1, while the third response was coded as 0. Higher scale score values indicate a higher level of school preparation for remote instruction after the start of the pandemic, while lower scale score values indicate less preparation for remote instruction after the start of the pandemic. Table 19.A.156 shows the item wording and item parameters for the items in this scale.

Preparedness for digital learning (DIGPREP)

School administrators were asked in SC155 to rate their agreement with statements about their school's capacity to use digital devices to enhance learning and teaching ("Teachers have sufficient time to prepare lessons integrating digital devices", "The school has sufficient qualified technical assistant staff"). The four response categories for the six items in the scale were "Strongly disagree", "Disagree", "Agree", and "Strongly agree". Higher scale score values indicate greater capacity for a school to use digital technology for learning and teaching, while lower scale score values indicate less capacity for a school to do so. Table 19.A.157 shows the item wording and item parameters for the items in this scale.

Teacher Questionnaire derived variables

The Teacher Questionnaire is an international option that countries/economies could choose to implement. Routing within the questionnaire was used to deliver specific questions to mathematics teachers and specific questions to all other, non-mathematics teachers. Other questions in the questionnaire were administered to all teachers. There were 45 variables derived from this questionnaire, including 13 simple DVs and 32 IRT scaled DVs. An overview of all DVs in this questionnaire is shown in Table 19.A.158 and each are described in the following sections.

Simple questionnaire indices

Originally trained teachers – Strict (OTT1) and broad (OTT2) definitions

The Teacher Questionnaire addressed two questions about teachers' initial education and professional development. The first question, TC014, asks if teacher education or training programme was completed, with response options "*1, Yes*, a programme of 1 year or less" "2, *Yes, a programme longer than 1 year"* and *"3, No"*. TC015 asked about how the teacher qualification was received. Response options included "1, I attended a standard teacher education or training programme at an <educational institute which is eligible to educate or train teachers>.", "2, I attended an in-service teacher education or training programme.", "3, I attended a work-based teacher education or training programme.", "4, I attended training in another pedagogical profession.", or "5, Other". These two questions (TC014, TC015) were used to build the DV OTT1 (Originally trained teachers, strict definition) and OTT2 (Originally trained teachers, broad definition). The strict definition implies that a teacher had intended to be trained as a teacher from the very beginning of his or her career and has finished a "standard teacher education or training programme at an <educational institute which is eligible to educate or train teachers>". In the less strict definition, the teacher has finished any of the following three programmes: either a "standard teacher education or training programme at an <educational institute which is eligible to educate or train teachers>" (option 1 in TC015), an "in-service teacher education or training programme" (option 2) or a "work-based teacher education or training programme" (option 3 in TC015).

Trained to teach certain subjects (NTEACH1-11)

TC018 asked about the specific subjects that were included in the teacher's education or training programme or other professional qualification and asked if the respondents taught these subjects to the national modal grade for 15-year-olds in the current school year. The DVs NTEACH1 to NTEACH 11 reflect whether the teacher was trained to teach a certain subject. A value of "1" indicate that teachers were trained to teach the subject in question, while "0" indicates they were not trained to teach the subject in question.

Subject-specific overlap between initial education and teaching the modal grade (STTMG1-11)

TC018 enquired about the specific subjects that were included in the teacher's education or training programme or other professional qualification and asked if the respondents taught these subjects to the national modal grade for 15-year-olds in the current school year. This question is used to build the DVs STTMG1 to STTMG11, indicating the subject-specific overlap between initial education and teaching the modal grade, i.e., whether a teacher currently teaches a certain subject combined with whether it was included in the teacher's initial training. A value of "0" indicates that teachers were neither trained nor teach the subject in question, "1" indicates they were trained to teach the subject in question but do not teach it, "2" indicates they were not trained to teach the subject in question but they do teach the subject, and "3" indicates they were trained to teach the subject in question and they teach the subject.

Country born (COBN_T)

COBN_T is based on question TC186, which asks about the country/economy a teacher is born in, coded into the following categories: (1) "Country of test" and (2) "Other country". Each country/economy adapts the items for this question to collect relevant country/economy of birth information for teachers in their country/economy, so the index also gives detailed categories of teachers' original countries/economies of birth within a country/economy that vary from country/economy to country/economy.

Content overlap between initial education and professional development (TC045Q01-TC045Q18)

TC045 asked about 10 content topics (e.g., "knowledge and understanding of my subject field(s)", "knowledge of the curriculum") that might have been included in the teachers' initial education and training and/or in professional development activities during the last 12 months. Teachers could select both if applicable. The DVs TC045Q01 to TC045Q18 reflect the content overlap between initial education and professional development.

Higher educational level attained (TCISCED)

Teachers' responses to TC210 regarding education were classified using ISCED 2011. An index on higher educational level attained was constructed by recoding educational qualifications into the following categories: Less than ISCED Level 3.3 (upper secondary with no direct access to tertiary education), ISCED Level 3.3 (upper secondary with no direct access to tertiary education), ISCED 3.4 (upper secondary with direct access to tertiary education), ISCED 4 (post-secondary non-tertiary), ISCED 5 (short-cycle tertiary), ISCED 6 (Bachelor's or equivalent level, ISCED 7 (Master's or equivalent level), and ISCED 8 (Doctoral or equivalent level). Scores are assigned as noted in Table 19.A.159.

Employment status (EMPLSTAT and EMPLSTATd)

TC211 asked about employment status in terms of the contract duration with three response options ("Permanent employment", "Fixed-term contract for a period of more than 1 school year", "Fixed-term contract for a period of 1 school year or less"). The corresponding DVs reflected the duration of employment, measured via TC211, a) on the original three-point scale (EMPLSTAT) and b) dichotomous, distinguishing a permanent position from fixed-term contracts (EMPLSTATd).

Study abroad (STABROAD)

Teachers' responses to the whether they had studied abroad in question TC188 were scaled into the index of "Study abroad". There are four response options (No; Yes, for less than three months; Yes, for three to twelve months; Yes, for more than a year). This question was previously included in the PISA 2018 Teacher Questionnaire, but with no index. Values on this index are 0 (Never studied abroad) and 1 (Studied abroad). This variable was skipped for teachers who reported that they taught mathematics to students in national modal grade for 15-year-olds this school year on question TC217.

Weekly teacher workload (TCWKLOAD)

Teachers were asked to enter how many hours a week they spend on various teaching tasks (e.g., “Marking/correcting of student work”, General administrative work”). Teachers’ responses to each of the eight fill-in items in question TC216 were summed to create the simple index “Weekly teacher workload”.

Use of digital resources for mathematics (ICTMATTC)

Teachers’ frequency ratings of how often they instruct their students to use digital resources for a range of mathematics tasks in class or for homework (e.g., “Use digital resources for simple calculations”, “Use digital resources for simulations and modelling, virtual laboratories”) in question TC222 were aggregated into a simple index “Use of digital resources for mathematics” as follows: A value of “1” is assigned if teachers select response options 4 (“About once or twice a week”) or 5 (“Every day or almost every day”) at least once across the 4 items in this question, “0” otherwise. The index captures whether teachers said that they use some digital resources for mathematics lessons at least on a weekly basis or not. This variable was skipped for teachers who reported that they did not teach mathematics to students in national modal grade for 15-year-olds this school year on question TC217.

Proportion of working years at this school (PROPWORK)

In TC007 teachers were asked to how many years of teaching experience they had at the school having them take the questionnaire and how many years of teaching experience they had in total. These responses were used to form the simple index “Proportion of working years at this school”, by dividing the number of years at their current school by total number of years teaching.

Derived variables based on IRT Scaling

The Teacher Questionnaire provided data for 32 DVs based on IRT scaling. The Cronbach’s alpha for each scale and group are presented in Table 19.A.160, the number of items with international parameters for each scale and group are presented in Table 19.A.161, the number of trend items with international parameters for each trend scale and group are presented in Table 19.A.162, the countries/economies for which the scale scores were suppressed for each scale are presented in Table 19.A.163, and the groups that did not administer each scale are presented in Table 19.A.164.

Proportion of professional development (PRPDT)

Teachers’ responses about their participation in different professional development activities in the last 12 months (e.g., “Individual or collaborative research on a topic of interest to you professionally”, “Course, workshop, or conference on teaching methods”) in question TC020 were scaled into the index of “Proportion of professional development”. Each of the 14 items included in this scale had two response options (“Yes”, “No”). Note that this scale was skipped for teachers who reported that they taught mathematics to students in the national modal grade for 15-year-olds in the current school year on question TC217. Table 19.A.165 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Exchange and co-ordination for teaching (EXCHT)

Teachers’ frequency ratings of how often they participate in teaching-related co-operation (e.g., Exchange teaching materials with colleagues”, “Attend team conferences”) in question TC046 were scaled into the index of “Exchange and co-ordination for teaching”. Note that this scale was linked to the EXCHT scale in PISA 2018. Each of the four items included in this scale had six response options (“Never”, “Once a year or less”, “2-4 times a year”, “5-10 times a year”, “1-3 times a month”, “Once a week or more”). Table

19.A.166 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Teaching ICT awareness (ICTOTL)

Teachers' responses about whether they taught various ICT awareness activities to their students (e.g., "How to decide whether to trust information from the Internet", "How to detect phishing or spam emails") in question TC166 were scaled into the index of "Teaching ICT awareness". Each of the seven items included in this scale had two response options ("Yes", "No"). Note that this scale was skipped for teachers who reported that they taught mathematics to students in the national modal grade for 15-year-olds in the current school year on question TC217. Table 19.A.167 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Teachers' use of specific ICT applications (TCICTUSE)

Teachers' frequency ratings of how often they used specific ICT applications while teaching (e.g., "Digital learning games", "Data logging and monitoring tools") in question TC169 were scaled into the index of "Teachers' use of specific ICT applications". Note that this scale was linked to the TCICTUSE scale in PISA 2018. Each of the 14 items included in this scale had four response options ("Never", "In some lessons", "In most lessons", "In every or almost every lesson"). Table 19.A.168 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Disciplinary climate in mathematics (TCDISCLIMA)

Teachers' frequency ratings of how often a range of situations occurred in their mathematics lessons (e.g., "There is noise and disorder.", "I have to wait a long time for students to quiet down.") in question TC170 were scaled into the index of "Disciplinary climate in mathematics", which measures how much discipline there is during mathematics lessons. Each of the seven items included in this scale had four response options ("Every lesson", "Most lessons", "Some lessons", "Never or almost never"). Note that this scale was skipped for teachers who reported that they did not teach mathematics to students in the national modal grade for 15-year-olds in the current school year on question TC217. Table 19.A.169 shows the item wording and item parameters for the items in this scale.

Need for professional development (DEVNEED)

Teachers' ratings of their need for professional development in various areas (e.g., "Knowledge of the curriculum", "Student behaviour and classroom management") in question TC185 were scaled into the index of "Need for professional development". Each of the 13 items included in this scale had four response options ("No need at present", "Low level of need", "Moderate level of need", "High level of need"). Table 19.A.170 shows the item wording and item parameters for the items in this scale.

Teachers' attitudes toward equal rights for immigrants (TCATTIMM)

Teachers' ratings of their agreement with statements about immigrants (e.g., "Immigrant children should have the same opportunities for education that other children in the country have.", "Immigrants should have the opportunity to continue their own customs and lifestyle.") in question TC196 were scaled into the index of "Teachers' attitudes toward equal rights for immigrants". This scale was linked to the TCATTIMM scale in PISA 2018. Each of the four items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Note that this scale was skipped for teachers who reported that they taught mathematics to students in the national modal grade for 15-year-olds in the current school year on question TC217. Table 19.A.171 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Satisfaction with current job environment (SATJOB)

Teachers' ratings of their agreement with various statements indicating their satisfaction with the current job environment (e.g., "I enjoy working at this school.", "I would recommend my school as a good place to work.") in question TC198 were scaled into the index of "Satisfaction with current job environment". Note that this scale was linked to the SATJOB scale in PISA 2018. Each of the four items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.172 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Satisfaction with teaching profession (SATTEACH)

Teachers' ratings of their agreement with various statements indicating their satisfaction with the teaching profession (e.g., "The advantages of being a teacher clearly outweigh the disadvantages.", "I regret that I decided to become a teacher.") in question TC198 were scaled into the index of "Satisfaction with teaching profession". Note that this scale was linked to the SATTEACH scale in PISA 2018. Each of the five items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.173 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling and which are trend items.

Teacher's self-efficacy in classroom management (SEFFCM)

Teachers' ratings of their classroom management skills (e.g., "Control disruptive behaviour in the classroom", "Get students to follow classroom rules") in question TC199 were scaled into the index of "Teacher's self-efficacy in classroom management". This scale was linked to the SEFFCM scale in PISA 2018. Each of the four items included in this scale had four response options ("Not at all", "To some extent", "Quite a bit", "A lot"). Note that this scale was skipped for teachers who reported that they taught mathematics to students in the national modal grade for 15-year-olds in the current school year on question TC217. Table 19.A.174 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Teacher's self-efficacy in maintaining positive relations with students (SEFFREL)

Teachers' ratings of their ability to maintain positive relations with students (e.g., "Help my students value learning", "Motivate students who show low interest in school work") in question TC199 were scaled into the index of "Teacher's self-efficacy in maintaining positive relations with students". This scale was linked to the SEFFREL scale in PISA 2018. Each of the four items included in this scale had four response options ("Not at all", "To some extent", "Quite a bit", "A lot"). Note that this scale was skipped for teachers who reported that they taught mathematics to students in the national modal grade for 15-year-olds in the current school year on question TC217. Table 19.A.175 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Teacher's self-efficacy in instructional settings (SEFFINS)

Teachers' ratings of their self-efficacy in instructional settings (e.g., "Craft good questions for my students", "Provide an alternative explanation for example when students are confused") in question TC199 were scaled into the index of "Teacher's self-efficacy in instructional settings". This scale was linked to the SEFFINS scale in PISA 2018. Each of the four items included in this scale had four response options ("Not at all", "To some extent", "Quite a bit", "A lot"). Note that this scale was skipped for teachers who reported that they taught mathematics to students in the national modal grade for 15-year-olds in the current school year on question TC217. Table 19.A.176 shows the item wording and item parameters for the items in this scale. It also indicates which items are trend items.

Teacher use of ICT (TCDIGRES)

Teachers’ frequency ratings of how often they use ICT for various teaching tasks (e.g., “Use <digital resources> to design tasks”, Use <digital resources> to provide feedback to students”) in question TC220 were scaled into the index of “Teacher use of ICT”. Each of the nine items included in this scale had five response options (“Never or almost never”, “About once or twice a year”, “About once or twice a month”, “About once or twice a week”, “Every day or almost every day”). Table 19.A.177 shows the item wording and item parameters for the items in this scale.

Emphasis on ICT competencies (ICTCOMP)

Teachers’ ratings of how much emphasis they place on teaching various ICT competencies (e.g., “Evaluating the credibility of digital information”, “Using digital tools to work collaboratively”) in question TC221 were scaled into the index of “Emphasis on ICT competencies”. Each of the five items included in this scale had four response options (“No emphasis”, “Little emphasis”, “Some emphasis”, “A lot of emphasis”). Table 19.A.178 shows the item wording and item parameters for the items in this scale.

Teaching of mathematical reasoning and 21st century mathematics topics (EXPO21TC)

Teachers’ frequency ratings of how often they had taught a range of different mathematics topics during the school year (e.g., “Extracting mathematical information from diagrams, graphs, or simulations”, “Using the concept of statistical variation to make a decision”) in question TC223 were scaled into the index of “Teaching of mathematical reasoning and 21st century mathematics topics”. Each of the 10 items included in this scale had four response options (“Frequently”, “Sometimes”, “Rarely”, “Never”). Note that this scale was skipped for teachers who reported that they did not teach mathematics to students in the national modal grade for 15-year-olds in the current school year on question TC217. Table 19.A.179 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Encouraging mathematical thinking (COGACMTC)

Teachers’ frequency ratings of how often they showed a range of behaviours indicative of encouraging mathematical thinking during the school year (e.g., “I encouraged students to "think mathematically".”, “I asked students how different topics are connected to a bigger mathematical idea.”) in question TC227 were scaled into the index of “Encouraging mathematical thinking”. Each of the nine items included in this scale had five response options (“Never or almost never”, “Less than half of the lessons”, “About half of the lessons”, “More than half of the lessons”, “Every lesson or almost every lesson”). Note that this scale was skipped for teachers who reported that they did not teach mathematics to students in the national modal grade for 15-year-olds in the current school year on question TC217. Table 19.A.180 shows the item wording and item parameters for the items in this scale.

Fostering reasoning (COGACRTC)

Teachers’ frequency ratings of how often they showed a range of behaviours indicative of fostering mathematics reasoning during the school year (e.g., “I asked students to explain their reasoning when solving a mathematics problem.”, “I asked students to defend their answer to a mathematics problem.”) in question TC228 were scaled into the index of “Fostering reasoning”. Each of the nine items included in this scale had five response options. (“Never or almost never”, “Less than half of the lessons”, “About half of the lessons”, “More than half of the lessons”, “Every lesson or almost every lesson”). Note that this scale was skipped for teachers who reported that they did not teach mathematics to students in the national modal grade for 15-year-olds in the current school year on question TC217. Table 19.A.181 shows the item wording and item parameters for the items in this scale.

Goals and views about teaching mathematics (TCMGOALS)

Teachers' agreement with statements about their views and goals when teaching mathematics (e.g., "Explaining why an answer is correct is just as important as getting a correct answer.", "Asking students to solve difficult problems in class helps them become good problem solvers.") in question TC230 were scaled into the index of "Goals and views about teaching mathematics". Each of the 11 items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Note that this scale was skipped for teachers who reported that they did not teach mathematics to students in the national modal grade for 15-year-olds in the current school year on question TC217. Table 19.A.182 shows the item wording and item parameters for the items in this scale.

Adaptation of instruction (ADAPTINSTR)

Teachers' frequency ratings of how often they adapt instruction for students (e.g., "I tailor my teaching to meet the needs of my students.", "I provide individual support for advanced students.") in question TC232 were scaled into the index of "Adaptation of instruction". Each of the four items included in this scale had four response options ("Never or almost never", "Some lessons", "Many lessons", "Every lesson or almost every lesson"). Note that this scale was skipped for teachers who reported that they taught mathematics to students in the national modal grade for 15-year-olds in the current school year on question TC217. Table 19.A.183 shows the item wording and item parameters for the items in this scale.

Feedback provided by the teachers (FEEDBINSTR)

Teachers' frequency ratings of how often they provide feedback to students (e.g., "I give students feedback on their strengths in my course.", "I tell students how they can improve their performance.") in question TC232 were scaled into the index of "Feedback provided by the teachers". Each of the five items included in this scale had four response options ("Never or almost never", "Some lessons", "Many lessons", "Every lesson or almost every lesson"). Note that this scale was skipped for teachers who reported that they taught mathematics to students in the national modal grade for 15-year-olds in the current school year on question TC217. Table 19.A.184 shows the item wording and item parameters for the items in this scale.

Openness to creativity (OPENCTTC)

Teachers' ratings of their agreement with statements about their openness to creative activities (e.g., "I enjoy projects that require creative solutions.", "I express myself through art.") in question TC234 were scaled into the index of "Openness to creativity". Each of the eight items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.185 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Creative values (CREATVAL)

Teachers' ratings of their agreement with statements about their values regarding creativity (e.g., "It is important that students are able to make creative works like drawing and painting.", "It is important for students to solve science problems creatively.") in question TC235 were scaled into the index of "Creative values". Each of the six items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.186 shows the item wording and item parameters for the items in this scale.

Teachers' use of creative pedagogies (CREATPED)

Teachers' ratings of how much importance they place on using creative pedagogies in class (e.g., "Finding ideas through brainstorming", "Debating ideas or current issues") in question TC236 were scaled into the index of "Teachers' use of creative pedagogies". Each of the seven items included in this scale had four response options ("No importance", "Very little importance", "Some importance", "A lot of importance"). Table 19.A.187 shows the item wording and item parameters for the items in this scale.

Teachers' capacity to concentrate at work (CAPCON)

Teachers' frequency ratings of how often they experienced various situations during the school day (e.g., "I was distracted.", "I felt focused.") in question TC237 were scaled into the index of "Teachers' capacity to concentrate at work". Each of the six items included in this scale had four response options ("Never", "Seldom", "Often", "Always"). Table 19.A.188 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Teachers' affect (AFFECT)

Teachers' frequency ratings of how often they felt various emotions during the school day (e.g., "I felt cheerful and in good spirits.", "I felt active and vigorous.") in question TC238 were scaled into the index of "Teachers' affect". Each of the five items included in this scale had four response options ("Never", "Seldom", "Often", "Always"). Table 19.A.189 shows the item wording and item parameters for the items in this scale.

Teachers' feeling of trust (TRUST)

Teachers' ratings of their agreement with statements about a climate of trust within the school (e.g., "Teachers can rely on the school's management for professional support.", "I feel that I can trust my colleagues.") in question TC241 were scaled into the index of "Teachers' feeling of trust". Each of the five items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.190 shows the item wording and item parameters for the items in this scale.

Teachers' work overload (OVERLOAD)

Teachers' ratings of their agreement with statements concerning work overload (e.g., "I am given enough time to do what is expected of me at work.", "I have too much work for one person to do.") in question TC243 were scaled into the index of "Teachers' work overload". Each of the six items included in this scale had four response options ("Strongly disagree", "Disagree", "Agree", "Strongly agree"). Table 19.A.191 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Teachers' work autonomy (AUTONOMY)

Teachers' ratings of how much control they have over various decisions at their school (e.g., "Determining course content", "Disciplining students") in question TC246 were scaled into the index of "Teachers' work autonomy". Each of the seven items included in this scale had four response options ("No control", "Some control", "A lot of control", "Full control"). Table 19.A.192 shows the item wording and item parameters for the items in this scale.

School leadership (LEADSHIP)

Teachers' frequency ratings of how often their school's principal took various actions (e.g., "My principal collaborated with teachers to solve classroom discipline problems.", "My principal observed instruction in the classroom.") in question TC253 were scaled into the index of "School leadership". Each of the seven items included in this scale had four response options ("Never or rarely", "Sometimes", "Often", "Very often"). Table 19.A.193 shows the item wording and item parameters for the items in this scale.

Occupational stress (OCSTRESS)

Teachers' ratings of their agreement with various statements regarding their stress at work (e.g., "I experience stress in my work.", "My job negatively impacts my mental health.") in question TC254 were scaled into the index of "Occupational stress". Each of the four items included in this scale had four response options ("Not at all", "To some extent", "Quite a bit", "A lot"). Table 19.A.194 shows the item wording and item parameters for the items in this scale. It also indicates which items were reverse-coded prior to scaling.

Sources of stress (STRESS)

Teachers' ratings of their agreement with various situations causing stress at work (e.g., "Having too many lessons to teach", "Being held responsible for students' achievement") in question TC255 were scaled into the index of "Sources of stress". Each of the nine items included in this scale had four response options ("Not at all", "To some extent", "Quite a bit", "A lot"). Table 19.A.195 shows the item wording and item parameters for the items in this scale.

Negative physical symptoms (NEGSYMPT)

Teachers' frequency ratings of how often they had various psychosomatic symptoms during the school day (e.g., "Headache", "Fatigue") in question TC256 were scaled into the index of "Negative physical symptoms". Each of the 10 items included in this scale had five response options ("Never or almost never", "About once or twice a year", "About once or twice a month", "About once or twice a week", "Every day or almost every day"). Table 19.A.196 shows the item wording and item parameters for the items in this scale.

References

Avvisati, F. (2020), "The measure of socio-economic status in PISA: A review and some suggested improvements", *Large-Scale Assessments in Education*, Vol. 8/1, pp. 1-37, https://doi.org/10.1186/s40536-020-00086-x. [14]

Bertling, J. and J. Weeks (2020), *Getting More Bang for Your Buck: Within-construct Questionnaire Matrix Sampling*, Paper presented to PISA Technical Advisory Group, September 2020, Princeton, NJ. [3]

Bertling, J. and J. Weeks (2018), *Plans for Within-construct Questionnaire Matrix Sampling in PISA 2021*, Paper presented to PISA Technical Advisory Group, August 2018, Princeton, NJ. [2]

Bertling, J. et al. (2020), "Comparison of within-construct matrix sampling with scale shortening: Impact on IRT scaling and population modelling"*, Memo prepared for PISA Technical Advisory Group, TAG (2020)2*. [4]

Birnbaum, A. (1968), "Some latent trait models and their use in inferring an examinee's ability", in Lord, F. and M. Novick (eds.), *Statistical Theories of Mental Test Scores*, Addison-Wesley. [5]

Cowan, C. et al. (2012), *Improving the measurement of socioeconomic status for the National Assessment of Educational Progress: A theoretical foundation*, National Center for Education Statistics, https://nces.ed.gov/nationsreportcard/pdf/researchcenter/socioeconomic_factors.pdf. [13]

Ganzeboom, H. and D. Treiman (2003), "Three internationally standardised measures for comparative research on occupational status", in Hoffmeyer-Zlotnik, J. and C. Wolf (eds.), *Advances in Cross-National Comparison: A European Working Book for Demographic and Socio-Economic Variables*, Kluwer Academic Press. [12]

Masters, G. (1982), "A Rasch model for partial credit scoring", *Psychometrika*, Vol. 47/2, pp. 149-174, https://doi.org/10.1007/BF02296272. [10]

Muraki, E. (1992), "A generalized partial credit model: Application of an EM algorithm", *Applied Psychological Measurement*, Vol. 16/2, pp. 159-177, https://doi.org/10.1002/j.2333-8504.1992.tb01436.x. [6]

OECD (2023), *PISA 2022 Assessment and Analytical Framework*, PISA, OECD Publishing, Paris, https://doi.org/10.1787/dfe0bf9c-en. [1]

OECD (2020), *PISA 2018 Technical Report*, PISA, OECD Publishing, Paris, https://www.oecd.org/en/about/programmes/pisa/pisa-data.html. [15]

OECD/Eurostat/UNESCO Institute for Statistics (2015), *ISCED 2011 Operational Manual: Guidelines for Classifying National Education Programmes and Related Qualifications*, OECD Publishing, Paris, https://doi.org/10.1787/9789264228368-en. [16]

Rasch, G. (1960), *Probabilistic Models for Some Intelligence and Attainment Tests*, Nielsen and Lydiche. [9]

Shin, H. et al. (2017), *Multidimensional Discrete Latent Trait Models (mdltm) [Draft manual]*, Educational Testing Service. [7]

von Davier, M. (2015), *Multidimensional discrete latent trait models (mdltm) (Version 1.965) [Computer software]*. [8]

Warm, T. (1989), "Weighted likelihood estimation of ability in item response theory", *Psychometrika*, Vol. 54/3, pp. 427-450, https://doi.org/10.1007/BF02294627. [11]

Notes

1. For the seven trend scales linked to PISA 2012, the Rasch model (Rasch, 1960) was used to scale the dichotomous items, while the partial credit model (PCM) was used to scale the polytomous items, in line with the models used in PISA 2012.

2. The International Standard Classification of Education (ISCED) is used in international educational statistics to classify levels in education systems worldwide. A link to the 2011 framework, ISCED 2011, used in PISA 2022 can be found at

https://uis.unesco.org/sites/default/files/documents/international-standard-classification-of-education-isced-2011-en.pdf

3. Separate simple DVs (ST250D06JA, ST250D07JA, ST251D08JA, ST251D09JA) combine all the customizations across countries/economies to ST250C06JA, ST250C07JA, ST251C08JA, and ST251C09JA, respectively. See Annex D.

4. A separate simple DV (ST330D10WA) combines all the customizations across countries/economies to ST330C10WA. See Annex D.

5. The random value was drawn from a normal distribution with a mean of 0 and a standard deviation equal to the standard deviation of the residuals of the regression model for the country/economy.

6. Due to missing data within the OECD countries, the senate weights of the individuals that were ultimately included in the calculation of the mean and standard deviation of each component did not sum to 5,000 for each OECD country. Therefore, the OECD countries were only approximately equally weighted in the calculation of the mean and standard deviation of each component.

7. Again, due to missing data within the OECD countries, the senate weights of the individuals that were ultimately included in the calculation of the mean and standard deviation of the preliminary ESCS scores did not sum to 5,000 for each OECD country. Therefore, the OECD countries were only approximately equally weighted in the calculation of the mean and standard deviation of the preliminary ESCS scores.

8. Due to missing data within the OECD countries, the senate weights of the individuals that were ultimately included in the calculation of the mean and standard deviation of each component in PISA 2018 did not sum to 5,000 for each OECD country. Therefore, the OECD countries were only approximately equally weighted in the calculation of the mean and standard deviation of each component in PISA 2018.

9. The random value was drawn from a normal distribution with a mean of 0 and a standard deviation equal to the standard deviation of the residuals of the regression model for the country/economy and cycle.

Annex 19.A. Methodology and Overview of Derived Variables in PISA 2022 Context Questionnaires

Annex Table 19.A.1. Chapter 19: Comprehensive Scales and Variables Analysis from PISA 2022 Questionnaires

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StatLink https://stat.link/v6uq1n

20 Questionnaire Design and the Computer-Based Questionnaire Platform

Introduction

Questionnaires are a critical component of the PISA survey, providing important information about the context in which students learn and live as well as demographics and other reporting information. When coupled with the cognitive results, the questionnaires can provide insights into relationships between background information and cognitive achievement. The mode of administration of the questionnaires has evolved across the PISA cycles. PISA administered questionnaires in paper-based format alone until PISA 2012 when an optional online School Questionnaire was introduced. Computer-based questionnaires were implemented more broadly in PISA 2015 and expanded in PISA 2018 to be the primary mode of questionnaire administration. In PISA 2022, computer-based questionnaires continued to be the primary mode of administration and all questionnaires except the Parent Questionnaire were available in computer-based format. The use of a computer-based questionnaire administration allowed for innovations to be introduced in the PISA 2022 cycle, as well as continuing to increase the data quality over PISA 2018.

Annex Table 20.A.2 shows the compulsory and optional questionnaires that were administered in PISA 2022 and their administration mode.

This chapter first explains the PISA 2022 questionnaire design for the Field Trial and the Main Survey, then provides an overview of the process used to author the international master and the national questionnaires, and finally provides an overview of the technical design of the questionnaire platform.

Questionnaire Design

PISA emphasizes the importance of collecting context information from students and schools along with the assessment of student achievement. A Student Questionnaire (STQ) and a School Questionnaire (SCQ) cover a broad range of contextual variables. The content of these questionnaires – especially the content of the Student Questionnaire – changes considerably between cycles based on the major domain of the assessment, but the administration has remained stable: every student participating in the PISA assessment completes the STQ, and every school principal of the participating schools, one per school, completes the SCQ.

PISA has also included several international questionnaire options, i.e. additional instruments that countries could administer as an international option. For PISA 2022, it included a Parent Questionnaire (PAQ), Teacher Questionnaire (TQ), and optional questionnaires for the students: the Financial Literacy Questionnaire (FLQ), ICT Familiarity Questionnaire (ICQ), and Well-Being Questionnaire (WBQ). Annex

Web Table 20.A.5. summarises the participation of countries/economies in the different international questionnaires.

The context questionnaires contribute to integral aspects of the analytical power of PISA as well as to its capacity for innovation. Therefore, the questionnaire design must meet high methodological standards, allowing for the collection of data that leads to reliable, precise and unbiased estimations of parameters for each participating country. In addition, the design also must ensure that important policy issues and research questions can be addressed in later analysis and reporting based on PISA 2022 data. Both the psychometric quality of the variables and indicators and the analytical power of the study must be considered when proposing and evaluating a questionnaire design. This is usually done by pre-testing all questionnaire content and innovations in the Field Trial one year prior to the Main Survey assessment. Accordingly, more material is tested in the Field Trial than will be implemented later in the Main Survey. Results are then discussed with the PISA expert groups and Main Survey material is selected.

The Field Trial and the Main Study questionnaire designs differ greatly in many respects. The goal of the PISA 2022 Field Trial questionnaires is to re-evaluate the quality of the context questionnaire items used in previous cycles as well as the quality of new items developed for this cycle and test methodological innovations, namely the within-construct matrix sampling of items. Moreover, the PISA 2022 Field Trial provided an opportunity for countries to test the questionnaire administration procedures. The main survey questionnaires must collect equivalent information across all students, schools, teachers, and parents to be used in reporting results, and thus the design needs to administer the same questions to all respondents.

The following sections discuss the main differences between the PISA 2022 Field Trial and the Main Survey design for both the paper-based and computer-based questionnaires.

Student-Administered Questionnaires

In the field trial, approximately 1 992 students per country respond to the computer-based questionnaires and approximately 900 students per country respond to the paper-based questionnaires. These numbers were increased in the main survey to approximately 6 300 students per computer-based country and 5 250 students per paper-based country. In addition, countries/economies opting to administer the Financial Literacy assessment sampled an additional 1 650 students. Because of the differences in the tools available for authoring a computer-based questionnaire versus a paper-based questionnaire, the design of the computer-based and paper-based questionnaires administered to students were different.

Computer-based design

Field trial

The field trial Student Questionnaire design for PISA 2022 allowed for the maximum number of potential items to be tested, experiments on different wording and formats of similar questions, and the piloting of the new within-construct matrix sampling methodology. To accomplish these goals, the Student Questionnaire design included two overlapping virtual booklets, each of which included a subset of items with different wording for version comparison experiments that were administered to only half the students taking that booklet. This resulted in four major paths through the questionnaire: Booklets 1a, 1b, 2a, and 2b. The field trial Student Questionnaire design is shown in Table 20.1. and describes the constructs included in each of the booklets and experiments. The field trial Student Questionnaire was authored as one form and each student had an equal chance of being assigned to one of the four paths.

Table 20.1. Field trial computer-based design for Student Questionnaire

<table>
<tr><th colspan="4">Student Questionnaire</th></tr>
<tr><td colspan="4">Common items: basic demographics (age, gender), educational career, migration and language exposure, ESCS (home possessions, guardians)</td></tr>
<tr><td colspan="2">Booklet 1:
ESCS: Parental education and occupation (parent/guardian versions)</td><td colspan="2">Booklet 2:
ESCS: Parental education and occupation
(trend mother/father versions)</td></tr>
<tr><td colspan="4">Common items: ESCS (food insecurity and subjective socioeconomic status)</td></tr>
<tr><td colspan="2">Booklet 1:
• Educational career
• School culture and climate
• Out-of-school experiences
• Parent/guardian involvement and support
• Social and emotional characteristics (vignettes)
• Health and well-being
• Postsecondary preparedness and aspirations
• Future aspirations</td><td colspan="2">Booklet 2:
• Migration and language exposure
• Organisation of student learning at school
• Mathematics teacher behaviours
• Exposure to mathematics content
• School culture and climate
• Subject-specific beliefs, attitudes, feelings, and behaviours
• Out-of-school experiences
• Organisation of student learning at school
• Creative Thinking</td></tr>
<tr><td>Booklet 1a:
Social and Emotional Characteristics (agreement)</td><td>Booklet 1b:
Social and Emotional Characteristics (frequency)</td><td>Booklet 2a:
1. Mathematics teacher behaviours (version A)
2. Exposure to mathematics content (version A)
3. Subject-specific beliefs, attitudes, feelings, and behaviours (version A)</td><td>Booklet 2b:
a) Mathematics teacher behaviours (version B)
b) Exposure to mathematics content (version B)
c) Subject-specific beliefs, attitudes, feelings, and behaviours (version B)</td></tr>
<tr><td colspan="4">Common items: Future aspirations, Global crises, PISA preparation and effort</td></tr>
<tr><th colspan="4">Financial Literacy Questionnaire</th></tr>
<tr><th colspan="4">ICT Familiarity Questionnaire</th></tr>
<tr><th colspan="4">WBQ</th></tr>
</table>

The PISA 2022 field trial Student Questionnaire also tested the implementation of the within-construct matrix sampling for the first time in PISA. Certain PISA constructs are measured using 6 to 12 individual items. However, to decrease the amount necessary for responding to ach instrument, each student is administered only 5 of the items from a construct. To this effect, each student received a random selection of the items in a construct, so students received different combinations of the items from the construct. The computer platform used a combination of the student questionnaire random number and the position of the question screen within the questionnaire form to determine which items to show so that students were shown items in different positions on each screen (e.g. a student did not always see items 1, 2, 4, 5, and 7 on every matrix-sampled screen). The master version of the field trial questionnaire identified all the questions for which the within-construct matrix sampling would be applied.

The optional questionnaires for students: Financial Literacy (FLQ), ICT Familiarity (ICQ), and Well-being (WBQ) were administered following the Student Questionnaire and were available only as computer-based instruments. These optional questionnaires each consisted of a single form with no virtual booklets or version comparisons. The field trial version of the FLQ and ICQ contained more content than needed for the main study in order to evaluate the quality of new items. The WBQ was administered unchanged from PISA 2018. Within-construct matrix sampling was not applied to any questions in the optional questionnaires.

The computer-based Student Questionnaire Une Heure (STQ-UH) booklet consisted of a subset of questions from the Student Questionnaire designed to take approximately 20 minutes for a student to complete. If students received the STQ-UH, they did not receive any of the other optional questionnaires even if their country had elected to participate in those options.

Main survey

For the main survey, the number of items in the Student Questionnaire was reduced significantly as decisions were made about which version of each of the experimental wording questions collected the highest quality data and which of the other new questions collected the highest quality data on other constructs. The main survey Student Questionnaire consisted of one booklet of items that was administered to all students. In creating the main survey questionnaire, the sequence of items was updated because items from two virtual booklets were combined into one virtual booklet. Within-construct matrix sampling was still applied to those questions where the field trial analysis showed that high-quality scales could be constructed from the matrix-sampled items.

The design for the optional Financial Literacy, ICT Familiarity, Well-being, and Parent Questionnaires remained the same as in the Field Trial with one form per questionnaire and the number of items administered in the FLQ and ICQ was reduced slightly to eliminate items that were deemed to not function well, and to reduce the time needed to complete each questionnaire.

Paper-based design

Field trial

A paper-based Student Questionnaire was administered in the four countries that chose the paper-based mode of delivery for both the questionnaires and the cognitive assessment. The paper-based Student Questionnaire took up to 41 minutes of assessment time and included a subset of the items from the field trial computer-based version. The paper-based version did not include questions on the creative thinking module, eliminated some of the version comparison experiments, and had a reduced item pool for a few modules. The field trial paper-based Student Questionnaire was administered in two overlapping booklets as shown in Table 20.2. Students were randomly assigned to one of the two booklets during the survey administration. Within-construct matrix sampling could not be applied to the paper-based version, so students taking the questionnaire on paper received all items in a question instead of a random subset. This did not increase the response time since the paper-based version had fewer items than the computer-based version.

Where possible, questions were administered in the same format and layout in both the paper-based and computer-based questionnaires. However, some questions had to be changed from drop-down or slider response format to open-ended format to accommodate data collection in the PBA mode.

Table 20.2. Field Trial Paper-based Design for Students

<table>
<tr><th colspan="2">Paper-based Student Questionnaire</th></tr>
<tr><td colspan="2">Common items:
Basic demographics (grade, age, gender)
Educational career
Migration and language exposure
Organisation of student learning at school
ESCS: home possessions</td></tr>
<tr><td>Booklet 1:
ESCS: Parental education and occupation (new parent/guardian versions)
Educational career
Mathematics teacher behaviours (version A)</td><td>Booklet 2:
ESCS: Parental education and occupation (trend mother/father versions)
ESCS: Food insecurity and subjective socioeconomic status
Migration and language exposure</td></tr>
</table>

• School culture and climate • Exposure to mathematics content • Subject-specific beliefs, attitudes, feelings, and behaviours • Participation in additional mathematics instruction • Social and emotional characteristics (agreement) • Social and emotional characteristics (vignettes)	• Educational Career • School Culture and Climate • Mathematics teacher behaviours (version B) • Out-of-school experiences • Parental involvement and support • Social and emotional characteristics (frequency) • Health and well-being • Post-secondary preparedness and aspirations
Common items: Global crises, PISA preparation and effort	

International option questionnaires were not available to paper-based countries, so the FLQ, ICQ, and WBQ were not administered to these participants. In addition, no paper-based country chose to administer the STQ-UH questionnaire.

Main Survey

For the main survey, items that were removed from the field trial version of the computer-based student questionnaire were also removed from the paper-based version. The paper-based student questionnaire content was combined into one 35-minute booklet that was administered to all students. When combining the questions from multiple booklets, the sequence was questions was updated as well.

School Questionnaire

Each school that participated in PISA completed one School Questionnaire to provide contextual information on the environment in which students learn. Since the school questionnaire was answered by approximately 28 schools in the Field Trial, there was not a large enough sample size to administer two booklets of material in the field trial and still evaluate the quality of the items in each country. Therefore, the design of the school questionnaire remained consistent between the field trial and the main survey for both computer-based and paper-based administration, with slightly more material administered in the field trial than in the main survey.

Field Trial

The School Questionnaire in the Field Trial included trend and new material and took approximately 60 minutes to complete. This questionnaire was designed as one form without virtual booklets or version comparisons, and the same questions were administered in both the paper-based and computer-based versions. Data quality was improved in the computer-based version by using automated checks and routing. Certain questions in the computer-based version contained automated range limits that would not permit unrealistic values to be entered and soft checks to encourage the respondent to confirm responses that were higher or lower than expected. In the computer-based version, when filter questions were implemented, routing rules could be used to automatically hide skipped questions from respondents. No automatic range checks are possible in the paper-based version, but rather these were implemented during data entry. Also, in the paper-based version, respondents saw printed instructions to skip certain questions based on their response to a filter question. The computer-based questionnaire also contained four questions where data was collected using sliders instead of the open-response format used in the paper-based questionnaire.

Main Survey

For the Main Survey, the number of questions administered in the School Questionnaire was reduced to 45 minutes of material. The same questions were administered to both the computer-based and paper-based administration countries. The same technical differences between the computer and paper-based administration were present for the main survey. As the main survey collects information from a larger sample than in the field trail, there are approximately 150 respondents per country to the School Questionnaire.

Teacher Questionnaire

The Teacher Questionnaire is administered to up to 25 respondents per school. The sampling process attempts to have an equal number of teachers of the major PISA domain (mathematics) and teachers of other subject areas. Additional information about the sampling of teachers can be found in Chapter 6. Certain items in the questionnaire are domain-specific and designed to collect information only from teachers of mathematics, and other questions are general and may be answered by all teachers of 15-year-olds. In the PISA 2018 cycle, the Teacher Questionnaire was designed as two independent forms (one for teachers of reading as a subject, and one for teachers of all other subject areas), and teachers were instructed to log into the version of the questionnaire that applied to them. Once a teacher logged into one form of the questionnaire, they were not able to switch to the other form, and so if a teacher was erroneously classified as a major-domain teacher then they would be asked to respond to questions that did not apply to them. This design was changed for the PISA 2022 cycle, and in this cycle the Teacher Questionnaire was administered as one form and teachers were asked to self-identify as teachers of mathematics or teachers of other subjects and questions were presented based on those responses. If a teacher erroneously marked that they were a teacher of mathematics, they could go back and change their answer and then route to the appropriate questions for their subject area.

Field Trial

The optional computer-based Teacher Questionnaire was designed as one form containing two overlapping virtual booklets of questions and took approximately 60 minutes to complete in the Field Trial. All teachers first answered a set of common questions about their background, including whether they are a maths teacher or not. Then teachers were routed to one of two blocks of questions: maths-specific questions for teachers who self-identified as teachers of mathematics or general questions for teachers who self-identified as not teaching mathematics. Table 20.3. shows the Teacher Questionnaire Field Trial design.

Table 20.3. Field trial computer-based design for Teacher Questionnaires (TCQ)

<table>
<tr><th colspan="2">Teacher Questionnaire</th></tr>
<tr><td colspan="2">Common items:
• Socio-demographic characteristics
• Education, Certification, Teacher qualifications
• Employment status, years of experience, grade assigned
• Workload
• Alignment of roles, content of teacher preparation
• Support for learning and development
• Self-reported impact of professional development on teaching practices
• Use of specific ICT applications
• Assessments, evaluation and feedback
• Emphasis on ICT competencies
• Job satisfaction</td></tr>
<tr><td>Administered to self-identified mathematics teachers:
• Teacher qualifications
• Mathematics curriculum
• Disciplinary climate in mathematics
• Use of digital devices in mathematics
• Exposure to formal and applied mathematics tasks
• Exposure to mathematics problems requiring reasoning
• Structure of mathematics instruction
• Cognitive activation in mathematics
• Mathematics teacher feedback
• Goals and views about teaching mathematics
• Support for learning and development</td><td>Administered to self-identified non-mathematics teachers:
• Teacher qualifications
• Self-efficacy
• Students' engagement with their learning
• Opportunity to learn
• Attitudes toward immigrants
• Teacher background – studying experience abroad</td></tr>
<tr><td colspan="2">Common items:
• Creative thinking
• Teacher well-being
• Classroom composition</td></tr>
</table>

Main Survey

The main survey Teacher Questionnaire design was unchanged from the field trial: it consisted of common questions administered to all teachers as well as distinct blocks of questions that were administered to either mathematics teachers or teachers of other subjects. The questionnaires in total still covered all policy modules proposed in the questionnaire framework for this cycle (see Chapter 5). However, the number of questions administered was reduced to eliminate those questions that had lower data quality and/or were not as critical for the purpose of PISA so that the total main survey questionnaire response time was approximately 45 minutes.

Parent Questionnaire

The optional Parent Questionnaire (PAQ) was administered on paper only. Only one Parent Questionnaire was completed for each student, and the questionnaire could be completed by either parent or the student's guardian. The PAQ included trend items as well as newly developed content. The field trial version of the Parent Questionnaire contained approximately 35 minutes of material so that new content could be considered for inclusion in PISA 2022. The main survey version of the Parent Questionnaire contained approximately 30 minutes of material.

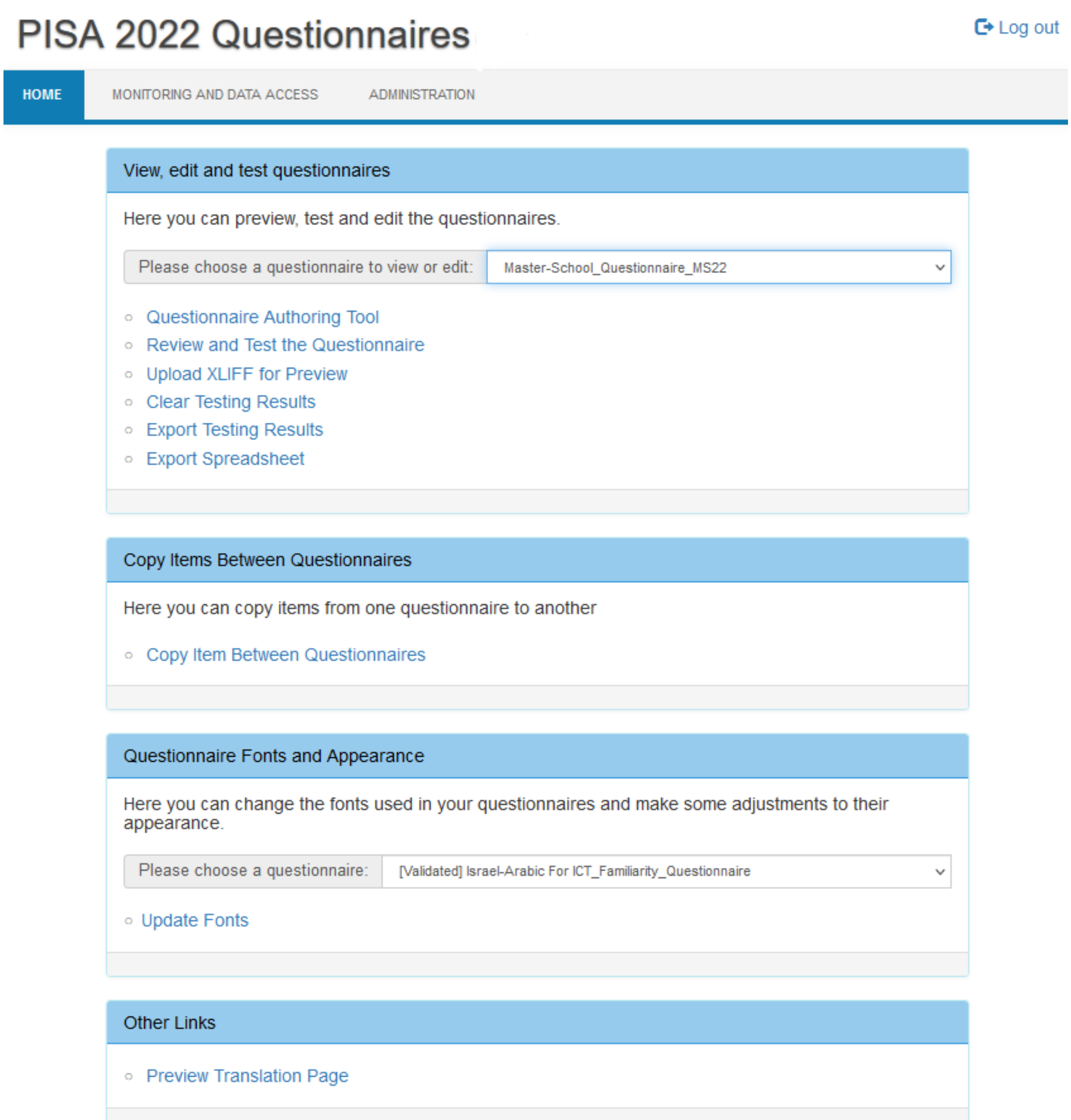
Computer-based Questionnaire Plaftorm

The computer-based questionnaires were designed and administered using the PISA Questionnaire Authoring Tool (QAT), a platform focused on the specific goal of production (i.e. the definition, authoring, testing, translation, adaptation, and validation) of the Master and National versions of the questionnaires, the delivery of these questionnaires to the appropriate respondents, and the management of all administrative tasks relating to questionnaire delivery.

The QAT editor is used to create the questions, routing logic, and consistency checks used in the computer-based questionnaires of PISA 2022. It is an online editor that allows administrators to create a profile for each questionnaire for each country, and then allows users to add, delete, or edit a questions and routings within those questionnaires. Users edit the content and question format of items in the questionnaire in the Editor, then that structure is transformed by the platform into the formatted screens presented to users (the runtime version), and finally the translation of the text is integrated into the runtime to show the questions in each national language.

When users log into the QAT, they are taken to the home page shown in Figure 20.1. This page gives users access to the many features of the tool.

Figure 20.1. Questionnaire Authoring Tool home page



QAT Questionnaire Editing Features

When users open the QAT Editor, they are presented with a view of the structure of an entire questionnaire. It is important to note, though, that this is not the view that respondents will see during the Field Trial and Main Study phases of PISA 2022, this tool is used to define the elements of the questions that will be displayed to respondents through a runtime. Figure 20.2 and Figure 20.3 show the main view of the QAT for a National Project Manager (NPM).

Figure 20.2. Questionnaire Authoring Tool: Main View (with a specific question example)

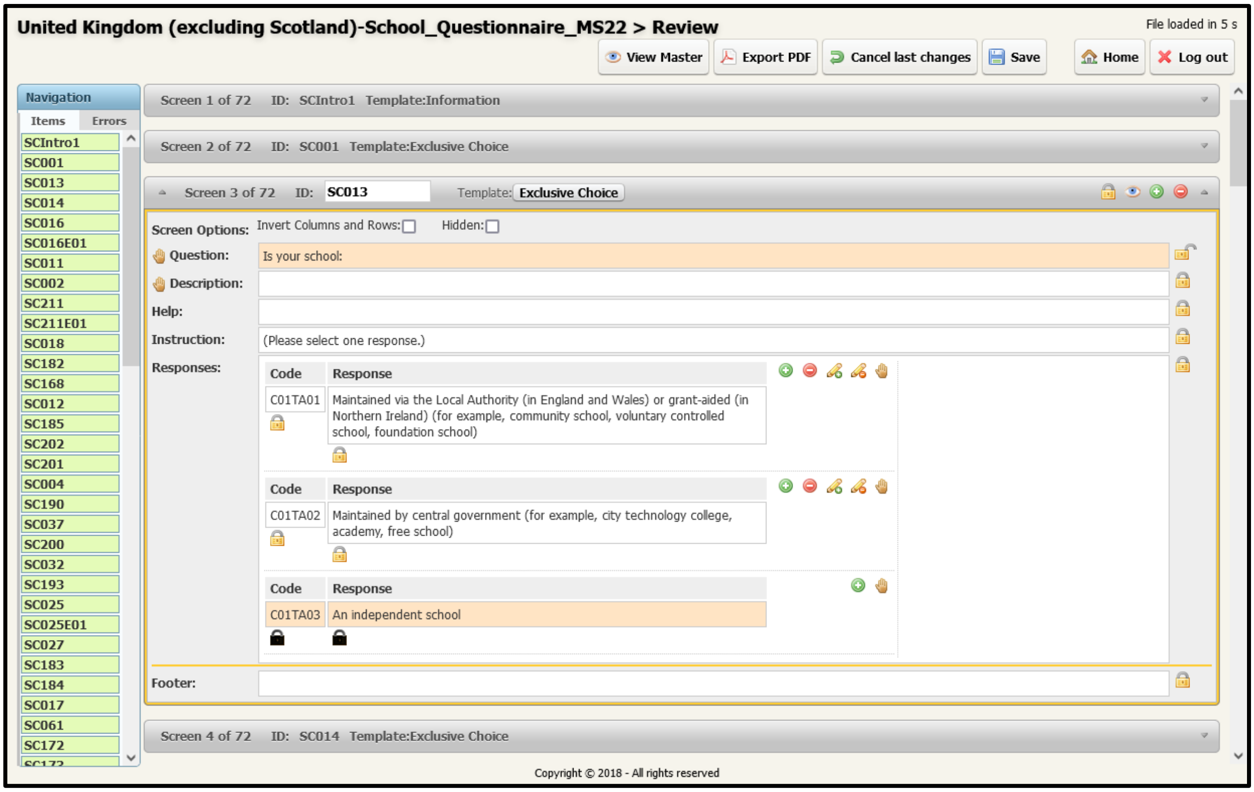
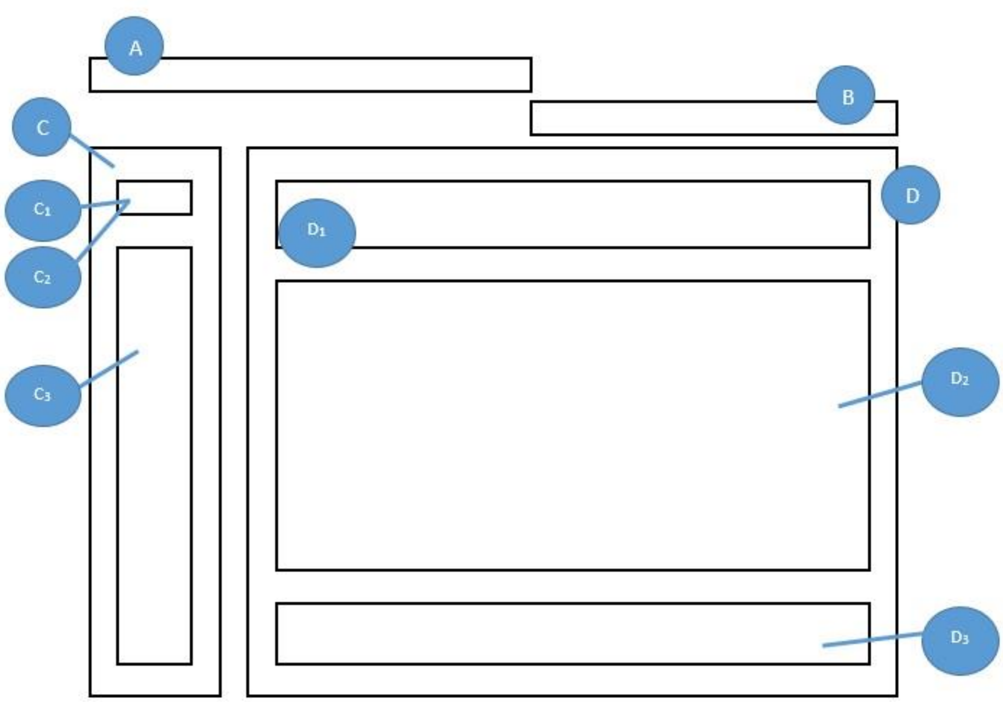


Figure 20.3. Questionnaire Authoring Tool: Organisation of Main View



The organisation of the main view is the following:

Panel A: The Questionnaire Title contains the questionnaire label (country and type of questionnaire) and the questionnaire mode (i.e. the modes of the QAT are important to note as they define the rights of a current user. Depending on the mode, the access for modifying questionnaires in the QAT editor is locked or unlocked, allowing users to work independently).

Panel B. The Questionnaire Toolbar provides the following options:

- View Master – Opens the Master English version of the current questionnaire.
- Export PDF – Generates a PDF file of the current version of the questionnaire.
- Cancel Last Changes – Undoes any changes since the last time the user has saved their work on the questionnaire.
- Save – Saves the questionnaire to the database. When clicked, this action also provides a check for whether routing rules and consistency checks are correctly formatted, in the questionnaire. If the test fails, the user will receive a notification that there are currently errors in the questionnaire.
- Home – Redirects the user to the QAT homepage.
- Log Out – Disconnects the user from the QAT platform.

Panel C: The Navigation Panel lists contains the following elements:

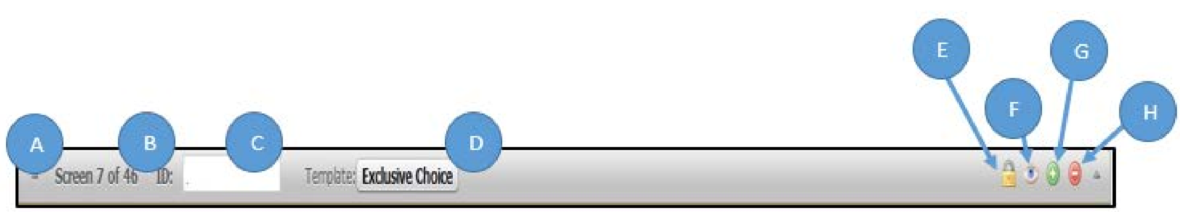
- The questionnaire items (C_1) or
- A list of errors currently present in the questionnaire (C_2)
- Quick access to questionnaire screens (C_3)

Panel D: The QAT Editor displays the list of all questions (referred to as “screens”) and rules (referred to as “rules headers”) available for a questionnaire. When clicked, each part will expand or collapse a specific screen or rule window (D, D_1, D_2, D_3).

Questions Expanded View

When a specific screen is expanded in the QAT editor, additional features are available. Figure 20.4 shows the Expanded View information. Inside the expanded view, the user can edit the different parts of a questionnaire screen using the QAT editor: the question text, description, instruction, help, and response categories/options.

Figure 20.4. The expanded view information



The features available for users in the questionnaire screens include:

A. Show/Hide Screen button can expand or collapse a specific questionnaire screen or rule header.

B. Screen Number label shows the location of the screen in the sequence of questionnaire items out of the total number of questionnaire screens.

C. Screen ID displays the technical identifier of the screen and rule headers (i.e. SC025).

D. Template label displays the name of the template used for editing the questionnaire screen (see section about questionnaire templates for additional information).

E. Lock/Unlock button makes a questionnaire screen editable or not for a National Project Manager. This button is not available to NPMs.

F. Preview button opens a preview of an item, giving the user a view of how the question stem, response options, helps and instructions will be displayed.

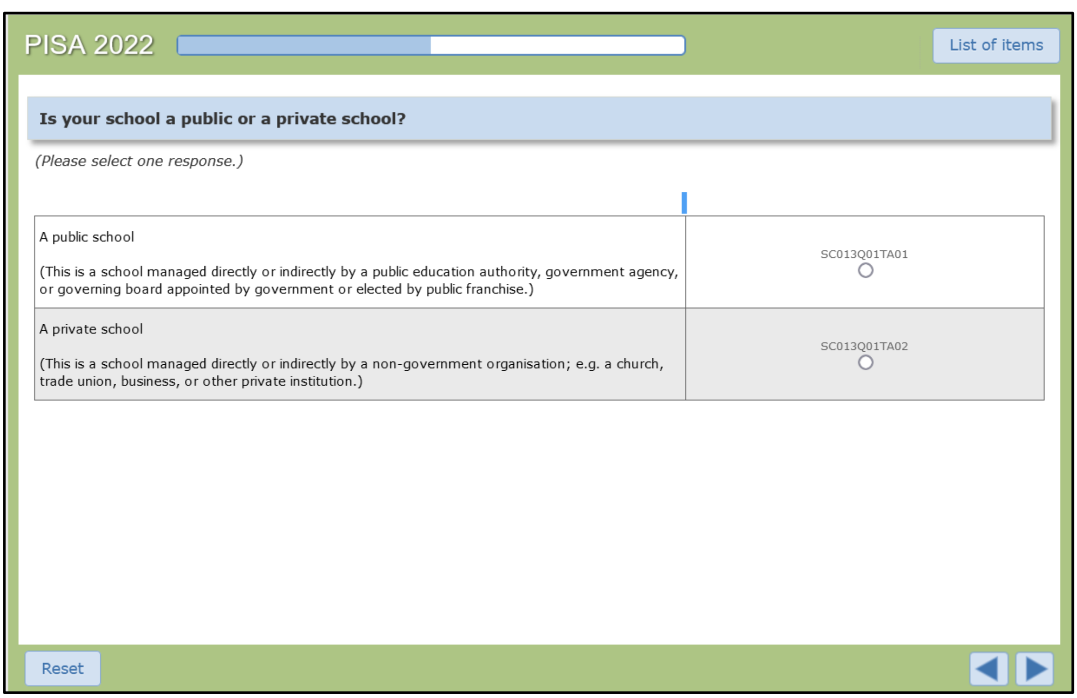
G. Add Screen button inserts a new question or rule in the questionnaire just below the currently expanded item.

H. Delete Screen button will remove the question or rule from the questionnaire. Users who click this button will first receive a notification asking for confirmation of deletion.

Previewing Questionnaire Items

The questionnaire platform offers three preview options for reviewing and checking the quality of the questionnaires. The first option is a question preview panel that can be accessed from within the QAT Editor using the Preview button available in the expanded view of each question. In this preview mode users see only the screen for the individual item selected *with the English source text*. This preview tool is helpful for reviewing the general layout of the question and the IDs for each response field to better understand how data will be labelled. This preview tool is shown in Figure 20.5.

Figure 20.5. Preview of a question in the QAT



The second option is to preview the full national adapted questionnaire in the English source version using the “Review and Test the Questionnaire” link on the QAT homepage. This option lets users navigate through the entire questionnaire in a test environment to confirm the agreed-upon adaptations and routing are working appropriately before beginning translation.

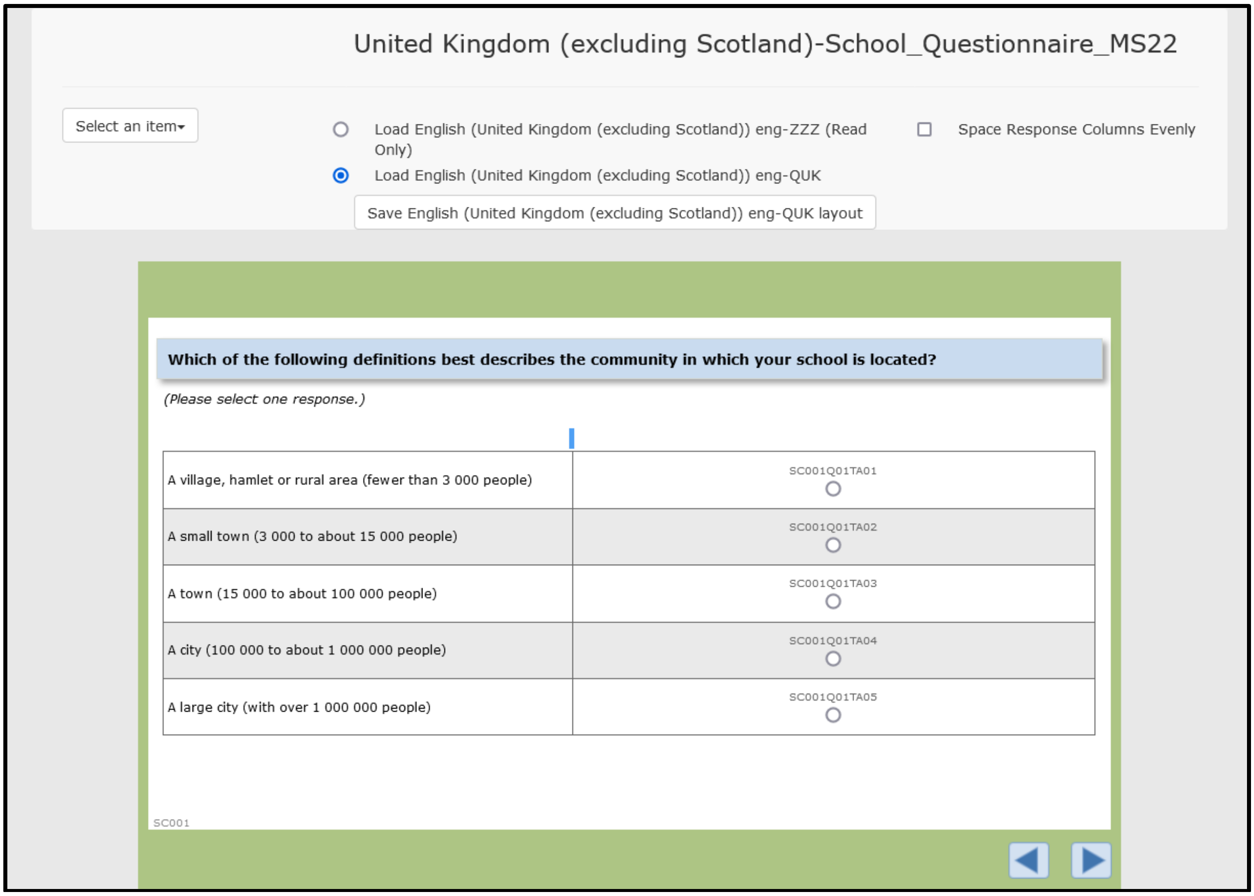
The third option is to preview the national questionnaire in the language of administration. This tool integrates the translation file that the country has worked on separately into the questionnaire runtime. This allows users to see the questionnaire as it will be administered to students, school administrators, or teachers. Users access this preview by clicking "*Upload XLIFF for Preview*" from the home page of the QAT and then upload their translation file (XLIFF format) as shown in Figure 20.6.

Figure 20.6. Upload XLIFF for Preview feature in the QAT

The Upload XLIFF for Preview feature allows NPMs to view their translated questionnaire materials in a runtime environment identical to what questionnaire respondents would see in either the Student Delivery System (SDS) or in the online questionnaire. NPMs may also use create a translated PDF version of their questionnaires.

The XLIFF previewer an "Edit Columns" tool that allows users to adjust the width of response columns for each question and language version individually to ensure that translated text is not truncated. The features of this tool are shown in Figure 20.7. For PISA 2022, this tool was used only by PISA Administrators.

Figure 20.7. Upload XLIFF for Preview – Edit Columns feature



Question templates

The QAT editor is a template-based questionnaire authoring system that supports the creation of multilingual content (this includes left-to-right and right-to-left texts, and extended character sets for Arabic, Chinese, Hebrew, Japanese, Korean, Russian, Thai, etc.), the design of the rules-based routings driving the questionnaire flow, and the enforcement of the quality of the answers via consistency checks. In PISA 2018, national centres entered their translations directly into the QAT and so the support of the languages within the system was important; however, in PISA 2022 the national centres entered the English back-translation of their agreed-upon content adaptations in the QAT and then the QAT generated electronic translation files (XLIFFs) that could be used by standard computer-assisted translation tools.

All PISA 2022 questions were authored using one of the following screen templates available through the QAT editor:

- Drop Down (Table)
- Drop Down
- Exclusive Choice
- Multiple Choice
- List of Text Inputs
- Free Text Input
- List of Exclusive Choice (Table)
- List of Multiple Choice (Table)

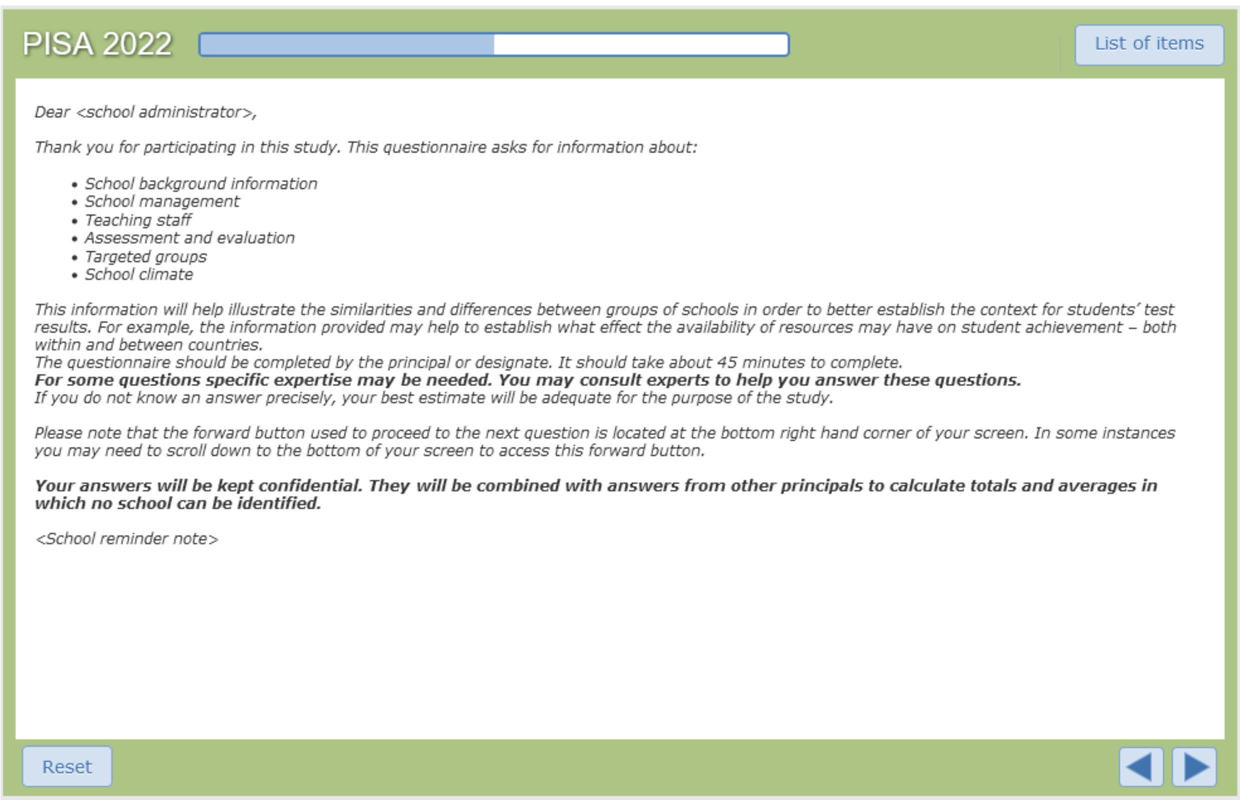
- Multiple List of Text Inputs (Table)
- Scale Question Type
- Information

Additionally, there were two templates for defining rules that were used within the questionnaires:

- Consistency Check Rule
- Routing Rule

A short description of each template is provided below, with examples in Figure 20.8 through Figure 20.21.

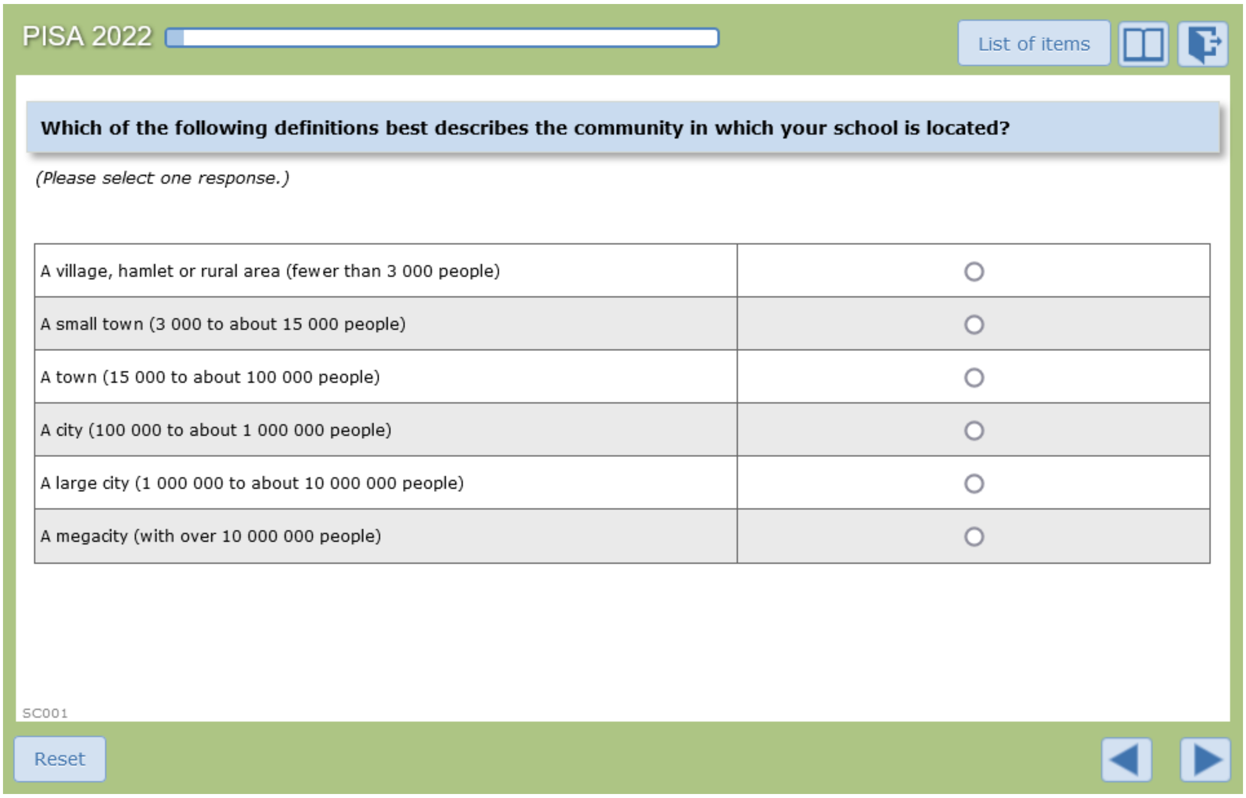
Figure 20.8. Information Template



The *Information* template shown in Figure 20.8 is used to insert an introduction, a transition, or a closing page into the questionnaire. The author can use this template to present the questionnaire (e.g. its goals, structure, general recommendations, and other instructions), introduce a new section of questions, and to thank the respondent at the end of the questionnaire for their participation.

The *Exclusive Choice* template shown in Figure 20.9 presents a question to the respondent as well as a set of mutually exclusive responses. Each response option receives an identifier. The data saved for this template is a pre-assigned response number assigned to each radio button (e.g. values 01, 02, 03, 04, 05, or 06 is assigned to each of the radio buttons shown in Figure 20.9). The presentation of this item type to the respondents uses a single set of standard radio buttons. Choosing one of the options will remove any previous choices.

Figure 20.9. Exclusive Choice Template



The *Multiple-Choice* template shown in Figure 20.10 presents a question to the respondent as well as a set of non-exclusive responses. Each response option receives an identifier. The data saved for this template includes a value, either 0 or 1, for each response option. The presentation of this template uses standard checkboxes. The checkboxes are selected when a user clicks on them and unselects if clicked a second time.

Figure 20.10. Multiple-Choice Template

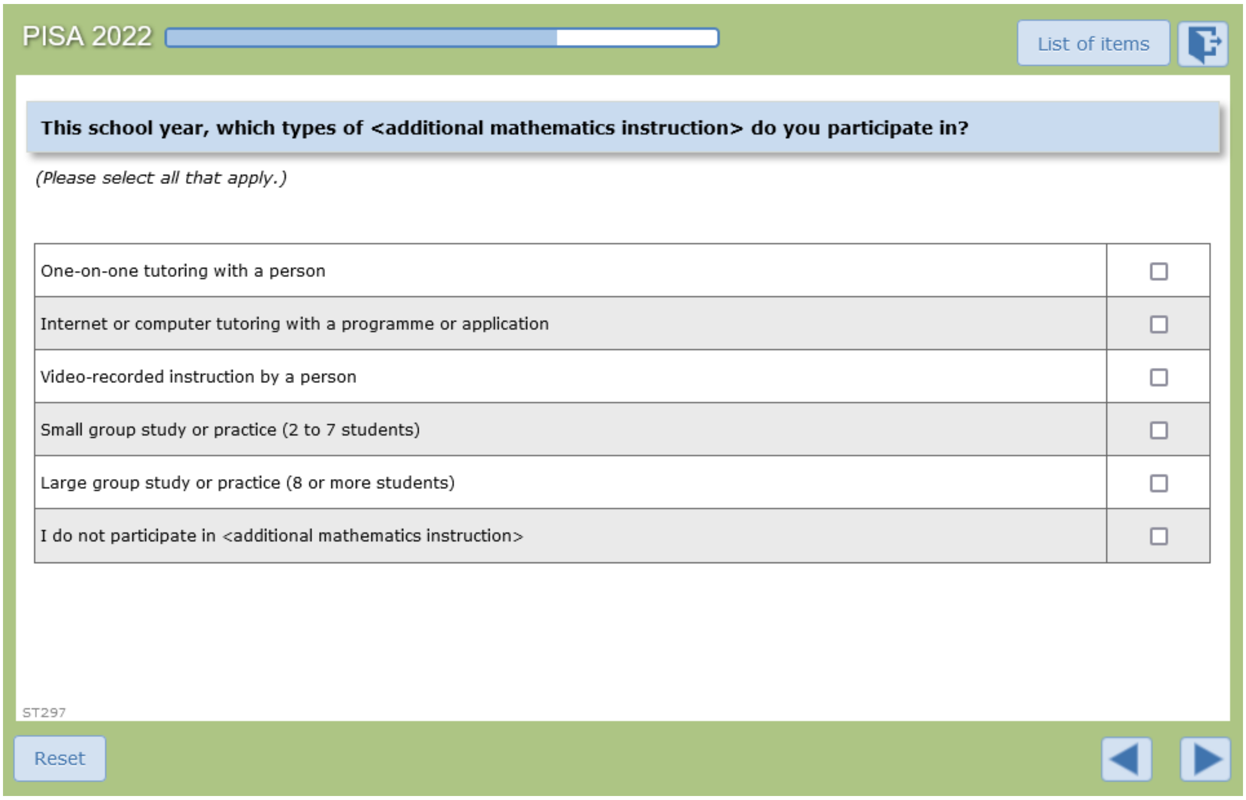
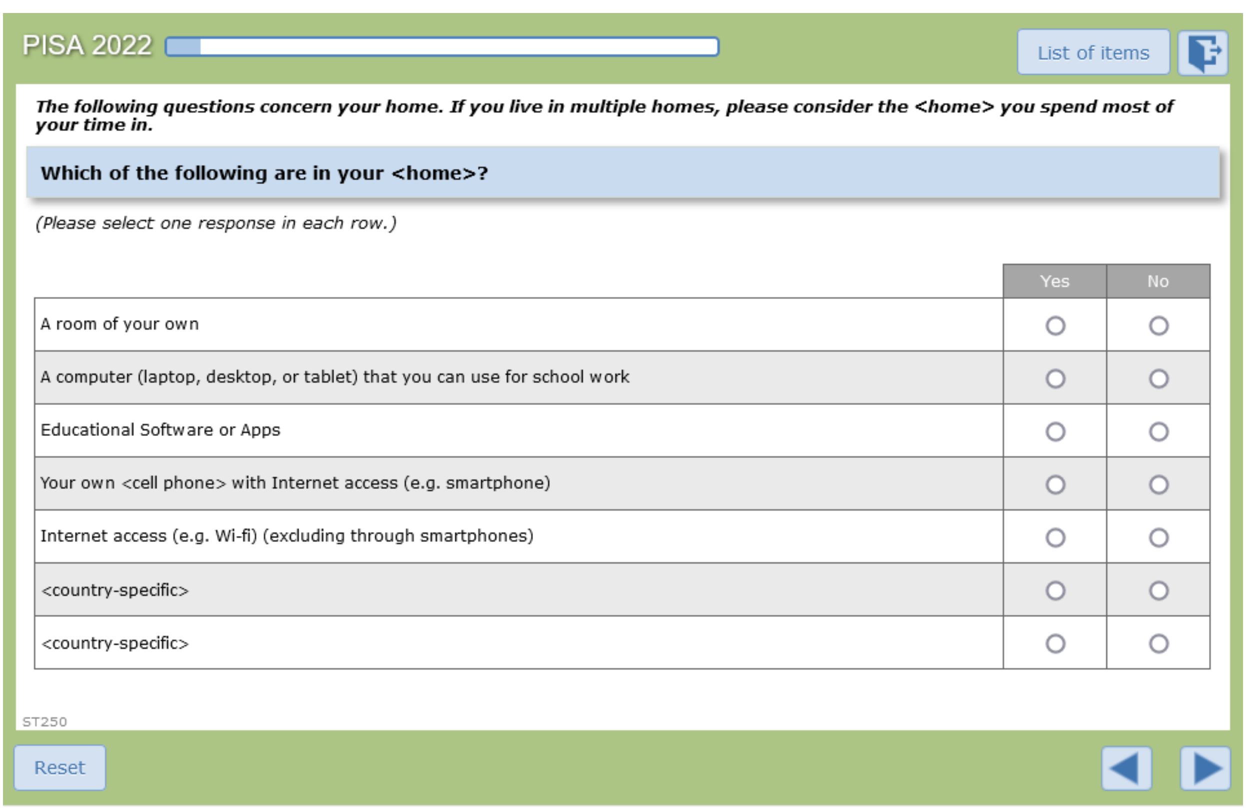


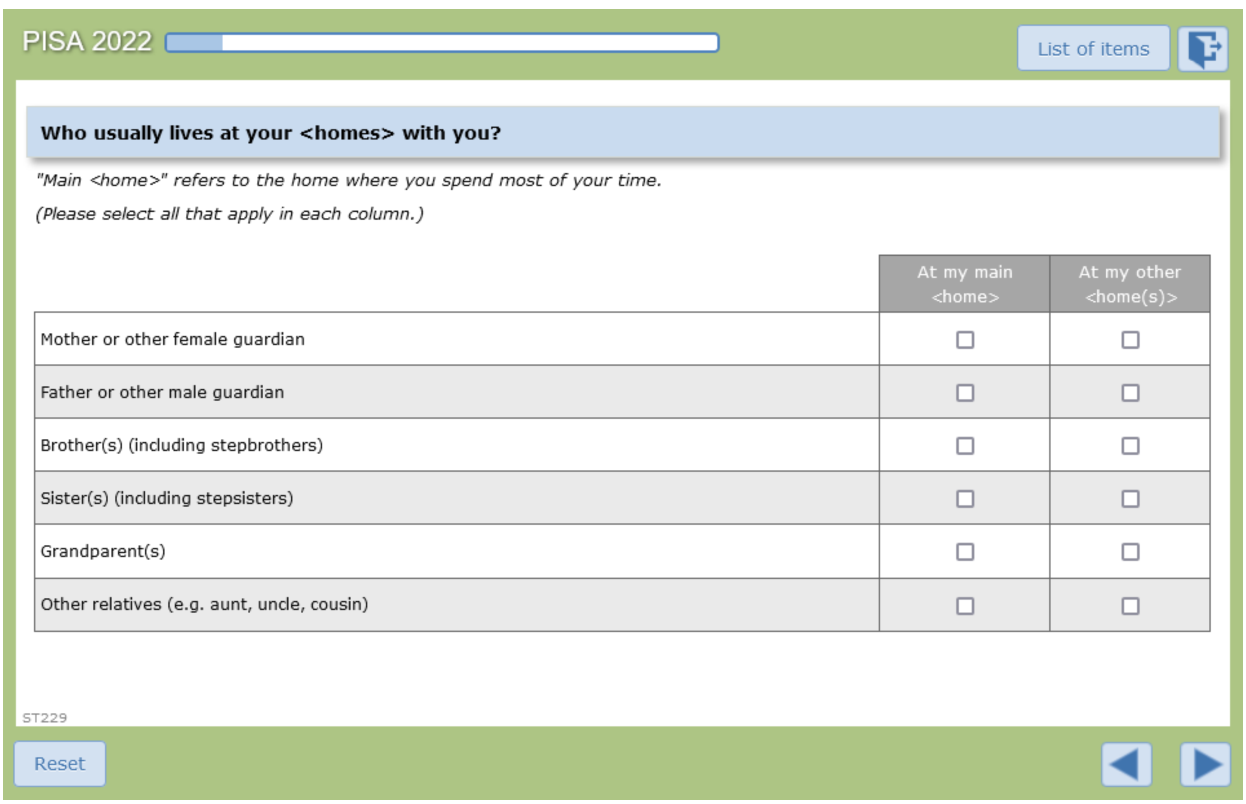
Figure 20.11. List of Exclusive Choice (Table) Template



The *List of Exclusive Choice (Table)* template shown in Figure 20.11 presents the user with a set of exclusive choice questions on a single screen in a tabular format. In the default format, each row of the table is a separate item, and the columns are the response options for each item. In addition, the QAT editor allows the author to invert the table, so that items are in the columns and the response options are in the rows. Typically, this template presents a single question text or stem in the blue box at the top of the screen, and the items that are part of that question are represented in each row.

The *List of Multiple Choice (Table)* template shown in Figure 20.12 presents the respondent with one or more non-exclusive choice questions on a single screen in a tabular format. It is like the previous template; however, it uses checkboxes so that more than one choice can be selected for each item (row), or column if the presentation is inverted. The data generated by this screen include a response of 0 (unchecked) or 1 (checked) for each response option for each question. In the example shown in Figure 20.15 the screen will generate 12 individual variables of data.

Figure 20.12. List of Multiple Choice (Table) Template



The *List of Text Inputs* template shown in Figure 20.13 is used for collecting short, open ended response data. The template presents the respondent with one or more areas to type a response, each with a label indicating the information to be entered, the responses can be unfiltered text, or they can be limited to numeric values. Constraints of a minimum/maximum numeric value or text length can be placed on the values entered in each case.

Figure 20.13. List of Text Inputs Template

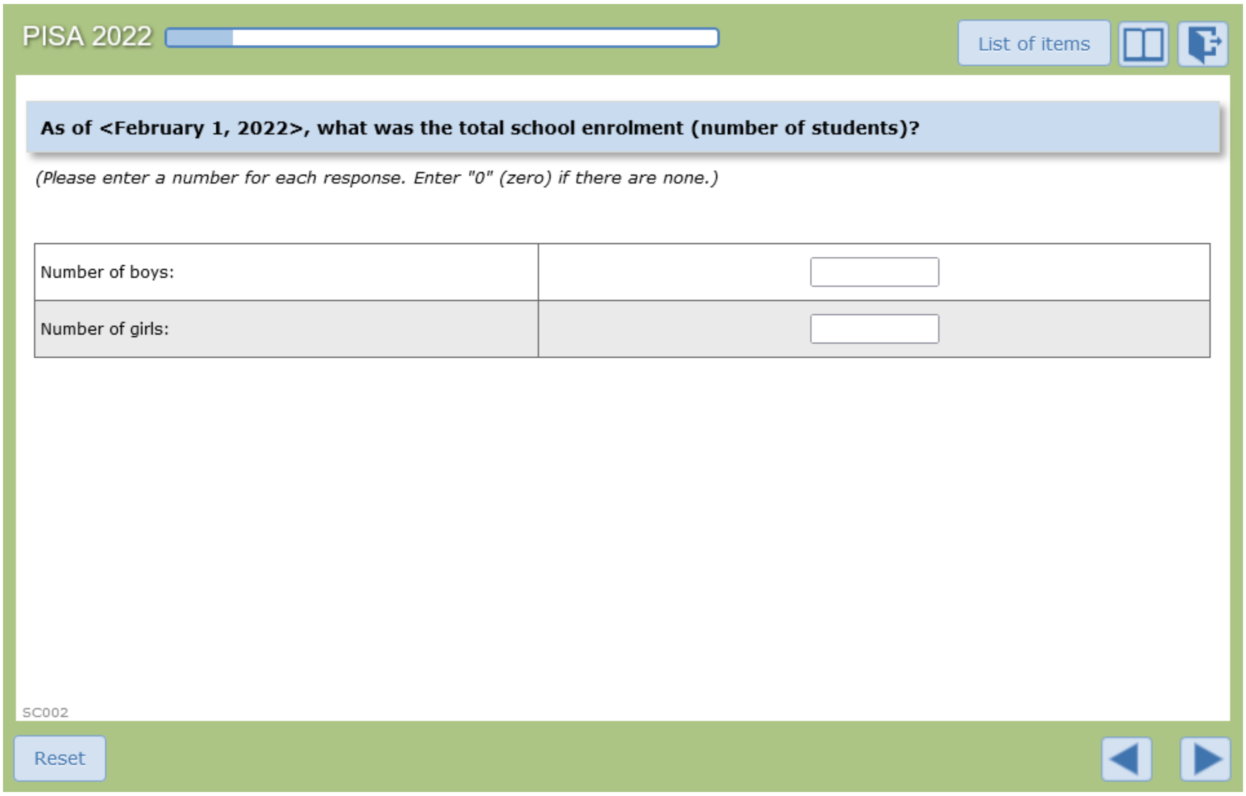
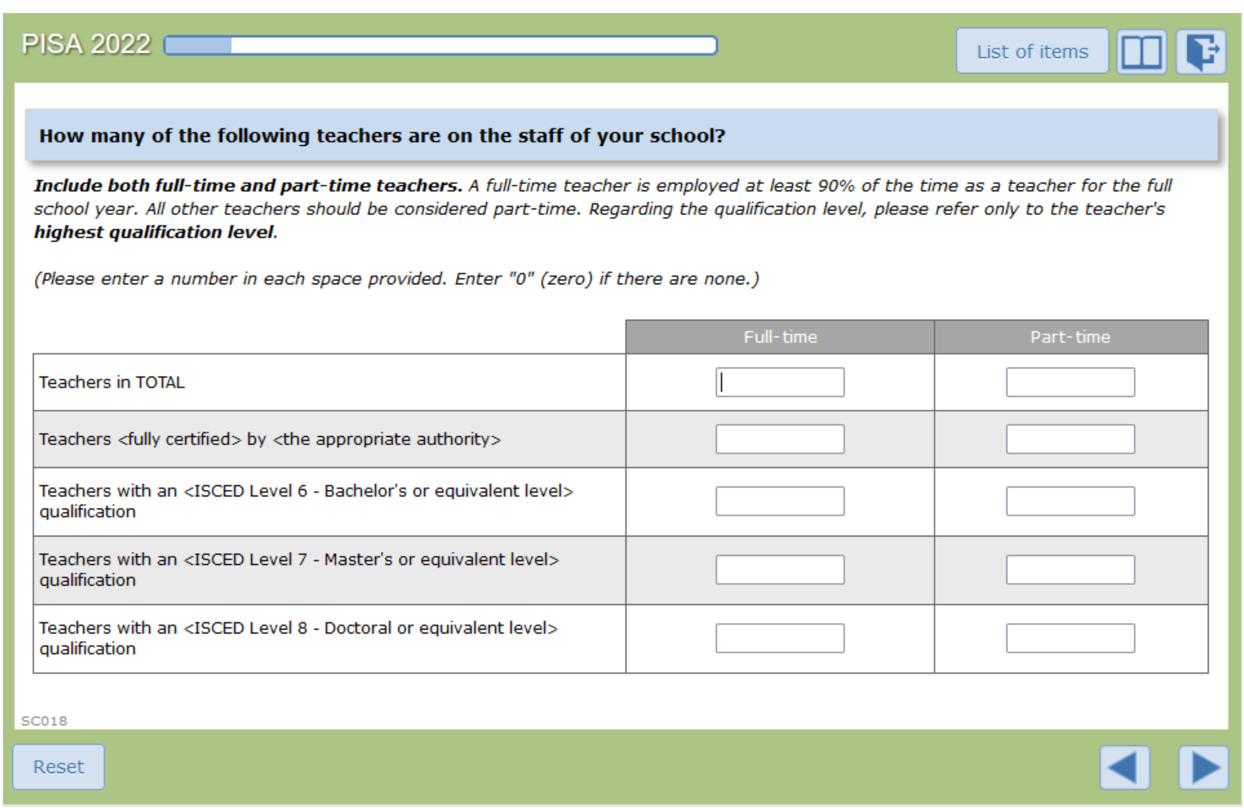


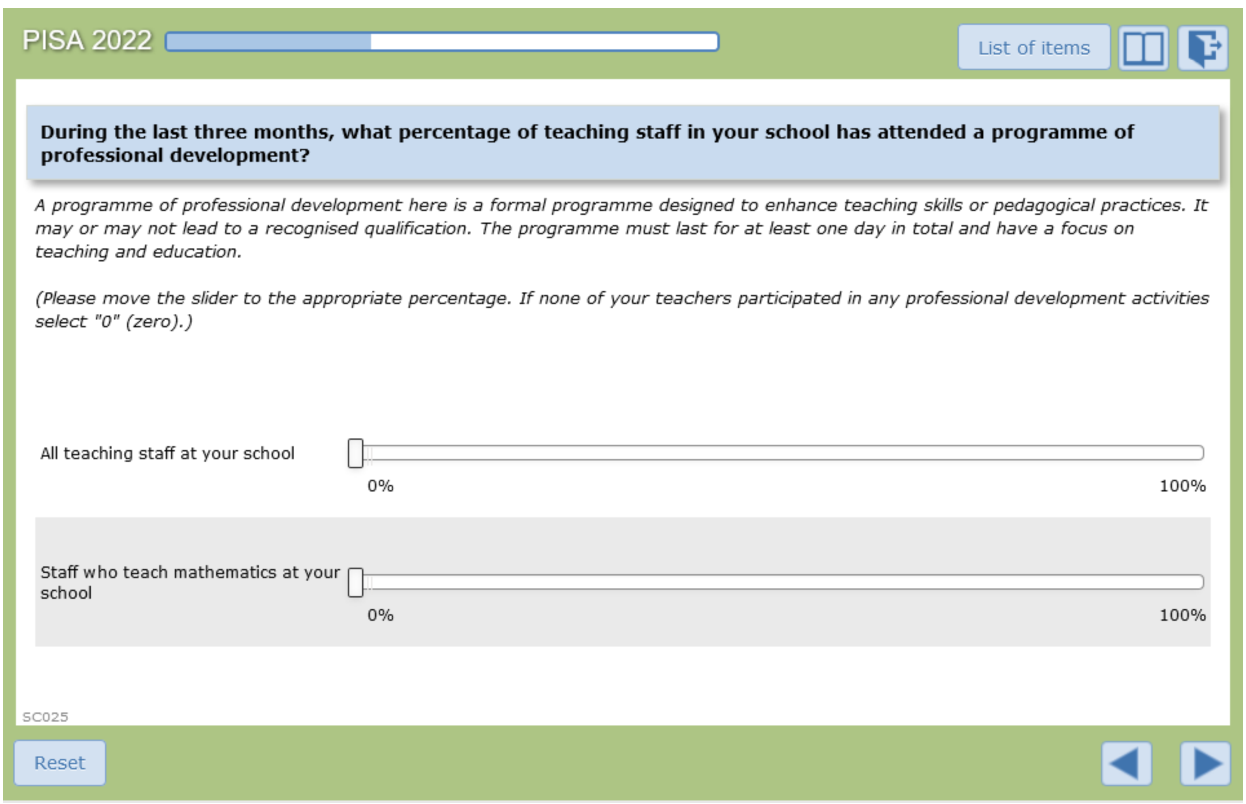
Figure 20.14. Multiple List of Text Inputs (Table) Template



The *Multiple List of Text Inputs (Table)* template, shown in Figure 20.14, is used for collecting short, open ended response data. However, in this case more than one response can be collected for each area of interest. The response areas are presented as a table. Like the previous template, the response values can be either text or numeric, and can be limited in their range.

The *Scale Question Type* (slider) template shown in Figure 20.15 is used to collect numeric information on a sliding scale. The respondent moves an indicator along a scale line to indicate where in the range their answer should be. The template allows the author to include one or more slider responses on a screen. Each slider has upper and lower limits. Step values for the sliders can be set, and the author may include labels for the left and right ends of the scale. The slider differentiates between no response (not moving the slider at all) and moving the slider to the "0" position. PISA 2022 did not use the scale question type template for new questions, but there were a handful of trend questions that still used this template this cycle.

Figure 20.15. Scale Question Type Template



The *Free Text Input* template shown in Figure 20.16 supports an open-ended text response. The respondent is presented with a large text box in which they can enter a long response with line breaks to provide multiple paragraphs. This template was only used in national questions in PISA 2022.

Figure 20.16. Free Text Input Template

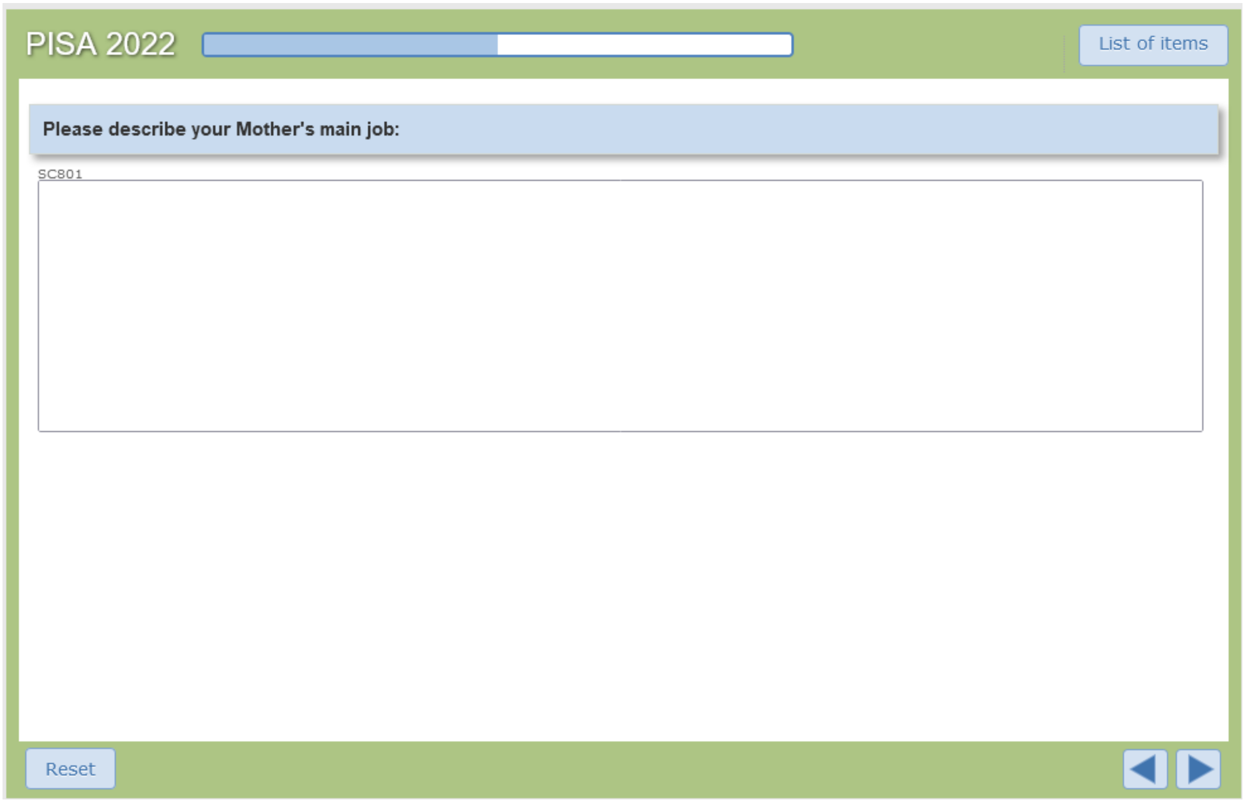
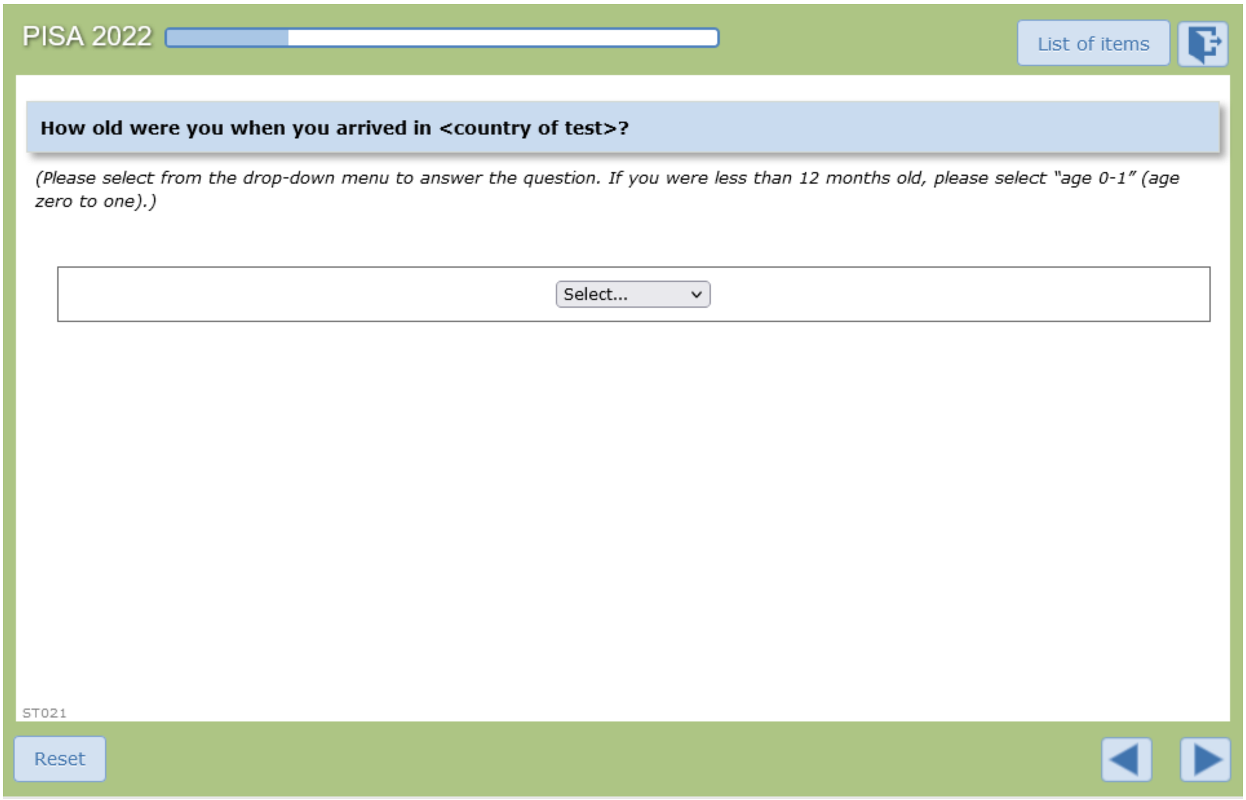


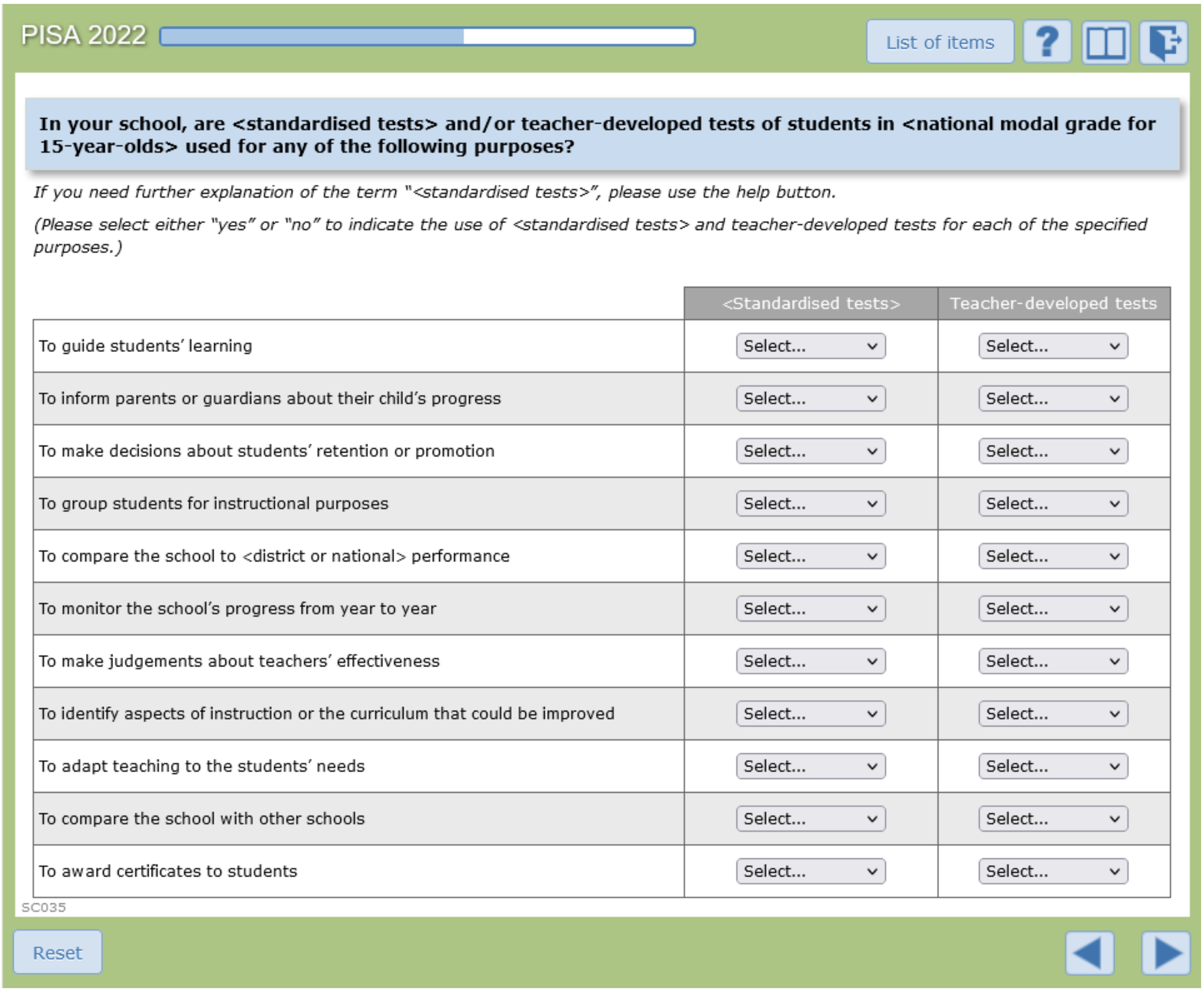
Figure 20.17. Drop-Down Template



The *Drop-Down* template shown in Figure 20.17 presents the respondent with one or more drop down menus from which to select their response to a question. Each menu can have a textual label to present a question or to indicate what type of information (e.g. age) the respondent should select from the menu. The contents of a menu are defined using a list with each text response in the list assigned a number in the QAT editor. The menus can share the same list of response values across items, or each item on the screen can have a unique list.

Like the Drop-Down template, the *Drop-Down (Table)* template shown in Figure 20.18 presents the respondent with one or more drop down menus for providing a response. In this template, the menus are organised into a table. The drop-down menu contents themselves are defined in one or more lists. In the standard layout, each menu in a row will contain the same list of response values. However, like the other table-based templates, it is possible for the author to invert the rows and columns so that columns contain the same menu values.

Figure 20.18. Drop-Down (Table) Template



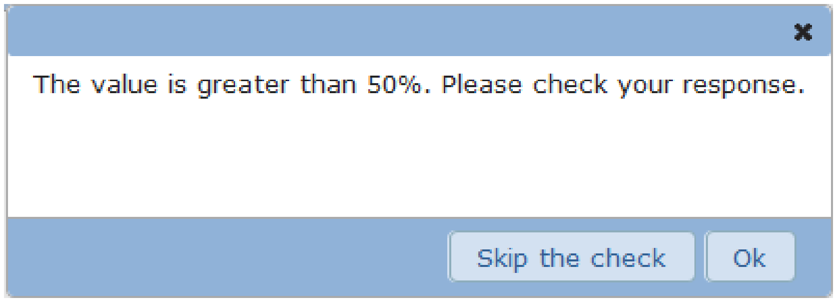
Consistency Check Rule

The *Consistency Check Rule* template supports a rule-based approach for validating the response provided by a user. The author provides a condition (i.e. "True" or "False") intended to represent the logic of the rule that checks the values of some response variables from different questions the respondent has answered. If the condition evaluates "True," a notification message is displayed to the user. The template for defining the consistency check rule appears in Figure 20.19.

Figure 20.19. Consistency Check Rule Template

The rule is evaluated when the respondent navigates away from the current question, by clicking either Forward, Back, or Log Out. When the condition is true, a message is shown like the one in Figure 20.20.

Figure 20.20. Consistency Check Message



The consistency check is a soft check and will not require the respondent to change their answer if the check appears. The respondent can click on "Ok" in the check and go back to the current question to change their response. If the respondent clicks the "Skip the Check" button, the questionnaire will proceed as normal.

Routing Rule

The *Routing Rule* template allows the author to use branching within a questionnaire to direct the question flow. Routing rules appear in between questions in the questionnaire, and they are executed after the completion of the question before the rule.

The routing rules are based on specific conditions, like the consistency checks. The rules are defined using IF—THEN--ELSE logic. If the condition evaluates "True" the "Then" portion is executed, otherwise the "Else" part is executed. The "Then" and "Else" parts can be either another IF--THEN--ELSE rule or GOTO commands, directing the questionnaire runtime to branch to a specific question in the questionnaire.

The routing rules are typically used for skipping questions that do not make sense given a specific initial response from the respondent. A simple case is an exclusive choice question, where the last response option is "Other". If the respondents select this option, they should be shown a question asking for more information about their answer. For example, an open response where they can type their answer. In PISA 2022 field trial routing rules were used for the first time to create virtual booklets to indicate that certain questions should be skipped if a student's random number was within a certain range. An example of a routing rule can be seen below in Figure 20.21.

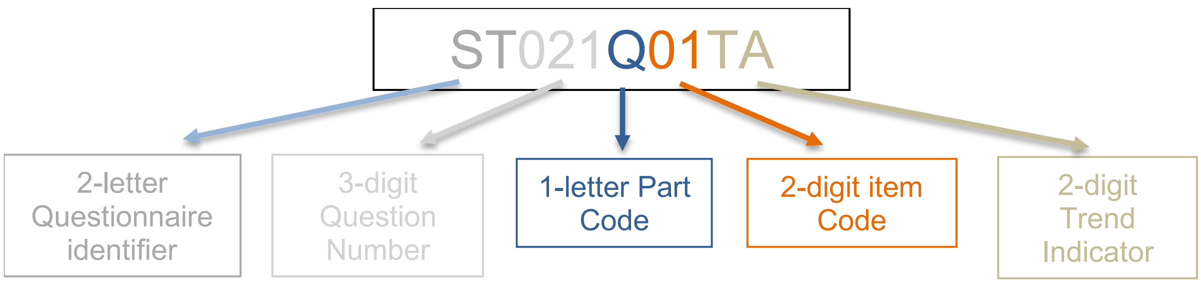
Figure 20.21. Routing Rule Template

Identifiers within the QAT

An identifier (or ID) is a tag attached to an object in the QAT. When authoring questionnaires, it is important that each question, item, and rule has an ID that follows a standard convention so that each object in the questionnaire can be appropriately identified and data from the questionnaires is generated in a standard format.

In the QAT editor, the types of objects receiving an ID are the various questions, helps, instructions, and response options. These IDs and tags are used when importing the translations used to display the questionnaires in each local language. IDs are also used for each rule and for each element designed to receive and store the data provided by the respondents (i.e. answers). IDs used for data capture within the questionnaires are at least 10 characters long and follow a set format shown in Figure 20.22

Figure 20.22. Question IDs



The interpretation of these IDs is as follows:

- The 2-letter questionnaire identifier indicates the questionnaire in which the item was administered. ST for Student Questionnaire or Student Questionnaire-UH, SC for School Questionnaire, FL for Financial Literacy Questionnaire, IC for ICT Familiarity Questionnaire, WB for WBQ, and PA for Parent Questionnaire.
- The 3-digit question number: this is a unique ID given to the particular screen on which the questions are administered and can be used for either a single question or a set of questions presented in a table on the screen. As much as possible, questions retain their numbers across cycles to allow for easier identification of trend variables. Question numbers beginning with 800 indicate a national question administered only in a particular country.
- The 1-letter part code introduces the item code and indicates whether the question is equivalent to the master (a code of Q) or is a country-adapted variable requiring harmonization to be compared to the master (a code of C). Consistency checks are labelled with the code E, and routing rules are labelled with the part code R.
- The 2-digit item code indicates the number of the individual question item administered on the screen. If a screen (such as a table screen) contains four items, these numbers will typically range from 01 to 04; however, due to trend IDs or elimination of items after the field trial, the item code is not always sequential on a screen.
- The 2-digit trend indicator is used to indicate the cycle in which the question was originally introduced in order to facilitate trend analysis. The trend indicators are shown Annex Table 20.A.3

The IDs are one of the key parts for the computer-based questionnaires and are the basis for the data analysis. A question (or part of a question) with an unexpected or inappropriate ID is unusable and can eventually not be analysed. Checking the consistency of IDs was one of the critical tasks performed by contractors when authoring and reviewing the computer-based questionnaires.

General questionnaire development process

The life-cycle of a questionnaire in PISA followed a process that can be split in ten major steps. These steps are described in Figure 20.23.

This sequence of steps took place twice: once for the Field Trial (FT) and again in an abbreviated process for the Main Survey (MS). During the Field Trial, the whole platform (i.e. the tools, computer servers, network access, etc.) and the material (i.e. the questionnaires) were tested with a limited sample of respondents. After the Field Trial, the results and feedback collected are analysed and reviewed. Then, for the Main Survey, the sequence was started for a second time and each step integrated all necessary adjustments in terms of process, questionnaires material, and tooling. This double-phase cycle provided better data quality.

The procedure for creating the paper-based questionnaires was the same, except step 2 (authoring the questionnaires in the computer platform) and step 5 (quality checks on the implementation of the adaptations in English) were omitted.

In the following sections, each step of this process is explained in more detail.

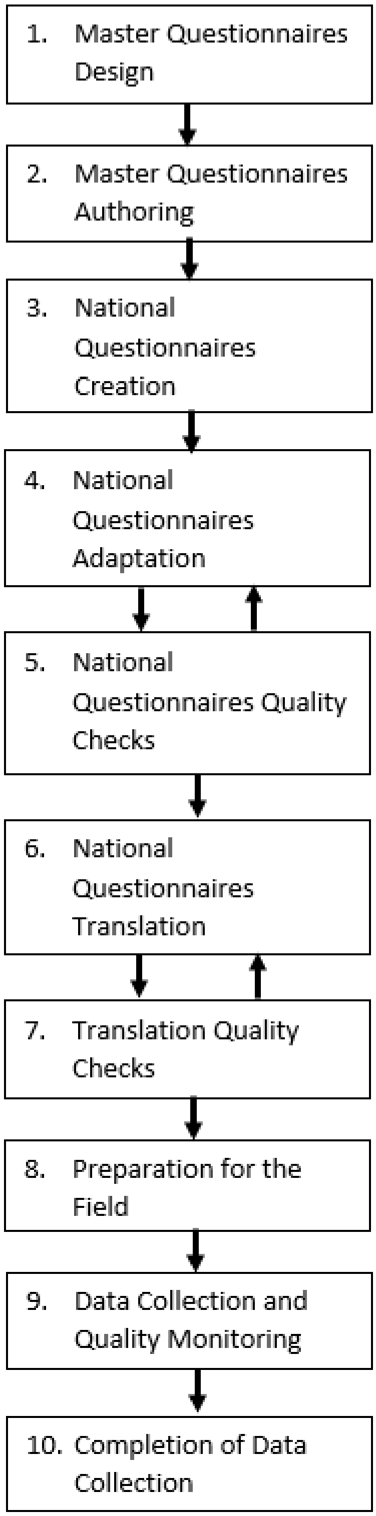
Step 1: Master questionnaires design

The master versions of the questionnaires were created as Word documents, and each contained information about the trend status, IDs, application of matrix sampling to the question, and routing instructions. Once the master versions of the questionnaires were finalized, the Word document was used to create the paper-based master version to be adapted and translated by paper-based participants. The contractors then used the master Word version to begin authoring the computer-based questionnaires in the platform.

Step 2: Master questionnaires authoring

Contractors first used the QAT to author the international master version of the questionnaires in English. Trend questions from the previous cycle were copied from the main survey PISA 2018 master questionnaire profiles to ensure across-cycle continuity of question templates, IDs, data format (e.g. string, numeric), and range limits. The appropriate question template was chosen for each of the new items and the appropriate data types, range restrictions, and consistency checks were added. Questions were ordered appropriately to follow the design-specified routing through the questionnaire and routing rules were inserted. Each questionnaire's formatting was reviewed to ensure it had the appropriate layout, and the master version of the questionnaires were tested extensively using testing scenarios to ensure a high-quality initial version.

Figure 20.23. PISA 2022 computer-based questionnaire life cycle



Step 1: The **Master Questionnaires were designed** in collaboration with the Questionnaire Expert Group. These questionnaires are first created as Microsoft Word documents that will become the Master Paper-Based Questionnaires.

Step 2: The computer-based versions of the **Master questionnaires were authored** using a unique authoring tool in the PISA questionnaire platform. They were produced in English then reviewed and tested.

Step 3: The Master Questionnaires were duplicated for the participating countries. These questionnaires, called **National Questionnaires, were created** and made available to countries for adaptation.

Step 4: The **adaptation of National Questionnaires** was performed by members of the national centres. The adaptations take the form of adding or suppressing questions or changing parts of questions as required by the national context.

Step 5: The **quality of the adapted National Questionnaires was checked** against the original Master Questionnaire. The quality of adaptation is important for guaranteeing that the collected results are comparable at the international level.

Step 6: The National Centre **translated the questionnaire text** into each national language version administered.

Step 7: The **quality of the translations of the National Questionnaires** was checked against the agreed-upon national adaptations.

Step 8: When a questionnaire had successfully passed all the quality and technical checks, it was **prepared for the field**. The deployment was either online (via a connection to Internet) or as part of the Student Delivery System.

Step 9: During the data collection periods, **data was collected** either online or in the schools, depending on the distribution method of the questionnaires.

Step 10: At the **end of the data collection**, the online National Questionnaires were deactivated, and respondents could no longer access them. Final data files were exported for data cleaning and analysis.

Step 3: Creation of national questionnaires

Once the master questionnaires were authored and finalized, they were used as the template to create the national questionnaires for each country. The contractors duplicated the master questionnaire for each country, so every participant started with the same set of questions in the same order. In order to maintain trend adaptations from the previous cycle, trend question screens were copied from the country's PISA 2018 main survey questionnaire instead of the master questionnaire. Since PISA 2018 questionnaires were translated in the profile, as part of this copy process, the English back-translation of the PISA 2018 adaptation was inserted into the QAT editor. The initial version of each national questionnaire contained a combination of new questions for PISA 2022 copied directly from the master questionnaire and trend questions from PISA 2018 copied from the country's final PISA 2018 computer-based Main Survey Questionnaire. This copy operation was performed by the contractors using several system scripts. These national questionnaires were then put into a mode that allowed the national centres to adapt the content.

Step 4: National questionnaire adaptation

At this step in the process, the National Centre first documented in a spreadsheet all the structural adaptations needed to the questionnaires, including adding or deleting questions and response options and all required content adaptations such as the specific names of study programmes. The contractors reviewed and approved the adaptations to ensure internationally comparable questionnaires. Once all adaptations were negotiated, the National Centre connected to the QAT to view and edit their national questionnaires in the platform and insert the agreed-upon adaptations. All adaptations were inserted into the QAT in English so that the text in the national questionnaire in the QAT became the nationally adapted English source text used later for translation. Much like for authoring the master questionnaires, the National Centre had access to the same functionalities in the QAT editor, such as adding new national questions and adapting existing questions, as well as the functionalities for previewing the questions.

When opening the questionnaire in the QAT, the National Centre could see and edit the questions for the new PISA 2022 content. Trend questions copied from the PISA 2018 cycle were locked so that the National Centre could not edit them, and any changes approved for these questions were implemented centrally by the contractors. Maintaining the quality and integrity of the trend questions over time is important to be able to analyse data across cycles.

Step 5: National questionnaire Quality Check

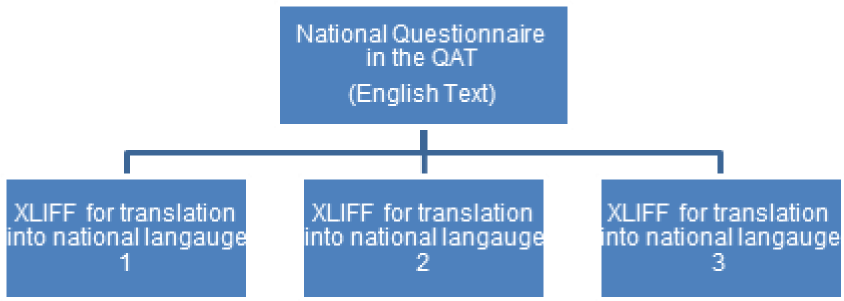
Once a country's adaptations and national questions were implemented in the QAT, the national centres tested the questionnaire using contractor-prepared testing scenarios to review all adaptation in English and confirm the routing of the questionnaire worked as expected. Then the contractors reviewed and approved the national version in the QAT to confirm all the agreed-upon adaptations had been correctly authored and to centrally insert any agreed-upon changes to trend adaptations. The contractor also carefully reviewed the questions to ensure all agreed national questions had been inserted in the questionnaire and reviewed questionnaire IDs to confirm that they were appropriately updated to conform to the ID conventions for the cycle. IDs are the key identification point for the data analysis and an error in this part might result in loss of data. After these quality checks, the questionnaires were locked in the QAT so that no further edits could be introduced by the national centre.

Step 6: National questionnaire Translation

With the national questionnaire adaptations finalized, the contractors then used the national questionnaire in the QAT to generate the country-specific English source XLIFF files for translation. Each country had one single structure and source text per questionnaire that then could be translated into multiple languages (see Figure 20.24). The XLIFF files were inserted into an OmegaT translation project which allowed the

National Centre to reference translations of similar text used in PISA 2018 to speed up translation. The OmegaT project also allowed the translation of items from PISA 2018 to be automatically filled with the trend translation and locked so the National Centre could not edit them.

Figure 20.24. Translation of Questionnaires into multiple national languages



Countries performed double translation of new questionnaire items and reconciled those translations. If updates were needed to trend translations, the country documented those changes for review by the contractors. During the translation process, countries could generate translated XLIFF files and upload them to the preview tool in the QAT to review the questionnaire as a respondent would. As part of the preview process, countries also noted any layout issues that needed to be fixed by the contractors.

Step 7: National questionnaires quality check

After the National Centre completed the translation of the questionnaires, the contractors checked the quality of the translations, the accuracy of the translation compared to the English master version, the routing and formatting, and the IDs and data generation of the questionnaires.

National translations were reviewed by verifiers under the direction of the translation contractors to confirm that all adaptations were appropriately translated, all translation notes from the questionnaire developers were followed, and to confirm the accuracy of any requests to update translations of trend items. Verifiers updated translations or provided notes to national translators to address issues as necessary. The translation and adaptation discrepancies were documented in a spreadsheet which was delivered to the National Centre for their review. The National Centre was able to accept or refuse these comments and could update their translations accordingly.

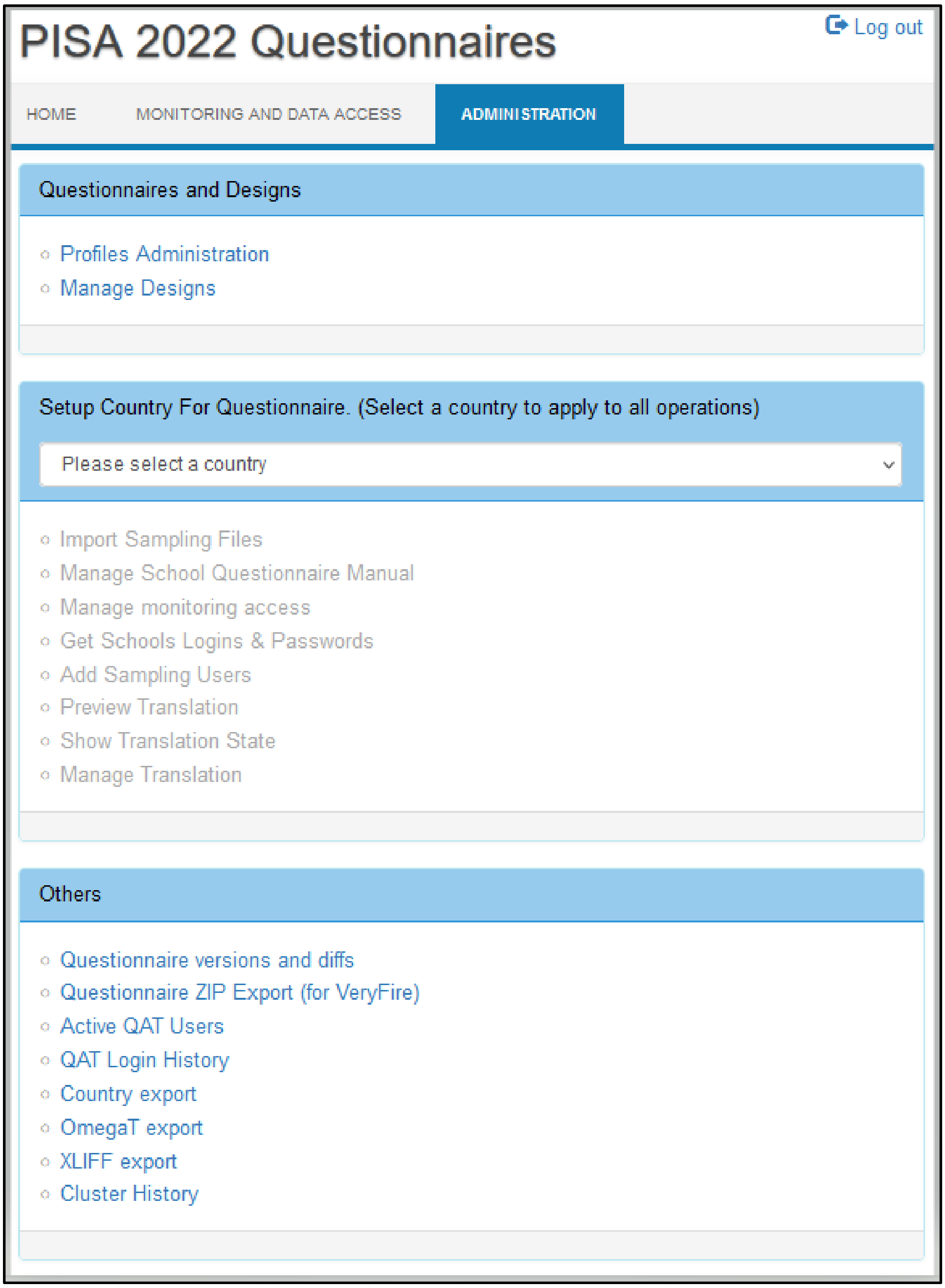
The contractors also reviewed the questionnaires to manually check that a user was able to go through the questionnaire from the beginning until the end without a software error due to, for instance, errors in routing rules; check if all questions and messages were translated; and check if all the parts of the interface were translated and well-integrated.

National centres were provided testing scenarios for each questionnaire to validate the accuracy of their translation and adaptation work. These testing scenarios defined different ways in which a respondent could answer a questionnaire following every possible routing. National centres were required to test the questionnaires in each language version following these scenarios and provide their test results to the technical team for review. The contractors reviewed the output files to confirm no technical problems were detected when saving data. Only once all these reviews were completed were questionnaires deemed ready for administration.

Step 8: Preparation of national questionnaires for delivery

To prepare the questionnaires for administration, the QAT administrators and technical team used the features of the questionnaire platform's administrative interface shown in Figure 20.25.

Figure 20.25. Questionnaire Platform – Administrative View



There were two modes of delivery used for the questionnaires in PISA 2022. The student questionnaires, including the optional ICT, Financial Literacy, and Well-Being questionnaires, were run as part of the PISA student delivery system (SDS). The School and Teacher questionnaires were delivered online over the

Internet. Both delivery modes shared a common code base and database structure, but the preparation for delivery followed different procedures.

For the student questionnaires, the preparation step involved uploading the final translation file for each national language version into the QAT and then exporting the completed national questionnaires for each country, as well as the questionnaire software and user interface translations, in a form that could be integrated into the Student Delivery System (SDS) and loaded onto and accessed from USB drives. The export only included the software components needed to run the questionnaire, so components such as the QAT, were not included in the export, and a database image with the national questionnaires was created. These exported files were directly integrated into the PISA SDS software for a country, and then tested and validated.

The online School and Teacher questionnaires required more steps to prepare for delivery. The first critical step in this process was to import the sampling information from each computer-based testing country into the questionnaire platform so that the selected schools and teachers would be known to the system and could be identified when they connected to complete the questionnaires online. To do this, the final approved “Sampling Task 5b” (specific to the Field Trial) or “Sampling Task 11” (specific to the Main Study) output files were taken from the PISA Portal and uploaded into the QAT. The content of these forms is described in Chapter 6 of this Technical Report. These files contained the list of schools selected from the sampling process, using anonymous ID codes. The QAT software used these files to generate logins and passwords for each sampled school. These logins and passwords were then sent to the national centre, which distributed them to the selected schools and teachers accordingly. The ACER Maple sampling software generated the IDs and passwords used for the Teacher Questionnaire, and Teacher Questionnaire authentication process was set up to recognize valid teacher IDs and passwords.

The countries participating in the online questionnaires in PISA 2022 were spread out across the world. To improve performance for end users, servers were set up on several continents, as shown below in Figure 20.26, following the same distribution used in the 2018 PISA cycle. The distribution of servers helped to reduce network latency and improved the performance. Server installations for this cycle of PISA were in Germany, Singapore, Australia, and the United States.

Figure 20.26. Distribution of the PISA 2022 servers

Participating countries/economies were routed to their nearest server locations. When respondents logged into the questionnaire using their user ID and password on the School or Teacher login website, they were automatically redirected to their assigned server based on their login ID to complete the questionnaire. One country, the United States, delivered the online questionnaires from their own national server. This server was completely standalone, so respondents connected to it directly, and were not rerouted through the central PISA server.

Step 9: Data collection and quality monitoring

During the field trial and main survey data collection periods, students, school principals, and teachers responded to the questionnaires. For the students, responses were captured as part of the PISA Student Delivery System, which ran from either a USB drive on a school computer, from laptops with the SDS software pre-loaded, or through Google Chromebooks. The system ran in full screen, locked down mode when running on Windows and Macintosh computers, and in “kiosk mode” (https://chromeos.dev/en/kiosk) on Chromebooks.

The questionnaire software ran offline, in a standalone mode on the school computer, and all results were saved back to the USB drive. The students did not need to login to start the questionnaire. Identification and authorization of the students was performed by the Student Delivery System.

For the online questionnaires for school principals and teachers, delivery was performed online over the Internet. Schools were assigned login IDs and passwords as part of the sampling process in Step 6. A set number of teacher questionnaire IDs were accepted for each school. When respondents first connected to the questionnaire platform, they entered their ID and password. The questionnaire software selected the appropriate national questionnaire based on this ID. In countries with multiple language versions of their questionnaires, users had to select which language they wanted to use before proceeding further.

As respondents completed the questionnaires, data was collected by the questionnaire platform. The original data saved was the response to each question or item. This data depended on the template used for each question. For questions that used radio buttons, the data value saved was the response ID associated with that radio button. For checkboxes, a data value was saved for each of these boxes on the screen and the value would be zero or one depending on whether the box was selected. For sliders, drop-down menus, and textual responses, the value selected or entered is saved. If no response is selected or entered, a value of “null” is saved. For questions where matrix sampling was applied, those items that were not presented to the student received a special code so that nonresponse could be distinguished from not administered.

Along with the response data, the questionnaire saved the final valid path taken by the respondent in the questionnaire. This allowed the contractors to easily identify which questions were presented to the respondent based on the routings so that not administered questions could be distinguished from not answered. Also, a log of actions by the respondent and the questionnaire system was saved. This log includes events such as those shown in Annex Table 20.A.4.

During data collection for the online questionnaires, National Project Managers and administrators of the questionnaire platform could monitor the activity of the questionnaire respondents. The monitoring showed which respondents had connected to the questionnaire platform and how far they had progressed through the questionnaire. The platform also supported generating a PDF file for a respondent showing the questionnaire including all the responses that had been saved. The overall status for each of the questionnaires could be exported to a spreadsheet for further sorting and filtering.

During the Main Study, the sampling process selected schools to participate in the PISA survey, along with replacement schools if the originally samples schools refused or were unable to participate. Through the monitoring tools available in the questionnaire platform, the NPMs were able to activate or disable school logins to control access to the questionnaire depending on the school’s status as selected or replacement.

The administrators of the questionnaire platform had additional tools available for monitoring the progress of the respondents. These included a view of all currently connected users, as well as a history of the logins, both successful and unsuccessful. These reports were important in supporting users who reported problems and in monitoring performance issues on the servers. Additionally, the questionnaire platform saved many different logs, which the administrators used for detecting problems and troubleshooting them. All the servers were monitored and active 24 hours a day during the entire field test and main survey administration dates.

Step 10: Completion of data collection

Access to the online access to the questionnaires closed four weeks after the country’s negotiated field trial or main study data collection period ended. Once access to the questionnaires was closed, national centres exported their results data for inclusion in their national database that they submitted to the contractors. After the questionnaires were closed, respondents who attempted to login received a message indicating that the questionnaires were currently not available and asking them to contact their National Centre for further information.

Each country’s result data was available throughout the data collection period and could be reviewed for completeness by the national centre. Once data collection was complete, the national centres were required to download the final results in a single compressed file and import it directly into the Data Management Expert system for data processing.

The access to the servers and the questionnaire software was available several weeks after the end of the data collection to allow some time for the NPMs to retrieve the data and ask the contractors questions about any issues in the data that they uncovered.

Overview of the technical infrastructure

This section describes the technical aspects of the software and hardware used to support the PISA 2022 computer-based questionnaires.

The PISA Questionnaire platform is a complex and relatively large software system. The development followed standard software development processes. A modified Agile process (see https://en.wikipedia.org/wiki/Agile_software_development) was used, implementing multiple releases during the course of developing and extending the platform.

The PISA Questionnaire platform is composed of two primary subsystems. One, the QAT supports authoring questionnaires and managing the many national versions of the questionnaires. The second, the QAT Runtime, implements the execution environment for the questionnaires, presenting questions to the respondents, implementing branching rules, and collecting data for later analysis. The QAT software was written primarily in PHP on the server side and JavaScript within the web browser. The QAT Runtime software was new for the PISA 2022 cycle. This subsystem was split off and built from scratch to address performance issues that arose in previous cycles, as well as to support the new Electron based PISA Student Delivery System (SDS) that was specifically developed for this cycle. The QAT Runtime was built using Node.JS and Express.JS on the server, and JavaScript within the web browser. The Apache web server was used for delivery of web content, and data was saved using the MySQL database system in the QAT, and MongoDB (for online questionnaires) and NEDB (for offline questionnaires) in the QAT Runtime. The questionnaire content was structured using custom XML markup.

The online questionnaire servers were Linux based, using Ubuntu 20.04 LTS. The student questionnaires were delivered as part of the PISA Student Delivery System, which was based on Electron. For the Main Study online questionnaires, multiple servers were deployed using the Amazon Web Services EC2 system.

Summary

Improvements made to the authoring and delivery of questionnaires in PISA 2022 provided several advantages over the PISA 2018 cycle. First, the updates to the QAT Runtime allowed for faster administration of the questionnaires and for significant increases in the amount of material that could be administered to students. PISA 2022 introduced the use of a random number to assign students to a path in the questionnaire, which allowed multiple version experiments to be conducted during the field trial to collect data that informed future development of PISA context questionnaire items. In addition, the field trial pilot and main study adoption of within-construct matrix sampling, allowed for greater efficiency in the instruments and an increase in the coverage of the constructs to be implemented for the first time in PISA 2022.

Second, including the English back-translation of questionnaire adaptations in the QAT instead of the translated text allowed for clearer documentation of the adaptations to ensure international comparability and higher-quality translation. Contractors and data users were able to clearly distinguish between adaptations to the content of the questions and linguistic adaptations necessary for the translation of certain terms into the local language, ultimately leading to more assurances of international comparability of the questionnaire data.

Third, due to the ability to export customised translation files for each country and language version, translators and translation verifiers were able to make use of translation tools to ensure repeated text used in multiple modules across questionnaires appeared consistently, ensuring that constructs appearing in multiple questionnaires were measured using as identical an instrument as possible. Also, translators and verifiers were able to clearly see in the translation files where national translations did not match the agreed-upon customised source text, reducing misunderstandings about whether adaptations were linguistic or content-related.

Annex 20.A. Evolution and Implementation of Questionnaire Administration in PISA 2022

Annex Table 20.A.1. Chapter 20: PISA 2022 Questionnaires and Participation Metrics

Table	Title
Table 20.A.5	List of Logged Events

StatLink https://stat.link/xk4dcw

Annex Table 20.A.2. The PISA 2022 Questionnaires

Questionnaire	Respondent	Mode of delivery	Compulsory
Student Questionnaire	Student	Computer (SDS) and paper	Yes
Student Questionnaire – Une Heure	Student	Computer (SDS)	No
School Questionnaire	School Principal or Administrator	Computer (Online) and paper	Yes
Financial Literacy Questionnaire	Student	Computer (SDS) only	No
ICT Questionnaire	Student	Computer (SDS) only	No
Teacher Questionnaire	Teacher	Computer (Online) only	No
Parent Questionnaire	Parent of selected student	Paper only	No
Well-Being Questionnaire	Student	Computer (SDS) only	No

Annex Table 20.A.3. Trend Indicator Values in PISA Question IDs

Trend Indicator	Cycle PISA Question Introduced
TA	In multiple cycles prior to and post 2009
IA	2009
WA	2012
NA	2015
HA	2018
JA	2022

Annex Table 20.A.4. List of Logged Events

Event	Description
SESSION_START	The user starts or resumes a questionnaire.
ITEM_START	The user starts an item.
HELP	The user clicks on the Help button.
RESET	The user clicks the Reset button to clear previously entered answers.
LIST_OF_ITEMS	The user clicks the List of Items button to see the questions that have already been visited in the questionnaire.
SELECTED_JUMP	The user clicks on one of the questions in the List of Items to jump to that item.
SELECTED_FORWARD	The user clicks the Next button to move forward in the questionnaire.
SELECTED_BACK	The user clicks the Back button.
SELECTED_LOG_OUT	The user clicks the Logout button to leave the questionnaire.

Event	Description
ANSWER_SELECTION	An answer is selected or entered.
ANSWER_UNSELECTION	The user unselects a checkbox item.
RANGE_CHECK	The answer entered triggered a range check.
RANGE_CHECK_FAILED	The range check warning message was shown letting the user know the permitted range of answers.
CONSISTENCY	A consistency error message is displayed.
CONSISTENCY_CANCEL	The user presses OK to return to the current screen and update their answer.
CONSISTENCY_SKIP	The consistency error is skipped and the move action proceeds.

21 The PISA 2022 Computer-based Platform

Introduction

The PISA 2022 computer-based platform was the primary mode of assessment of student skills. While paper-and-pencil versions of the assessments remained available to participating countries/economies, development of new content to represent and measure the constructs defined in the updated assessment frameworks was done for all cognitive domains and most questionnaires in the computer-based assessments. The vast majority of countries/economies chose to implement and deliver the survey on a computer-based platform to make the most of the opportunities for reporting that this option provided. All cognitive domains were delivered via computer, including the innovative domain (Creative Thinking) and the optional Financial Literacy assessment. The Student Questionnaire, including any international options, was delivered via computer to all students who took the computer-based cognitive assessments. The computer-based assessments were delivered in over 120 different language versions across the participating countries/economies.

This chapter focuses on the functionality and technical implementation of the computer-based assessments. It also describes the functionality and technical requirements of the PISA student delivery system (SDS) and the Chromebook student delivery system (CDS) used for delivery of the PISA survey in schools. Finally, it concludes with a discussion of the open-ended coding system (OECS), used for coding of student responses to open-ended questions in the cognitive assessments that required human coding.

Item rendering

The items for PISA 2022 were implemented using the web-based technologies HTML, CSS and JavaScript®. Modern web browsers, such as the bundled Chromium™ browser v94 used in the PISA SDS, provide a suite of features and functionalities that enable attractive presentations and facilitate engaging interactivity. At the beginning of the development work, an overall user interface was designed with a common set of elements such as navigation, help and progress indicators. Items were implemented in such a way that these common elements were shared, so that the same elements were used across all items in each language version.

PISA 2022 items are generally grouped into units consisting of one or more common stimuli and a set of items, which are also referred to as questions in this chapter. Each unit was constructed independently; with the stimulus and questions components developed first in English, then translated into French to create the two harmonized source language versions. The development was done by experienced web user-interface (UI) developers using standard HTML components and adding custom functionality via JavaScript. Each unit could be viewed on its own or grouped with other units into a test form for delivery to students as part of the assessments.

In some cases, such as the interactive mathematics and scientific literacy units, common functionalities were split out into shared programming libraries that could be reused in multiple units. For example, in the scientific literacy units the experimental data tabling and management functionality was built as a shared library. The library also managed the recording of data and supported scoring of the student's performance based on unit-specific criteria. Likewise, in reading literacy, the management and display of multiple sources in tabs was encapsulated into a shared library. In mathematics literacy the spreadsheet management functionality was built and used as a shared library.

The visual aspects of the PISA 2022 items and the automated coding of student responses were both implemented using JavaScript®. Shared libraries were created to implement this coding in a common way. The libraries targeted the various response modes used within PISA. These were:

- Form: for all responses using common web form elements such as radio buttons, checkboxes, dropdown menus and textboxes.
- Drag and Drop: for items using drag and drop as the response mode.
- Selection: for items where the response is given by clicking on an object or region of the screen. This can be, for instance, clicking on part of an image, a cell in a table or a segment of text.
- Ad hoc: A general catch all that uses custom JavaScript® code to implement the coding. This was used for unique situations, such as coding for interactive mathematics and scientific literacy items.

In all cases, except the ad hoc coding, the coding for a specific item was specified using rules composed of conditional expressions and Boolean operators. Each library implemented appropriate conditional expressions (e.g. a CONTAINS operator in the Drag and Drop library to test if a drop target held a particular drag element).

Translation and online item review

Given the need to support translations and adaptations for over 120 different national language versions of each unit, an automated process for the integration of these translations adaptations was critical. This process commenced with the initial development of the units. The HTML files that implement the display of the unit contained only HTML mark up and the text to be shown on the screen. Layout and formatting specifications were stored separately in CSS stylesheets. The text of the units was then extracted from these HTML files and saved to a standard file format, XLIFF (http://docs.oasis-open.org/xliff/v1.2/os/xliff-core.html), used for computer supported translation. Once a translation was completed, the XLIFF file was injected into the original source version of the unit, resulting in HTML files with the translated text of the unit.

One of the guiding principles of the platform development was that the quality of a translation is enhanced when translators can view their translation in the context of the stimulus and items they are translating. In an ideal world, translators would work in a completely WYSIWYG (what-you-see-is-what-you-get) mode, so that they enter their translations directly within an interface that displays the items as they would be seen by the students, but this was not technically feasible with the current authoring method. Furthermore, the visual aspects of the items, which are tightly controlled for comparability, may distract translators from the text to be translated. A good compromise was to provide translators with an easy-to-use preview feature to view their translations as functioning items from the PISA 2022 Portal. Users were able to upload a locally prepared and saved XLIFF file, with either partial or complete translation, and in a matter of seconds be able to preview the given unit in exactly the same design and layout as a student would view and interact with it within the student delivery system. This was an important factor, particularly for the more complex and interactive units across the domains. This preview also allowed countries to test and identify potential problems with their translated units before receiving the final versions packaged within the software to be used in schools. Therefore, reported problems were fixed as early in the schedule as possible.

School computer requirements

The goal for the PISA 2022 computer-based administration was, to the extent possible, to use the computers available in the sampled schools with no modifications or upgrades to existing hardware. The PISA 2022 system supported both Windows based and Macintosh computers and offered a Chromebook administration option as a pilot study in the field trial. All these options were also supported for the main survey. The following minimum technical requirements were established for the main survey:

	Windows	Macintosh	Chromebook
CPU Speed	1000MHz (1500 MHz Recommended)	1000MHz (1500 MHz Recommended)	N/A
Operating System	Windows 7, 8, 10 or 11	Mac OS X version 10.11 or later	Chrome OS with Google Chrome web browser version 57.0 or later
Installed Memory	1280 MB	2048 MB	N/A
Available Memory	774 MB (878 MB Recommended)	774 MB (878 MB Recommended)	N/A
Screen Resolution	1024 x 768 pixels	1024 x 768 pixels	1024 x 768 pixels
USB Transfer Rate	7.5MB/s (12MB/s Recommended)	7.5MB/s (12MB/s Recommended)	Download and Upload speed of 0.5 MB/s (2.0MB/s Recommended)

Computers with higher capabilities would obviously perform better (e.g. respond faster) when delivering the survey, but the requirement listed above were the minimum settings that would provide adequate performance.

System diagnostic

In order to verify that the available school computers met the minimum requirements, a system diagnostics application was provided to the national PISA centres within the participating countries/economies. The System Diagnostics is a version of the delivery system without the tests and questionnaires. It was intended to be given to schools to check the compatibility of the school computers with the PISA software. It checked the computer's hardware and software setup and reports results of this check back to the user, typically the test administrator or technical support staff in the school. Additionally, the user was given the option to run a modified version of the assessment using publicly available items to verify performance.

The system diagnostics was provided to countries approximately six months prior to the start of the field trial and main study. This allowed PISA centres to review the results in advance of the data collection period with time for an alternative solution to be implemented if minimum requirements were not met. Additionally, it was recommended that test administrators run the system diagnostics on the day of the test prior to conducting the assessment.

For cases where schools did not have adequate quality or quantity of computers, PISA centres arranged for test administrators to bring laptops into schools to augment the available infrastructure. In a few cases, countries chose to administer the PISA tests in all sampled schools on external laptops brought into the schools. This avoided "surprises" on the day of the test, where computers were not available or not functioning properly.

Test delivery system

The PISA 2022 test delivery system, called the student delivery system or SDS (CDS for the Chromebook delivery system), integrated the PISA computer-based assessments and questionnaires for a country/economy, along with a number of components packaged together to run as a standalone application on a USB drive. The SDS did not require network connectivity or external resources to operate.

All software and data were on a single USB drive, and results were saved back to the USB drive. The SDS could also be deployed from the computer's local hard drive or a network file server or terminal server if desired. The components which made up the SDS included the following:

- Electron framework (https://www.electronjs.org/)
- No SQL database engine (NeDB for the SDS version and MongoDB for the CDS version)
- Chromium™ open-source project web browser (https://www.chromium.org/chromium-os/).

The actual test and questionnaire content were included together with these open-source applications. The PISA 2022 test delivery system was implemented to display this content to the students and collect their responses. Using components of the open-source TAO test delivery system (http://www.taotesting.com/) as a basis, the system was custom built for the needs of PISA 2022. This included implementation of the test flow, which assigned the designated test form and questionnaires to a student, then sequences through the test units and questionnaires in the appropriate order. It also included the functionality for collecting the survey results and exporting them when the tests were completed. The PISA test delivery system was built using Electron, an open source, cross platform framework for creating applications using Chromium and Node.JS.

The system was launched by running a single executable program written for controlling the delivery of the PISA tests. Custom builds were developed for Windows and Macintosh operating systems. From this program, a test administrator could launch the PISA tests, launch the system diagnostics, or manage exported data files. These exported files are described below. Launching either the PISA tests or system diagnostics would start a local web server and in memory database, then launch a browser window to begin the process.

The Google Chromium browser used for the PISA tests was configured to run in “kiosk mode”, so that it filled the full screen of the computer, making it difficult for users to access external applications when running the PISA test mode. A keyboard filter was also installed so that students could not easily leave or terminate the browser window, e.g. by pressing Alt-Tab, and switch to another program during the test. The keyboard filter did not completely block such attempts, though. For example, it was also not possible to block the Ctrl-Alt-Delete sequence under Windows, as this required installation of a custom software driver at the system level. The goal was not to install any software on the school computers, so this driver was not used. It was expected that the test administrator would monitor the students during the test and watch for cases of students trying to break out of the system.

The first screen a student would see after the test was started was the option to select one of two sessions: Session 1 – The PISA Tests and Session 2 – The PISA Questionnaires. After selecting the appropriate session (which usually was done by the test administrator before the students arrived), the student was prompted for a login ID and password. The login ID was the 13-digit student ID assigned by the ACER Maple software as part of the sampling process. The password was also assigned by the ACER Maple software and was a 10-digit number. The first few digits comprised a checksum of the student ID, guarding against input errors. The next three digits encoded the test form which should be used for the student. The last few digits were a checksum of the three-digit test form number.

While the SDS was built with all the national languages available for a given country, it could be configured to support only one language. This was the recommended method of operation, where the test administrator chose the language configuration when starting the SDS, based on the school where the testing occurred. However, in some countries/economies, it was necessary to allow the students to choose the language of assessment. The typical reason for allowing student choice for the language was for countries and schools with mixed language environments. In these cases, In this situation, once logged in, the student would be shown a screen asking to select a language they wanted to use for the session. The test administrator would then guide students through the login and language selection process where applicable.

An important facet of the system setup was protecting the test content on the USB drives. The PISA tests contain secure test materials, and people who obtain a USB drive should not have access to the test items except during the administration of the assessment. To accomplish this, the files for rendering all test materials were stored in a NoSQL database on each USB drive. The files were stored in an encrypted format, and access to these was controlled via the web server. When a testing session was first started, the program would prompt for the password used to encrypt the files. Each country was assigned a unique password. This password was validated against known encrypted content in the database and then saved for the duration of the testing session. When a request was made to the web server for some part of the test content (e.g. one of the web pages or graphic images), the web server retrieved the content from the database and decrypted it on the fly.

One advantage of the SDS architecture implemented for PISA 2022 was that it could be run without administrator rights to the local computer. This was a big improvement over earlier PISA cycles, thus significantly reducing greatly the amount of technical support needed within the schools.

Data capture and scoring student responses

Student responses and other process data from the PISA tests and questionnaires were stored on the USB drives. Data was saved as the students answered each question, then exported at key intervals during the sessions. At the end of a session, the results from that session were exported in a single password protected ZIP file. For the PISA tests from Session 1 (the cognitive PISA domains, including the optional financial literacy domain), the ZIP files contained XML formatted data including logs of the students' actions going through the tests and files with the "variables" exported from the test. The following set of variables were exported for each item in the tests:

- Response: A string representing the raw student response.
- Scored Response: The code assigned to the response when the item was coded automatically.
- Number of Actions: The number of actions taken by the student during the course of interacting with the item. Actions counted were clicks, double-clicks, key presses and drag/drop events.
- Total Time: The total time spent on the item by the student.
- Time to First Action: The time between the first showing of the item and the first action recorded by the system for the item.

In addition to these five standard variables, some more complex items had custom variables that were of interest to the test development and psychometric teams. For instance, for the science simulations, the system exported counts of the number of experiments performed and the final set of results from each of these experiments.

An important task in PISA 2022 was coding of student responses. For computer delivered tests, many of the item responses could be coded automatically. In PISA 2022, this included multiple-choice items, drag-and-drop items, numeric-response items, and complex responses to mathematics or science simulations.

For standard response modes, such as multiple choice or numeric entry, automated coding was done using a rule-based system. The correct answer (or partially correct answers in the case of partial-credit items) were defined based on Boolean rules defined in a custom syntax. Simple conditional statements were possible, e.g. to support different combinations of checkboxes in a multiple selection item where two out of three correct options should be selected. For numeric response items, the rules could check for string matches, which required an exact match against a known correct answer, or numeric matches, which used numeric equivalence to check an answer. For numeric equivalence, for instance, 34.0 would match 34, but they would not match when using string matching.

A challenging part of evaluating numeric responses in an international context like PISA is how to parse the string of characters typed by the student and interpret it as a number. There are differences in decimal and thousands separators that must be taken into account, based on conventions used within countries and local usage. Use of these separators is not always consistent within a country/economy. For PISA 2022, the coding rules tried multiple interpretations of the student response to see if one of them could be coded as correct. The numbers were parsed in different ways, changing the decimal and thousands separators, testing each option to see if a correct response could be granted full or partial credit. Only if no alternate interpretation of the response resulted in a correct answer would the answer be coded as incorrect.

Open-ended coding system

While automatically coded items formed a significant portion of the units for PISA 2022, approximately 30% of the items required a response that needed to be coded by a human scorer or coder. On paper, this would be done directly on the test booklets. On the computer, a procedure was necessary to extract the responses provided by the students and present them to human coders. It was important to present these responses in a way that reflected the students' intent. This task is complicated by the fact that these responses could be more than just text. For example, for some items, a student would be required to select an option from a multiple-choice part, then type in an explanation for why they chose that option. Additionally, in mathematics, students could use an equation editor to insert complex mathematics notation into their response.

For PISA 2022, the coding of these responses was done using the open-ended coding system (OECS). The OECS is a computer tool that was developed to support the coders in their work to code the responses according to the coding guides. All PISA 2022 open-ended responses collected with the computer-based platform were coded using the OECS.

The OECS works online so it required coders to have a reliable network connection. The OECS organizes responses according to the coding designs for each of the assessment domains. The system gives coders access to all the responses assigned to the coder. For each response, the coder will have access to part of the question for reference, the individual response to be coded, and the acceptable codes for each question. The coder selects the appropriate code and clicks on the "Record Code" button to saves the selected code. It should be noted that this system was only used for response data from the computer-based assessment.

Also included on each page of the OECS were two checkboxes labelled "recoded" and "defer." The recoded box was used when the response had been recoded by another coder. The defer box was used when the coder was not sure what code to assign to the response. These deferred responses were typically reviewed and coded by the Lead Coder, or by the coder after consultation with the Lead Coder. When deferring a code for a response, coders were encouraged to enter comments into the box labelled "comment" to indicate the reason for deferring.

The OECS included the necessary features to support the monitoring of reliability. It organized all anchor, multiple and single coding of responses. According to a predetermined design, some responses were single coded – coded by one person only – while others will be multiple coded – coded by more than one coder. Anchor responses (in English) were used to assess reliability across countries. Since the OECS gives coders only those responses that are assigned to them, coders do not know whether they are single or multiple coding. Once coding was complete for each item, the data was integrated across coders and the OECS generated reliability reports that included multiple sections such as i) a summary, ii) item overview, iii) coders overview, iv) proportion agreement, v) coding category distributions, and vi) deferred and uncoded report.

22 International data products

Following the data processing and data analysis, data products were delivered to the OECD. These included public-use data files and codebooks, compendia tables, and the PISA Data Explorer, a data analysis tool. These data products are available on the OECD website (http://www.oecd.org/pisa/). The IEA IDB Analyzer was configured to work with PISA data and can be downloaded from www.iea.nl.

Public-use files

The public-use files (PUF) contain response records from all participating countries/economies that are part of the approved PISA sample. Student-level files contain over 6000 variables that include responses to the background questionnaire and the cognitive assessments, as well as sampling weights, proficiency estimates and variables derived from responses to the background questionnaire. The student and teacher files contain over 1000 variables. The variables included in the PUF represent a common subset of the variables that were collected across all participating countries/economies and are available on the OECD website at http://www.oecd.org/pisa/.

Variables excluded or suppressed for some or all countries

The PUF include only a subset of the variables included in the individual country files. The PUF do not include any data collected using national adaptations and extensions. Rather, they include common data that were collected or derived across all countries. Additionally, variables were also excluded after consultation with the OECD Secretariat because they i) have little or no analytical utility, ii) were intended for internal or interim purposes only, iii) relate to secure item material, or iv) include personally identifiable data, or at least data that may increase the risk of unintended or indirect disclosure.

The groups of variables excluded from the PUF are:

- direct, indirect, and operational identifiers for respondents;
- certain background questionnaire (BQ) or process variables such as free text entry responses and random numbers used by the SDS to determine routing;;
- all national adaptations and extensions in the BQ;
- original scale score values (theta) before standardisation to an international metric.

Countries were given the option of suppressing variables in the PUF. Suppression of variables was approved when data presented a risk to student, school, and/or teacher anonymity. Suppressed data are represented in the database by means of missing codes.

Data files

Data files are provided in both SAS and SPSS formats. The files include:

- Student questionnaire data file: This file includes ID variables, all student questionnaire response data, parent-questionnaire response data, student and parent background questionnaire scale and derived variables, plausible values for the core domains (Reading, Math, and Science), and overall and replicate student weights.

- School questionnaire data file: The school questionnaire data file includes ID variables, school questionnaire response data, school questionnaire scale and derived variables, and an overall school weight.
- Teacher questionnaire data file: The teacher questionnaire data file includes ID variables, teacher questionnaire response data, and teacher questionnaire scale and derived variables, and overall and replicate teacher weights.
- Cognitive item data file[1]: The cognitive data file includes ID variables, raw and coded item responses, item log data for the computer-based assessment (e.g., total time and number of actions) for the core domains (Mathematics, Reading, Science).
- Creative Thinking cognitive data file: The cognitive data file includes ID variables, Creative Thinking raw and coded item responses, computer-based assessment (CBA) item log data (total time and number of actions); and Creative Thinking plausible values including the Maths, Reading, and Science plausible values that were created as part of the population model with the Creative Thinking cognitive data.
- Financial Literacy student questionnaire data file[2]: This file includes ID variables, all student questionnaire response data, parent-questionnaire response data, student and parent background questionnaire scale and derived variables, plausible values for the domains assessed (Financial Literacy, Reading, and Maths), and overall and replicate student weights for the optional financial literacy sample.
- Financial Literacy cognitive item data file[1 2]: The cognitive data file includes ID variables, raw and coded item responses, item log data for the computer-based assessment (e.g., total time and number of actions) for the domains assessed in the Financial Literacy sample (Financial Literacy, Maths, Reading).
- Questionnaire timing data file: The questionnaire timing data file includes CBA questionnaire log data (i.e., total time on a unit/screen).

The Creative Thinking datasets and Financial Literacy datasets are scheduled to be published in 2024.

Variables used in sampling, weighting and merging

The variable STRATUM is included to identify sampling strata. The variable is created as a concatenation of a three-letter country code and a two-digit original stratum identifier.

The variables W_FSTUWT and W_FSTURWT1 - W_FSTURWT80 represent the full student sampling weight, and the 80 replicate weights used for estimation of sampling variance.

The variable SENWT is a normalised weight variable typically used for analyses of student performance across a group of countries where contributions from each of the countries in the analysis is desired to be equal regardless of their population or sample size. The senate weight adds to a constant of 5 000 across all cases within each country/economy in the file. This weight adds to 5000 within each country/economy only when there is no missing data for the variable of interest. The relative contribution of each country/economy is affected by the incidence of missing data.

The student and teacher data files can be merged to the school data file using the variable CNTSCHID. CNTSCHID is the combination of the three-digit country code and a randomised five-digit school ID number, making it unique across all countries.

Codebooks for the PISA 2022 public-use data files

Included with the PISA 2022 Main Survey data products is a set of data codebooks in Excel format. The data codebook is a printable report containing descriptive information for each variable contained in a corresponding data file. The codebooks report frequencies and percentages for all categorical variables from the cognitive and background questionnaire variables, as well as those that have been derived and/or added during data processing. The codebooks are available from the OECD website (https://www.oecd.org/pisa/data/).

The information is displayed with variable names, variable labels, values and value labels. Other metadata are provided, such as variable type (e.g., string or numeric) as well as precision/format. Additionally, the codebooks contain the range of valid values (minimum and maximum) for non-categorical numeric variables.

Codebooks for the main files are contained in separate worksheets within the file made available at the OECD website. Each worksheet corresponds to one of the eight public-use data files described above.

Data compendia tables

Using the PUF as the source data, the compendia are sets of summary tables that provide percentages for both cognitive and background items. The compendia support public-use file users so that they can gain knowledge of the contents of the data files and use the compendia results to confirm that they are performing analyses on the PUF correctly. The compendia are available on the OECD website (http://www.oecd.org/pisa/).

Questionnaire compendia provide the distribution of students according to the variables collected with the questionnaires. Cognitive compendia provide the distribution of student responses for each test item. Results are provided in Excel format, separately for background questions and test items, and are further broken out by type of questionnaire and by domain (and by gender for cognitive items). Each Excel file contains multiple worksheets, with each worksheet corresponding to a single variable. The first worksheet in each file is a table of contents that contains a hyperlink to each variable so users can see at a glance which variables are available and can click to go directly to the desired data.

Separate tables are provided with percentage and percentile data for continuous background variables across all questionnaires.

All statistics including in the compendia are calculated using weighted data and are presented with their corresponding standard error that take into account both the sampling and measurement uncertainty. The OECD average is created as the simple average of the 38 current OECD member countries.

Data analysis and software tools

Standard analytical packages for the social sciences and educational research do not readily recognise or support handling the complex PISA sample and assessment design. This gap is filled by the two software tools made available to assist database users to access and analyse PISA data and produce basic outputs: The PISA Data Explorer (PDX) and the IEA's International Database Analyzer (IDB Analyzer). Each of these two software tools address a slightly different set of needs. While the PDX is a web-based application that allows relatively easy and publication-ready access to basic estimates of means, totals and proportions, the IEA's IDB Analyzer used in conjunction with the PUFs allows unit record access to the public-use database and the opportunity to conduct analysis offline, derive additional variables, and produce various estimates for further use and reporting. The PDX and IEA's IDB Analyzer are described in turn in the remainder of this chapter.

PISA Data Explorer (PDX)

The PDX is a web-based application that allows the user to query an OECD hosted, secure, PISA International Database via a web browser. In addition to the PISA 2022 data, the PDX database contains data from previous cycles of PISA. The PDX is available on the OECD website (https://pisadataexplorer.oecd.org/ide/idepisa/) and provides access to a secure PISA database that is protected by the OECD firewalls and security mechanisms. The PDX allows the user to navigate, analyse, and produce report quality tables and graphics.

The database underlying the PDX is populated using the PUF to import more than 3.5 million unique student records across eight PISA cycles. About 8,700 variables across eight assessment cycles and over 100 countries, economies, and adjudicated subregions are available for analysis. Because certain variables that are included in the public-use file (PUF) for secondary analysis are not informative as part of the PDX, they are not included in the PDX database. The majority of variables included in the PUF but not the PDX relate to the individual cognitive item responses and process information.

The PDX can be used to compute a diverse range of statistics including, but not limited to, means, standard deviations, standard errors, percentages by subgroup, percentages by performance levels, and percentiles. All statistics are computed taking into account the sampling and assessment design. In addition, the PDX has the capability of conducting significance testing between statistics from different groups and displaying the results in graphical form.

In the PISA Data Explorer, the International Average (OECD) includes all OECD member countries for which data are available for the corresponding subject and year (38 OECD Member countries as of PISA 2022). Depending on data availability, the countries contributing to this average might vary by cycle and subject.

Because it is web-based, and processing takes place on a central server, the PDX can be accessed and used with computers that meet fairly simple requirements. The user’s computer is used only to create a request or data query, deliver the request to a central server where processing takes place, and then receive and display back the results in a user-friendly format.

A typical query consists of the user selecting the domain(s), jurisdiction(s), and variable(s) of interest. Then the user proceeds to select the statistics of interest and format the table. Statistics are calculated for each of the subgroups defined by the variables selected, for one variable at a time or in cross-tabulation mode. In addition, the user is able to collapse categories for each of these variables and used the collapsed categories in the analysis. All statistics are calculated using weighted data, with their corresponding standard errors taking into account sampling and measurement uncertainty. The user has the option to select whether the standard errors are displayed in the table or not, as well as the precision with which the statistics are displayed. The results can then be displayed in a table or in a graphic.

Regardless of whether the results are displayed in a table or graphic mode, the results can be saved or exported for further post processing or for inclusion in an external document. Export formats currently available include MS Word, MS Excel, PDF and HTML.

A significance test module allows the user to specify significance testing to be done between subgroup means, percentages and percentiles, within and across cycles, while implementing necessary adjustments that take into account the sample and test design, as well as adjustment for multiple comparisons. Significance test results can be displayed in table or in graphic format.

Table results can be easily exported and manipulated using spreadsheet software, allowing the user to customise the titles and legends of the tables, and to do any required post processing. Likewise, the graphic results can also be exported to be included in documents and used in reports and presentations.

The web application is compatible with many widely used browsers including Microsoft Edge, Firefox, Google Chrome, and Safari.

Import of trend data

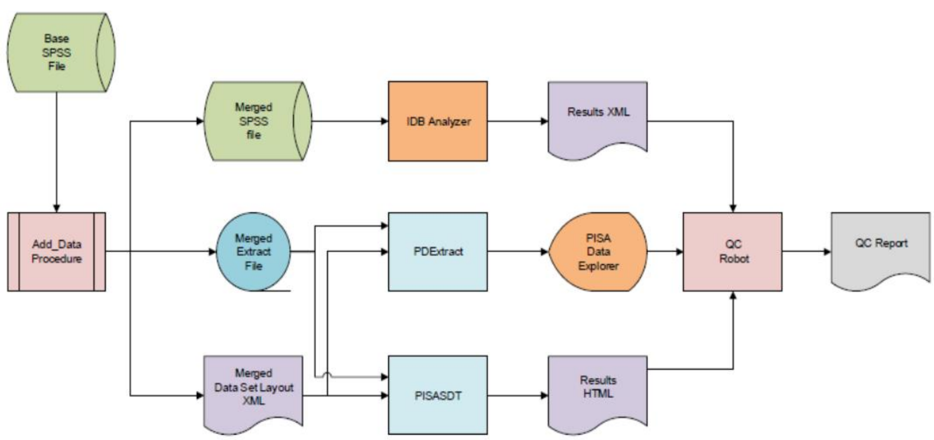
The PISA trend data from 2000 to 2012 were imported into the PDX directly from a database that had been established earlier by the United States Department of Education to develop and support a Data Explorer for PISA and other international studies. These data were taken from all PUF that were available for those cycles and were updated with all subsequent releases of modified or additional data. This approach ensured that all calculated results were consistent with all available OECD reports.

An important outcome of this prior work was the establishment of a naming convention for all data variables to ensure that valid trend comparisons could be made across cycles, even though the variable names as used in the public-use file data were not consistent across cycles. This naming convention was extended and applied to all of the variables in subsequent PISA cycles (2015, 2018, and 2022) in order to ensure continuity and comparability with previous cycles.

Population and quality check of the PISA Data Explorer

The process to populate the PISA Data Explorer database and confirm the results it produces is summarised in Figure 22.1 below. This process was applied separately to the data from each country.

Figure 22.1. PISA database population and quality control



The Base SPSS file contained the data as forwarded to the appropriate country for its analysis and reporting.

The Add_Data procedure performed two functions. The first was conditional on whether a country provided supplemental data that was collected or derived and merged these data with the Base file. The second function created two files from the enhanced Base file: an ASCII text rectangular file containing the data values extracted from the Base file and an XML file containing information about the extracted data variables (location, format, labels). This Data Set Layout (DSL) XML is structured in a proprietary ETS schema.

The PDExtract program used the information from an input parameter file to process the data from the Extract file and metadata from the DSL file to produce a series of text files suitable for loading into the appropriate tables in the PISA Data Explorer (PDX) database. The program also produced a SQL script that is customised for performing the loading of these tables and contains a procedure for forming the data tables used by the PDX.

The PISASDT program also used the information from an input parameter file as well as a list of data variable names to calculate and produce summary data tables (SDT) – one analysis for each scale score. Each table in the analysis was a one-way tabulation of various statistics for each category of a given variable. The statistics pertained to a scale score and include percentage, average score and percentages within the benchmark levels. Each statistic was accompanied by the standard error estimate, degrees of freedom, number of cases on which the statistic is based and number of strata on which the standard error was based. All of these results were stored in an HTML document in full precision. This document may be viewed with any of the popular Internet browsers when accompanied by the appropriate Cascading Style Sheet (CSS) document, which ETS has produced and is available upon request to the OECD Secretariat[3]. The document may also be parsed or translated to produce Excel workbooks and report quality tables, among others.

In the QC Robot procedure, the Results HTML document from the PISASDT program was used to generate analysis requests for the PDX, one for each variable, and the results returned from the PDX were compared with those in the HTML document. The results of these comparisons were posted to the QC Report document where differences above specified criteria were flagged and subsequently examined.

The only statistics that can be reported in the PDX which cannot be calculated by the PISASDT program are the percentiles. Because the calculation of the percentiles within the PDX uses more resources than the other statistics, only a subset of critical variables was selected for quality-assurance analysis. The Analyzer reads data from the Base SPSS file, uses SPSS macros to calculate the desired percentile statistics, and writes the results to an XML file. The QC Robot procedure processed this XML file in the same way as the HTML file from the PISASDT program and added the comparison results to the QC Report file.

Prior to the first execution of the procedure described above, the Analyzer and the PISASDT programs were extensively calibrated with each other to ensure that the Merged SPSS and Merged Extract files were isomorphic and produced identical results for the statistics common to both programs.

IEA’s International Database Analyzer

The IEA International Database Analyzer (IDB Analyzer) is an application developed by the International Association for the Evaluation of Educational Achievement (IEA) that can be used to analyse data from most major large-scale assessment surveys, including those conducted by the OECD, such as PISA. Originally designed for international large-scale assessments, it is also capable of working with national assessments such as the United States National Assessment of Educational Progress (NAEP).

The IDB Analyzer creates SPSS, SAS, or R syntax that can be used to perform analysis with these international databases. The syntax considers information from the sampling design in the computation of sampling variance and handles the multiple plausible value imputations. The code generated by the IDB Analyzer enables the user to compute descriptive statistics and conduct statistical hypothesis testing among groups in the population without having to write any programming code.

The IDB Analyzer is licensed free of cost, not sold, and is for use only in accordance with the terms of the licensing agreement. While anyone can use the software for free, users do not have ownership of the software itself or its components, including the SPSS, SAS, or R macros, and users are only authorised to use the SPSS, SAS, and R macros in combination with the IDB Analyzer, unless explicitly authorised by the IEA. The software and license expire at the end of each calendar year, when the user will again have to download and reinstall the most current version of the software and agree to the new license. A complete copy of the licensing agreement is included in the Appendix of the Help Manual of the IDB Analyzer.

The analysis module of the IDB Analyzer provides procedures for the computation of means, percentages, standard deviations, correlations, and regression coefficients for any variable of interest overall for a country, and for specific subgroups within a country. It also computes percentages of people in the population that are within, at, or above benchmarks of performance or within user-defined cut points in the proficiency distribution, percentiles based on the achievement scale, or any other continuous variable.

The analysis module can be used to analyse data files from PISA. The following analyses can be performed with the analysis module:

- Percentages and means: Computes percentages, means, design effects and standard deviations for selected variables by subgroups defined by the user. The percent of missing responses is included in the output. It also computes t-test statistics of group mean differences taking into account sample dependency.
- Percentages only: Computes percentages by subgroups defined by the user.
- Linear regression: Computes linear regression coefficients for selected variables predicting a dependent variable by subgroups defined by the user. The IDB Analyzer has the capability of including plausible values as dependent or independent variables in the linear regression equation. It also has the capability of contrast coding categorical variables (dummy or effect) and including them in the linear regression equation.
- Logistic regression: Computes logistic regression coefficients for selected variables predicting a dependent dichotomous variable, by subgroups defined by the user. The IDB Analyzer has the capability of including plausible values as independent variables in the logistic regression equation. It also has the capability of contrast coding categorical variables and including them in the logistic regression equation. When used with SAS, the user can also specify multinomial logistic regression models.
- Benchmarks: Computes percent of the population meeting a set of user-specified performance or achievement benchmarks by subgroups defined by the user. It computes these percentages in two modes: cumulative (percent of the population at or above given points in the distribution) or discrete (percent of the population within given points of the distribution). It can also compute the mean of an analysis variable for those at a particular achievement level when the discrete option is selected as well as the computation of group mean and percent differences between groups taking into account sample dependency.
- Correlations: Computes correlation for selected variables by subgroups defined by the grouping variable(s). The IDB Analyzer is capable of computing the correlation between sets of plausible values.
- Percentiles: Computes the score points that separate a given proportion of the distribution of scores by subgroups defined by the grouping variable(s).

When calculating these statistics, the IDB Analyzer has the capability of using any continuous or categorical variable in the database or make use of scores in the form of plausible values. When using plausible values, the IDB Analyzer generates SPSS, SAS, or R code that takes into account the multiple imputation methodology in the calculation of the variance for statistics, as it applies to the corresponding study.

All procedures offered within the analysis module of the IDB Analyzer make use of appropriate sampling weights and standard errors of the statistics that are computed according to the variance estimation procedure required by the design as it applies to the corresponding study.

Notes

[1] After the analysis phased completed, it was determined that 4 students in Iceland's grade-based sample were analysed along with Iceland's main sample data. As a result, the public use data for Iceland excludes these 4 students, yet they are still included in PISA 2022 technical report tables where Iceland data are referenced.

[2] For Financial Literacy, only a subset of participants for Canada and Belgium received the Financial Literacy assessment and it is not a nationally representative sample. Only the Belgium Flemish community as well as the Canadian provinces British Columbia, Manitoba, New Brunswick, Newfoundland and Labrador, Nova Scotia, Ontario and Prince Edward Island, participated in Financial Literacy for PISA 2022.

3. Please send your request via email to EDU.Pisa@oecd.org.

Annex A. Item Pool Classification tables

Tables A. Item Pool Classification tables

Table	Title
Table A.1	Math - Computer-based assessment
Table A.2	Math - New paper-based assessment
Table A.3	Math - Old paper-based assessment
Table A.4	Reading - Computer-based assessment
Table A.5.	Reading Fluency - Computer-based assessment
Table A. 6	Reading - New computer-based assessment
Table A.7	Reading Components - New paper-based assessment
Table A.8	Reading - Old paper-based assessment
Table A.9	Science - Computer-based assessment
Table A.10	Science - New paper-based assessment
Table A.11	Science - Old paper-based assessment

StatLink https://stat.link/5nw02t

Annex B. Contrast Coding Tables

Tables B. Contrast Coding Tables

Table B.1	Contrast Coding BQ Variables

StatLink https://stat.link/

Annex C. Student and School Sample Size Tables

Tables C. Student and School Sample Size Tables

Table C.1	Main Sample Sizes by Country Domain
Table C.2	Financial Literacy Sample Sizes by Country and Domain

StatLink https://stat.link/

Annex D. National Household Possession Items Tables

Table D. National Household Possession Items

	Variable Name			
	ST250Q06JA	ST250Q07JA	ST251Q08JA	ST251Q09JA
OECD Countries				
Australia	a home theatre	a pool or outdoor spa bath/jacuzzi	air conditioning unit	solar panels
Austria	swimming pool/pond	n/a	n/a	n/a
Belgium	a room where you can study quietly	n/a	antiques	n/a
Canada	gaming console (e.g. Nintendo Switch™, XBox®, PlayStation®)	your own sports equipment	Smart home devices (e.g. smart thermostat, Google Home™, Amazon Echo Dot™)	Television or video subscription service (e.g. cable TV, Netflix®, Apple TV®)
Chile	printer	scanner	digital video camera	exercise machines that are working
Colombia	television	refrigerator	video game console	n/a
Costa Rica	your own TV	3D screen	cars	3D screens
Czech Republic	n/a	n/a	n/a	n/a
Denmark	your own game console	your own headphones	boat	n/a
Estonia	game console (PlayStation®, Xbox®)	your own table for studying	n/a	n/a
Finland	alarm system	garage or carport	n/a	n/a
France	a paying televison program (e.g. Neflix, Canal Plus, OCS)	an action camera (e.g. GoPro)	n/a	n/a
Germany	a desk to study at	a quiet place to study	n/a	n/a
Greece	iPod and MP3 players	digital games (e.g Playstation4 ®)	dishwasher	home alarm system
Hungary	air conditioner	dishwasher in kitchen	Video game console (e.g. Sony PlayStationTM)	Digital camera (not built in a cell phone)
Iceland	n/a	n/a	n/a	n/a
Ireland	n/a	n/a	Subscription to TV or streaming services	n/a
Israel	4WD car	membership to the theater, gym, or swimming pool	n/a	n/a
Italy	printer	n/a	air conditioners	n/a
Japan	game console (e.g. PlayStation 4®, Nintendo Switch™)	passport	air conditioner	rooms for visitors
Korea	a desk to study at	massage chair	air conditioner	air cleaner
Latvia	bicycle	scooter	antique things	textile works
Lithuania	n/a	n/a	n/a	n/a
Mexico	a desk to study at	a quiet place to study	n/a	n/a
Netherlands	your own personal computer or laptop	your own tablet (e.g. iPad, Samsung Galaxy)	a subscription to a newspaper	an electric car

	Variable Name			
	ST250Q06JA	ST250Q07JA	ST251Q08JA	ST251Q09JA
New Zealand	your own snowboard or skis	your own musical instrument (e.g. guitar, keyboard)	heat pumps	Large outdoor recreation items (e.g. tent, boat, mountain bike, surfboard)
Norway	a good place to do school work	n/a	electric bicycles	n/a
Poland	n/a	n/a	washer	n/a
Portugal	cable TV or satellite TV	n/a	poetry books	n/a
Slovak Republic	n/a	n/a	n/a	n/a
Slovenia	multifunction printer	sports equipment (e.g. skis, bike, tennis racket, etc.)	external hard disk	sauna
Spain	media services for TV series and films (HBO, Netflix)	pay TV (Movistar+, Orange TV)	dishwashers	parking places
Sweden	n/a	n/a	home cinema	n/a
Switzerland	n/a	n/a	dishwashers	mowing machines
Türkiye	TV subscriptions with payment (e.g. Digiturk, Tivibu and Teledunya)	helper for houseworks (part time or full time)	Air conditioning type heating-cooling system	LCD, LED TV or Plasma TV
United Kingdom (Excl. Scotland)	a games console such as a PlayStation 4® or Nintendo Wii®	a smart speaker such as Amazon Echo or Google Home	a study desk or table for your use	computers (e.g. desktop, latop or tablet)
United Kingdom (Scotland)	your own bicycle	your own smartwatch	spaces to park cars	outdoor spaces attached to your home (e.g. garden)
United States	n/a	n/a	n/a	n/a
Partner Countries/Economies				
Albania	n/a	n/a	n/a	n/a
Argentina	a quiet place to study	n/a	n/a	n/a
Baku (Azerbaijan)	washing machine	n/a	n/a	n/a
Brazil	cable TV	your own desk to study at	game console with internet access	refrigerator
Brunei Darussalam	bedroom with an air conditioner	Video or online games (e.g. used with game consoles such as a PlayStation 4®)	rooms with marble floor	surveillance camera or CCTV
Bulgaria	digital camera	air conditioning	n/a	n/a
Cambodia	books to help with your school work	a dictionary	refrigerator	smart televisions (Internet connectivity)
Croatia	n/a	n/a	dishwasher	air conditioner
Cyprus	a swimming pool	a home security alarm system	Cable or Satelite TV (e.g. Cablenet, Cytavision, Nova, PrimeTel)	Game Platforms (e.g. Playstation, Nintendo Switch/Wii)
Dominican Republic	wrist watch	your own car	air conditioner	smart TV
El Salvador	typewriters	microwave	trees	pets
Georgia	video games	n/a	n/a	n/a
Guatemala	stereo	blender	books to help with your school work	n/a
Hong Kong (China)	storeroom	Newspaper or Educational Magazine (e.g. National Geographic Magazine)	television	air conditioning unit
Indonesia	your own Personal Computer/ Desktop / Laptop	your own tablet/iPad	refrigerator	oven
Jamaica	cable TV	portable Wi-fi	TV	car
Jordan	iWatch	digital books	antiques	office rooms

	Variable Name			
	ST250Q06JA	ST250Q07JA	ST251Q08JA	ST251Q09JA
Kazakhstan	bicycle	digital photo camera	bicycle	digital photo camera
Kosovo	n/a	n/a	n/a	n/a
Macao (China)	safe box	electric massage chair	air purifier	hi-fi audio set
Malaysia	printer	refrigerator	pressure cooker	air conditioning unit
Malta	n/a	n/a	smart TV with internet access	n/a
Mongolia	n/a	n/a	silver bowl	carpet
Montenegro	n/a	n/a	n/a	n/a
Morocco	swimming pool	electric water heater tank	smart TV	fishing boat
North Macedonia	n/a	n/a	n/a	n/a
Palestinian Authority	iWatch	digital books	antiques	office rooms
Panama	all in one printer	simulation tools	cable or satellite TV	internet TV
Paraguay	n/a	n/a	n/a	n/a
Peru	Playstation, Nintendo, Wii	bike	refrigerator	stereo
Philippines	means of transportation (e.g. motorcycle, tricycle, jeepney, car, etc.)	air conditioning unit	Video game console (PlayStation, Xbox, etc.).	smart TV
Qatar	office	cinema	digital video camera	video game console
Republic of Moldova	n/a	n/a	n/a	n/a
Romania	air conditioning	cable/satellite TV	n/a	n/a
Saudi Arabia	gaming system	n/a	n/a	n/a
Serbia	n/a	n/a	LED/LCD/Plasma TV	digital camera
Singapore	air conditioner	domestic helper (e.g. full/part-time maid)	n/a	n/a
Chinese Taipei	n/a	n/a	n/a	n/a
Thailand	smart television	air purifier	air conditioner	refrigerator
Ukraine	your own desk to do hometasks	reference books	family heirlooms	smart TV
United Arab Emirates	cinema	electronic games with internet access	luxury cars (e.g. Bentley, Rolls-Royce)	domestic Workers (e.g. housemaid, Drivers, etc.)
Uruguay	a desk to study	n/a	smart TV	n/a
Uzbekistan	TV with an access for international channels	bookshelf	bicycle	refrigerator
Viet Nam	a dictionary	a set of chair and table for learning	air conditioner	n/a

Annex E. Final Distribution of RMSD Values Across Groups for Each Scale Tables

Tables E. Final Distribution of RMSD Values Across Groups for Each Scale

Table E.1	RMSD values for BQ scales

StatLink https://stat.link/

Annex F. Common and Unique Item Parameters in Each Domain, by Country and Language Tables

Tables F. Common and Unique Item Parameters in Each Domain, by Country and Language

Table F.1	Consolidated common unique item parameters
Table F.2	Summary
Table F.3	Unique item parameters

StatLink https://stat.link/efz3n0

Annex E. Equated P Tables

Tables G. Equated P Tables

Table G.1	Guidelines
Table G.2	Equated P+
Table G.3	Equated P+ (Standard error)
Table G.4	Country economy code 3-character

StatLink https://stat.link/5dqy71

Annex F. Testing Periods Tables

Tables H. Testing Periods

Table H.1	Testing periods worksheet

StatLink https://stat.link/

Annex G. PISA 2022 Contractors, Staff and Consultants

PISA is a collaborative effort, bringing together experts from the participating countries, steered jointly by their governments based on shared, policy-driven interests.

A PISA Governing Board, on which each country is represented, determines the policy priorities for PISA, in the context of OECD objectives, and oversees adherence to these priorities during the implementation of the programme. This includes setting priorities for the development of indicators, for establishing the assessment instruments, and for reporting the results.

Experts from participating countries also serve on working groups that are charged with linking policy objectives with the best internationally available technical expertise. By participating in these expert groups, countries ensure that the instruments are internationally valid and take into account the cultural and educational contexts in OECD member and partner countries and economies, that the assessment materials have strong measurement properties, and that the instruments place emphasise authenticity and educational validity.

Through National Project Managers, participating countries and economies implement PISA at the national level subject to the agreed administration procedures. National Project Managers play a vital role in ensuring that the implementation of the survey is of high quality, and verify and evaluate the survey results, analyses, reports and publications.

The design and implementation of the surveys, within the framework established by the PISA Governing Board, is the responsibility of external contractors. For PISA 2022, the overall management of contractors and implementation was carried out by the Educational Testing Service (ETS) in the United States as part of its responsibility as the Core A contractor. The OECD Secretariat worked closely with the International Project Director and Project Manager, to co-ordinate all aspects of implementation. In addition to overall management, Core A was responsible for the computer-delivery platform, instrument development, scaling and analysis, and all data products. As the lead of Core A, ETS worked in co-operation with Westat in the United States for survey operations, cApStAn for translation and verification of the assessment instruments, the International Association for Evaluation of Educational Achievement (IEA) in the Netherlands for the data management software,

The additional tasks related to the implementation of PISA 2022 were carried out by three additional contractors – Cores B1, B2, B3, C, D, and E.

The Research Triangle Institute (RTI) in the United States facilitated the development of the mathematics assessment framework as the Core B1 contractor. ETS also facilitated the development of the background questionnaire frameworks as the Core B2 contractor. ACT in the United States and Cito in the Netherlands performed the test development for the innovative domain as the Core B3 contractor. Core C focused on sampling and was implemented by Westat in the United States in co-operation with the Australian Council for Educational Research (ACER). Core D was managed by cApStAn Linguistic Quality Control in Belgium for linguistic quality control in co-operation with BranTra in Belgium. Core E focused on country preparation and implementation support and was managed by the Australian Council for Educational Research (ACER) in Australia.

The OECD Secretariat has overall managerial responsibility for the programme, monitors its implementation daily, acts as the secretariat for the PISA Governing Board, builds consensus among countries and serves as the interlocutor between the PISA Governing Board and the international Consortium charged with implementing the activities. The OECD Secretariat also produces the indicators and analyses and prepares the international reports and publications in co-operation with the PISA Consortium and in close consultation with member and partner countries and economies both at the policy level (PISA Governing Board) and at the level of implementation (National Project Managers).

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Annex H. PISA 2022 Technical Standards and guidelines

Purpose of document

The purpose of this document is to list the set of standards upon which the PISA 2022 data collection activities will be based, as was the case for previous PISA. In following the procedures specified in the standards, the partners involved in the data collection activities contribute to creating an international dataset of a quality that allows for valid cross-national inferences to be made.

The standards for data collection and submission were developed with three major, and inter-related, goals in mind: consistency, precision and generalisability of the data. Furthermore, the standards serve to ensure a timely progression of the project in general.

- *Consistency*: Data should be collected in an equivalent fashion in all countries, using equivalent test materials that were translated and/or adapted as appropriate. Comparable samples of each country's student population should perform under test conditions that are as similar as possible. Given consistent data collection (and sufficiently high response rates), test results are likely to be comparable across regions and countries. The test results in different countries will reflect differences in the performance of the students measured, and will not be caused by factors which are un-related to performance.
- *Precision*: Data collection and submission practices should leave as little room as possible for spurious variation or error. This holds for both systematic and random error sources, e.g. when the testing environment differs from one group of students to another, or when data entry procedures are questionable. An increase in precision relates directly to the quality of results one can expect: The more precise the data, the more powerful the (statistical) analyses, and the more trustworthy the results to be obtained.
- *Generalisability*: Data are collected from specific individuals, in a specific situation, and at a certain point in time. Individuals to be tested should be selected, and test materials and tasks etc. be developed in a way that will ensure that the conclusions reached from a given set of data do not simply reflect the setting in which the data were collected but hold for a variety of settings and are valid in the target population at large. Thus, collecting data from a representative sample of the population, for example, will be essential, but not sufficient, for the results to accurately reflect the level of literacy of fifteen-year-old students in a country.
- *Timeliness*: Consistency, precision and generalisability of the data can be obtained in a variety of ways. However, the tight timelines and budgets in PISA, as well as the sheer number of participating countries, preclude the option of developing and monitoring local solutions to be harmonised at a later stage in the project. Therefore, the standards specify one clear-cut path along which data collection, coding and data submission should progress.

This document strives to establish a collective agreement of mutual accountability among countries, and of the international contractor towards the countries. This document details each standard, and the quality assurance and quality management plan to demonstrate that the standard has been met. While the terms quality assurance and quality control are sometimes used interchangeably, they relate to different aspects of quality. Quality assurance is most often associated with the processes and procedures that are put in place to make sure the survey is likely to meet its intended goals. Quality control, on the other hand, relates to the set of judgements that are made with regard to the suitability of the quality assurance procedures and the suitability of the survey results in terms of their intended uses or applications.

Where standards have been fully met and data quality of the final databases judged as appropriate, the international contractors will recommend to the OECD Secretariat that the data be included in the PISA 2022 database. Where standards have not been fully met or data quality has been questioned, an adjudication process will determine the extent to which the quality and international comparability of the data have been affected or whether additional analysis or evidence are necessary. The result of data adjudication will determine whether the data will be recommended for inclusion in the PISA 2022 dataset.

In principle each dataset should be evaluated against all standards jointly. Also, it is possible that countries' proposed plans for implementation are not, for various and often unforeseen circumstances, actually implemented (e.g. national teacher strike affecting not only response rates but also testing conditions; unforeseen National Centre budget cuts which impact on testing, printing and data management quality). Therefore, the final evaluation of standards needs to be made with respect to the data as submitted since this is the definitive indication of what may appear in the released international dataset.

If any issues with attaining standards or data quality are identified, the International Survey Director initiates communication with the National Centre as soon as possible to give advice on resolving problems.

The PISA standards serve as benchmarks of best practice. As such, the standards are designed to assist National Centres and the international contractors by explicitly indicating the expectations of data quality and study implementation endorsed by the PISA Governing Board, and by clarifying the timelines of the activities involved. The standards formulate levels of attainment, while timelines and feedback schedules of both the participating countries and the contractors are defined in the PISA operations manuals.

As specified in the contracts for the implementation of the eighth cycle of the OECD Programme for International Student Assessment, the international contractors take responsibility for developing and implementing procedures for assuring data quality. Therefore, the internaitonal contractors mediate, and monitor the countries' activities specified in this document, while the adherence to the standards by all international contractors is monitored by the participating countries via the OECD Secretariat. The international contractors must communicate timelines and tasks well in advance to National Centres and give reasonable deadlines for National Centres to respond to tasks.

Where the technical standards stipulate that variations from the standards require agreement between participating countries and the international contractors, National Project Managers are asked to initiate the process of negotiation and to undertake everything possible to facilitate an agreement. Where agreement between National Project Managers and the international contractors cannot be reached, the OECD will adjudicate and resolve the issues. The OECD will also adjudicate any issues resulting from non-compliance with the technical standards that cannot be resolved between participating countries and the contractors.

There are three types of standards in this document; each with a specific purpose:

- Data Standards refer to aspects of study implementation that directly concern the data qualityand its assurance.
- These standards have been reviewed by the Technical Advisory Group, and their comments and suggestions have been taken into careful consideration in finalising the standards.
- Management Standards are in place to ensure that all PISA operational objectives are met in a timely and coordinated manner.
- National Involvement Standards reflect the expectations set out in the PISA 2022 Terms of Reference that the content of the PISA tests is established in consultation with national

representatives with international content expertise. In particular, these standards ensure that the internationally developed instruments are widely examined for cross-national, cross-cultural and cross-linguistic validity and that the interests and involvement of national stakeholders are considered throughout the study.

Format of the document

The standards are grouped into sections that relate to specific tasks in the PISA data collection process. For every section, a rationale is given explaining why standard setting is necessary. The standards in each section consist of three distinct elements. First, there are the Standards themselves that are numbered and are shown in shaded boxes. Second, there are Notes that provide additional information on the standards directly. The notes are listed after the standards in each section. Third, there are the quality control measures that will be used to assess if a standard has been met or not. These are listed at the end of each section. In addition, the standards contain words that have a defined meaning in the context of the standards. These words are shown in italics throughout the document and are clarified in the Definitions section at the end of the document, where the terms are listed alphabetically.

Scope

The standards in this document apply to data from adjudicated entities that include both PISA participants and additional adjudicated entities. The PISA Governing Board will approve the list of adjudicated entities to be included in a PISA cycle.

Data standards

Target population and sampling

Rationale: Meeting the standards specified in this section will ensure that in all countries, the students tested come from the same target population in every country, and are in a nearly equivalent age range. Therefore, the results obtained will not be confounded by potential age effects. Furthermore, to be able to draw conclusions that are valid for the entire population of fifteen-year-old students, a representative sample shall be selected for participation in the test. The size of this representative sample should not be too small, in order to achieve a certain precision of measurement in all countries. For this reason, minimum numbers of participating students and schools are specified. In PISA 2022, a teacher questionnaire will be offered as an international option. The response-rate standard for teachers specified in this section applies only to countries that participate in this international option, and will ensure that the analysis and reporting goals for this option can be met.

The procedures for drawing the samples used in the study are crucial to data quality. The goal of the project is to collect data that are representative for the population at large, in such a way that the results are comparable, reliable and valid. To reach these goals the sampling procedures must follow established scientific principles for drawing samples from finite populations.

Standard 1.1 The PISA Desired Target Population is agreed upon through negotiation between the National Project Manager and the international contractors within the constraints imposed by the definition of the PISA Target Population. The Target Population for PISA starts with students attending all educational institutions located within the country, and in grade 7 or higher. The "standard" PISA target population is further refined to its age basis: students between 15 years and 3 (completed) months and 16 years and 2 (completed) months at the beginning of the testing period.

Standard 1.2 Unless *otherwise agreed upon* only *PISA-eligible students* participate in the test.

Standard 1.3 Unless otherwise agreed upon, the testing period:

- is no longer than eight consecutive weeks in duration for computer-based testing participants,
- is no longer than six consecutive weeks in duration for paper-based testing participants,
- does not coincide with the first six weeks of the academic year, and

begins exactly three years from the beginning of the testing period in the previous PISA cycle

Standard 1.4 Schools are sampled using agreed upon, established and professionally recognised principles of scientific sampling.

Standard 1.5 Student lists should not be collected more than 8 weeks prior to the start of data collection, unless otherwise agreed upon.

Standard 1.6 Students are sampled using *agreed upon*, established and professionally recognised principles of scientific sampling and in a way that represents the full population of *PISA-Eligible students*.

Standard 1.7 The PISA Defined Target Population covers 95% or more of the PISA Desired Target Population. That is, school-level exclusions and within-school exclusions combined do not exceed 5%.

Standard 1.8 The student sample size for the **computer-based mode** is a minimum of 6 300 assessed students, and 2 100 for additional adjudicated entities, or the entire PISA Defined Target Population where the PISA Defined Target Population is below 6 300 and 2 100 respectively. The student sample size of assessed students for the **paper-based mode** is a minimum of 5 250.

Standard 1.9 The school sample size needs to result in a minimum of 150 participating schools, and 50 participating schools for additional adjudicated entities, or all schools that have students in the PISA Defined Target Population where the number of schools with students in the PISA Defined Target Population is below 150 and 50 respectively. Countries not having at least 150 schools, but which have more students than the required minimum student sample size, can be permitted, if agreed upon, to take a smaller sample of schools while still ensuring enough sampled PISA students overall.

Standard 1.10 The minimum acceptable sample size in each school is 25 students per school (all students in the case of school with fewer than 25 eligible students enrolled).

Standard 1.11 The final weighted school response rate is at least 85% of sampled eligible and non-excluded schools. If a response rate is below 85% then an acceptable response rate can still be achieved through *agreed upon* use of replacement schools.

Standard 1.12 The final weighted student response rate is at least 80% of all sampled students across responding schools.

Standard 1.13 The final weighted teacher response rate is at least 75% of all sampled teachers across responding schools.

Standard 1.14 The final weighted sampling unit response rate for any optional cognitive assessment is at least 80% of all sampled students across responding schools.

Standard 1.15 Analyses based on questionnaire data that do not link to a weighted 75% of the target population shall be flagged or replaced by a missing code in OECD reports.

Standard 1.16 Unless *otherwise agreed upon*, the international contractors will draw the school sample for the Main Survey.

Standard 1.17 Unless *otherwise agreed upon*, the National Centre will use the sampling contractor's software to draw the student sample, using the list of eligible students provided for each school.

Note 1.1 Standards 1.1 through 1.17 apply to the Main Survey but not the Field Trial.
Note 1.2 Data from schools where the (unweighted) student response rate is greater than 33% will be included in the PISA dataset and the school counted as a respondent. Otherwise, the school will be a non-respondent, and no student, school or teacher data will be retained.
Note 1.3 A PISA-eligible student recorded in the database as not doing the minimum required number of questions of the main cognitive part of the PISA assessment will be counted as a nonparticipant.
Note 1.4 Acceptable response rates obtained through the use of replacement schools are described in detail in the School Sampling Preparation Manual.
Note 1.5 Guidelines for acceptable exclusions that do not affect standard adherence, are as follows:
School level exclusions that are exclusions due to geographical inaccessibility, extremely small school size, administration of PISA would be not feasible within the school, and other agreed upon reasons and whose students total to less than 0.5 % of the PISA Desired Target Population,
School level exclusions that are due to a school containing only students that would be within-school exclusions and that total to less than 2.0 % of the PISA Desired Target Population, and
Within-school exclusions that total to less than 2.5 % of the PISA Desired Target Population – these exclusions could include, for example, students not able to do the test because of a functional disability.
Note 1.6 Principles of scientific sampling include, but are not limited to:
The identification of appropriate stratification variables to reduce sampling variance and facilitate the computation of non-response adjustments.
The incorporation of an agreed target cluster size of PISA-eligible students from each sampled school: The recommended target cluster size is 42 and 25 is the minimum. In determining the target cluster size for a given country, or stratum within a country, it is necessary to ensure that the minimum sample size requirements for both schools and students will be met.
Note 1.7 Any exceptional costs associated with verifying a school sample taken by the National Centre, or a student sample selected other than by using the sampling contractor's software will be borne by the National Centre.
Note 1.8 Agreement with the international contractor on alternative methods of drawing samples will be subject to the principle that the sampling methods used are scientifically valid and consistent with PISA's documented sampling methods. Where a PISA participating country chooses to draw the school sample, the National Centre provides the international contractor with the data and documentation required for it to verify the correctness of the sampling procedures applied. Where a PISA participating country chooses not to use the sampling contractor's software to draw the student sample, the National Centre provides the international contractor with the data and documentation required for it to verify the correctness of the sampling procedures applied.
Note 1.9 Teachers recorded in the database as completing at least one valid response will be counted as respondents.

Quality assurance

- Sampling procedures as specified in the PISA operations manuals
- School sample drawn by the international contractors (or if drawn by the National Centre, then verified by the international contractors)
- Student sample drawn through the sampling contractor's software (or if drawn by other means, then verified by the international contractors)
- Sampling forms submitted to the international contractors
- Main Survey Review Form

Language of testing

Rationale: Using the language of instruction will ensure analogous testing conditions for all students within a country, thereby strengthening the consistency of the data. It is assumed that the students tested have reached a level of understanding in the language of instruction that is sufficient to be able to work on the PISA test without encountering linguistic problems (see also the criteria for excluding students from the potential assessment due to insufficient experience in the language of assessment: within-school exclusions). Thus, the level of literacy in reading, mathematics and science can be assessed without interference due to a critical variation in language proficiency.

Standard 2.1 The PISA test is administered to a student in a language of instruction provided by the sampled school to that sampled student in the major domain (Mathematics) of the test.

If the language of instruction in the major domain is not well defined across the set of sampled students, then, if *agreed upon*, a choice of language can be provided, with the decision being made at the student, school, or National Centre level. Agreement with the international contractor will be subject to the principle that the language options provided should be languages that are common in the community and are common languages of instruction in schools in that *adjudicated entity*.

If the language of instruction differs across domains, then, if *agreed upon,* students may be tested using assessment instruments in more than one language on the condition that the test language of each domain matches the language of instruction for that domain. Information obtained from the Field Trial will be used to gauge the suitability of using assessment instruments with more than one language in the Main Survey.

In all cases the choice of test language(s) in the assessment instruments is made prior to the administration of the test.

Field Trial participation

Rationale: The Field Trial gives countries the opportunity to try out the logistics of their test procedures and allows the contractors to make detailed analyses of the items so that only suitable ones are included in the Main Survey.

Standard 3.1 PISA participants participating in the PISA 2021 Main Survey will have successfully implemented the Field Trial. Unless otherwise agreed upon:

A Field Trial should occur in an assessment language if that language group represents more than 5% of the target population.

For the largest language group among the target population, the Field Trial student sample should be a minimum of 200 students per item.

For all other assessment languages that apply to at least 5% of the target population, the Field Trial student sample should be a minimum of 100 students per item.

For additional adjudicated entities, where the assessment language applies to at least 5% of the target population in the entity, the Field Trial student sample should be a minimum of 100 students per item.

Note 3.1 The PISA Technical Standards for the Main Survey generally apply to the Field Trial, except for the Target Population standard, the Sampling standard, and the Quality Monitoring standard. For the Field Trial a sampling plan needs to be agreed upon.

Note 3.2 The sample size for the Field Trial will be a function of the test design and will be set to achieve the standard of 200 student responses per item.

Note 3.3 Consideration will be given to reducing the required number of students per item in the Field Trial where there are fewer than 200 students in total expected to be assessed in that language in the Main Survey.

Adaptation of tests, questionnaires and school-level materials

Rationale: In order to be able to assess how the performance in a country has evolved from one PISA cycle to the other, the same instruments have to be used in all assessments. If instruments differ, then it is unclear whether changes in performance reflect changes in competencies or whether they just mirror the variation in the test items. The same holds true for the assessment instruments that are used within a PISA cycle: To validly compare performance across countries, all assessment instruments and other survey materials have to be as equivalent as possible. In fact, it is of utmost importance to provide equivalent information to the students in all countries that take part in the study. Therefore, not only the assessment instruments, but also the instructions given to the students and the procedures of data-collection have to be equivalent. To achieve this goal, other individuals who play a key role in the data-collection process, i.e. the test administrators, school coordinators, and school associates, should receive equivalent information and training in all participating countries.

Standard 4.1 The majority of test items used in previous cycles will be administered unchanged from their previous administration, unless amendments have been made to source versions, or outright errors have been identified in the national versions.

Standard 4.2 All assessment instruments are equivalent to the source versions. Agreed upon adaptations to the local context are made if needed.

Standard 4.3 National versions of questionnaire items used in previous cycles will be administered unchanged from their previous administration, unless amendments have been made to source versions, outright errors have been identified in the national versions, or a change in the national context calls for an adjustment.

Standard 4.4 The questionnaire instruments are equivalent to the source versions. Agreed upon adaptations to the local context are made if as needed.

Standard 4.5 School-level materials are equivalent to the source versions. Agreed upon adaptations to the local context are made as needed.

Note 4.1: The quality assurance requirements for this standard apply to instruments that are in an assessment language used as a language of instruction for more than 10% of the target population.

Quality assurance

- Agreed upon adaptation to school-level materials using methods specified by the international contractors
- Questionnaire Adaptation Spreadsheet (QAS)
- Test Adaptation Spreadsheets (TAS, for paper and computer instruments) or other agreed upon monitoring tool in which adaptations to assessment units, orientation and help files and coding guides are documented. For languages that are the languages of instruction for 10% or more of the target population, adaptations will be checked for compliance with the PISA Translation and Adaptation Guidelines by international verifiers, and the verifiers' recommendations will be vetted by the translation referee.
- For languages that are the languages of instruction for 10% or more of the target population: Verifier Reports (verification statistics generated by the monitoring tool, in combination with a short qualitative report)
- Field Trial and Main Survey Review Forms.
- Item and scale statistics generated by the international contractors (assessment materials and questionnaires).

Translation of assessment instruments, questionnaires and school-level materials

Rationale: To be able to compare the performance of students across countries, and of students with different instruction languages within a country, the linguistic equivalence of all materials is central. While Standards 4.1 to 4.4 serve to ensure that equivalent information is given to the students in all countries involved, in general, the following Standards 5.1 and 5.2 emphasise the importance of language. Again the goal is to ensure that competencies will be assessed, and not variations of information caused by differences in the translation of materials.

Standard 5.1 The following documents are translated into the assessment language in order to be linguistically equivalent to the international source versions.

- All administered assessment instruments
- All administered questionnaires
- The Test Administrator script from the Test Administrator (or School Associate) Manual
- The Coding Guides (unless otherwise agreed upon)

Standard 5.2 Unless otherwise agreed upon, school-level materials are translated/ adapted into the assessment language to make them functionally equivalent to the international source versions.

Note 5.1: The quality assurance requirements for this standard apply to instruments that are in a language that is administered to more than 10% of the target population.

Quality assurance

- Agreed upon Translation Plan, developed in accordance with the specifications in the PISA operations manuals, that requires double translation by independent translators followed by reconciliation for any newly translated questionnaires and cognitive instruments; and a thoroughly documented adaptation process for any materials adapted from one of the source versions, from a common reference version, or from verified materials borrowed from another country.
- Agreed Upon Questionnaire Adaptation Spreadsheet (QAS)
- Test Adaptation Spreadsheets (TAS) or other agreed upon monitoring tool in which adaptations to assessment units, orientation and help files and coding guides are documented. Adaptations will be checked for compliance with the PISA Translation and Adaptation Guidelines by international verifiers, and the verifiers' recommendations will be vetted by the translation referee.
- Verifier Reports (verification statistics generated by the monitoring tool, in combination with a short qualitative report)
- Submitted final materials as used in the study
- Field Trial and Main Survey Review Forms
- Item and scale statistics generated by the international contractors (assessment materials and questionnaires)

Testing of national software versions

Rationale: Countries must thoroughly test and validate the national software releases that are used to deliver the PISA computer-based instruments in schools, as well as the online questionnaires that are delivered via the Internet.

Standard 6.1 The international contractors must test all national software versions prior to their release to ensure that they were assembled correctly and have no technical problems.

Standard 6.2 Once released, countries must test the national software versions following testing plans to ensure the correct implementation of national adaptations and extensions, display of national languages, and proper functioning on computers typically found in schools in each country. Testing results must be submitted to the international contractors so that any errors can be promptly resolved.

Quality assurance

- Detailed testing plans
- Review of testing results

Technical support

Rationale: Countries participating in the computer-based delivery mode will be primarily responsible for resolving PISA-related operational issues in their countries, including hardware issues and provision of technical support to schools and test administrators.

Standard 7.1 Each country should have a designated PISA helpdesk with contact information provided to each of its *test administrators* and school coordinators.

Standard 7.2 In countries that administer the computer-based version of PISA, the helpdesk staff must:

- be familiar with the PISA computer system requirements applications and training materials,
- be familiar with all national software standards and procedures; and
- attend the *test administrator* training sessions to become familiar with the computer-based assessments and appreciate the challenges faced by schools and test *administrators*.

Quality assurance

- National Centre Quality Monitoring
- Field Trial and Main Survey Review Forms

Test administration

Rationale: Certain variations in the testing procedure are particularly likely to affect test performance. Among them are session timing, the administration of test materials and support material like blank papers and calculators, the instructions given prior to testing, the rules for excluding students from the assessment, etc. A list of these and other relevant test conditions is given in the school-level materials. To ensure that the data are collected consistently, and in a comparable fashion, for all participants, it is therefore very important to keep the chain of action in the data collection process as constant as possible.

Furthermore, the goal of the assessment is to arrive at results which cover a wide range of areas. Given the time constraints, any one student is presented only with a certain portion of the test items. Moreover, to preclude sources of random error unforeseen by the test administrators and the test designers, the students taking part in the survey have to be selected a-priori, in a statistically random fashion. Only then will the students participating in the study mirror the population of fifteen-year-old students in the country. The statistical analysis will take this sampling design into account, thereby arriving at results that are representative for the population at large. For these reasons, it is of utmost importance to assign the proper instruments (tests and questionnaires) to the participants specified beforehand. The student tracking form is central in monitoring whether this goal has been achieved.

The test administrator plays a central role in all of these issues. Special consideration is therefore given to the training of the test administrators, ensuring that as little variation in the data as possible is caused by random or systematic variation in the activities of test administrators.

An important part of the testing situation relates to the relationship between test administrators and test participants. Therefore, any personal interaction between test administrators and students, either in the past or in the testing situation, counteracts the goal of collecting data in a consistent fashion across countries and participants. Strict objectivity of the test administrator, on the other hand, is instrumental in collecting data that reflect the level of literacy obtained, and that are not influenced by factors un-related to literacy. The results based on these data will be representative for the population under consideration.

Standard 8.1 All test sessions follow international procedures as specified in the PISA school-level materials, particularly the procedures that relate to:

- test session timing,
- maintaining test conditions,
- responding to students' questions,
- student tracking, and
- assigning assessment materials.

Standard 8.2 The relationship between Test Administrators and participating students must not compromise the credibility of the test session. In particular, the Test Administrator should not be the reading, mathematics, or science instructor, a relative, or a personal acquaintance of any student in the assessment sessions he or she will administer for PISA.

Standard 8.3 National Centres must not offer rewards or incentives that are related to student achievement in the PISA test to students, teachers, or schools.

Note 8.1 Test Administrators should preferably not be school staff.

Note 8.2 This does not apply to incentives or rewards intended to improve participation, and that are unrelated to student achievement in the PISA test.

Quality assurance

- Session Report Forms
- PISA Quality Monitors feedback and Data Collection Forms (only for Main Survey)
- Field Trial and Main Survey Review Forms

Training support

Rationale: NPMs or their designees shall participate in a train-the-trainer session conducted by qualified contractor staff. This facilitates standardisation of training delivery to test administrators, allows trainers to become familiar with PISA materials and procedures, and informs trainers of their responsibilities for overseeing the PISA testing.

Standard 9.1 Qualified contractor staff will conduct trainer training sessions with NPMs or designees on PISA materials and procedures to prepare them to train PISA test administrators.

Standard 9.2 NPMs or designees shall use the comprehensive training materials and approach developed by the contractors and provided on the PISA Portal to train PISA test administrators.

Standard 9.3 All test administrator training sessions should be scripted to ensure consistency of presentations across training sessions and across countries. Failure to do so could cause errors in data collection and make results less comparable.

Standard 9.4 In-person and/or web based test administrator trainings should be conducted by the NPMs or designees, unless a suitable alternative is *agreed upon*.

Standard 9.5 PQMs need to successfully complete self-training materials, attend webinars to review and enhance the self-training, and attend the *test administrator* training, *unless otherwise agreed upon*.

Quality assurance

- Participation in trainer training sessions in standardised procedures by qualified contractor staff
- National Centre Quality Monitoring
- Field Trial and Main Survey Review Forms
- Standard training of PQMs
- Review of Test Administrator Training Observation Forms

Implementation of national options

Rationale: These standards serve to ensure that for students participating both in the international and the national survey, the national instruments will not affect the data used for the international comparisons. Data are therefore collected consistently across countries, and potential effects like test fatigue, or learning effects from national test items, are precluded.

Standard 10.1 Only *national options* that are *agreed upon* between the National Centre and the international contractors are implemented.

Standard 10.2 Any *national option* instruments that are not part of the core components of PISA are administered after all the test and questionnaire instruments of the core component of PISA have been administered to students that are part of the international PISA sample, unless otherwise agreed upon.

Security of the material and test preparation

Rationale: The goal of the PISA assessment is to measure the literacy levels in the content domains. Prior familiarisation with the test materials, or training to the test, will heavily degrade the consistency and validity of the data. In the extreme case, the results would only reflect how well participants are able to memorise the test items. In order to be able to assess the competencies obtained during schooling rather than short-term learning success, and to make valid international comparisons, confidentiality is extremely important. As high levels of student and school participation in PISA are very important, it is appropriate for national centres to prepare communication materials for participants with the intent to raise awareness, to set out what is involved in participating in PISA and to encourage participation in the survey. These materials may include general information about the survey, what students and schools might expect on the test day, as well as an OECD set of released test materials prepared for this purpose. The use of sample test items in informational materials could also serve to prepare students for the format of the PISA test in order to reduce potential test anxiety and help the students focus on the subject-matter content when taking the test.

Standard 11.1 PISA materials designated as secure are kept confidential at all times. Secure materials include all test materials, data, and draft materials. In particular:

- no-one other than approved project staff and participating students during the test session is able to access and view the test materials,
- no-one other than approved project staff will have access to secure PISA data and embargoed material, and
- formal confidentiality arrangements will be in place for all approved project staff.

Standard 11.2 Participating schools, students and/or teachers should only receive general information about the test prior to the test session, rather than formal content-specific training. In particular, it is inappropriate to offer formal training sessions to participating students, in order to cover skills or knowledge from PISA test items, with the intention to raise PISA scores.

Note 11.1: It is unnecessary to train students for interacting with the student interface, with different item types or response formats prior to the testing session. All PISA test materials and procedures are accompanied by detailed instructions as well as by orientation modules at the beginning of each test session to ensure that participants are familiarised with the interface and with all the question formats that they will encounter.

Note 11.2: “Formal training sessions” refers to training that relies on standardised instructional material and involves feedback provided by an instructor, machine, or other training participants. Formal training sessions may include (but are not limited to) lectures, practice tests, drills or online instruction modules.

Note 11.3. The general information about the survey shared with participants may include information about the length of the test, the general scoring principles applied to missing and incorrect answers, data protection and confidentiality of results. It may include an OECD set of released test materials prepared for this purpose but should not assemble sample items in PISA-like test forms with the intent to teach or prepare students for participation in PISA.

Quality assurance

- Security arrangements as specified in the PISA operations manuals or agreed upon variation
- National Centre Quality Monitoring
- PISA Quality Monitor feedback and Data Collection Forms (only for Main Survey)
- Field Trial and Main Survey Review Forms

Quality monitoring

Rationale: To obtain valid results from the assessment, the data collected have to be of high quality, i.e. they have to be collected in a consistent, reliable and valid fashion. This goal is implemented first and foremost by the test administrators, who are seconded by the quality monitors. The quality monitors provide country-wide supervision of all data-collection activities for the Main Survey.

Standard 12.1 PISA Main Survey test administration is monitored using site visits by trained independent quality monitors.

Standard 12.2 Fifteen site visits to observe test administration sessions are conducted in each PISA participating country/economy, and five site visits in each adjudicated region.

Standard 12.3 Test administration sessions that are the subject of a site visit are selected by the international contractors to be representative of a variety of schools in a country/economy.

Note 12.1 A failure to meet the Quality Monitoring standards in the Main Survey could lead to a significant lack of reliable and valid quality assurance information.
Note 12.2 The Quality Monitoring standards apply to the Main Survey but not to the Field Trial.
Note 12.3 The National Centre provides the international contractors the assistance required to implement the site visits effectively. This includes nominating sufficient qualified individuals to ensure that the required number of schools is observed. It also includes timely communication of school contact information and test dates.

Quality assurance

- The process of selecting the PISA Quality Monitor nominees .
- PISA Quality Monitor feedback and Data Collection Forms (only for Main Survey)

Assembling and printing paper-based materials

Rationale: Variations in assembly and print quality may affect data quality. When the quality of paper and print is very poor, the performance of students is influenced not only by their levels of literacy, but also by the degree to which test materials are legible. To rule out this potential source of error, and to increase the consistency and precision of the data collection, paper and print quality samples are solicited from National Centres participating in paper-based components in their first cycle of participation.

Standard 13.1 All paper-based student assessment material will be centrally assembled by the international contractors and must be printed using the final print-ready file and *agreed upon* paper and print quality. New countries/entities must submit a printed copy of all Field Trial instruments (booklets and questionnaires) for approval of the printing quality for the Main Survey. The same printing standard must be used for both the Field Trial and the Main Survey.

Standard 13.2 The cover page of all national PISA test paper-based materials used for students and schools must contain all titles and approved logos in a standard format provided in the international version.

Standard 13.3 The layout and pagination of all test paper-based material is the same as in the *source versions*, *unless otherwise agreed upon.*

Standard 13.4 The layout and formatting of the paper-based questionnaire material is equivalent to the source versions, with the exception of changes made necessary by national adaptations.

Note 13.1 The cover page of all PISA PBA instruments used in schools should contain all information necessary to identify the material as being part of the data-collection process for PISA, and for checking whether the data collection follows the assessment design, i.e. whether the mapping of the student on the one hand, and test booklets and questionnaires, on the other, have been correctly established The features of the cover page referred to in Standard 13.2 are specified in the PISA operations manuals.

Quality assurance

- Agreement that quality will be similar to Field Trial versions
- For new countries/economies, materials submitted to the international contractors, as described in Standard 13.1 above.
- Field Trial and Main Survey Review Forms

Response coding

Rationale: To ensure the comparability of the data, the responses from all test participants in all participating countries have to be coded following approved coding designs that are presented to both the Field Trial and the Main Survey. Therefore, all coding procedures have to be standardised, and coders have to complete training sessions to master this task.

Standard 14.1 The coding scheme described in the coding guides is implemented according to instructions from the international contractors' item developers.

Standard 14.2 Representatives from each National Centre attend the international PISA coder training session for both the Field Trial and the Main Survey.

Standard 14.3 Both the single and multiple coding procedures must be implemented as specified in the *PISA operations manuals* (see Note 14.1). These procedures are implemented in the coding software that countries will be required to use.

Standard 14.4 Coders are recruited and trained following *agreed procedures*.

Note 14.1 Preferred procedures for recruiting and training coders are outlined in the PISA operations manuals.
Note 14.2 The number of Coder Training session participants will depend on factors such as the expertise of National Centre staff, and resource availability.

Quality assurance

- Indices of inter-coder agreement
- Field Trial and Main Survey Review Forms

Data submission

Rationale: The timely progression of the project, within the tight timelines given depends on the quick and efficient submission of all collected data. Therefore, one single data submission format is proposed, and countries are asked to submit only one database to the international contractors. Furthermore, to avoid potential errors when consolidating the national databases, any changes in format that were implemented subsequent to the general agreement have to be announced.

Standard 15.1 Each *PISA participant* submits its data in a single complete database, unless otherwise *agreed upon.*

Standard 15.2 All *data* collected for PISA will be imported into a national database using the Data Management Expert (DME) data integration software provided by the international contractors following specifications in the corresponding operational manuals and international/national record layouts (codebooks). Data are submitted in the DME format.

Standard 15.3 Data for all *instruments* are submitted. This includes the assessment data, questionnaires data, and tracking data as described in the *PISA operations manuals*.

Standard 15.4 Unless *agreed upon*, all data are submitted without recoding any of the original response variables.

Standard 15.5 Each PISA participating country's database is submitted with full documentation as specified in the *PISA operations manuals.*

Management standards

Communication with the international contractors

Rationale: Given the tight schedule of the project, delays in communication between the National Centres and the international contractors should be minimised. Therefore, National Centres need continuous access to the various resources provided by the contractors.

Standard 16.1 The international contractors ensure that qualified staff are available to respond in English to requests by the National Centres during all stages of the project. The qualified staff:

- Are authorised to respond to National Centre queries,
- Acknowledge receipt of National Centre queries within one working day,
- Respond to coder queries from National Centres within one working day,
- Respond to other queries from National Centres within five working days, or, if processing the query takes longer, give an indication of the amount of time required to respond to the query.

Standard 16.2 The National Centre ensures that qualified staff are available to respond to requests in English by the international contractors during all stages of the project. The qualified staff:

- Are authorised to respond to queries,
- Are able to communicate in English,
- Acknowledge receipt of queries within one working day,
- Respond to queries from the international contractors within five working days, or, if processing the query takes longer, give an indication of the amount of time required to respond to the query.

Note 16.1 Response timelines and feedback schedules for the National Centres and the international contractor are further specified in the Tasks section of the PISA Portal.

Notification of international and national options

Rationale: Given the tight timelines, the deadlines given in the following two standards will enable the international contractors to progress with work on time.

Standard 17.1 National options are agreed *upon* with the international contractors before 1 December in the year preceding the Field Trial and confirmed before 1 November in the year preceding the Main Survey.

Standard 17.2 The National Centre notifies the OECD Secretariat of its intention to participate in specific international options three months prior to the start of the translation period. International options can only be dropped between the Field Trial and the Main Survey, not added.

Schedule for submission of materials

Rationale: To meet the requirements of the work programme, and to progress according to the timelines of the project, the international contractor will need to receive a number of materials on time.

Standard 18.1 An *agreed upon Translation Plan* will be negotiated between each National Centre and the international contractors.

Standard 18.2 The *following* items are submitted to the international contractors in accordance with *agreed timelines*:

- the Translation Plan
- a print sample of booklets prior to final printing, for new countries/entities using the paper-based instruments (where this is required, see Standard 13.1),
- results from the national checking of adapted computer-based assessment materials and questionnaires,
- adaptations to school-level materials,
- sampling forms (see Standard 1),
- demographic tables,
- completed Field Trial and Main Survey Review Forms,
- documents related to PISA Quality Monitors: nomination information, Test Administrator training schedules, translated school-level materials, school contact information, test dates, and
- other documents as specified in the PISA operations manuals.

Standard 18.3 *Questionnaire* materials are submitted for linguistic verification only after all adaptations have been *agreed upon*.

Standard 18.4 All adaptations to those elements of the school-level materials that are required to be functionally equivalent to the source as specified in Standard 5.2, need to be *agreed upon.*

Quality assurance

- Agreed upon Translation Plan
- International contractors' records from communications, forms, or documents
- Assessment materials submitted for linguistic verification with corresponding adaptation spreadsheets filled in by the National Centre

Management of data

Rationale: Consolidating and merging the national databases is a time-consuming and difficult task. To ensure the timely and efficient progress of the project, the international contractors need continuous access to national resources helping to rule out uncertainties and to resolve discrepancies. This standard aims to prevent substantial delays to the whole project which could result from a delay in processing the data of a small number of participating countries.

Standard 19.1 The timeline for submission of national databases to the international contractors is within eight weeks of the last day of testing for the Field Trial and within eight weeks of the last day of testing for the Main Survey, unless otherwise *agreed upon*.

Standard 19.2 National Centres execute data checking procedures as specified in the *PISA operations manuals* before submitting the database.

Standard 19.3 National Centres make a data manager available upon submission of the database. The data manager:

- is authorised to respond to international contractor data queries,
- is available for a three-month period immediately after the database is submitted unless otherwise *agreed upon*,
- is able to communicate in English,
- is able to respond to international contractor queries within three working days, and
- is able to resolve data discrepancies.

Standard 19.4 A complete set of PISA paper-based instruments as administered and including any *national options*, is forwarded to the international contractors on or before the first day of testing. The submission must include the: electronic PDF and/or Word versions of all instruments

Standard 19.5 To enable the *PISA participant* to submit a single dataset, all instruments for all *additional adjudicated entities* will contain the same variables as the primary *adjudicated entity* of the *PISA participant*.

Note 19.1: Each participating country/economy will receive its own national micro-level PISA database (the "national database"), in electronic form and delivered as agreed upon a pre-specified timeline that varies based on their data submission. The national database will contain the complete set of responses from the students, school principals and surveyed participants (parents, teachers) in that country/economy.
Each participating country/economy has access to and can publish its own data after a date that is established by the PISA Governing Board for the publication of the initial OECD publication of the survey results (the "initial international OECD publication").
The OECD Secretariat will not release national data to other countries/economies until participating countries/economies have been given an opportunity to review and comment on their own national data and until the release of such data has been approved by the national authorities.
A deadline and procedures for withdrawing countries/economies' national data from the international micro-level PISA database (the "international database") will be decided upon by the PISA Governing Board. Countries/economies can withdraw data only prior to obtaining access to data from other countries/economies. Withdrawn data will not be made available to other countries/economies.
The PISA Governing Board will discuss with participating countries/economies whose data manifests technical anomalies as to whether the data concerned can be included in the international database. The decision of the PISA Governing Board will be final. Participating countries/economies may, however, continue to use data that are excluded from the international database at the national level.
The international contractors will then compile the international database, which will comprise the complete set of national PISA databases, except those data elements that have been withdrawn by participating countries/economies or by the PISA Governing Board at the previous stage. The international database will remain confidential until the date on which the initial international OECD publication is released.
National data from all participating countries/economies represented in the international database will be made available to all participating countries/economies from the date on which the initial international OECD publication is released.
After release of the initial international OECD publication, the international database will be made publicly available on a cost-free basis, through the OECD Secretariat. The database may not be offered for sale.
The international database will form the basis for OECD indicator reports and publications.
The international contractors will have no ownership of instruments or data nor any rights of publication and will be subject to the confidentiality terms set in this agreement.
The OECD establishes rules to ensure adherence to the above procedure and to the continued confidentiality of the PISA data and materials until the agreed release dates. These include confidentiality agreements with all individuals that have access to the PISA material prior to its release.
As guardian of the process and producer of the international database, the OECD will hold copyright in the database and in all original material used to develop, or be included in, the PISA Field Trial and PISA Main Survey (among them the assessment materials, school-level materials, and coding guides) in any language and format.

Quality assurance

- International contractors’ records of communications, forms, or documents

Archiving of materials

Rationale: The international contractors will maintain an electronic archive. This will provide an overview of all materials used and ensure continuity of materials available in participating countries across PISA survey cycles, therefore building upon the knowledge gained nationally in the course of the PISA cycles. This will also ensure that the international contractors have the relevant materials available during data cleaning, when they are first required.

Standard 20.1 The international contractors will maintain a permanent electronic archive of all assessment materials, school-level materials and coding guides, including all national versions. For documents that are finalised by countries, they are required to upload the latest version to the PISA Portal.

Standard 20.2 The National Project Manager must submit one copy of each of the following adapted and translated Main Survey materials to the international contractors:

- electronic versions (Word and/or PDF) of all administered Test Instruments, including international and *national options*
- electronic versions (Word and/or PDF) of all administered Questionnaires, including international and *national options* (paper-based countries only);
- electronic versions of the school-level materials; and
- electronic versions of the Coding Guides.

Standard 20.3 Unless otherwise requested, National Centres will retain (1) all Field Trial materials until the beginning of the Main Survey, and (2) all Main Survey materials until the end of the calendar year, two years after the year when the Main Survey is conducted, (i.e. when the last international reports containing the results of the Main Survey will have been published). Materials to be archived include:

- all respondents’ paper-based test booklets and questionnaires (PBA countries or whenever paper-based materials are used in CBA countries)
- all respondents’ SDS result files and all associated data obtained from USB drives or other delivery mode (CBA countries)
- all sampling forms,
- all respondent lists,
- all tracking instruments, and
- all data submitted to the international contractors.

After completion of a survey, the National Centre will transfer final versions of all national materials to the international contractors who will compile the national archives from all participants and transfer them to OECD after completion of the Main Survey.

Note 20.1. Archiving applies to all materials from the Field Trial or Main Survey, including student, school, parent and teacher materials, as applicable.

Note 20.2. Should national legislation or other circumstances require that the Field Trial or Main Survey materials be deleted/erased before the timeline in Standard 20.3, countries must nevertheless retain these records, at a minimum, until the publication of the PISA dataset (and publication of the related international reports).

Note 20.3. It is recommended to retain the original USB drives (if used) and all paper-based booklets and questionnaires for all respondents until certified data has been released to the National Centres. Original USB drives are not required for long-term archiving purposes as long as there are copies of SDS result files for all respondents.
Note 20.4. Sampling forms for each sampling task for the Field Trial and Main Survey data collections must be retained for the periods outlined in Standard 20.3.
Note 20.5. "Respondent lists" refers to the student list (and teacher list if applicable) used for within-school sampling, and must be retained until, at a minimum, the period set out in Note 20.2.
Note 20.6. "Tracking instruments" refers to the Student Tracking Form (and Teacher Tracking Form if applicable) completed in each school, and must be retained until, at a minimum, the period set out in Note 20.2.

Data protection and the processing of personal data

Rationale: The OECD is committed to protecting the personal data it processes, in accordance with its Personal Data Protection Rules. The OECD, countries and contractors must protect the personal data of participants collected during PISA, ensuring that all data is stored and processed in a secure and standardised manner. This standard aims to ensure that National Centres process personal data securely, that participants are provided with clear information on data protection in PISA and the data rights of participants to access, rectify or erase their data are facilitated by countries and contractors.

Standard 21.1 Each National Centre must make data protection information available to all participants, that at least includes:

- Contact details of the National Centre
- Contact details of the OECD's Data Protection Officer and Data Protection Commissioner
- The purpose of the processing of the data
- Recipients of the data, including any international organisations or third party (this includes the PISA contractors and any national contractors)
- The storage and retention period of data
- The existence of the rights of data subjects, including the timeline for facilitating these requests.

Standard 21.2 Each National Centre will process the additional information related to a participant/data subject (e.g. link files with records of student names) securely and separately from the datasets during data processing and archiving (i.e. the data collected during the assessment will be categorised and treated as pseudonymised).

Standard 21.3 National Centres must inform schools or other holders of PISA forms that include student and/or teacher names, to delete, confidentially shred or return these materials to the National Centre. Materials to be deleted/shredded/returned include:

- All tracking instruments
- All respondent lists.

Standard 21.4 Each National Centre must facilitate requests from participants to exercise their data rights.

- Data access requests will be possible using the raw data from the assessment. No scaled data will be provided in breach of the PISA data embargo.
- Data erasure requests will be possible for a limited period before submission to the Contractors. This is to be decided by each National Centre, with two options, up to the submission of ST12 or to upload of student data files to the OECS.

- Each National Centre will retain and update a log of completed data requests for data erasure, to facilitate quality control processes. This information must be submitted to the PISA contractors in a timely manner to comply with the requests and for the purpose of data management and sampling processes.

Note 21.1. It Is best practice to make data protection information available to participants at the time of the data collection.
Note 21.2. National Centres may communicate data protection information to participants in the most effective way for their national context. The information may be provided in several ways, e.g. a video, an information sheet, a data protection notice, on a National Centre website or a link to the OECD's PISA 2022 data protection notice.
Note 21.3. "Tracking instruments" refers to the Student Tracking Form (and Teacher Tracking Form if applicable) completed in each school. "Respondent lists" refers to a list of students (or list of teachers if applicable) used during the implementation of PISA.
Note 21.4. The records of student/teacher names in link files or on PISA forms are permitted but must be stored separately and securely to the data collected during the assessment.
Note 21.5. The data collected from students as part of PISA is categorised as pseudonymised, as identifying characteristics in the data have been replaced with a number or value that does not allow the data subject to be directly identified. This also pertains to data from parents or teachers, if applicable. Pseudonymisation means that the data collected remains personal data, but can no longer be assigned to a natural person without additional information (e.g. record of a student name and the PISA student ID number).
Note 21.6. After the archiving period for the Field Trial and Main Survey materials, National Centres may choose and are encouraged to anonymise the data by breaking the link between the name of the student and the data from the cognitive and questionnaire sessions. Anonymisation of the data requires deleting and/or confidentially shredding all files and records that connect the PISA student ID number to identifying information (name, date of birth, national student ID, etc.). Once anonymisation is complete and all records of a participant's name are removed, it will be no longer possible to facilitate access and erasure requests.
Note 21.7. The timeline of disposal/return of PISA forms retained by schools is to be decided by the National Centre and communicated to schools. This should be before the end of the archiving period set out in Standard 20.3.

Quality assurance:

- Adherence to this standard is a National Centre responsibility.
- Retain a copy of national data protection information for PISA 2022 made available to respondents.
- Agreement that additional information related to data subject (e.g. linking information) will be stored separately and securely from datasets.
- Agreement to set and follow-up on the timeline and procedures for the deletion and/or shredding of data in PISA forms held by other parties involved in the implementation of PISA.
- Retain a record of completed data access or erasure requests and submit requests to international contractors in a timely manner.

National involvement standards

National feedback

Rationale: National feedback in areas such as test development is important in maintaining the dynamic and collaborative nature of PISA. National feedback ensures that instruments achieve cross-national, cross-cultural and cross-linguistic validity. It also promotes the inclusion of the interests and involvement of national stakeholders.

Standard 22.1 National Centres develop appropriate mechanisms in order to promote participation, effective implementation, and dissemination of results amongst all relevant national stakeholders.

Standard 22.2 National Centres provide feedback to the international contractors on the development of instruments, domain frameworks, the adaptation of instruments, and other domain-related matters that represent the perspectives of the relevant national stakeholders.

Note 22.1 As a guideline, feedback might be sought from the following relevant stakeholders: policy makers, curriculum developers, domain experts, test developers, linguistic experts and experienced teachers.

Quality assurance

- National Centre Quality Monitoring
- List of committees and groups of stakeholders
- Membership records of representative groups and/or committees
- Meeting records of representative groups and/or committees

Meeting attendance

Rationale: Attendance at National Project Managers and training meetings is required as these represent a key component of participating in PISA. Important information is shared and discussed and training in data management, sampling, computer systems, and coding is conducted at these international meetings. These also allow for individual consultation and communication with the international contractors, which is often very helpful.

Standard 23.1 Representatives from each National Centre are required to attend all PISA international meetings including National Project Manager meetings, coder training, and any separate within-school sampling training, and data management training, as necessary. Up to 6 international meetings are planned per cycle.

Standard 23.2 Representatives from each National Centre who attend international meetings must be able to work and communicate in English.

Note 23.1 The length of these meetings vary from 3 to 5 days.
Note 23.2 Based on the meeting type and hotel arrangements, the OECD Secretariat may, on the request of the international contractors, set a limit to the number of representatives per country that can attend NPM meetings. Countries/economies with separate participating entitites will have the possibility to send teams from all entitites.

Quality assurance

- Meeting attendance records

Definitions

Adjudicated Entity – a country, geographic region, or similarly defined population, for which the international contractors fully implements quality assurance and quality control mechanisms and endorses, or otherwise, the publication of separate PISA results. A PISA participant may manage more than one adjudicated entity.

Agreed procedures – procedures that are specified in the PISA operations manuals, or variations that are mutually agreed upon between the National Project Manager and the international contractors.

Agreed timelines – timelines that are specified in the PISA operations manuals, or variations that are mutually agreed upon between the National Project Manager and the international contractors.

Agreed upon – variations that are mutually agreed upon between the National Project Manager and the international contractors

Anonymisation - personal data is rendered anonymous, by irreversibly removing the link between each respondent's personal identifier (e.g. respondent name) and the data in the PISA dataset. This is achieved by deleting and erasing all additional information sources containing the link, so respondents in the PISA dataset cannot be personally identified. This applies to students, parents or teachers, if these options are administered, and anonymisation can be pursued after the archiving period in Standard 20.3.

Centrally produced reference documents – documents provided in English (and, for some documents, French and/or Spanish) by the international contractors according to contractual specifications.

Common reference version – a language version of assessment instruments that is used by countries sharing that language as a starting point to produce their respective national versions.

International options – optional additional international instruments or procedures sponsored by the OECD and fully supported by the international contractors.

National Centre quality monitoring – the procedures by which the international contractors monitor the quality of all aspects of the implementation of the survey by a National Centre.

National option – a national option occurs if:

d) National Centre administers any additional instrumentation, for example a test or questionnaire, to schools or students that are part of the PISA international sample. Note that in the case of adding items to the questionnaires, an addition of five or more items to either the school questionnaire or the student questionnaire is regarded as a national option.

OR

e) National Centre administers any PISA international instrumentation to any students or schools that are not part of an international PISA sample (age-based or grade-based) and therefore will not be included in the respective PISA international database.

OR

f) National Centre administers any PISA international option only in some, not all, jurisdictions. The country will in this case sign up for the international option with the OECD, as if it was administered in the entire jurisdiction, and the additional work involved with administering the international option to part of the jurisdiction only is considered a national option.

PISA-eligible students – students who are in the PISA target population. Also see PISA Target Population.

PISA National Project Manager (NPM) – The NPM is responsible for overseeing all national tasks related to the development and implementation of PISA throughout the entire cycle. The NPM is responsible for ensuring that tasks are carried out on schedule and in accordance with the specified international standards.

PISA Operations Manuals – all manuals provided by the international contractors. The preparation of the PISA operations manuals will be carried out by the international contractors and will describe procedures developed by the international contractors. The manuals will be prepared following consultation with participating countries/economies, the OECD Secretariat, the Technical Advisory Group and other stakeholders.

PISA Participant – an administration centre, commonly called a National Centre, that is managed by a person or persons, usually the National Project Manager, who is/are responsible for administering PISA in one or more adjudicated entities. The National Project Manager(s) must be authorised to communicate with the international contractor on all operational matters relating to the adjudicated entities for which the National Project Manager is responsible.

PISA Portal – the PISA 2022 project website can be accessed through the following address: http://pisa.ets.org/portal.

PISA Quality Monitor (PQM) – a person nominated by the National Project Manager and employed by the international contractors to monitor test administration quality in an adjudicated entity.

PISA Target Population – students aged between 15 years and 3 (completed) months and 16 years and 2 (completed) months at the beginning of the testing period, attending educational institutions located within the adjudicated entity, and in grade 7 or higher. The age range of the population may vary up to one month, either older or younger, but the age range must remain 12 months in length. That is, the population can be as young as between 15 years and 2 (completed) months and 16 years and 1 (completed) month at the beginning of the testing period; or as old as between 15 years and 4 (completed) months and 16 years and 3 (completed) months at the beginning of the testing period.

- **PISA Desired Target Population** – the PISA Target Population defined for a specific adjudicated entity. It provides the most exhaustive coverage of PISA-Eligible students in the participating country/economy as is feasible.
- **PISA Defined Target Population** – all PISA-Eligible students in the schools that are listed on the school sampling frame. That is, the PISA Desired Target Population minus school-level exclusions.

Pseudonymisation – personal data where the personal identifier (e.g. respondent's name) is replaced by an artificial identifier (pseudonym). In PISA, the respondent's name is not included in the data collected on the assessment day, but is replaced with an ID number. Therefore the data is categorised as pseudonymised, as long as the additional information linking the respondent name and the PISA respondent ID is stored in a different file or location.

School Associate (SA) – a person at a school who acts as a liaison between the school and the National Centre to prepare for the assessment and who administers the assessment to students on the day of the assessment.

School Co-ordinator (SC) – a person at a school who acts as a liaison between the school and the National Centre to prepare for the assessment in the school.

School-level exclusions – contractors' approved exclusion of schools from the sampling frame because:

- of geographical inaccessibility (but not part of a region that is omitted from the PISA Desired Target Population),
- administration of the PISA assessment within the school would not be feasible,
- all students in the school would be within-school exclusions, or
- of other reasons as agreed upon.

School-level materials – the key materials include:

- School Co-ordinator Manual and Test Administrator Manual (or School Associate Manual)
- Test administration scripts
- Key forms – Student Tracking Form, Session Attendance Form, and Session Report Form

Source versions assessment instruments provided in English (and, for some documents, in French) by the international cntractors according to contractual specifications.

Target cluster size – the number of students that are to be sampled from schools where not all students are to be included in the sample.

Test administrator – a person who is trained by the National Centre to administer the PISA test in schools. This person may be a Test Administrator or a School Associate (a School Co-ordinator who also has the role of a Test Administrator).

Testing period – the period of time during which data is collected in an adjudicated entity.

Translation plan – documentation of all the processes that are intended to be used for all activities related to translation and languages.

Within-school exclusions – potential exclusion of students from assessment because of one of the following:

- They are functionally disabled in such a way that they cannot take the PISA test. Functionally disabled students are those with a moderate to severe permanent physical disability.
- They have a cognitive, behavioural or emotional disability confirmed by qualified staff, meaning they cannot take the PISA test. These are students who are cognitively, behaviourally or emotionally unable to follow even the general instructions of the assessment.
- They have insufficient assessment language experience to take the PISA test. Students who have insufficient assessment language experience are those who meet all the following three criteria:
- they are not native speakers of the assessment language,
- they have limited proficiency in the assessment language, and
- they have received less than one year of instruction in the assessment language.
- There are no materials available in the language in which the student is taught.
- They cannot be assessed for some other reason as agreed upon.

PISA 2022 Technical Report

The Programme for International Student Assessment (PISA) is one of the largest and most comprehensive comparative education studies in the world. A wide variety of countries and economies worldwide collect information on student performance, school environment, and other relevant variables using standardized, uniform procedures that assure the results are comparable and meaningful. This Technical Report has been prepared by those who implemented PISA during its 2022 cycle to provide transparency to these procedures and to the statistical and mathematical methods that underpin the comparability and validity of PISA 2022 results.

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